



GERBER TECHNOLOGY
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GT 5250/7250 GERBERcutter[®]

System Service Manual

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Safety Precautions

Dangerous Procedure Warnings

Warnings precede potentially dangerous procedures throughout this manual. Instructions contained in warnings must be followed. You also should employ all other safety precautions that you deem necessary for safe operation of the equipment in your operating environment.

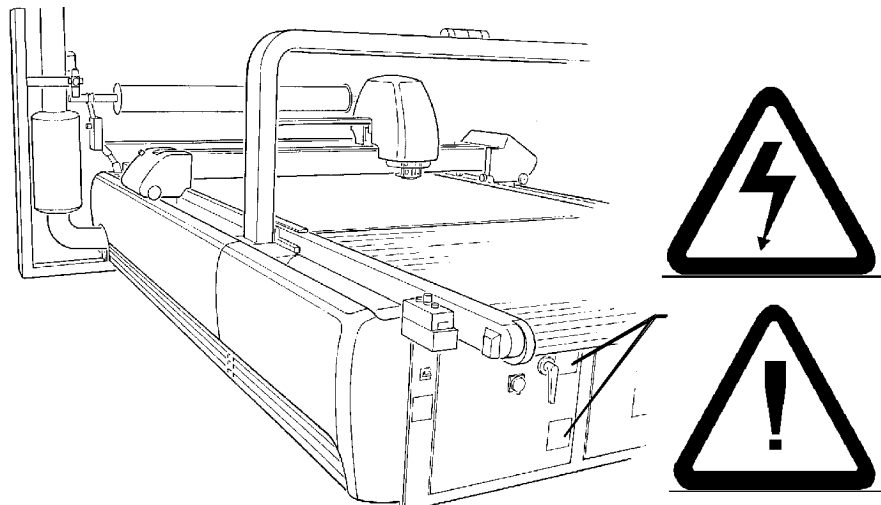
WARNING!  Failure to follow these safety precautions could result in injury or death.

Do Not Perform Maintenance Alone

Do not perform any maintenance on the equipment unless another person that can render first aid and resuscitation is present.

Live Circuits

High voltages exist inside the C-200 controller and vacuum and electric enclosures. **There are no operator-serviceable parts in these enclosure(s).** Two power cables enter the primary vacuum/electric enclosure: a three-phase cable for the vacuum generator system and a single-phase power cable for the C-200 controller. **Both** power cables must be disconnected before it is safe for trained, maintenance personnel to work within the enclosure. There must be a main disconnect that meets local electrical codes for each power supply.

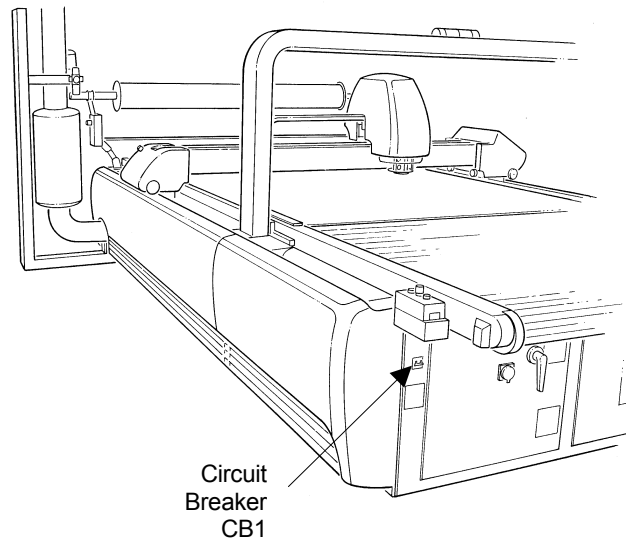


Turn Off Power Before Maintenance

To prevent accidental start-up while performing any maintenance, always turn off power as directed in the Maintenance Information chapter of this manual.

Circuit breakers located outside of the C-200 controller and vacuum and electric enclosures allow you to shut down power to the FEP (or PC), vacuum system, C-200 cutter controller, servomotors, and conveyor. See the Maintenance Information chapter for exact location(s).

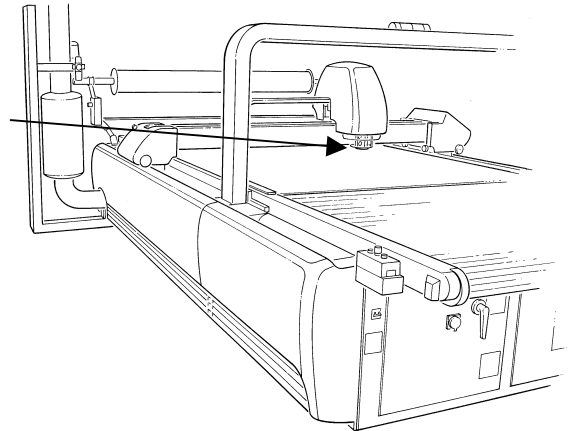
Use circuit breaker 1 to turn power on or off to the controller, servomotors, and conveyor as needed during normal operation of the GERBERcutter.



Laser Origin Light

The origin light in the cutter head uses a laser light that points toward the cutting surface. When viewed as intended (by looking at the dot projected on the material to be cut), the laser should pose no hazard to operators. However, the origin light is a Class IIIa laser product, with a peak power of 5 mW at a wavelength of 670 nm. Never look straight into the light source of this laser beam; doing so may cause temporary, or even permanent, damage to your eyes. Direct, bright reflections from the laser may also be harmful. Remove all highly reflective material (such as jewelry and mirrors) from the cutting surface before turning on the origin light. Turn off the origin light before performing any maintenance on the GERBERcutter.

WARNING!  Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.



Additional Precautions

Moving Machinery

Keep hands away from the Resealer and other moving machine parts to avoid personal injury. Make sure there is no one else near the table when you are operating the GERBERcutter.

Replacing Parts

Report any broken or missing parts to your Field Service representative. Because of the danger of introducing additional hazards, do not install substitute parts or perform any unauthorized modification of the equipment.

Knife Blade

The knife blade is very sharp. Be extremely careful when working near the knife blade.

Clothing

Loose-fitting clothing, dangling neck chains and ties, loose jewelry, and long hair can be a hazard when working around machinery.

Compressed Air

Always use caution when using compressed air. Never use compressed air to blow lint and dust off clothing or skin.

Lint

Avoid breathing in lint when changing the table and stack filters. Wear a mask suitable for this purpose.

Equipment Training

Never attempt to operate this machinery unless you have received formal training to do so.

Do Not Operate in an Explosive Atmosphere

Use extreme caution when handling flammable liquids such as cleaning solvents. Do not operate the equipment in the presence of flammable gases or fumes.

Use Caution When Handling a CRT Monitor

The monitor on the operator console is a flat panel screen. On other cutters the monitor may be a cathode ray tube (CRT). Breaking a CRT could cause it to implode and scatter glass fragments at a high velocity. To prevent CRT implosion, avoid rough handling or jarring of the equipment.

Information You Need to Know

What information is available?

A GT 5250 or GT 7250 GERBERcutter system includes hardware, electrical, and software components. Documentation for operating and maintaining the GERBERcutter is available on your computer or from CD in the form of online help for the C-200MT software and in PDF format. This means that information is available to you whenever you need it. The next sections explain:

- Where information can be found
- How to access and print this information
- How to get customer support

Where is it?

GT 5250/7250 GERBERcutter System Service Manual

To locate specific information, see the following sections inside this manual.

- Safety Precautions – **Read this section before you start cutting.**
- Maintenance Information – Safety warnings before beginning maintenance; tools and lubricants; illustrated drawings of the GERBERcutter.
- Maintenance Schedules – Maintenance Manager; schedules for maintenance sorted both by cutter component and by interval between required servicing.
- Preventive Maintenance – Procedures for routine maintenance.
 - Beam Maintenance – Procedures for maintaining the beam.
 - Cutter Head Maintenance – Procedures for maintaining the cutter head and changing the knife. Separate sections are included for the GT 5250 and GT 7250 cutters.
 - Drill Maintenance – Procedures for maintaining the drill. Separate sections are included for the GT 5250 and GT 7250 cutters.
 - Cutting Table Maintenance – Procedures for maintaining components of the table including cutting surface and some optional equipment.

- Troubleshooting – Where to find information about diagnostics, error messages, and specific cutting applications.
- Glossary – Terms you need to know
- Appendix A – Description of the vacuum system and other vacuum features.

C-200MT Online Help and Release Info

The C-200MT software program provides several ways to get help.

Note: Information in the C-200MT online help may be more current than in the *C-200MT Software User's Guide*.

To get help on a C-200MT function

1. Choose a C-200MT menu and use the pointing device or Down arrow key to highlight the command.
2. Press the **F1** key to display the appropriate Help topic.

To get help on a specific parameter

1. In Parameter Manager, highlight a parameter.
2. Press **F1**.

To view information for current release

1. From the Help menu, click **Index**.
2. Make sure the Contents page is displayed.
3. Then select the New Features for (current version number) or Limitations topics.

Available Documentation on Desktop or CDs

Documentation for your GERBERcutter hardware and software is now delivered in online format on CD. This means that specific information can be viewed on-screen, searched for, and printed as you need it. Some documentation for your cutter was installed on your computer during the C-200MT software installation and is available by double-clicking the Gerber Documents icon on your Windows desktop.

Additional documentation available on C-200MT software CD and/or GERBERservice Documentation CD:

- *GT Series GERBERcutters Getting Started Manual*
- C-200MT online help
- *GERBERcutter Application Manual*
- *NT Quick Reference Guide*
- Service drawings
- Parts information

Note: Both CDs allow you to view and print information at another PC. For hints on printing see a later section in this chapter.

Viewing Documentation

To view documents from Windows desktop

1. In the C-200MT program, click on the middle button (marked by two boxes) in the top left corner of the main cutter control screen. This does **not** shut down the C-200MT program.
2. On the Windows desktop, double-click on the Gerber Documents icon.
3. Double-click on the icon for the PDF file you want to view. The reader opens first and then the document displays.
4. From the reader's tool bar, use the buttons marked with arrows to turn pages. Click the button marked with binoculars to search for a specific item.

To view documents from CD

1. Follow instructions on the CD or use Windows Explorer to locate the PDF file you want to view.
2. Double-click on the file. The reader opens first and then the document displays.
3. From the reader's tool bar, use the buttons marked with arrows to turn pages. Click the button marked with binoculars to search for a specific item.

Acrobat Reader and PDF Format

GERBER Technology, Inc. now provides many documents and service drawings in PDF format. A document in PDF format is displayed in Adobe Acrobat Reader. If not already installed on your cutter, this viewer can be installed from the C-200MT Software CD.

Documents can be viewed on any PC that has the Acrobat Reader installed.

Printing Documentation

If you prefer to print out information and your cutter is not networked to a printer, you can take the CD to another PC that does have access to a printer. The Acrobat Reader must be installed on that PC.

To print from a PDF document

1. Follow the instructions for opening a PDF document from CD.
2. From the File menu, select **Print**. You can print one page or all pages.

Conventions

The following conventions are used throughout this manual:

- Bullets at the beginning of a section tell you what is covered in that section. Bullets also list a group of related items.

Bold type highlights important information. In procedures, selections to make with a pointing device or keyboard keys to press are bolded. In text, information that you need to consider first is bolded.

Text that you need to type or message prompts look like this.

Note: Notes contain important or helpful information.

CAUTION: Cautions appear before procedures that could cause damage to equipment if not performed correctly.

WARNING!  **Warnings tell you when you must follow a specific procedure or practice to avoid personal injury or death.**

Customer Support

If a problem still exists after troubleshooting, call Gerber Field Service Engineering for assistance. In North America, call Customer Support 800-321-2448. Outside North America, call your local service office.

Have the following information ready when you call:

- GERBERcutter you are using
- Any special equipment or software you are using, such as the drill, resealer, or bar code reader option
- Software packages you are using; include the software version numbers
- Equipment number, if available

Maintenance Information

About This Chapter

Proper maintenance is important because it improves cut quality and extends the life of the system. This chapter has information to help you do routine maintenance on your GT 5250 or GT 7250 GERBERcutter. It does not include highly-technical procedures that should be done by your Gerber qualified technician. This chapter includes:

- Safety issues before maintenance
- Maintenance Manager
- Maintenance summary and schedules
- Reference drawings

Safety Precautions

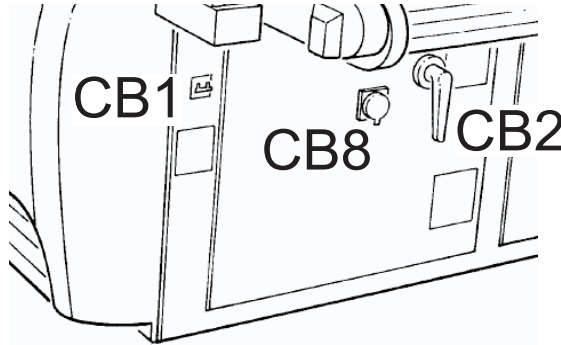
WARNING!   High voltages exist in the C-200 controller. There are no operator-serviceable parts inside the C-200 controller enclosure. Only qualified personnel should perform maintenance. Contact your service technician in case of a problem. Failure to follow these instructions could cause injury or death.

WARNING!  You may get seriously hurt if you do not turn off the C-200 controller and vacuum generator when instructed. When turned ON, the C-200 controller applies power to the cutting table and vacuum generator. Parts could move suddenly.

WARNING!   The origin light on the cutter head uses a Class IIIa laser product. Do not look along the axis of the laser light without protective eyewear. Do not place reflective material on the cutting table because the reflection from the laser light may cause injury. Unplug the cable connector on the origin light before doing any maintenance on the system. Failure to follow these warnings can cause injury.

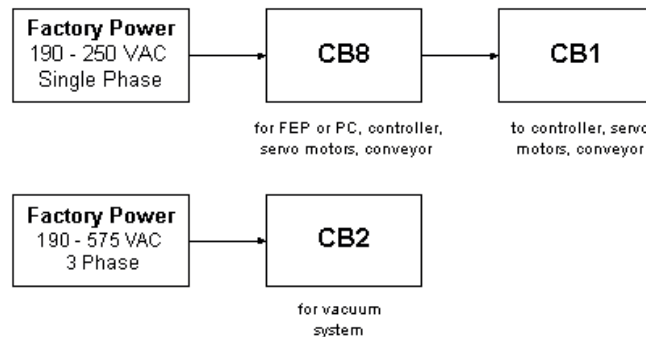
Circuit Breakers

The GERBERcutter has three main circuit breakers on the cutting table. Circuit breaker 1 provides power for the C-200 controller, servo motors, and conveyor. It is located in the upper left corner of the electrics enclosures at the take-off end. Use this circuit breaker to turn on the GERBERcutter.



WARNING!   Do not turn circuit breakers inside of the electrics enclosure on or off. High voltages exist within this enclosure and inside the controller board. Never perform any work inside the vacuum electrics enclosure. There are no operator-serviceable parts inside of this enclosure.

For information about the circuit breakers on your GERBERcutter, see the chart below.



WARNING!   Three-phase power for the vacuum generator and single-phase power for the control unit both enter the C-200 controller. BOTH power cords must be disconnected for a safe condition to exist within the C-200 enclosure. A single-switch main disconnect for both of these power supplies must be provided according to local electrical codes.

Ordering Spare Parts

If you find a broken or worn part while performing preventive maintenance, see the Customer Support section in the Troubleshooting chapter. This section tells how to order any spare parts you may need.

Tools Available

Listed below are some tools and lubricants that are available from Gerber. For more complete information, contact your Gerber sales representative.

Tool	Gerber Part No.
50-lb. spring scale	945500252
belt deflection gauge	45412000
belt tensioner	62031000
wrench ½ in. combo, 6 pt	944002506
metric combination wrench set	945500093
metric right-angle and ball combination hex key set	945500094
metric T-handle hex key set	945500089
hex key, T-handle 9 in., 9/64	944200406
Allen hex wrench 7/64	945500046
Allen wrench 1/16	945500255
slot screw driver 1/8 x 4 in.	945500298
internal/ external snap ring pliers	944291503
pin pusher	340500867B
split collar	324500026
knife alignment box	27014000
Y-axis belt adjustment kit	79099000
belt span fixture	78606000
0 – 30 lb. spring scale	945500016
vise grip	945500002
grease gun	944233101
oiler	653000000

Lubricant	Gerber Part No.
Mobil 1, 5w-30 synthetic oil	596118001
GSG-44	72395000
Cleaner - Wolf	308010101
Loctite #222-21	120050216
Loctite #7471	120050209
Petro-lon multipurpose grease	596500005

Maintenance Manager

The C-200MT program provides this utility to monitor routine, preventive maintenance procedures for your GERBERcutter. Maintenance Manager, which is available in newer versions of C-200MT software, keeps track of all routine maintenance and notifies you on-screen whenever a task, or maintenance procedure, needs to be performed. These procedures correspond to tasks in the Maintenance Manager. For availability, check with your Gerber Field Service representative.

Information about individual tasks is also available on-screen, including the steps required to perform the task. Once a maintenance procedure is completed, clearing the status for that task resets information stored in the database. Tracking continues automatically, without your attention being required until Maintenance Manager notifies you.

For information on how to use this utility, refer to the C-200MT online help. For safety precautions and corrective maintenance, use this manual.

Maintenance Summary

The table below summarizes where in this manual the preventive maintenance procedures can be found. The maintenance schedules, which begin on the next page, conveniently lists procedures by how often they must be done.

Note: The reference figures, beginning on page 16, can help you find various parts on your GT 5250 or GT 7250 GERBERcutter.

Major Assembly	Assembly/ Option	Page
beam		25
cutter head GT 5250		45
cutter head GT 7250		77
drill GT 5250		111
drill GT 7250		121
cutting table		131
	operator console	135
	conveyor	137
	air filters	149, 162
	vacuum	151
	film dispenser	167
	resealer	169
	CTOT	174
	lateral Drive	177

Maintenance Schedule

CAUTION: In addition to the maintenance recommended in the Schedule tables below and on the next pages, you must inspect all belt tensions and adjust gear backlash after the first 48 hours of operation of a newly-installed GERBERcutter or belt. This inspection and adjustment ensures the proper break-in of your GERBERcutter. Failure to do this maintenance may result in damage to your GERBERcutter.

Major Assembly	Assembly	Component	Action	Page
Each Shift				
C-200 controller	Electrics enclosure	MCC air filter	inspect	162
GT 5250 cutter head	Drill	Drill bit	inspect	115
	Elevator roundways	N/A	inspect	72
	C-axis assembly	Knife blade	inspect	56
		Presser foot	inspect	61
	Sharpener	Drive belt	inspect	66
		Grinding pulley	inspect	68
GT 7250 cutter head	Drill	Drill bit	inspect	125
	Elevator flatway	N/A	inspect	104
	C-axis assembly	Knife blade	inspect	87
		Presser foot	inspect	92
	Sharpener	Drive belt	inspect	97
		Grinding pulley	inspect	101
Cutting table	Flatways	N/A	inspect	133
	Racks	N/A	inspect	132
Mast	Warning light	N/A	inspect	166
Daily				
Beam	Roundways	N/A	inspect	28
			clean/ lubricate	28
Cutting table	Conveyor	BRISTLE SQUARES	inspect	137
	Table air filter	N/A	inspect	149
Resealer	Plastic film	N/A	inspect	170

Major Assembly	Assembly	Component	Action	Page
Weekly				
Beam	Beam	N/A	inspect alignment	26
	Cables	N/A	inspect	41
	Cutter head drive belt	N/A	inspect	29
		N/A	adjust	30
	X-axis drive belts	N/A	inspect	31
	X-axis gears	N/A	adjust	36
	Y-axis drive belts	N/A	inspect	33
GT 5250 cutter head	C-axis assembly	C-axis assembly	inspect	49
		C-axis driven gear	inspect	50
		Knife drive belt	inspect	52
	Drill	Drive belt	inspect	112
	Lancaster balancer	Belt	inspect	54
GT 7250 cutter head	C-axis assembly	C-axis assembly	inspect	78
		C-axis drive belt	inspect	79
		Knife drive belts	inspect	82
	Drill	Drive belt	inspect	122
	Lancaster balancer	Belt	inspect	85
Cutting table	Conveyor	Slats	inspect	142
Mast	Power/ air lines	N/A	inspect	165
Each Six Months				
CTOT	Drive chain	N/A	lubricate	175
Cutting table	Conveyor	Drive assembly	lubricate	147
		Drive chains	lubricate	148
Vacuum generator	N/A	Drive belt	inspect	155
	Motor	N/A	lubricate	154
	Generator	N/A	lubricate	161
Each Year				
Lateral drive	Wheels	N/A	lubricate	177

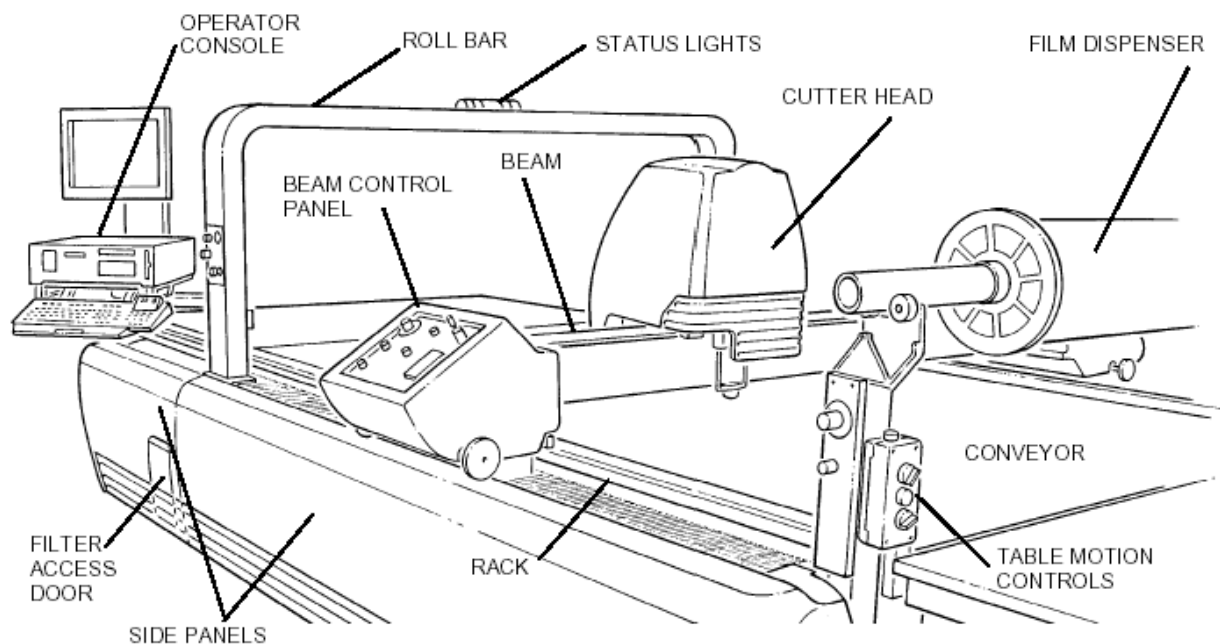
Major Assembly	Assembly	Component	Action	Page
As Needed				
Beam	Beam	N/A	adjust alignment	26
	BCP cover	N/A	remove	42
			install	42
	Remote end cover	N/A	remove	43
			install	43
	X-axis drive belts	N/A	adjust	32
	Y-axis drive belts	N/A	adjust	35
CTOT	N/A	N/A	open	174
			close	175
GT 5250 cutter head	C-axis assembly	C-axis driven gear	adjust	51
		Knife blade	remove	57
			install	58
		Knife drive belt	adjust	53
		Presser foot	clean/ lubricate	62
	Drill	Drill bit	remove	116
			install	117
		Drive belt	remove	113
			install	113
			adjust	114
	Elevator roundways	N/A	clean/ lubricate	72
	Cover	Back	remove	48
			install	48
		Front	open	46
			close	46
			remove	47
			install	47

Major Assembly	Assembly	Component	Action	Page
As Needed (cont'd)				
GT 5250 cutter head (cont'd)	Lancaster balancer	Belt	adjust	55
	Sharpener	Drive belt	remove	67
			install	68
		Grinding pulley	remove	69
			clean	70
			install	71
	Up/down stops	N/A	inspect	73
			adjust	76
GT 7250 cutter head	C-axis assembly	C-axis drive belt	remove	79
			install	80
			adjust	81
		Knife blade	remove	88
			install	90
		Knife drive belts	adjust	83
		Presser foot	clean/ lubricate	93
	Drill	Drill bit	remove	126
			install	126
		Drive belt	remove	123
			install	123
			adjust	124
	Elevator flatway	N/A	clean/ lubricate	104
	Cover	Back	remove	48
			install	48
		Front	remove	47
			install	47
			open	46
			close	46
	Lancaster balancer	Belt	adjust	85

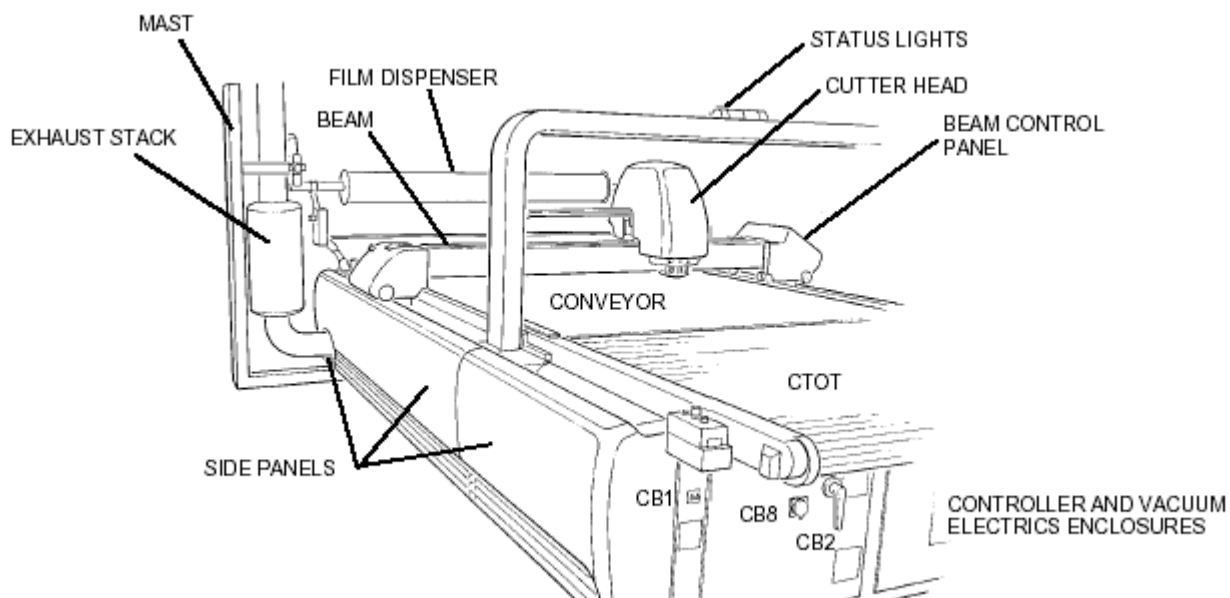
Major Assembly	Assembly	Component	Action	Page
As Needed (cont'd)				
GT 7250 cutter head (cont'd)	Lancaster balancer	Belt	adjust	86
	Sharpener	Drive belt	remove	98
			install	99
		Grinding pulley	remove	101
			clean	102
			install	103
	Up/down stops	N/A	inspect	105
			adjust	108
C-200 controller	Electrics enclosure	MCC air filter	clean	163
Cutting table	Conveyor	BRISTLE SQUARES	clean	138
			rearrange	140
		Slats	remove	144
			install	145
	Flatways	N/A	clean	133
	Racks	N/A	clean	132
	Side panels	N/A	remove	135
			install	135
			clean	135
	Table air filter	N/A	clean	150
	Take-on ramp	N/A	remove	136
			install	136
Film dispenser	Plastic film roll	N/A	remove	167
			install	168
Operator console	Monitor/ keyboard	N/A	inspect/clean	135
Resealer	Plastic film	N/A	remove	170
			install	171
Vacuum Generator	N/A	Drive belt	remove	159
			install	160
			adjust	157
	Motor	N/A	inspect	151
			adjust	153

Reference Drawings

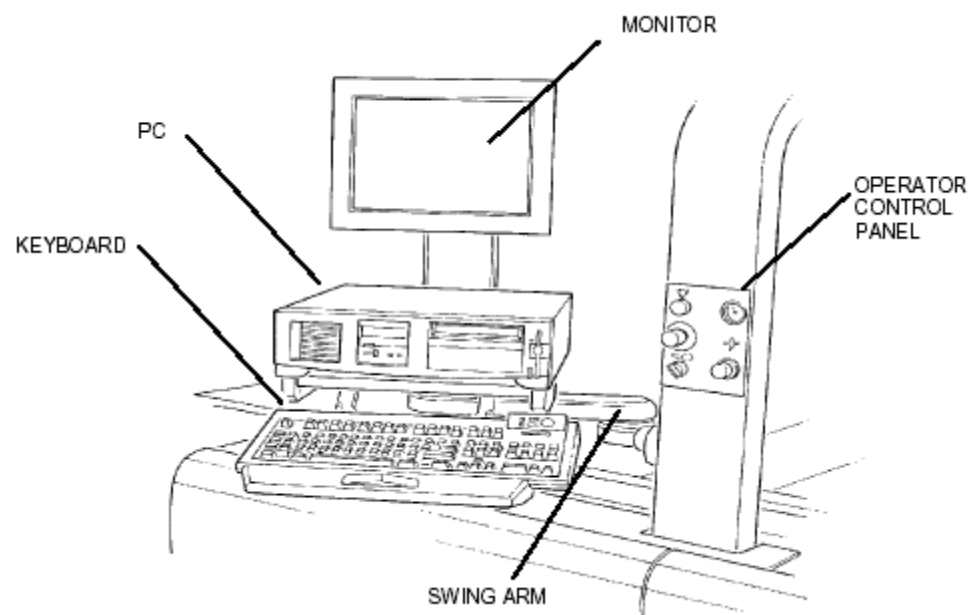
The following drawings can be used before performing maintenance to find various parts of your GT 5250 or GT 7250 GERBERcutter.



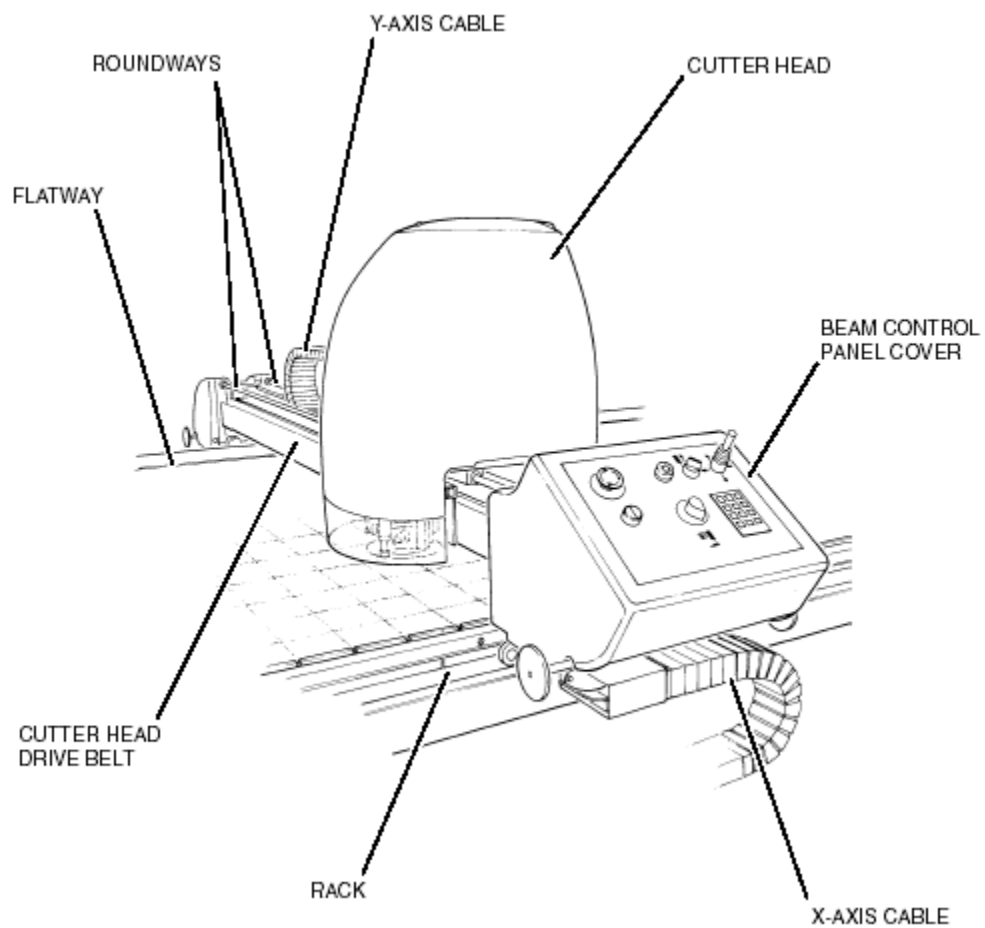
Operator Side from Take-on End



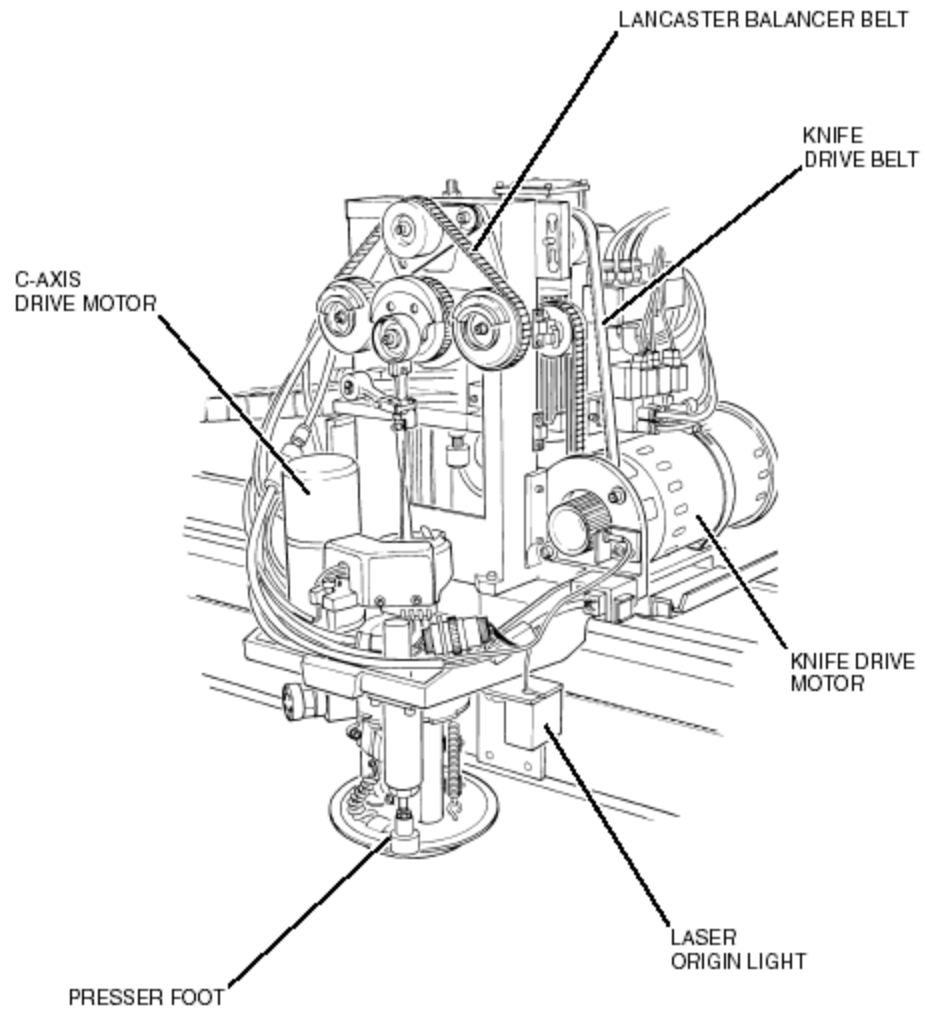
Remote Side from Take-off End



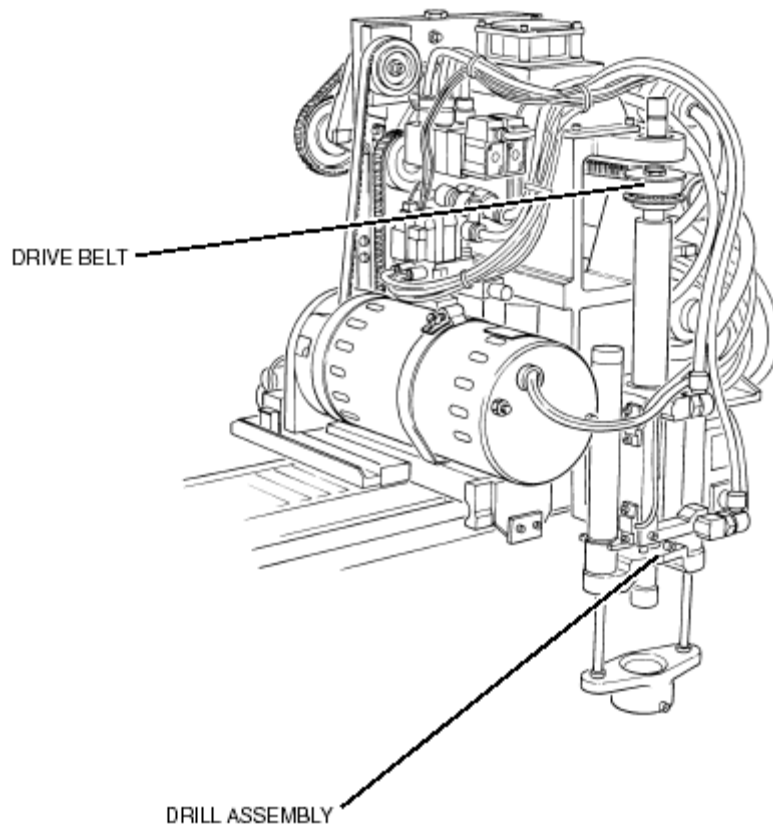
Operator Console



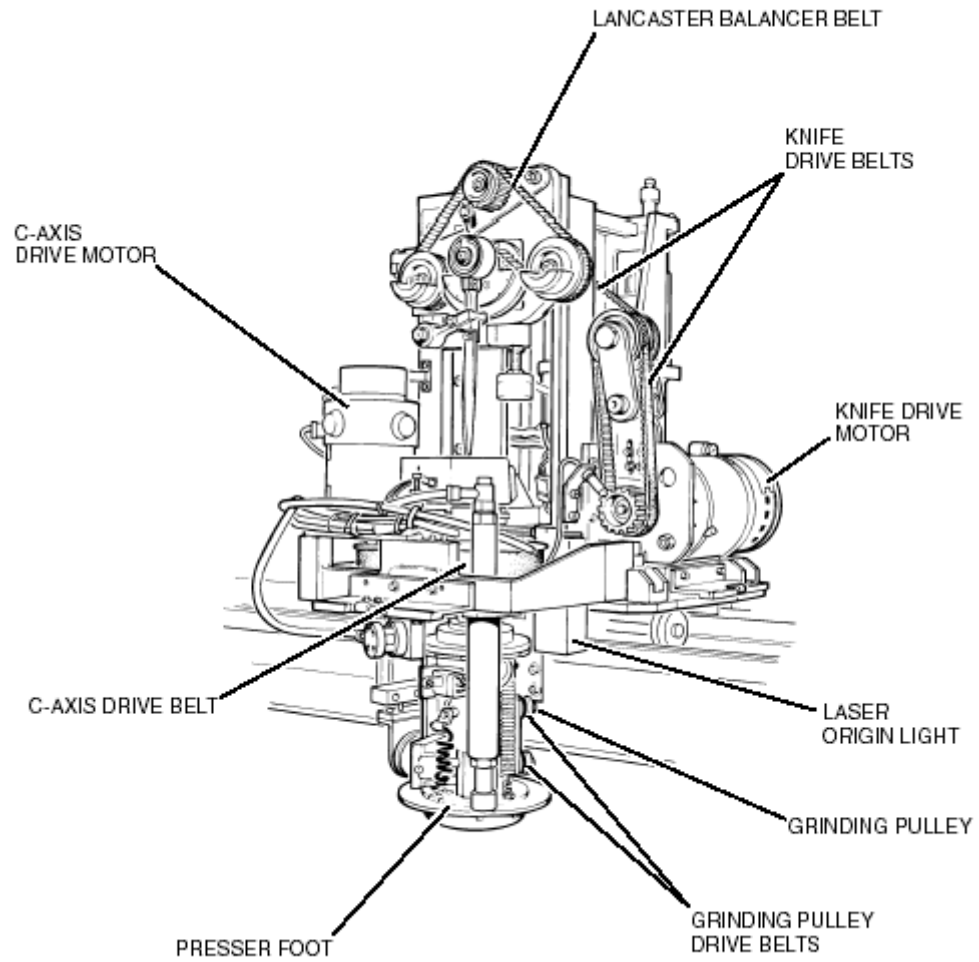
Cutter Head and Beam



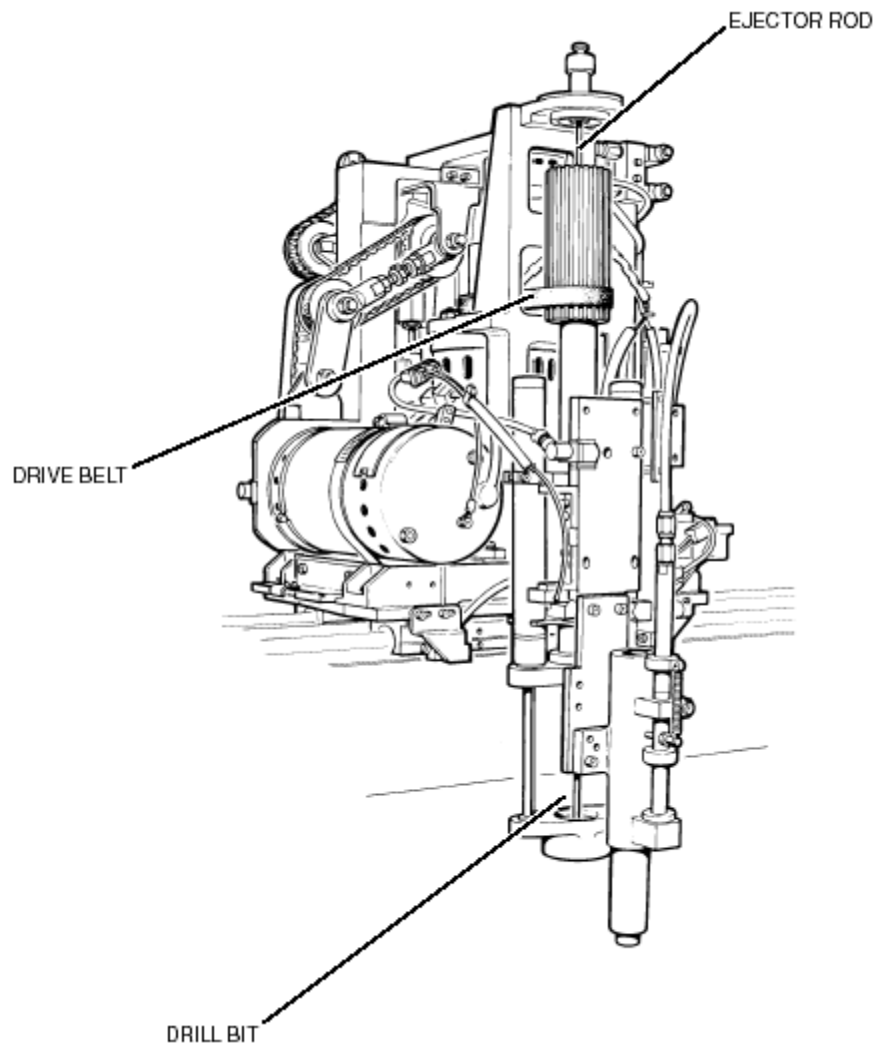
GT 5250 Cutter Head



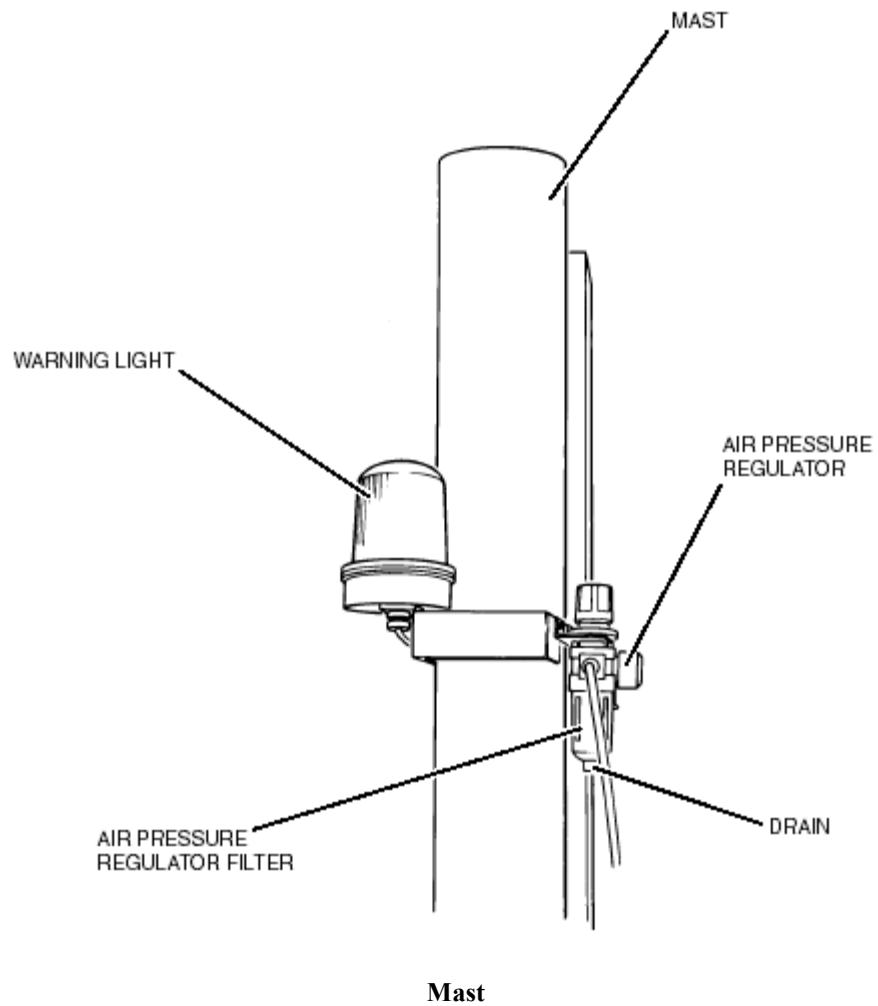
Drill Assembly on GT 5250

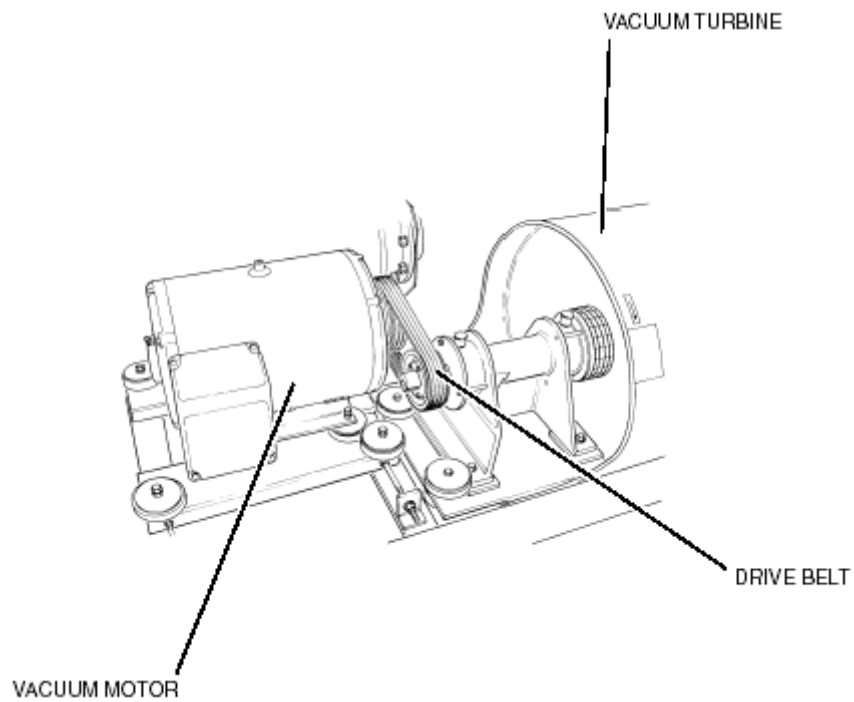


GT 7250 Cutter Head

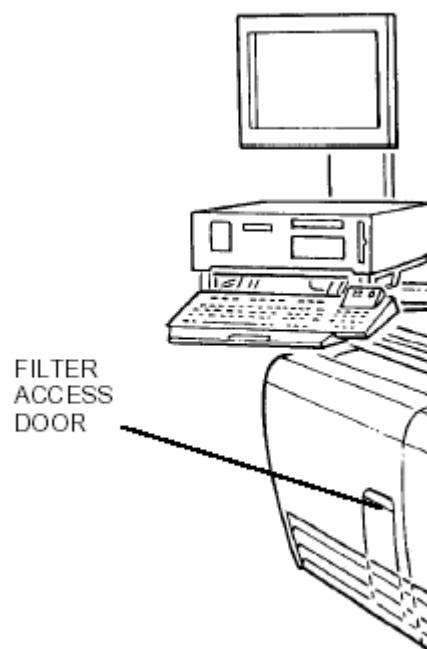


Drill Assembly on GT 7250





Vacuum Generator



Access to Table Filters

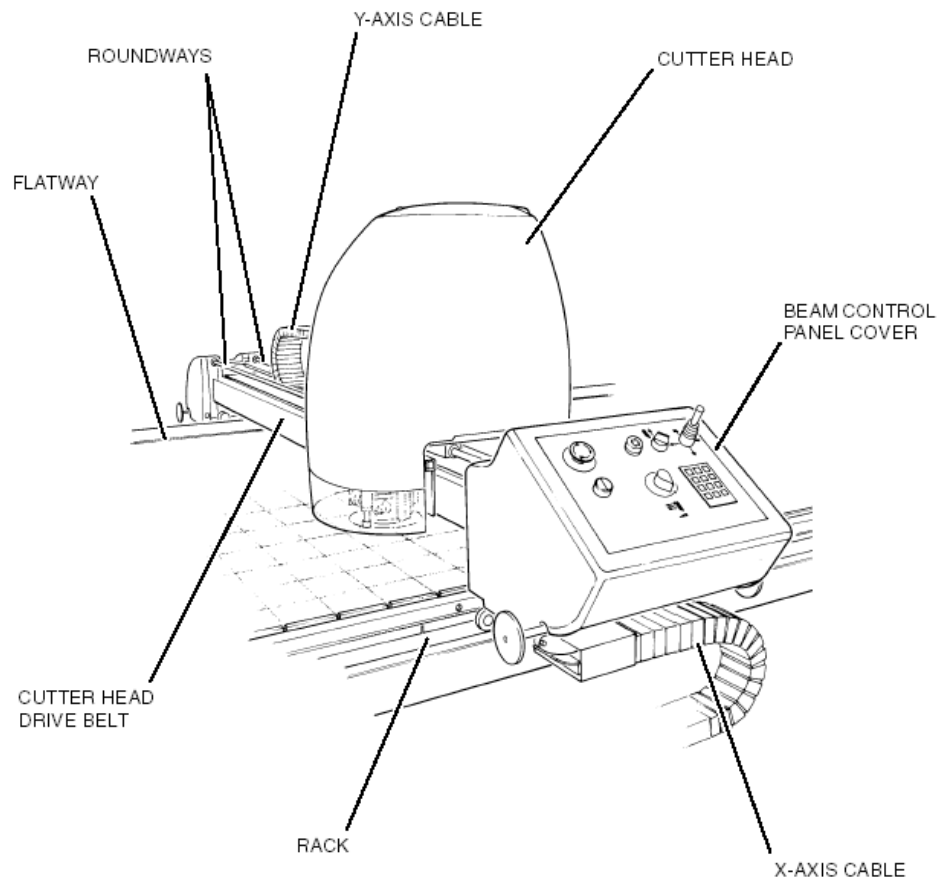
Preventive Maintenance

Proper maintenance is important because it improves cut quality and extends the life of the system. This chapter describes routine maintenance on your GERBERcutter. Preventive maintenance tasks include inspecting, cleaning, and lubricating moving parts such as gears and bearings.

If routine maintenance indicates the need to perform corrective maintenance procedures, alignments, or adjustments, these are best referred to a Gerber qualified technician (someone who has attended a training class for GERBERcutter operation and maintenance).

Beam Maintenance

The beam, shown below, carries the cutter head on two roundways. Electric motors drive the beam along racks on the cutting table and move the cutter head along the beam. The combined motions of the beam and cutter head let the knife cut complex patterns.



Cutter Head and Beam

Beam Alignment

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
paper, 1.5 m x 1.5 m (60 in. x 60 in.)	N/A

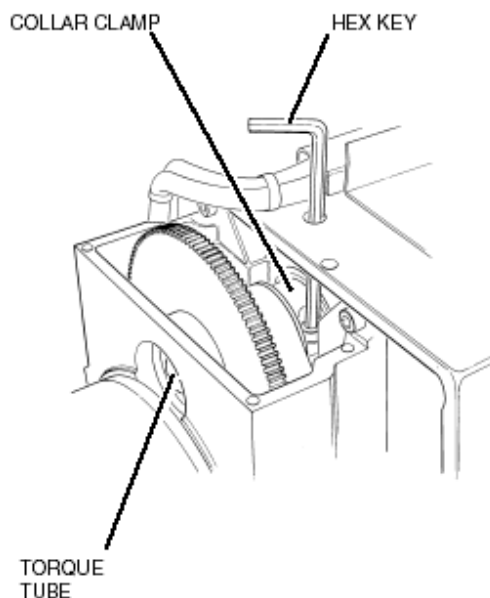
Inspect

1. Turn on the C-200 controller.
2. Run the C-200MT program.
3. Run the Beam Square Test (see the C-200MT online help or the *C-200MT Software User's Guide*). If the beam is not square, adjust the beam alignment.
4. Exit the C-200MT program.
5. Turn off the C-200 controller.

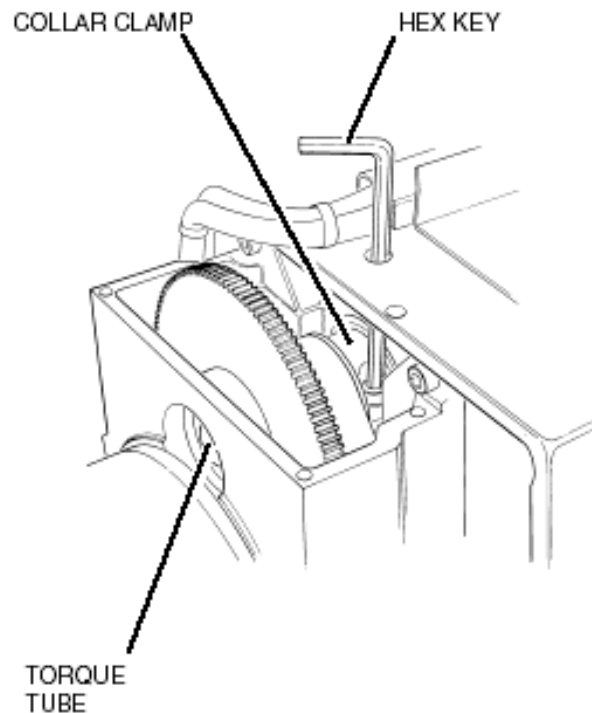
Adjust

1. Turn off the C-200 controller.
2. Remove the remote end cover (page 43).

3. Loosen the collar clamp on the right end torque tube.
 - a) Put the end of a hex key into the access hole.
 - b) Insert the hex key into the screw on the collar clamp. You may need to move the beam forward or backward so that the collar clamp screw is in the correct position.
 - c) Loosen the collar clamp screw by turning it counterclockwise.
 - d) Remove the hex key from the access hole.



4. Align the beam by moving the right side of the beam. The right side of the beam rides on the right-side rack. To align the beam, move it only slightly in either direction.
 - a) If the +Y end of the beam leads the -Y end, move the right end of the beam back towards the take-on end of the table.
 - OR**
 - b) If the -Y end of the beam leads the +Y end, move the right end of the beam forwards towards the take-off end of the table.
5. Tighten the collar clamp screw with the T-handled hex key.
6. Install the remote-end cover (page 43).
7. Turn on the C-200 controller.
8. Run the C-200MT program.
9. Run the Beam square test.
(see the C-200MT online help or the *C-200MT Software User's Guide*).
10. If the beam is not square, repeat steps 1 through 9 until the beam is square.



Roundways

The roundways support and guide the cutter head as it travels along the beam. Each roundway is a round shaft that runs the length of the beam. Each roundway must be clean, lightly lubricated, and free of scratches so the cutter head can move freely.

Required Tools and Supplies

Item	Gerber Part No.
clean shop rag	N/A
SAE 10 non-detergent oil	N/A

Inspect

1. Turn off the C-200 controller.
2. Inspect the roundways for dirt and built-up lint. Clean and lubricate the roundways, if needed.

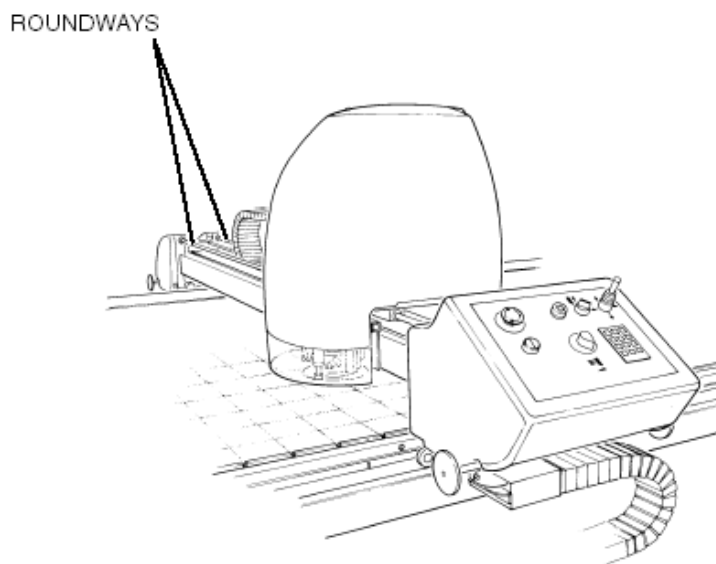
3. Inspect the roundways for scratches, nicks, and gouges. These defects can damage the roundway bearings in the cutter head. If you find any of these defects, have them repaired or replaced by a Gerber qualified technician.
4. Turn on the C-200 controller.

Clean/Lubricate

1. Turn off the C-200 controller.
2. Wipe the roundways with a lightly-oiled shop rag. This cleans and lubricates the roundways.

CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently than it is to apply one heavy coat.

3. Turn on the C-200 controller.



Cutter Head Drive Belt

The cutter head drive belt moves the cutter head along the beam. The drive belt must be properly tensioned and free of defects to ensure accurate cutting.

Required Tools and Supplies

Item	Gerber Part No.
belt tensioner	78606000
right-angle metric hex key set	945500094
spring scale, 0 - 23 kg (0 - 50 lb.)	945500252
sheet of paper	N/A

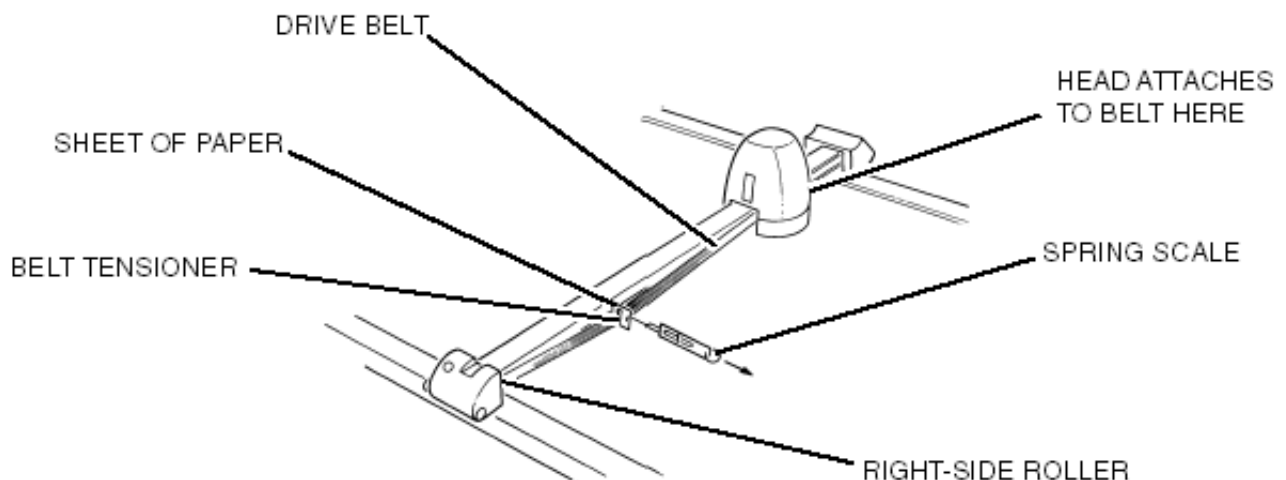
Inspect

1. Inspect the drive belt for nicks, cuts, gouges, or fraying.
 - a) Turn on the C-200 controller.
 - b) Slew the cutter head so you can see the entire drive belt.
 - c) If you see any defects in the belt, have it replaced by a Gerber qualified technician.
2. Check drive belt tension.

- a) Using the slew keys, move the cutter head to the left end of the beam. Move the cutter head so that the left side of the clamp that attaches the cutter head to the drive belt is 190.5 cm (75 in.) from the place where the drive belt touches the pulley on the right side of the beam.
- b) Turn off the C-200 controller.
- c) Place the belt tensioner on the drive belt halfway between the cutter head cover and the remote-end cover.
- d) Put a small piece of paper between the belt tensioner and beam. The belt tensioner should hold the paper against the beam.
- e) Attach a spring scale to the belt tensioner.
- f) Pull the spring scale until the paper falls from between the beam and belt tensioner. The spring scale should read 0.91 to 1.81 kg (2 to 4 lb.).
- g) If tension is incorrect, adjust the belt tension.

3. Remove the belt tensioner.

4. Turn off the C-200 controller.



Adjust

1. Turn on the C-200 controller.
2. Using the slew keys, move the cutter head to the left end of the beam. Move the cutter head so that the left side of the clamp that attaches the cutter head to the drive belt is 190.5 cm (75 in.) from the place where the drive belt touches the pulley on the right side of the beam.
3. Turn off the C-200 controller.
4. Remove the remote-end cover (page 43).
5. Adjust drive belt tension:

CAUTION: Make the same adjustment to both adjusting screws. Otherwise, you may misalign the tracking of the drive belt.

- a) Turn the adjustment screws clockwise to increase tension.

OR

- b) Turn the adjustment screws counterclockwise to decrease tension.

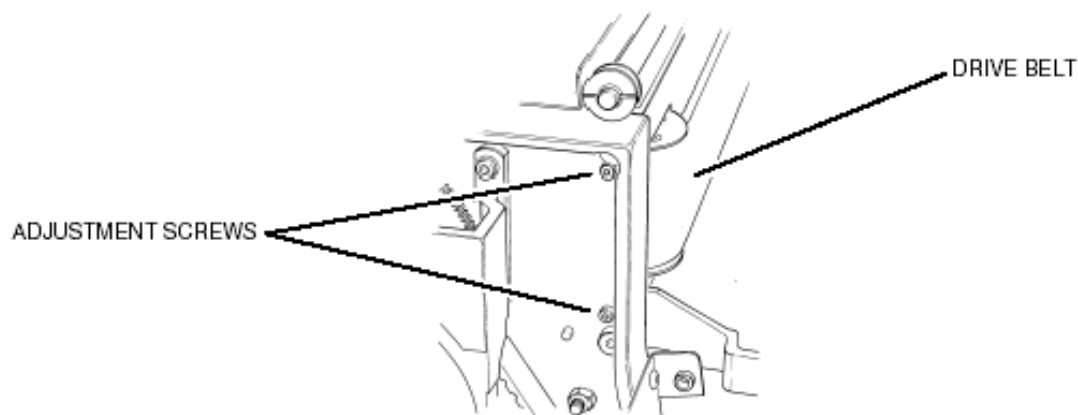
6. Check drive belt tension.

- a) Put a small piece of paper between the belt tensioner and beam. The belt tensioner should hold the paper against the beam.
- b) Attach a spring scale to the belt tensioner.
- c) Pull the spring scale until the paper falls from between the beam and belt tensioner. The spring scale should read 0.91 to 1.81 kg (2 to 4 lb.).
- d) Remove the belt tensioner and spring scale from the drive belt.
- e) Repeat steps 5 and 6 until belt tension is correct.

7. Check drive belt tracking.

- a) Turn on the C-200 controller.
- b) Slew the cutter head back and forth along the beam several times.
- c) If the drive belt rides up onto one edge of the pulley, adjust the adjusting screw closest to that edge.
- d) Repeat steps 7b and 7c until the belt tracks properly.
- e) Turn off the C-200 controller.

8. Install the remote-end cover (page 43).



X-axis Drive Belts

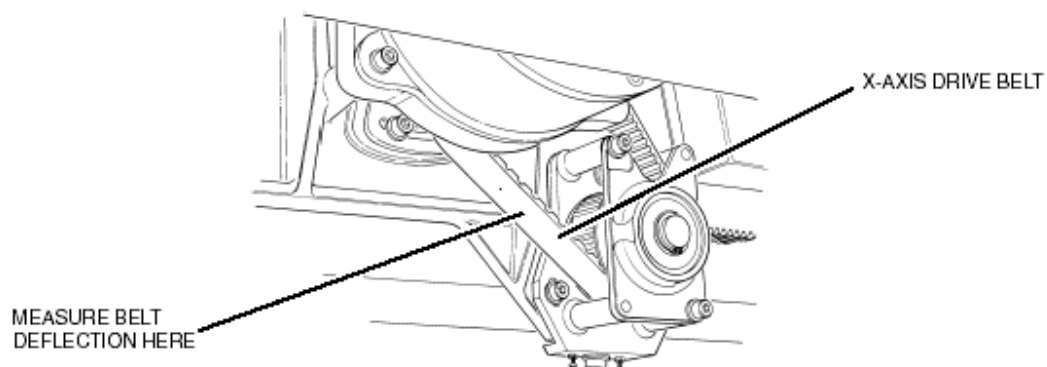
The X-axis drive belts let an electric motor in the beam turn the gears and pulleys that move the beam along the cutting table. The drive belts must be in good condition for the beam to move properly.

Required Tools and Supplies

Item	Gerber Part No.
belt tensioner	62031000
right-angle metric hex key set	945500094
belt deflection gauge	45412000
6 in. adjustable wrench	N/A
split collar	324500026

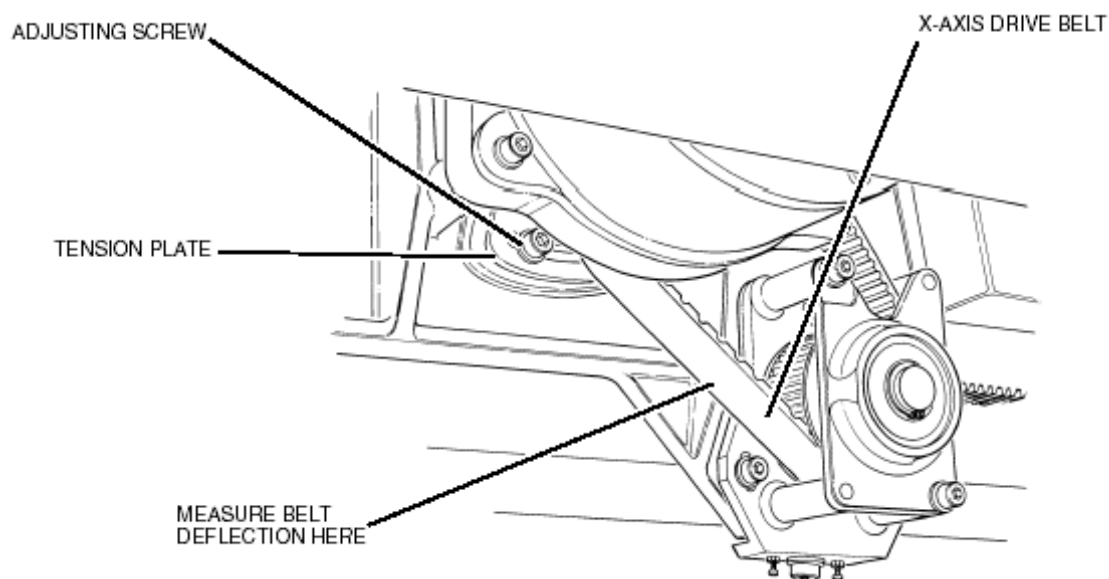
Inspect

1. Turn off the C-200 controller.
2. Remove the BCP cover (page 42).
3. Inspect the left-side drive belt.
 - a) Look for nicks, cuts, gouges, glazing, or fraying.
 - b) Move the beam as needed to bring hidden sections of the belt into view.
 - c) If you see any defects in the belt, have it replaced by a Gerber qualified technician.
- d) Measure the tension of the drive belt with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 3.18 to 4.09 kg (7 to 9 lb.).
- e) If the deflection is incorrect, adjust the drive belt tension.
4. Install the BCP cover (page 42).
5. Remove the remote-end cover (page 43).
 - a) Look for nicks, cuts, gouges, glazing, or fraying.
 - b) Move the beam as needed to bring hidden sections of the belt into view.
 - c) If you see any defects in the belt, have it replaced by your Gerber qualified technician.
 - d) Measure the tension of the drive belt with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 3.18 to 4.09 kg (7 to 9 lb.).
 - e) If the deflection is incorrect, adjust the drive belt tension.
6. Install the remote-end cover (page 43).
7. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
2. Remove the BCP cover (page 42).
3. Adjust left-side drive belt tension.
 - a) Loosen the adjusting screw for the left-side tension plate.
 - b) Move the tension plate to adjust belt tension.
 - c) While keeping tension on the tension plate, tighten the adjusting screw.
 - d) Measure the tension of the drive belt with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 3.18 to 4.09 kg (7 to 9 lb.).
 - e) Repeat steps 3a through 3d until the belt tension is correct.
4. Install the BCP cover (page 42).
5. Remove the remote-end cover (page 43).
6. Adjust the right-side drive belt tension.
 - a) Loosen the adjusting screw for the right-side tension plate.
 - b) Move the tension plate to adjust belt tension.
 - c) While keeping tension on the tension plate, tighten the adjusting screw.
 - d) Measure the tension of the drive belt with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 3.18 to 4.09 kg (7 to 9 lb.).
 - e) Repeat steps 6a through 6d until the belt tension is correct.
7. Install the remote-end cover (page 43).
8. Turn on the C-200 controller.



Y-axis Drive Belt

The Y-axis drive belt lets an electric motor in the beam turn the pulleys that move the cutter head along the beam. The drive belt must be in good condition for the cutter head to move properly.

Required Tools and Supplies

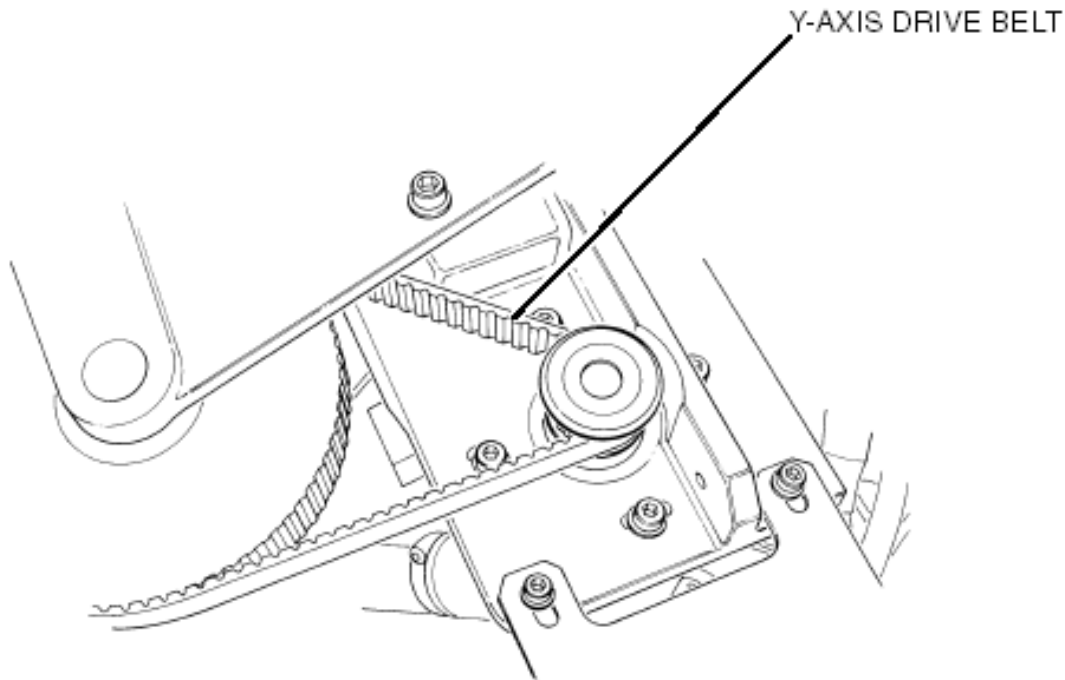
Item	Gerber Part No.
belt deflection gauge	45412000
right-angle metric hex key set	945500094

Inspect

1. Turn off the C-200 controller.
2. Remove the BCP cover (page 42).
3. Remove the BCP mounting screws.

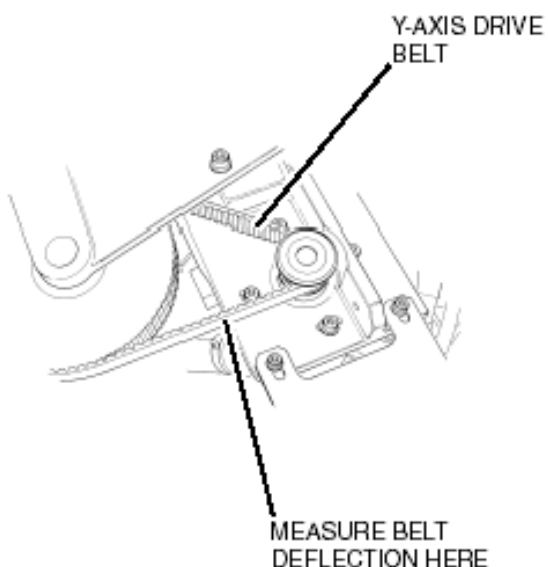
Note: Do not disconnect any electrical connections from the BCP.

4. Move the BCP out of the way.
5. Remove the metal belt guard from over the drive belt.
6. Inspect the drive belt for nicks, cuts, gouges, glazing, and fraying. If you see any of these defects, have it replaced by a Gerber qualified technician.



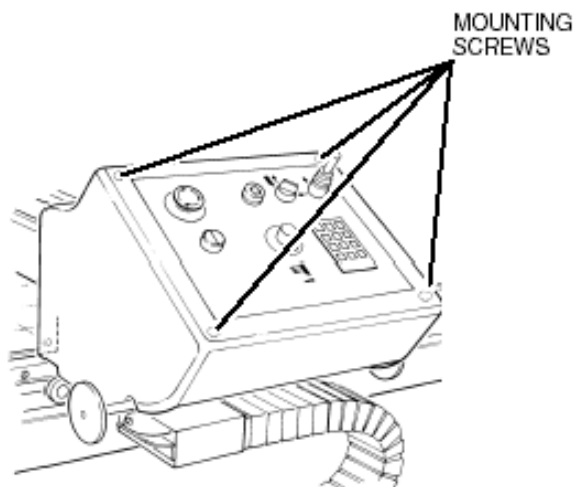
7. Check drive belt tension.

- a) Set the belt deflection gauge to measure a deflection of 3.8 mm (0.15 in.) at 1.36 to 1.82 kg (3 to 4 lb.).
- b) Measure the drive belt deflection halfway between the Y-axis motor pulley and the cutter head drive belt pulley.
- c) If the belt deflection is not correct, adjust the drive belt tension.
- d) Install the metal belt guard over the drive belt.



8. Install the BCP.

- a) Align the screw holes in the BCP with the screw holes in the BCP mounting bracket.
 - b) Install the BCP mounting screws.
9. Install the BCP cover (page 42).
10. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
2. Remove the BCP cover (page 42).
3. Remove the BCP mounting screws.

Note: Do not disconnect any electrical connections from the BCP.

4. Move the BCP out of the way.
5. Adjust drive belt tension.

- a) Loosen the three Y-axis motor retaining screws.
- b) Loosen the Y-axis motor pivot screw.
- c) To increase Y-axis drive belt tension, move the Y-axis motor away from the cutter head drive belt pulley.

OR

- d) To decrease Y-axis drive belt tension, move the Y-axis motor towards the cutter head drive belt pulley.

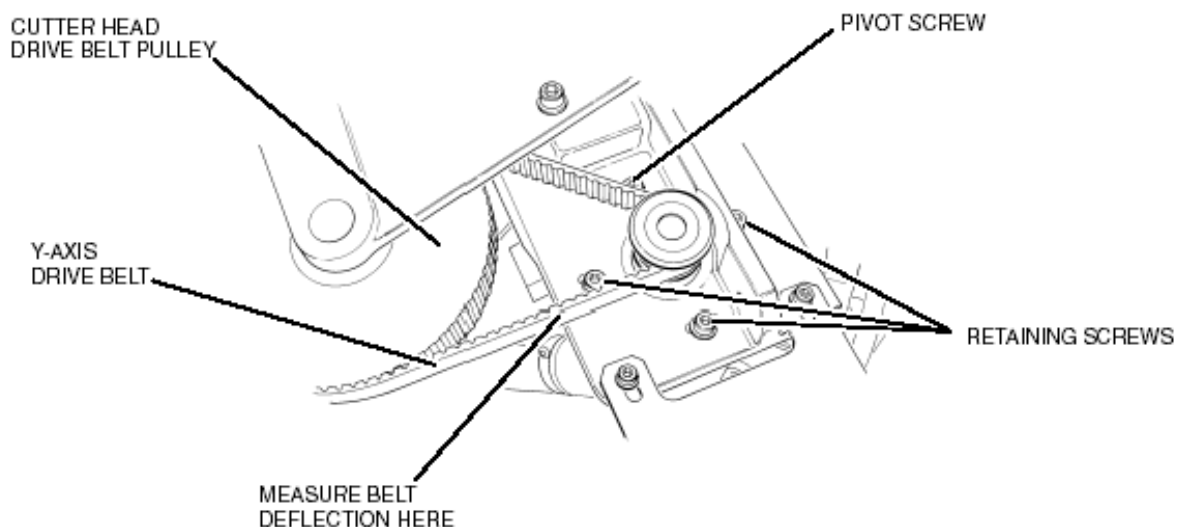
- e) Tighten the three Y-axis motor retaining screws.
- f) Tighten the Y-axis motor pivot screw.
- g) Set the belt deflection gauge to measure a deflection of 3.8 mm (0.15 in.) at 1.36 to 1.82 kg (3 to 4 lb.).
- h) Measure the drive belt deflection halfway between the Y-axis motor pulley and the cutter head drive belt pulley.
- i) If the belt deflection is not correct, repeat steps 5a through 5h until the tension is correct

6. Install the BCP.

- a) Align the screw holes in the BCP with the screw holes in the BCP mounting bracket.
- b) Install the BCP mounting screws.

7. Install the BCP cover (page 42).

8. Turn on the C-200 controller.



X-axis Gears

The X-axis gears need to be adjusted so that the beam moves quietly and smoothly without binding. X-axis gears include the four stabilizer gears, the two pinion gears, and the X-axis motor gear.

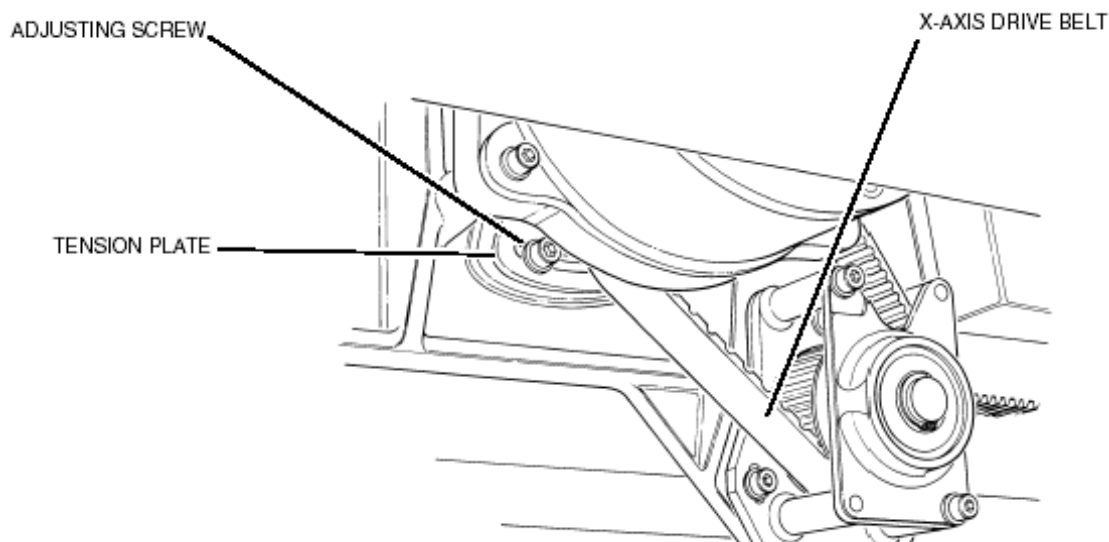
At each end of the beam, the gear that is closest to the take-on end and meshes with the rack is a stabilizer gear. Also, the gear that is closest to the take-off end and meshes with the rack is a stabilizer gear. The center gear that meshes with the rack is a pinion gear. The X-axis motor gear is on the right side of your GERBERcutter, behind the X-axis servo motor.

Required Tools and Supplies

Item	Gerber Part No.
metric combination wrench set	945500093
right-angle metric hex key set	945500094
screwdriver	945500298
steel shims (4) 25.4x25.4x.127-.254 mm (1x1x.005 -.010 in.)	N/A

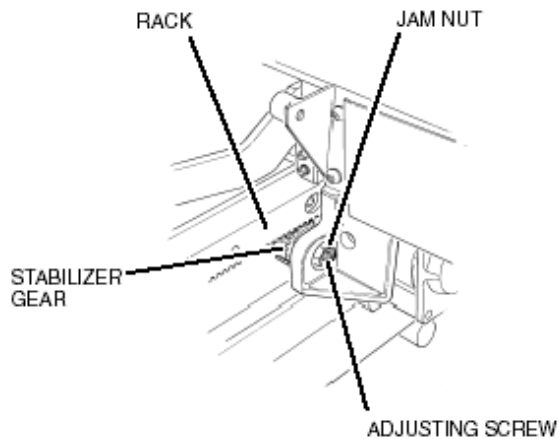
Adjust

1. Turn off the C-200 controller.
2. Remove the BCP cover (page 42).
3. Remove the remote-end cover (page 43).
4. Loosen the left-side X-axis drive belt tension.
 - a) Loosen the adjusting screw for the left-side tension plate.
 - b) Move the tension plate to loosen the left-side X-axis drive belt.
 - c) Tighten the adjusting screw.
5. Loosen the right-side X-axis drive belt tension.
 - a) Loosen the adjusting screw for the right-side tension plate.
 - b) Move the tension plate to loosen the right-side X-axis drive belt.
 - c) Tighten the adjusting screw.



6. Loosen both left-side stabilizer gears:

- a) Use an open-end wrench to loosen the jam nut on each stabilizer gear.
- b) Use a screw driver to turn the adjusting screw clockwise and increase gear backlash on each stabilizer gear.

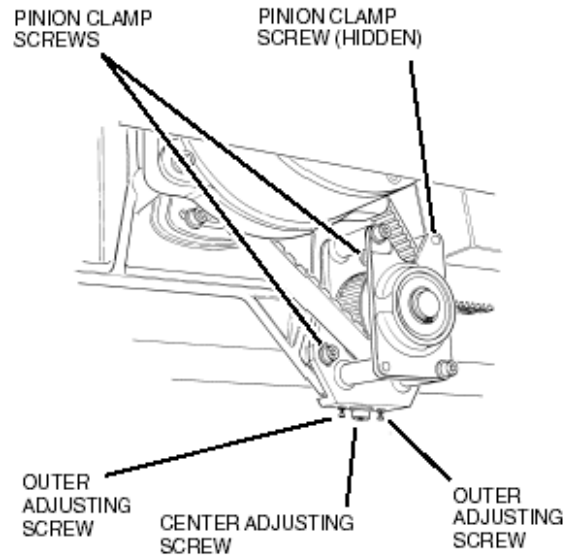


7. Repeat steps 6a and 6b to loosen the right-side stabilizer gears.

8. Loosen the right-side pinion gear:

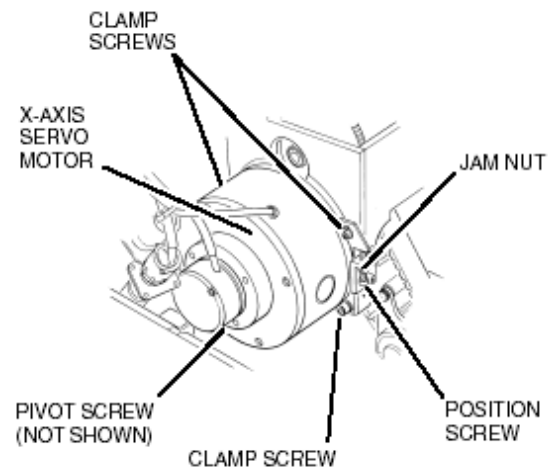
- a) Loosen the three pinion clamp screws.
- b) Loosen the locknuts on the outer adjusting screws.
- c) Turn the outer adjusting screws three full turns counterclockwise.
- d) Turn the center adjusting screw three full turns counterclockwise.

9. Repeat steps 8a through 8d for the left-side pinion gear.



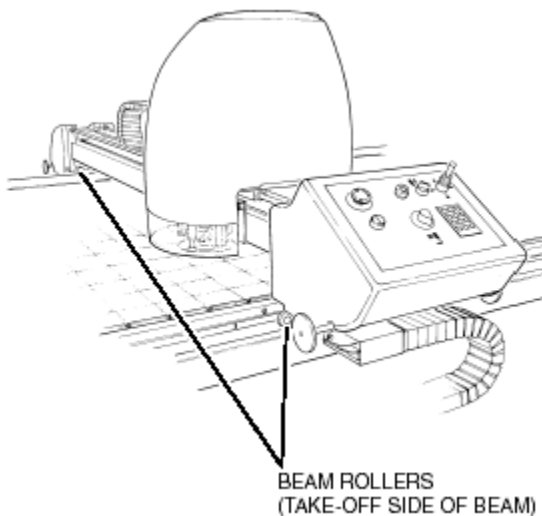
10. Loosen the X-axis servo-motor gear.

- a) Loosen the X-axis servo-motor clamp screws by turning them counterclockwise.
- b) Loosen the X-axis servo-motor pivot screw.
- c) Loosen the jam nut on the X-axis servo motor position screw.
- d) Turn the position screw six full turns counterclockwise.
- e) Move the X-axis servo motor down.



Note: There are four beam rollers. Two are on the BCP-end of the beam and two are on the remote-end of the beam. Two are on the take-on side of the beam and two are on the take-off side of the beam.

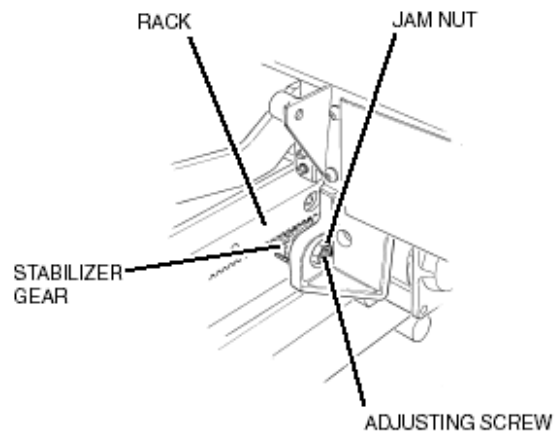
11. Put one steel shim (0.127 to 0.254 mm or 0.005 to 0.010 in. thick) on the top surface of the rack next to each beam roller. Make sure all shims are either on the take-on side of all beam rollers or all shims are on the take-off side of all beam rollers.



12. Push the beam so that each beam roller is on top of a shim.

13. Adjust both right-side stabilizer gears.

- a) Use a screwdriver to turn both adjusting screws clockwise until each stabilizer gear is snug with the rack.
- b) Use an open-end wrench to tighten both jam nuts.



14. Repeat steps 13a through 13b to adjust the left-side stabilizer gears.

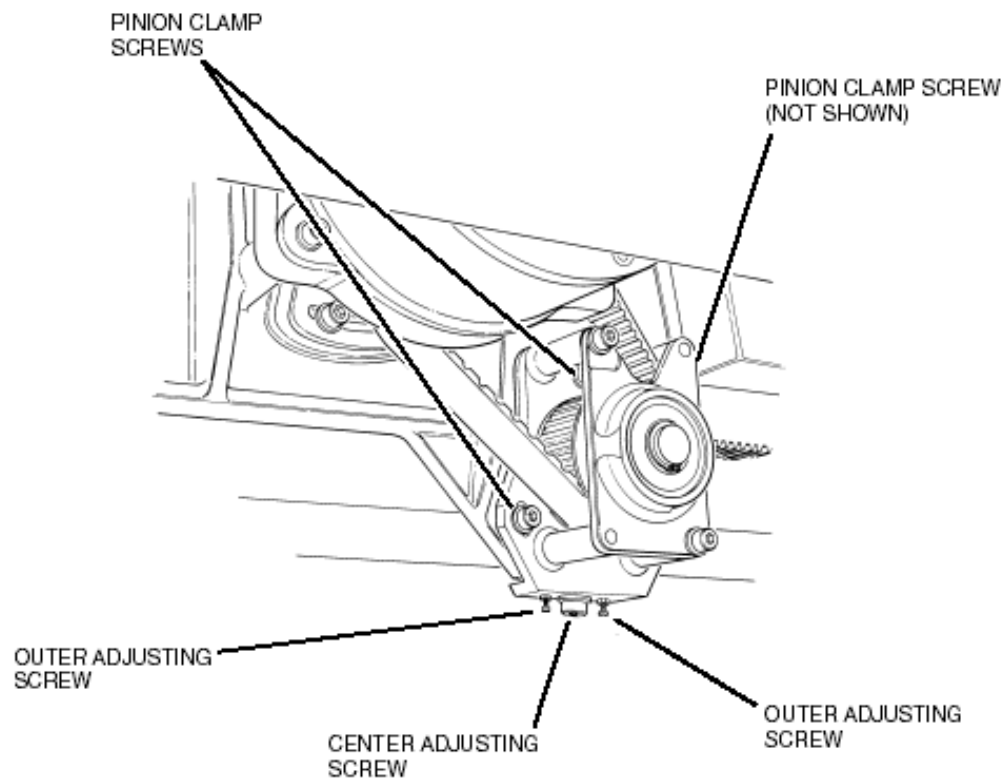
15. Adjust the left-side pinion gear.

CAUTION: To prevent damage to the pinion gear, do not over-tighten the center adjusting screw.

- a) Using very light pressure, push the pinion gear up until it just meshes with the rack.
- b) Tighten the three pinion clamp screws.
- c) Turn the center adjusting screw clockwise until the lock washer flattens out.

- d) Turn the outer adjusting screws counterclockwise until they stop.
- e) Tighten the locknuts on the outer adjusting screws.

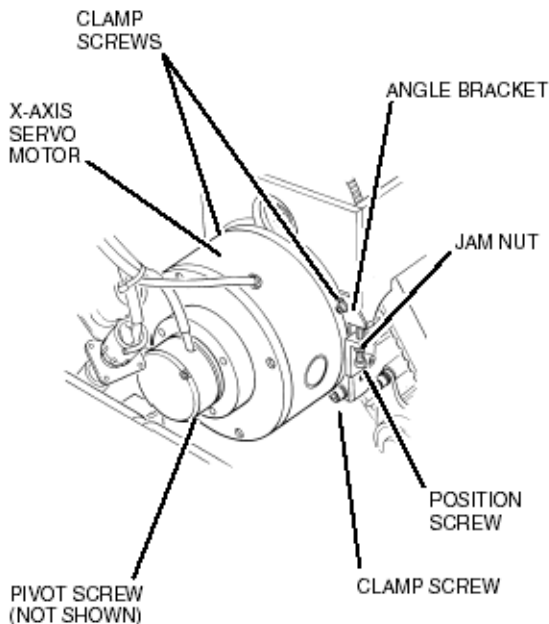
16. Repeat steps 15a through 15e for the right-side pinion gear.



17. Move the beam and rollers off the shims.
18. Remove the shims. There should be 0.076 to 0.102 mm (0.003 to 0.004 in.) of backlash between each pinion gear and the rack and between each stabilizer gear and the rack.
19. Adjust the X-axis drive belts (page 32).
20. Adjust the X-axis servo-motor gear.
 - a) Hook a spring scale onto the angle bracket.
 - b) Pull up on the spring scale until the scale measures 4.55 kg (10 lb.).

CAUTION: You must hold the servo-motor with the spring scale while you tighten the clamp screws. Otherwise, the servo-motor may move out of position.

- c) Tighten the pivot screw and the clamp screws while holding the servo-motor with the spring scale.



- d) Remove the spring scale.

CAUTION: Do not over-tighten the servomotor position screw. If you do you may damage the servomotor gear and servo motor.

- e) Turn the X-axis servo-motor position screw until it barely touches the angle bracket.
- f) Loosen the position screw 1/8 to 1/4 turn counterclockwise.
- g) Tighten the position screw jam nut.
- h) Loosen the pivot and clamp screws. This lets the angle bracket for the X-axis servo motor rest on the position screw.
- i) Tighten the pivot and clamp screws by turning them clockwise.
- j) Manually move the beam along the entire rack (to both the take-off and take-on ends of the cutting table). The beam should move smoothly, without binding.

21. If the beam does not move smoothly:

- a) Repeat step 10a through 10d to loosen the servo motor.
- b) Repeat steps 20a through 20j to adjust and check the X-axis servomotor gear.

OR

22. If the beam does move smoothly:

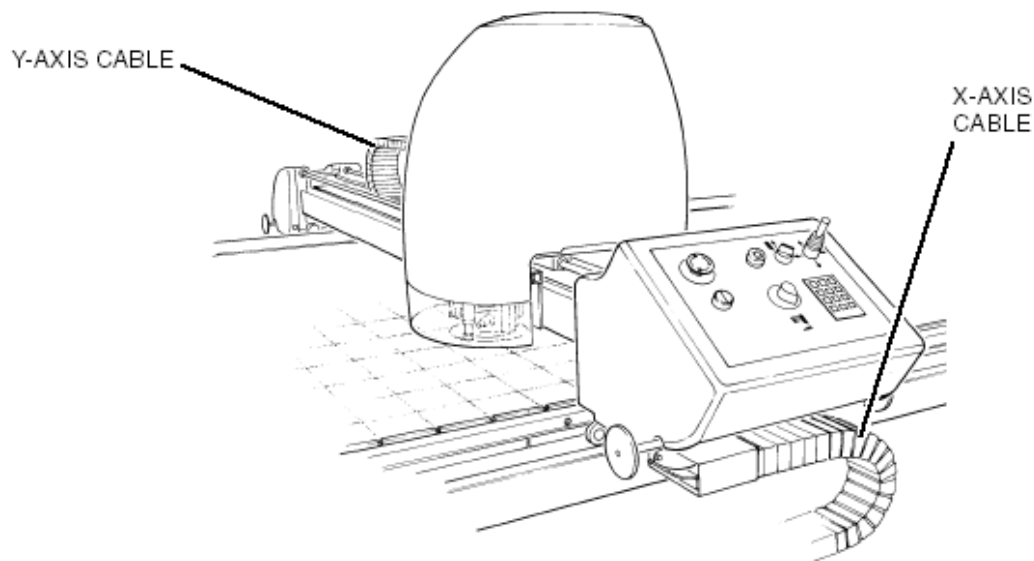
- a) Install the BCP cover (page 42).
- b) Install the remote-end cover (page 43).
- c) Turn on the C-200 controller.

Cables

The cables provide power and cutting commands for the cutter head and beam. They are protected with plastic link armor.

Inspect

1. Move the beam to the take-off end of the cutting table.
2. Remove left side cover panels as needed to see the entire X-axis cable (page 135).
3. Move the beam to the take-on end of the cutting table.
4. Turn off the C-200 controller.
5. Inspect the X and Y-axis cables for fraying or breaks and the armor for broken links. If you see any defects, have them repaired or replaced by a Gerber qualified technician.
6. Install any side cover panels that you removed (page 135).
7. Turn on the C-200 controller.



BCP Cover

The beam control panel (BCP) cover encloses the moving parts and electronics inside the BCP.

Required Tools and Supplies

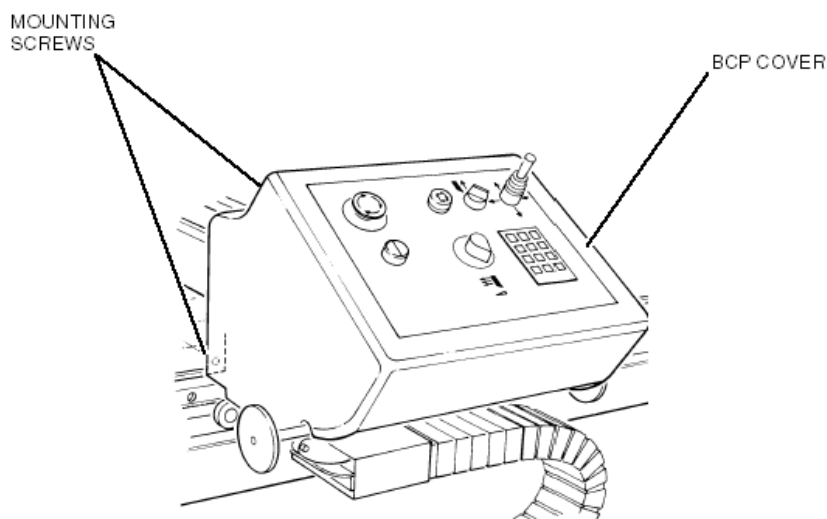
Item	Gerber Part No.
right-angle metric hex key set	945500094

Remove

1. Turn off the C-200 controller.
2. Remove the screws that retain the BCP cover.
3. Pull the BCP cover up and sideways away from the GERBERcutter.

Install

1. Slide the BCP cover down and sideways over the beam control panel. Be careful not to pinch any wires or tubing.
2. Install the screws that retain the BCP cover.
3. Turn on the C-200 controller.



Remote-End Cover

This cover encloses the moving parts and electrical components on the remote-end of the beam.

Required Tools and Supplies

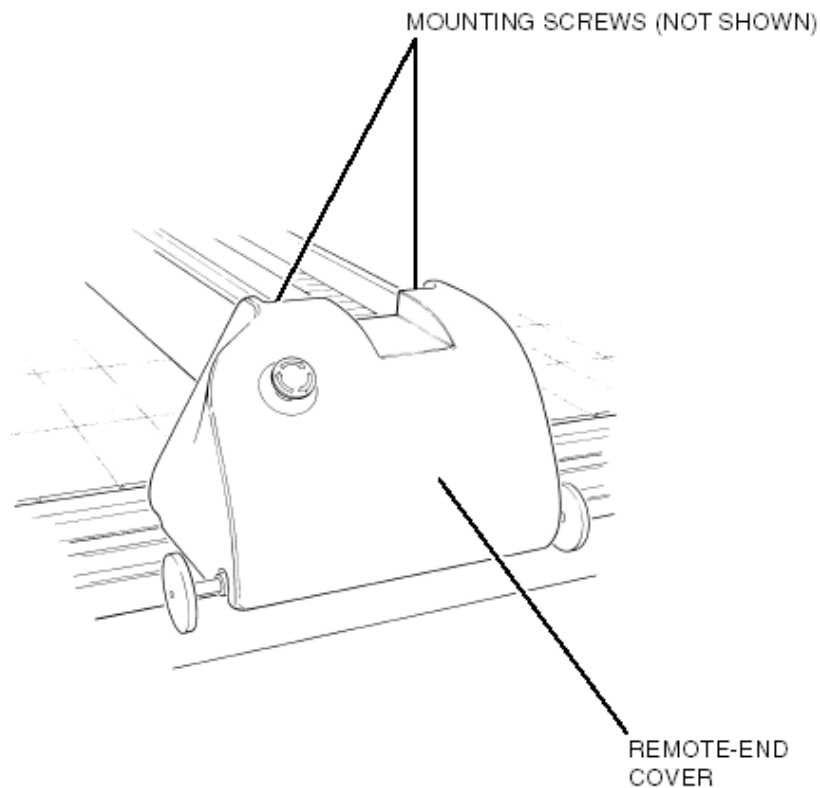
Item	Gerber Part No.
right-angle metric hex key set	945500094

Install

1. Slide the remote-end cover straight down over the beam control panel. Be careful not to pinch any wires.
2. Install the remote-end cover mounting screws.
3. Turn on the C-200 controller.

Remove

1. Turn off the C-200 controller.
2. Remove the remote-end cover mounting screws.
3. Remove the remote-end cover by lifting it straight up.



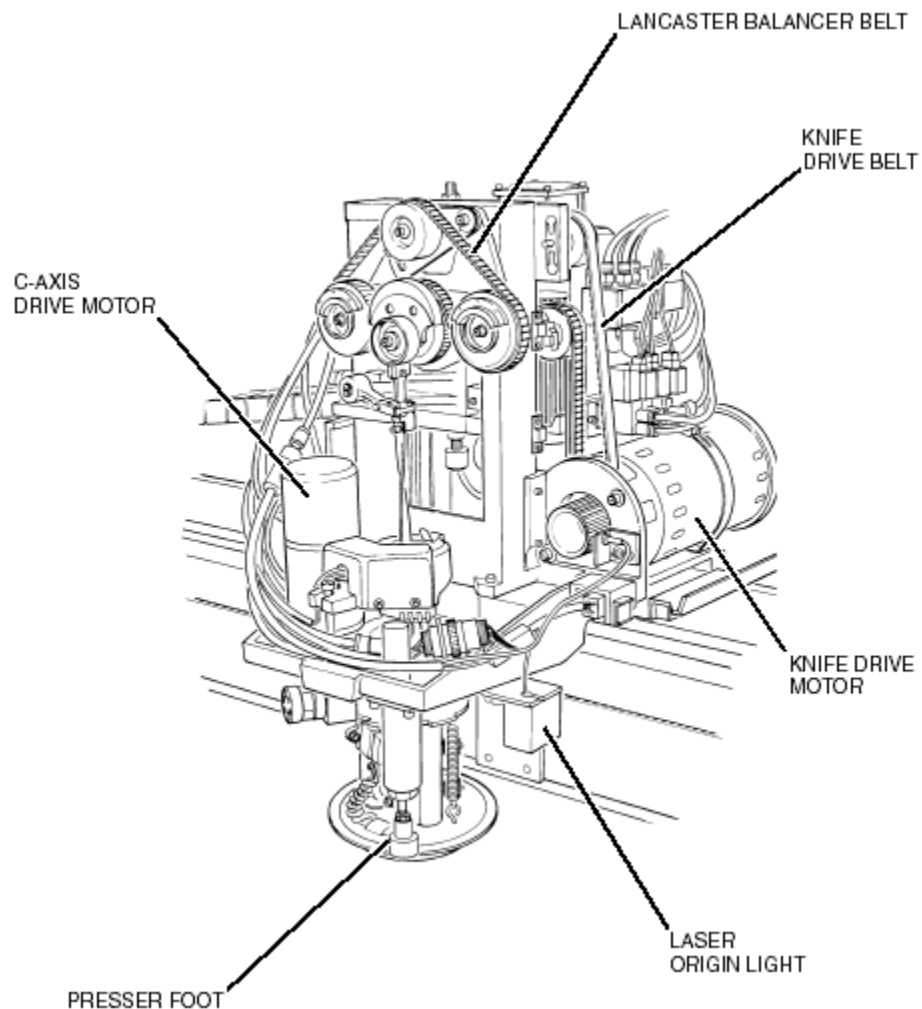
Cutter Head Maintenance

The cutter head contains the knife and does the actual cutting of material. It moves across the cutting table on the beam.

Cutter Head for GT 5250

Maintenance procedures specifically for the GT 5250 cutter head begin on page 49. Sections describing how to remove the front and back covers for both the GT 5250 and GT 7250 systems begin on page 46.

Note: Maintenance procedures for the GT 7250 cutter head begin on page 77.



GT 5250 Cutter Head

Front Cover

The cutter head front cover encloses the cutter head. It protects the cutter head from damage and the operator from injury. The cutter head front cover is hinged so that you can lift the front of the cover out of the way to do maintenance. For some procedures, though, you may want to remove the entire cover.

Required Tools and Supplies

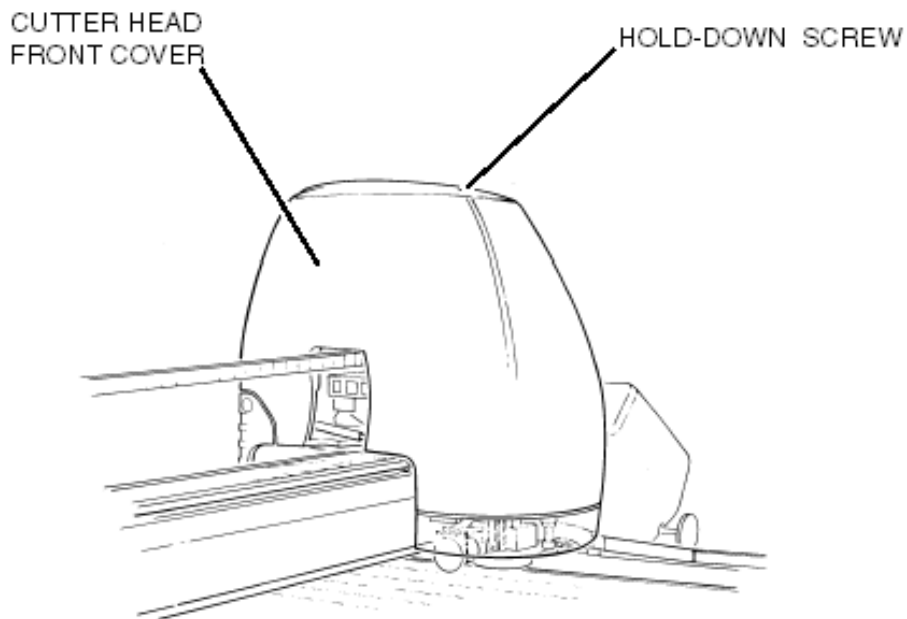
Item	Gerber Part No.
metric T-handle hex key set	945500089
appropriate standard T-handle hex key	944200406

Open

1. Turn off the C-200 controller.
2. Remove the hold-down screw from the top of the cutter head front cover.
3. Lift the cutter head front cover.

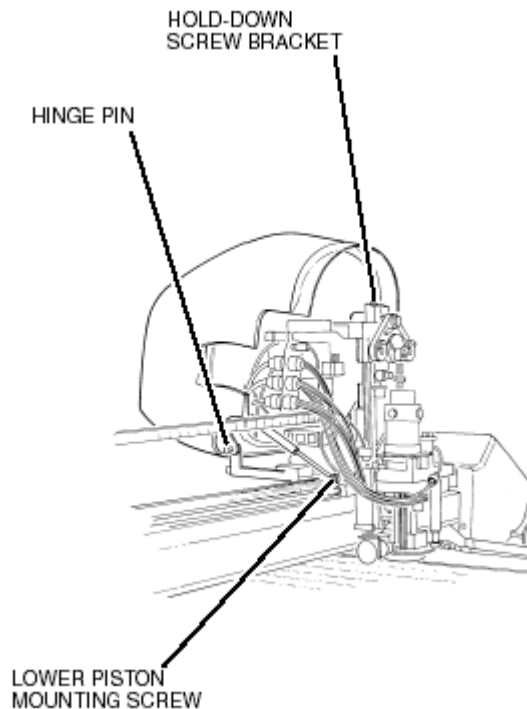
Close

1. Lower the cutter head front cover.
2. Install the cutter head cover hold-down screw.
3. Turn on the C-200 controller.

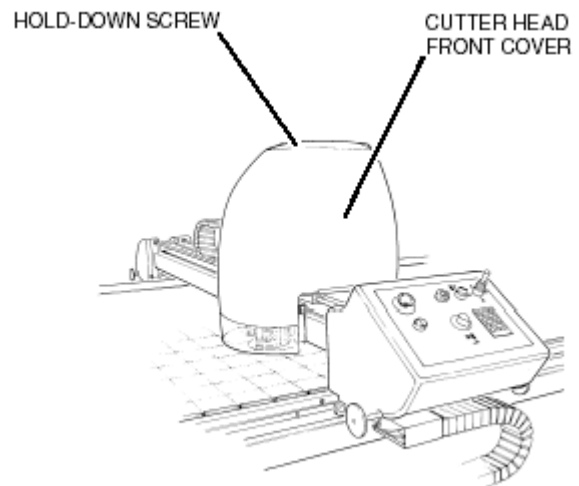


Remove

1. Turn off the C-200 controller.
2. Remove the hold-down screw from the top of the cutter head cover.
3. Open the cutter head cover (page 46).
4. Remove the lower piston mounting screw.
5. Remove the two hinge pins.
 - a) Push in the release button in the center of the hinge-pin head.
 - b) While holding in the release button, pull the hinge pin from the cutter head cover.
 - c) Repeat steps 5a and 5b for the second hinge pin.
6. Lift the head cover from the cutter head.

**Install**

1. Lower the cutter head cover onto the cutter head.
2. Install the two hinge pins.
 - a) Push in the release button in the center of the hinge-pin head.
 - b) While holding in the release button, push the hinge pin into the hinge pin holes in the cutter head cover and its mounting bracket.
 - c) Repeat steps 2a and 2b for the second hinge pin.
7. Install the lower piston mounting screw.
8. Close the cutter head cover (page 46).
9. Install the hold-down screw into the top of the cutter head cover.
10. Turn on the C-200 controller.



Back Cover

The cutter head back cover encloses the part of the cutter head that contains the drill. It protects the drill from damage and the operator from injury. You may need to remove this cover to do maintenance on the cutter head and drill.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
appropriate standard T-handle hex key	944200406

Remove

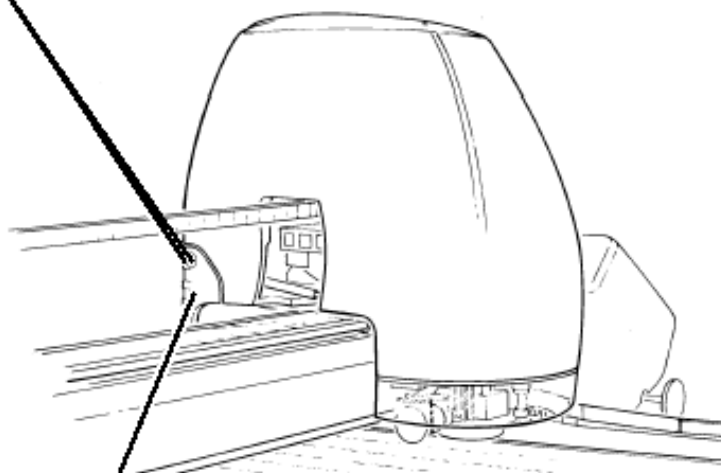
1. Turn off the C-200 controller.
2. Remove the mounting screws from the cutter head back cover.
3. Remove the cutter head back cover.

Install

1. Align the screw holes in the cutter head back cover with those in the mounting bracket.
2. Install the cutter head back cover mounting screws.
3. Turn on the C-200 controller.

MOUNTING SCREWS

CUTTER HEAD
BACK COVER



GT 5250 C-axis Assembly

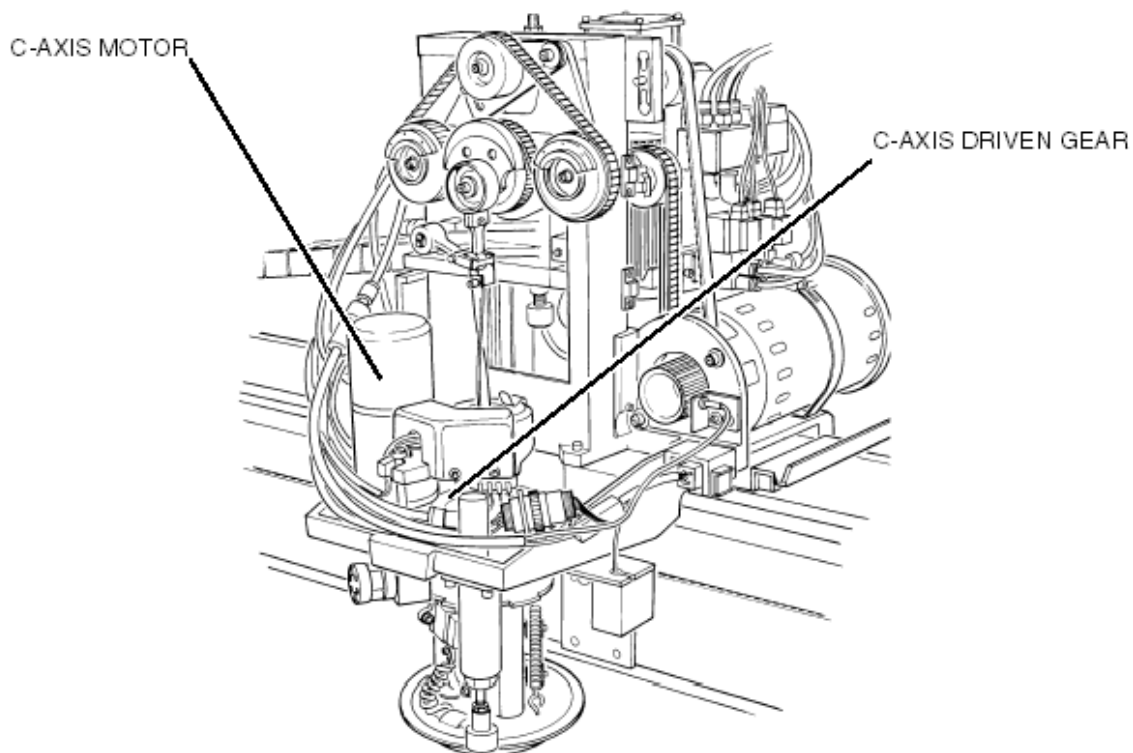
The C-axis assembly includes the C-axis motor, motor gear, and C-axis driven gear. The C-axis assembly turns the knife blade to cut corners, notches, curves, and other complex shapes.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
appropriate standard T-handle hex key	944200406

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Manually turn the C-axis assembly several times to the right and left. It should turn freely without binding or noise.
4. If you find any defects, have the assembly repaired or replaced by a Gerber qualified technician.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



GT 5250 C-axis Driven Gear

The C-axis driven gear is driven by the C-axis motor gear. The C-axis driven gear turns the knife blade about the C-axis. This gear must be in good condition and synchronized with the C-axis motor gear for accurate cutting.

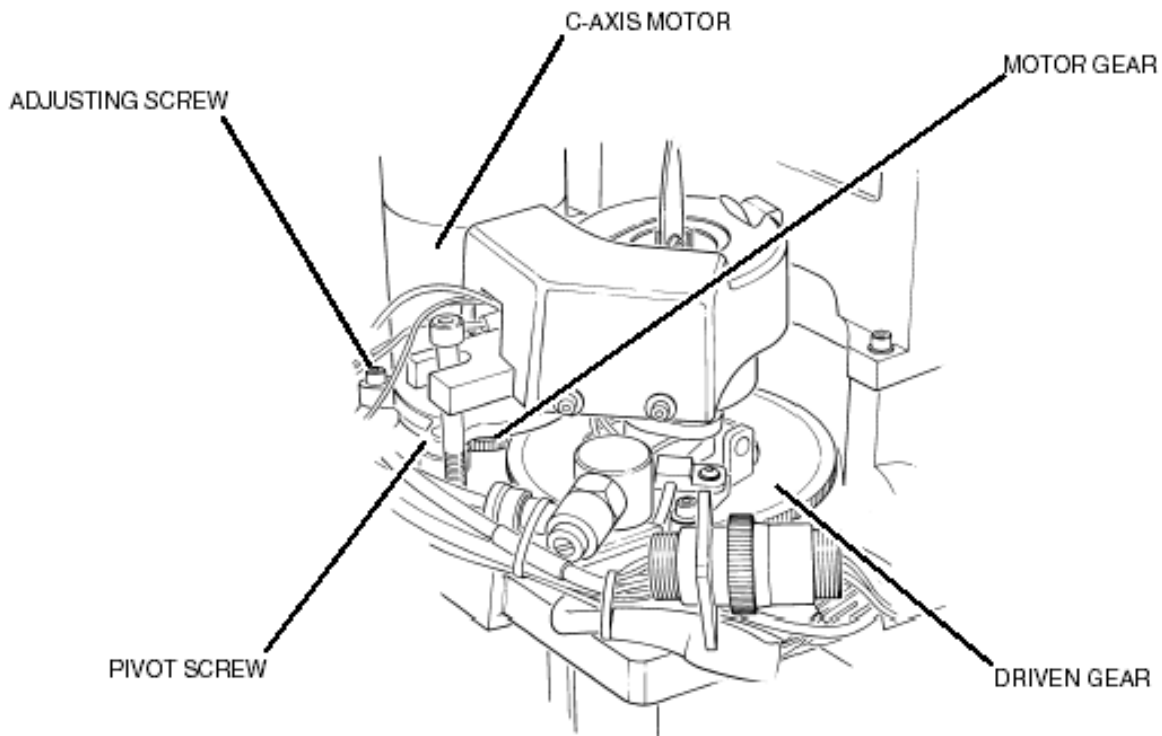
Required Tools and Supplies

Item	Gerber Part No.
right-angle metric hex key set	945500094
spring scale, 0 - 23 kg (0 - 50 lb.)	945500252
appropriate standard right-angle hex key	

Inspect

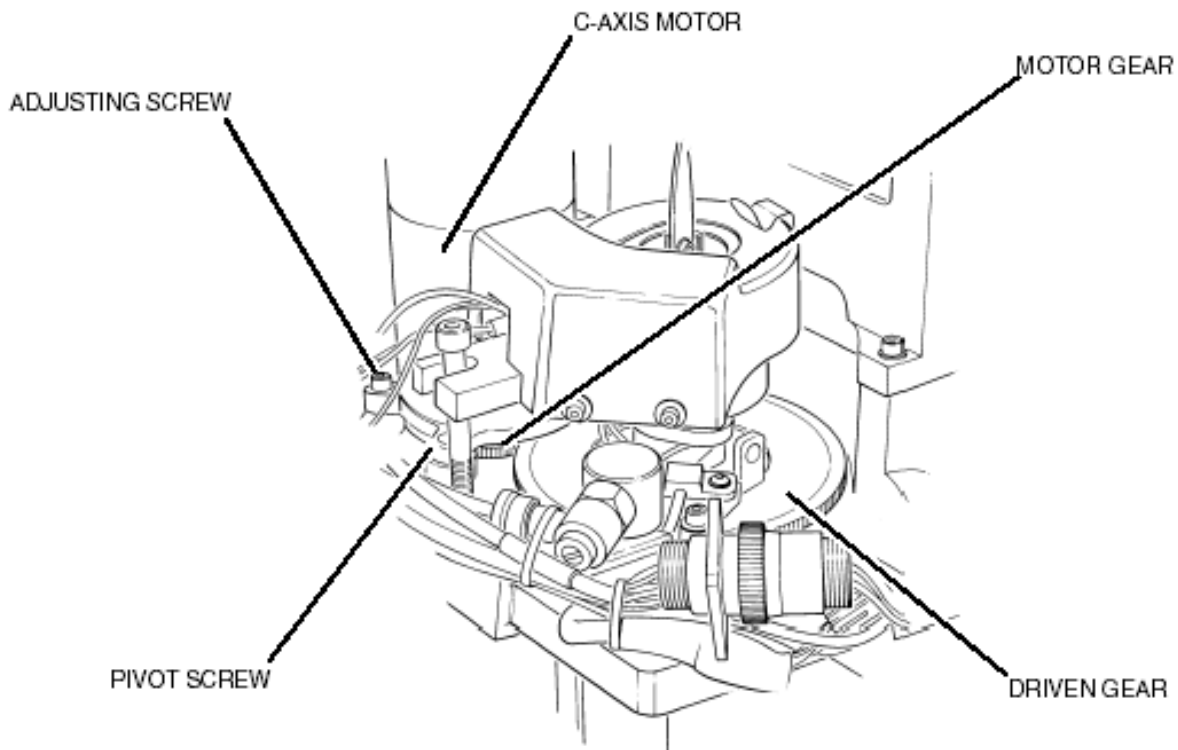
1. Turn off the C-200 controller.

2. Open the cutter head front cover (page 46).
3. Inspect the driven gear for damage and wear. If you see any defects, have it replaced by a Gerber qualified technician.
4. Check the driven gear for backlash. There should be a small amount of backlash. If there is insufficient backlash, too much backlash, or if the amount of backlash is different than usual, adjust the backlash.
5. Close the cutter head cover (page 46).
6. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Loosen the two adjusting screws for the C-axis motor.
4. Loosen the pivot screw for the C-axis motor.
5. Slide the C-axis motor toward the driven gear to decrease backlash.
6. Slide the C-axis motor away from the driven gear to increase backlash.
7. Tighten the two C-axis motor adjusting screws.
8. Tighten the C-axis motor pivot screw.
9. Check the backlash of the driven gear.
10. If the driven gear backlash is not correct, repeat steps 3 through 9 until the backlash is correct.
11. Close the cutter head front cover (page 46).
12. Turn on the C-200 controller.



GT 5250 Knife Drive Belt

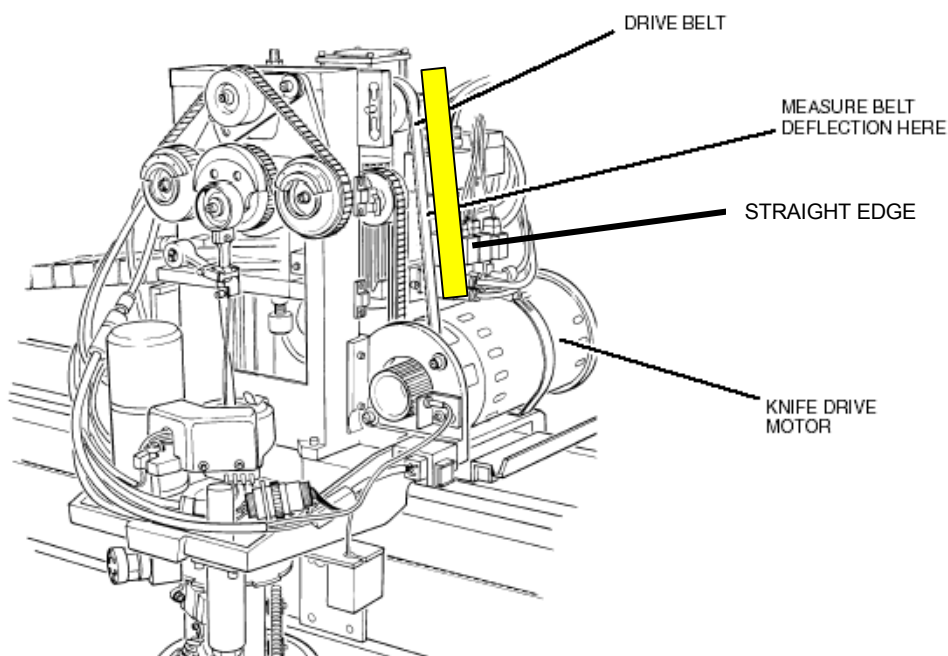
The knife drive belt connects the knife drive motor to the knife crankshaft through a series of pulleys. As the knife drive motor turns, the drive belt transfers its motion to the knife crankshaft. The drive belt must be in good condition and properly tensioned to give the best cutting performance.

Required Tools and Supplies

Item	Gerber Part No.
belt deflection gauge	45412000
right-angle metric hex key set	945500094
appropriate standard T-handle hex key	
knife drive belt	Refer to Service CD

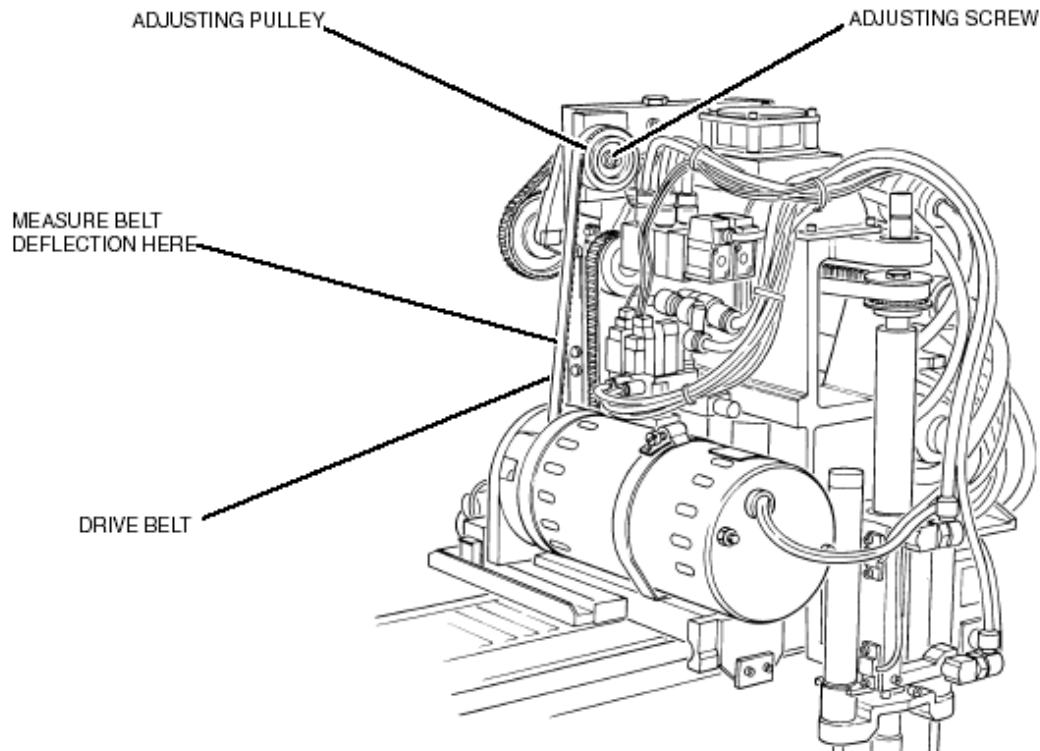
Inspect

- Turn off the C-200 controller.
- Open the cutter head front cover (page 46).
- Check the knife drive belt for cracks, nicks, cuts, and glazing. If you see any of these defects, have it replaced by a Gerber qualified technician.
- Check tension for the knife drive belt.
 - Set the collar on the belt deflection gauge for 6.35 mm (0.25 in.).
 - Align a straight edge parallel with the knife drive belt.
 - Holding the belt deflection gauge perpendicular to the straight edge, push the belt deflection gauge until the collar meets evenly on the straight edge.
 - Measure the tension of the knife drive belt with the belt deflection gauge at this point. The deflection should be 6.35 mm (0.25 in.) at 2.73 to 3.64 kg (6 to 8 lb).
- If the belt is tensioned incorrectly, adjust the belt tension (page 53).
- Close the cutter head front cover (page 46).
- Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
 2. Open the cutter head front cover (page 46).
 3. Adjust the drive belt.
 - a) Loosen the adjusting screw on the adjusting pulley.
 - b) To loosen the drive belt, slide the adjusting pulley towards the knife drive motor.
 - d) Measure the tension of the drive belt with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 2.73 to 3.64 kg (6 to 8 lb.).
 - e) If the belt tension is incorrect, repeat steps 3a through 3d until the belt is tensioned correctly.
 - f) Close the cutter head front cover (page 46).
 4. Turn on the C-200 controller.
- OR**
- c) To tighten the drive belt, slide the adjusting pulley away from the knife drive motor.



GT 5250 Lancaster Balancer

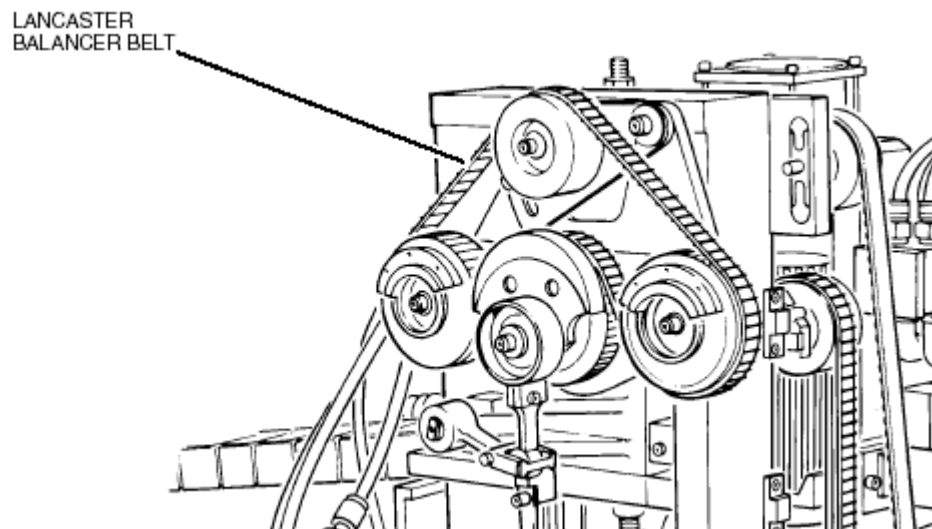
The Lancaster balancer reduces vibration and noise made by the knife drive. As the knife drive motor turns, the Lancaster balancer belt and pulleys also turn. This belt must be in good condition and properly tensioned for the Lancaster balancer to work properly.

Required Tools and Supplies

Item	Gerber Part No.
spring scale, 0 - 23 kg (0 - 50 lb.)	945500252
right-angle metric hex key set	945500094
appropriate standard T-handle hex key	
Lancaster Balancer belt	Refer to Service CD

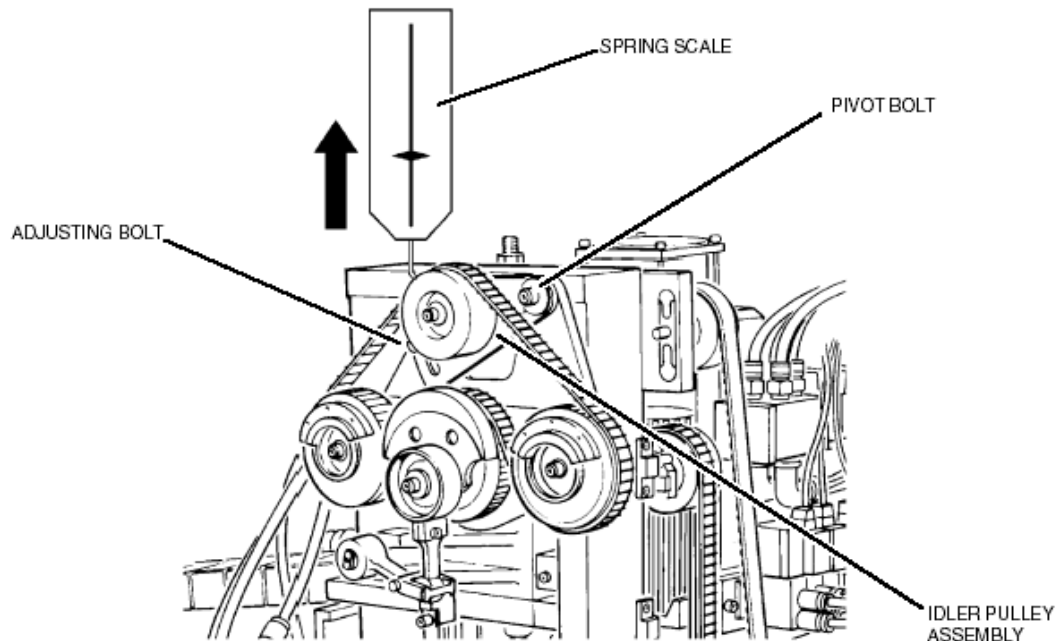
Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Check both sides of the Lancaster balancer belt for cracks, nicks, cuts, and glazing.
4. If you see any of these defects, have it replaced by a Gerber qualified technician.
5. Visually inspect the belt for proper tension.
6. If the belt is tensioned incorrectly, adjust the belt tension (page 55).
7. Close the cutter head front cover (page 46).
8. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Loosen the adjusting bolt and the pivot bolt on the idler pulley assembly.
4. Push the idler pulley assembly down to release tension on the Lancaster balancer belt.
5. Hook the end of the spring scale to the adjustable end of the idler pulley assembly.
6. Pull up on the spring scale until it reads 2.27 kg (5 lbs.).
7. Tighten the adjusting bolt and pivot bolt.
8. Close the cutter head front cover (page 46).
9. Turn on the C-200 controller.



GT 5250 Knife Blade

The knife blade cuts material. It must be sharp and within wear limits to cut properly.

WARNING!  The knife blade is very sharp. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
Petro-Lon multipurpose grease	596500005
appropriate standard T-handle hex key	944200406
screwdriver	945500298
knife blade	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Check the knife blade for excessive wear.
4. If the leading edge of the knife blade is damaged, replace the knife blade (page 57).
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



OK



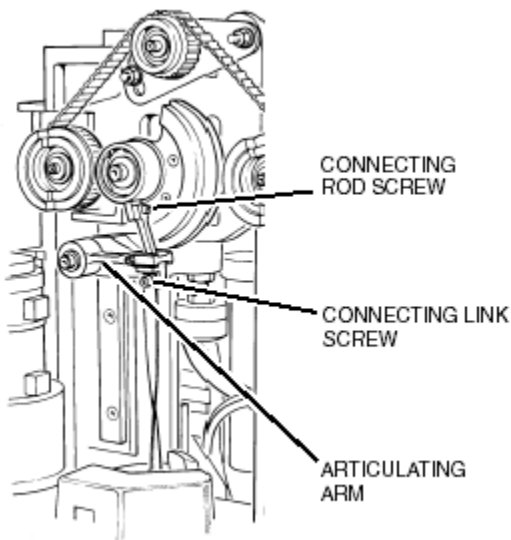
UNACCEPTABLE

Remove

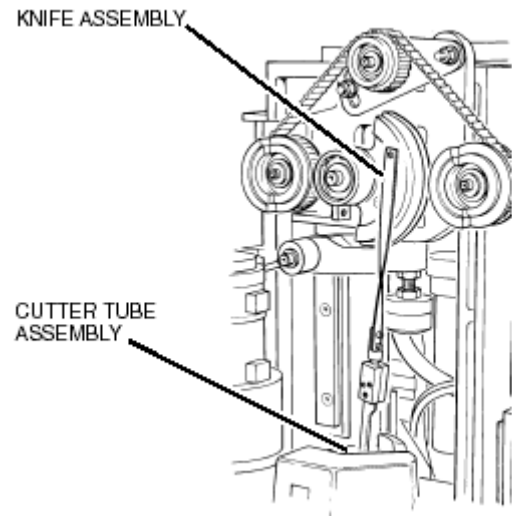
1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

WARNING!  The knife blade is very sharp and can cause injury. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

3. Remove the connecting link screw.
4. Remove the connecting rod screw.



5. Push the knife assembly down.
6. Move the articulating arm out of the way.
7. Pull the knife blade assembly up from the cutter tube assembly.



WARNING!



The knife blade is very sharp and can cause injury. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

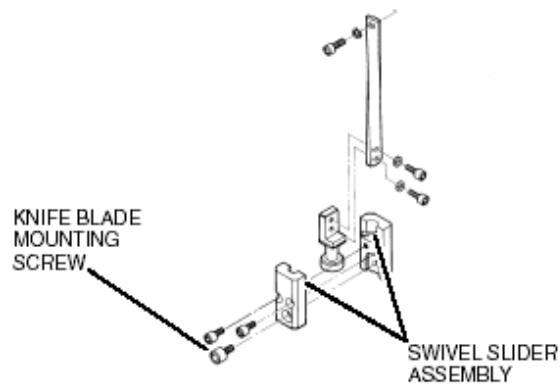
8. Remove the knife blade from the swivel slider assembly.
 - a) Remove the knife blade mounting screw.
 - b) Pull the knife blade from the swivel slider assembly.

Install

1. Fill the inside of the swivel slider assembly with Petro-Lon multipurpose grease. Put enough Petro-Lon into the assembly so that, when you install the knife blade mounting screw, a small amount of Petro-Lon is squeezed out of the assembly.

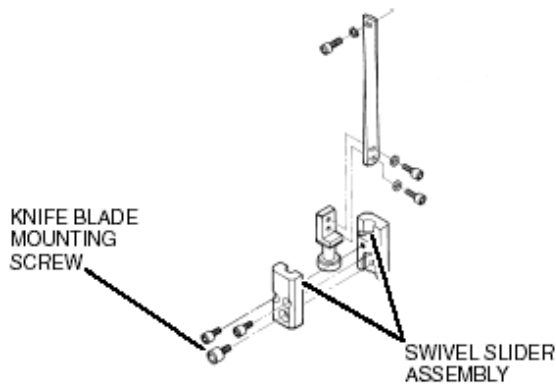
Note: You can also fill the inside of the swivel assembly after you install the knife blade. To do this, use a grease gun with a needle fitting to put Petro-Lon through the greasing hole in the swivel. Stop adding Petro-Lon as soon as the Petro-Lon starts to leak out of the assembly.

2. Put the knife blade into the swivel slider assembly.



WARNING!

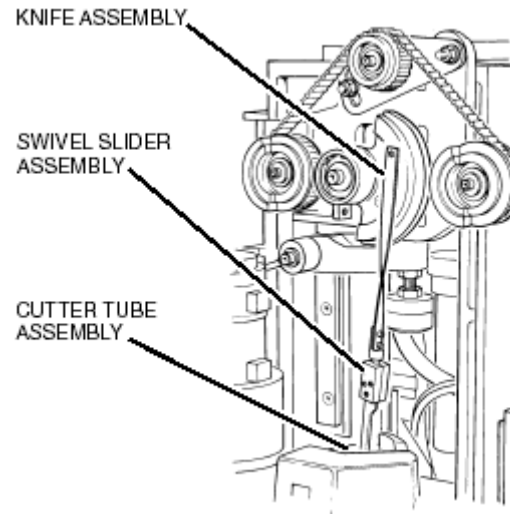
The knife blade is very sharp and can cause injury. Handle the knife blade carefully. Never hold the knife blade by its sharp end.



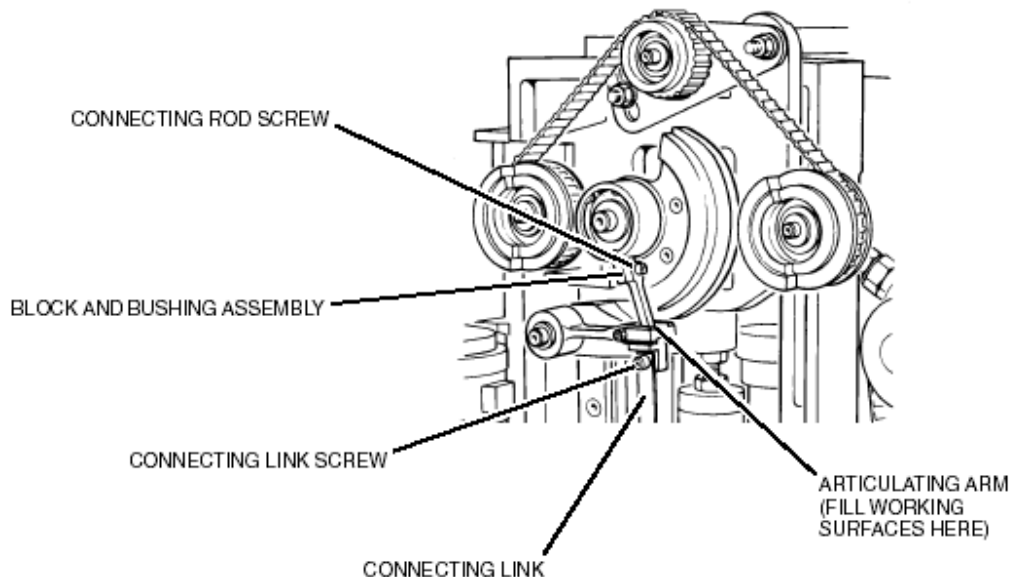
3. Install the knife blade mounting screw.

Note: The hole in the knife blade for the mounting screw is offset. You can only mount the knife blade one way.

4. Put a thin (1 mm or 0.040 in.) coat of Petro-Lon onto the outside of the swivel slider assembly.
5. Insert the knife assembly into the cutter tube assembly. Make sure the knife is straight.



6. Align the screw hole in the connecting rod with the screw hole in the block and bushing assembly.
7. Install the connecting rod screw.
8. Align the screw hole in the connecting link with the screw hole in the block and bushing assembly.
9. Install the connecting link screw.
10. Put Petro-Lon into the working surface of the articulating arm. Put in enough Petro-Lon to fill all the space in the arm. Do not overfill the arm so that Petro-Lon is squeezed onto the outside of the articulating arm.
11. Close the cutter head front cover (page 46).
12. Turn on the C-200 controller.
13. Check knife operation.
 - a) At the BCP, set **KNIFE** to **ON**.
 - b) Check that the knife moves smoothly and with no unusual noise.
 - c) At the BCP, set **KNIFE** to **OFF**.
14. Use the New Knife command from the C-200MT program to reset knife wear logic.
15. Sharpen the knife blade several times to check the position of the knife blade.



GT 5250 Presser Foot

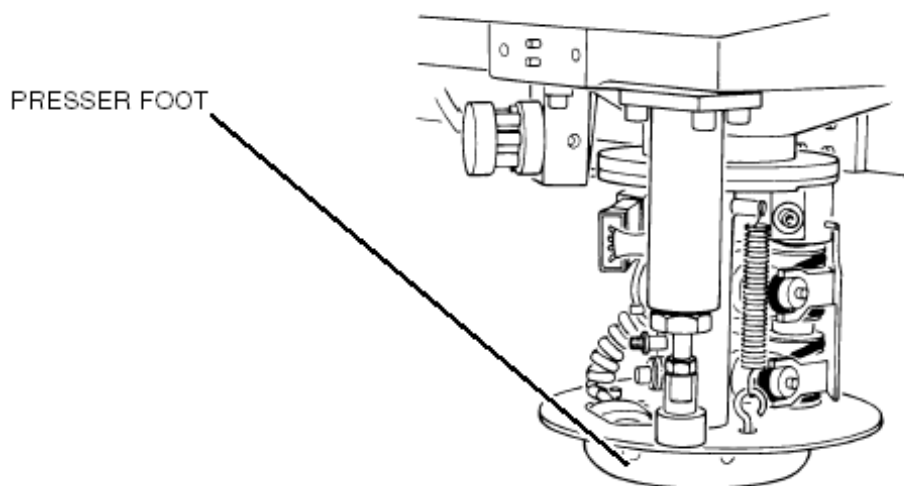
The presser foot flattens the top of the spread, or lay, as the knife blade cuts it.

Required Tools and Supplies

Item	Gerber Part No.
grease gun	944233101
clean shop rag	N/A
Mobil 1 5w-30 synthetic oil	596118001
right-angle metric hex key set	945500094
appropriate standard right-angle hex keys	945500046, 945500255

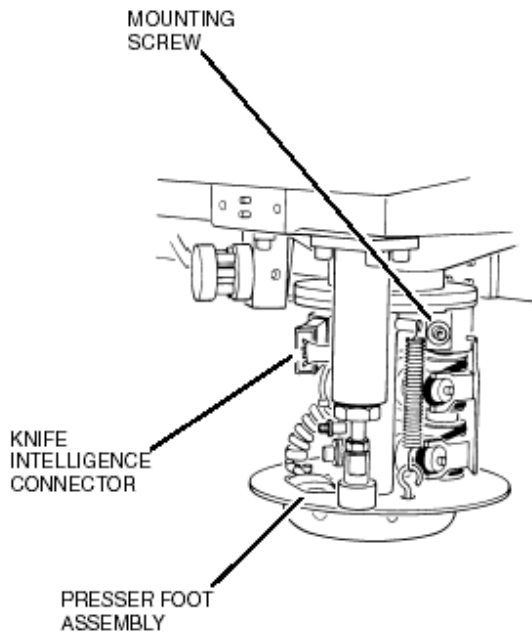
Inspect

- Turn off the C-200 controller.
- Open the cutter head front cover (page 46).
- Check the presser foot shafts for dirt and debris. Clean them if needed.
- Check the shafts for bends, nicks, and gouges. If you find any of these defects, have them repaired or replaced by a Gerber qualified technician.
- Check the knife intelligence wire for any breaks, nicks, and gouges. If you find any of these defects, have it replaced by a Gerber qualified technician.
- Check the presser foot for binding.
 - At the operator console, decrease the presser foot bowl pressure to 0 kgf/cm² (0 psi).
 - Slide the presser foot straight down. It should move smoothly without binding. Binding indicates a problem with the shafts and/or alignment of the lower blade guide.
 - If you notice any binding, have the assembly repaired or replaced by a Gerber qualified technician.
 - At the operator console, restore the presser foot bowl pressure to its original level.
- Close the cutter head front cover (page 46).
- Turn on the C-200 controller.



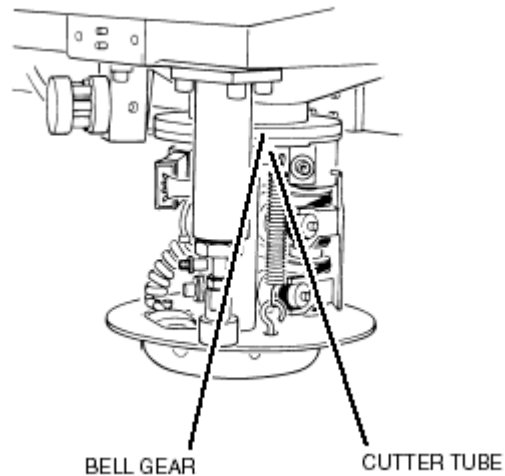
Clean/Lubricate

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife assembly (page 57).
4. Unplug the Knife Intelligence® (KI) sensor.

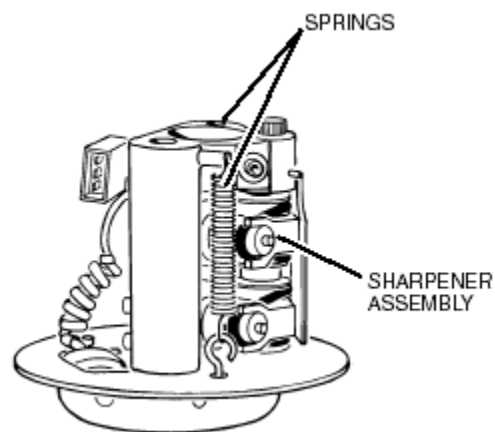


5. Loosen the mounting screw that holds the presser foot assembly on the cutter tube.
6. Lower the presser foot assembly and remove it from the cutter head.

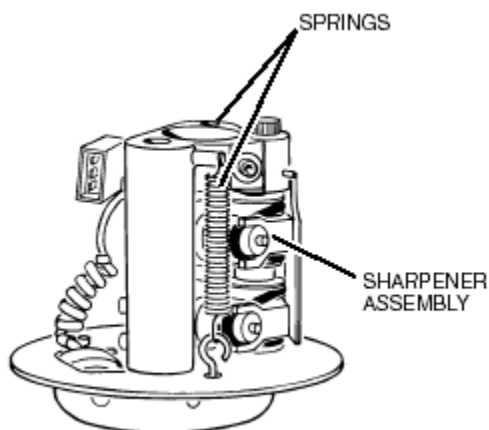
7. Use a clean rag dampened with denatured alcohol to clean the inside of the cutter tube and the bell gear.



8. Remove the Knife Intelligence connector bracket screws from the presser foot assembly.
9. Remove the two springs.

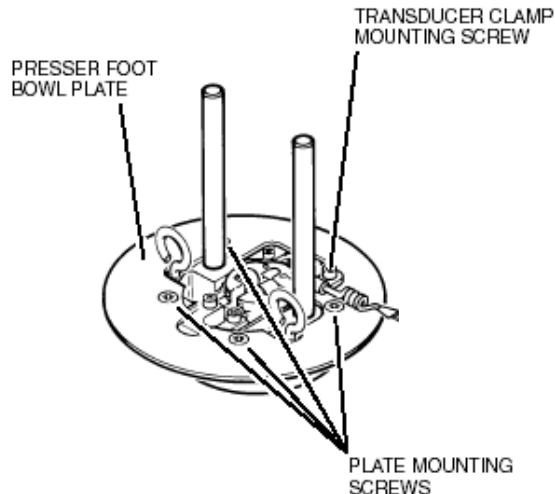


10. Lower the sharpener assembly off the presser foot assembly.



11. Remove the transducer cable clamp mounting screw.

CAUTION: Do not loosen the nuts on the back side of the sharpener assembly.

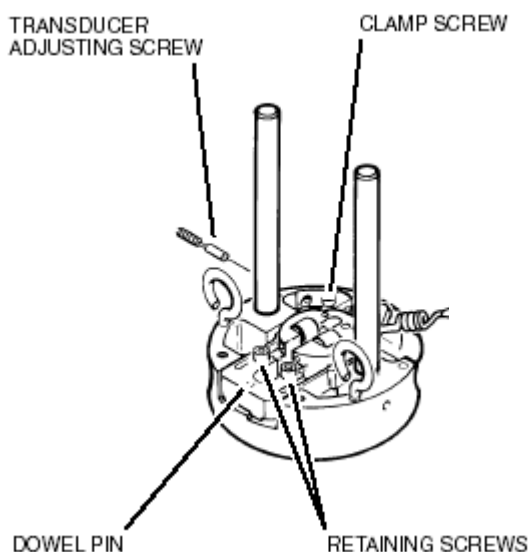


12. Remove the presser foot bowl plate mounting screws.

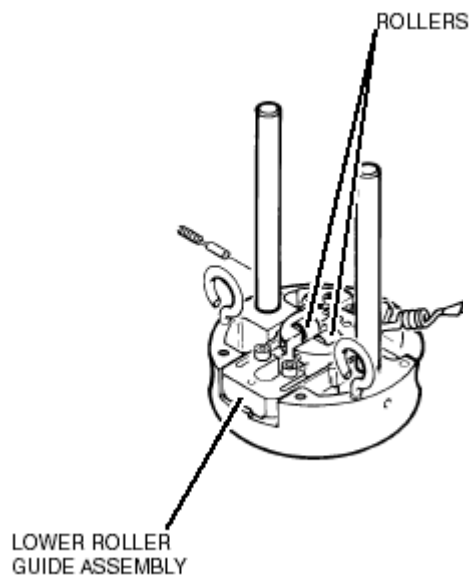
13. Remove the presser foot bowl plate.

14. Loosen the clamp screw by turning it counterclockwise.

15. Remove the transducer adjusting screw.



16. Lift out the lower roller guide assembly.
17. Dampen a lint-free clean cloth with denatured alcohol.
18. Clean the lower roller guide assembly with denatured alcohol.
19. Let the lower roller guide assembly dry completely.

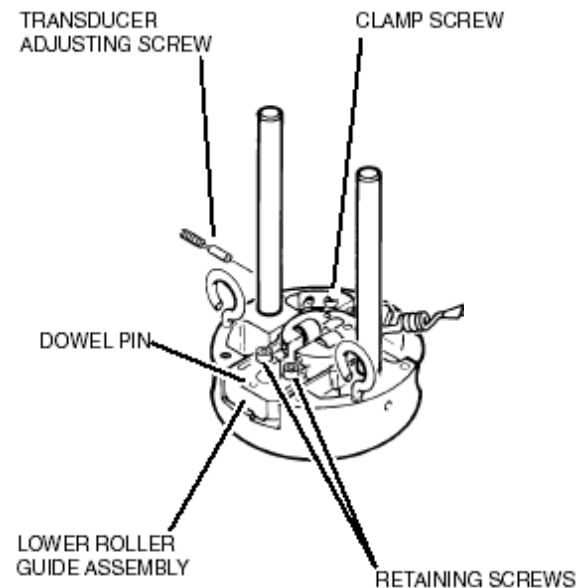


20. Inspect the rollers for wear and damage. Make sure they spin freely. If the rollers are worn or damaged, have them replaced by a Gerber qualified technician.

CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently, than to apply one heavy coat.

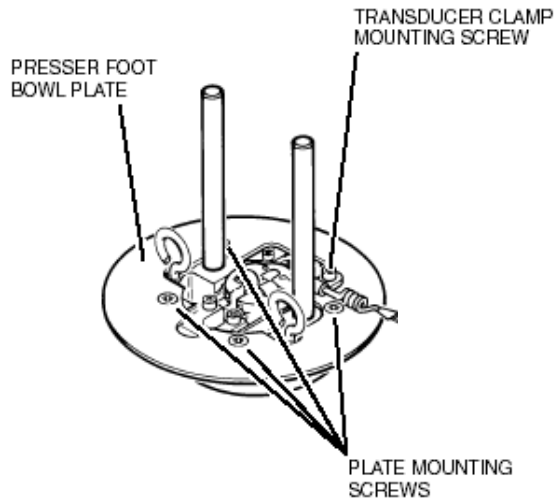
CAUTION: Only use synthetic oil. Non-synthetic oil can damage seals and bearings.

21. Lubricate the pins of the rollers with a drop of Mobil 1 5W-30 synthetic oil.
22. Put the lower roller guide assembly into the presser foot assembly.

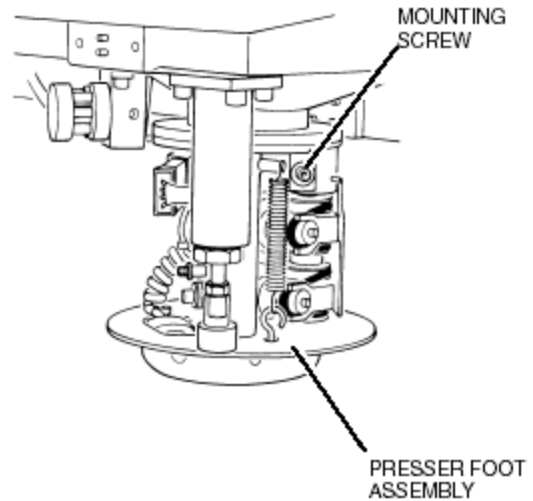


23. Install the lower roller guide onto the dowel pin.
24. Install the adjusting core.
25. Install the transducer adjusting screw.
26. Tighten the clamp screw.

27. Install the presser foot bowl plate mounting screws.



28. Install the transducer clamp mounting screw.
29. Put the sharpener assembly onto the presser foot bowl plate.
30. Install the two springs.
31. Install the Knife Intelligence connector bracket screws.
32. Install the presser foot assembly into the cutter head.



33. Tighten the mounting screw that holds the presser foot assembly on the cutter tube.
34. Plug in the Knife Intelligence sensor.
35. Install the knife blade (page 58).
36. Close the cutter head front cover (page 46).
37. Turn on the C-200 controller.
38. Run the C-200MT software.
39. Run the KI Calibration transducer alignment diagnostic. See C-200MT online help or the *C-200MT Software User's Guide* for more information about this procedure.

GT 5250 Grinding Pulley Drive Belts

The grinding pulley drive belts connect the sharpener motor to the grinding pulleys. These belts must be in good condition for the sharpener to work properly.

Required Tools and Supplies

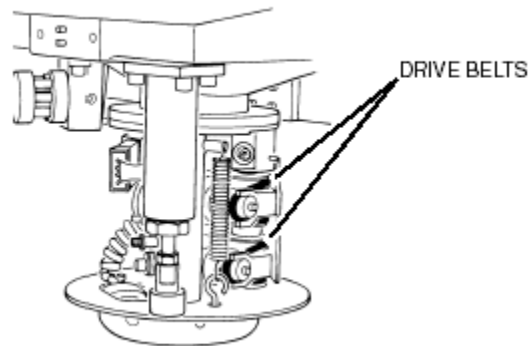
Item	Gerber Part No.
cleaning solvent	308010101
right-angle metric hex key set	945500094
appropriate standard right-angle hex key	945500046
grinding pulley drive belt (O-ring)	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Check both drive belts for nicks and cuts. Replace a drive belt if it has any of these defects.

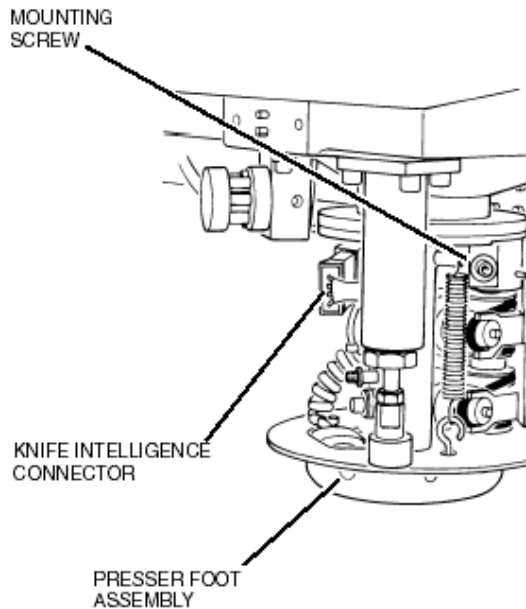
CAUTION: Do not use solvents on the drive belts. Solvents will damage the drive belts.

4. Make sure that there is no lubricant, oil, or grease on the drive belts. Replace a drive belt if it has a lubricant on it.
5. Close the cutter head cover (page 46).
6. Turn on the C-200 controller.



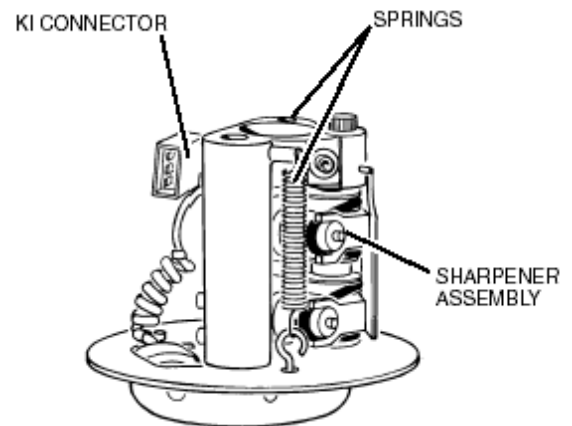
Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife assembly (page 57).
4. Unplug the Knife Intelligence sensor.
5. Loosen the mounting screw that holds the presser foot assembly on the cutter tube.



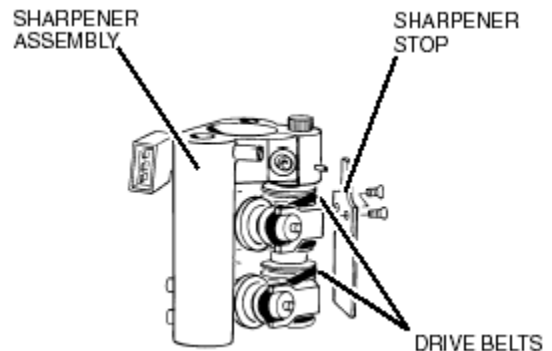
6. Lower the presser foot assembly and remove it from the cutter head.

7. Remove the two springs from the presser foot assembly.



8. Lift the sharpener assembly off the presser foot assembly.

9. Remove the KI connector.



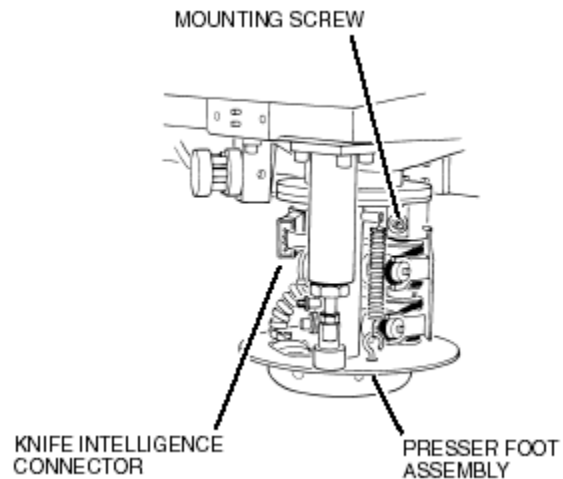
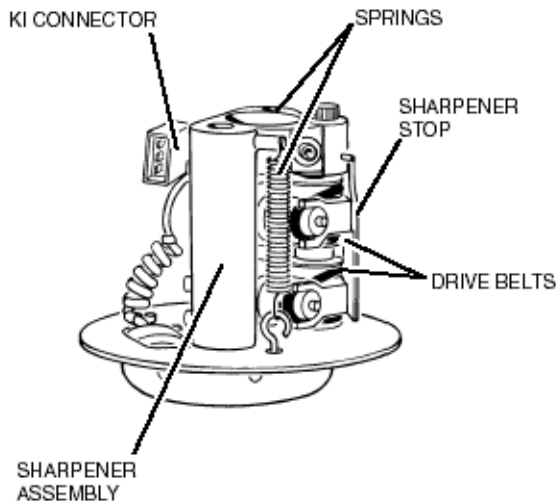
10. Remove the mounting screws from the sharpener stop.

11. Remove the sharpener stop.

12. Gently stretch each drive belt and slip it off the grinding pulleys.

Install

1. Slip each drive belt onto the grinding pulleys.
2. Install the sharpener stop plate with its mounting screws.
3. Place the sharpener assembly onto the presser foot assembly.
4. Install the Knife Intelligence (KI) connector mounting screw.
5. Install the two springs on the presser foot assembly.
6. Install the presser foot assembly into the cutter head.
7. Tighten the screw that holds the presser foot assembly on the cutter tube.
8. Plug in the Knife Intelligence Sensor connector.
9. Install the knife blade (page 58).
10. Close the cutter head front cover (page 46).
11. Turn on the C-200 controller.



GT 5250 Grinding Pulley

A grinding pulley is a grit-covered metal wheel that sharpens the knife blade. Grinding pulley condition affects sharpening which, in turn, affects cutting performance.

Required Tools and Supplies

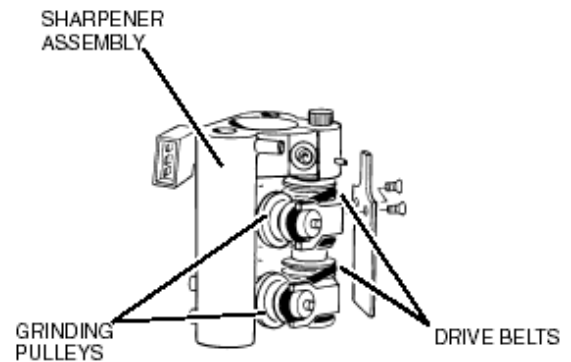
Item	Gerber Part No.
cleaning solvent	308010101
small brass brush	N/A
grinding pulley 80 grit	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Inspect the grinding pulleys for missing grit, debris, and chips. Replace or clean the grinding pulley, as needed.
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.

Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife blade (page 57).
4. Remove the grinding pulley drive belts (page 67).
5. Slide the four grinding pulleys from their supports.

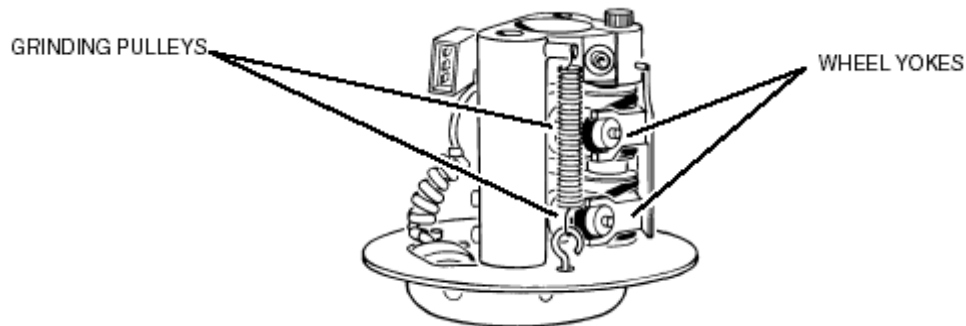


Clean

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife blade (page 57).
4. Remove the grinding pulley drive belts (page 67).
5. Remove the grinding pulleys (page 69).
6. Spray the grinding pulley thoroughly with cleaning solvent.
7. Using a stiff, brass brush, scrub the grinding pulley until you remove all deposits.
8. Air dry the grinding pulley until all solvent is gone.
9. Install the grinding pulley.
10. Check the wheel yokes for wear by trying to move them with your hand. If you can move the yokes easily, they are worn. Have them replaced by a Gerber qualified technician.

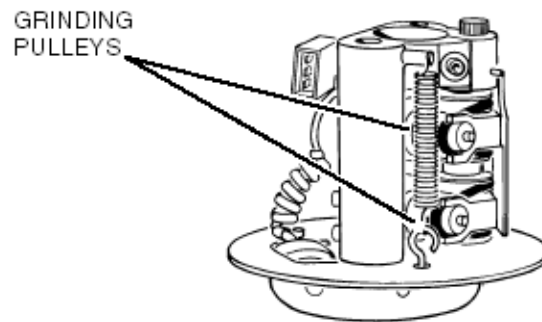
WARNING!  Cleaning solvent is flammable. If it ignites, it may cause injury or death. Do not smoke or have other ignition sources near the solvent.

CAUTION: Do not get cleaning solvent on any bearings because it will damage them.



Install

1. Slide the four grinding pulleys into their supports.
2. Install the grinding pulley drive belts (page 68).
3. Install the knife assembly (page 58).
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.



GT 5250 Elevator Roundways

The elevator roundways are shafts on which the elevator rides up and down. As the elevator moves, it plunges and raises the knife.

Required Tools and Supplies

Item	Gerber Part No.
Mobil 1 5w-30 synthetic oil	596118001
clean shop rag	N/A

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Inspect the roundways for dirt and lint build-up. Clean and lubricate them, if necessary.
4. Inspect the roundways for scratches, nicks, and gouges. If you see any of these problems, have them repaired or replaced by a Gerber qualified technician.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.

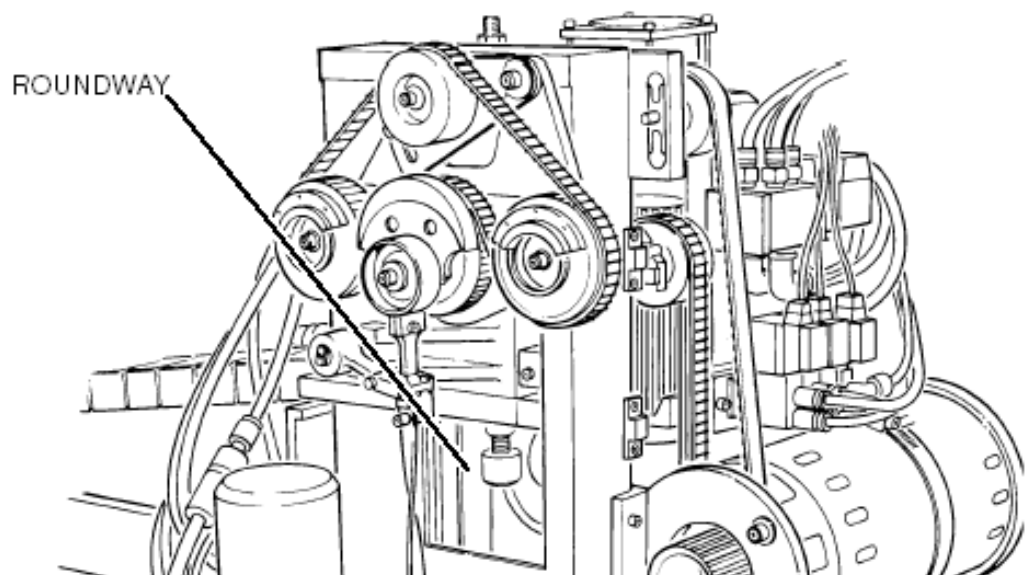
Clean/Lubricate

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

CAUTION: Only use synthetic oil to lubricate the roundways. Non-synthetic oil can damage seals and bearings.

CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently than to apply one heavy coat.

3. Use the shop rag to wipe a light coat of synthetic oil onto both roundways.
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.



GT 5250 Up/Down Stops

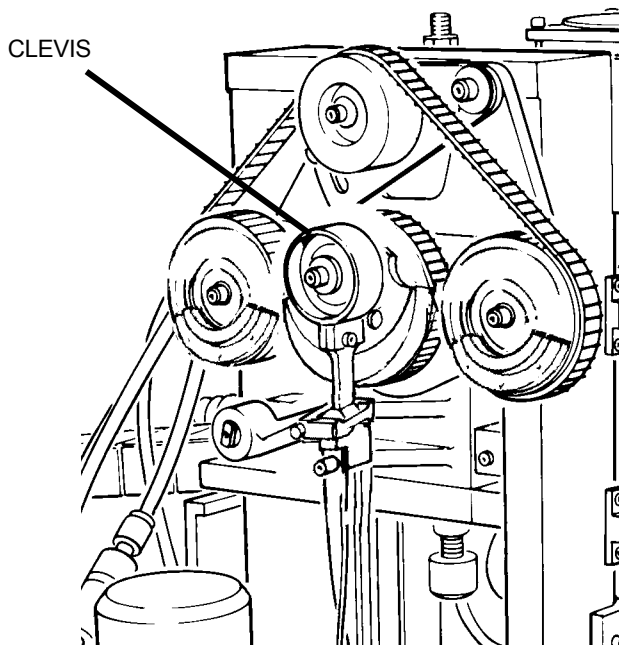
The up/down stops control the depth of knife cuts. If the down stop is set low, the knife cuts off the bristles in the BRISTLE SQUARES. If the down stop is set high, the knife does not cut bottom plies cleanly. Improper setting of the up stop causes poor sharpening.

Required Tools and Supplies

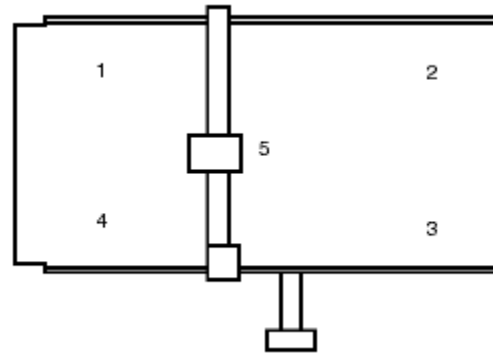
Item	Gerber Part No.
metric combination wrench set	945500093
screwdriver	945500298
appropriate standard combination wrench	944002506

Inspect

1. Turn on the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Move the clevis to top dead center (TDC).



4. Cover the work surface with paper and then with plastic film.
5. You must check the knife penetration at five points on the cutting surface. Move the cutter head to test point 1 on the work surface. Test points 1 through 4 are about 30.5 cm (1 ft) from each corner. Test point 5 is at the center of the work surface.



6. Plunge the knife and presser foot bowl in the following manner:
 - a) At the BCP, turn on the **KNIFE** switch.
 - b) From the Diagnostics menu, select Input Output Test (DG23). See C-200MT online help or the *C-200MT Software User's Guide*.
The monitor displays the DG 23 menu.
 - c) Use the arrow keys to highlight **Vacuum Enbl.**

- d) Hold down the CTRL key on your keyboard and press the SPACE bar. This turns on table vacuum.
- e) Use the arrow keys to highlight **Knife Sol.**
- f) Hold down the CTRL key on your keyboard and press the SPACE bar. This plunges the knife and the presser foot bowl. Note that the number in the KNIFE SOL line changes from 1 to 0.

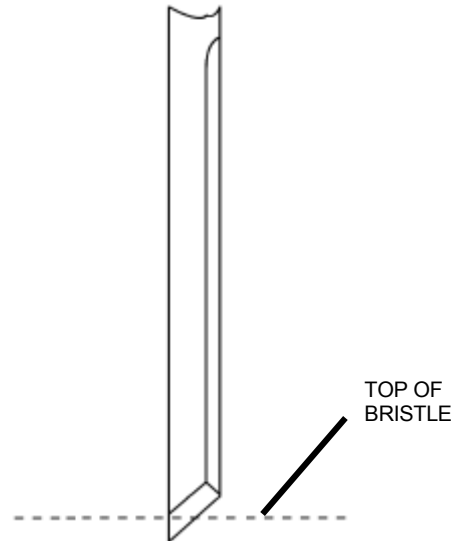
Note: Make a note of the air pressure reading on the presser foot air gauge at the operator console. You will need to reset the regulator to that pressure when you are through with this procedure.

- g) Turn off air to the presser foot bowl by releasing pressure for the regulator at the operator console. Do this by turning the regulator counterclockwise. You can now manually raise the presser foot bowl to see the knife.

7. Check the down stop setting as follows:

- a) With a pencil, mark the knife blade at the point it enters the plastic sealer.
- b) Hold down the CTRL key on your keyboard and press the SPACE bar. The knife rises.

- c) The pencil line should be 1.6 to 3.2 mm (1/16 to 1/8 inches) above the tip of the knife blade.



- d) Turn on air to the presser foot bowl by returning pressure at the regulator at the operator console. Do this by turning the regulator clockwise.
- 8. Move the cutter head to the next test point and repeat steps 6 through 7 for test points 2 through 5. If needed, adjust the down stop.
 - 9. On the BCP, press **VAC** to turn off table vacuum.
 - 10. Press **Esc** to exit the DG 23 menu.

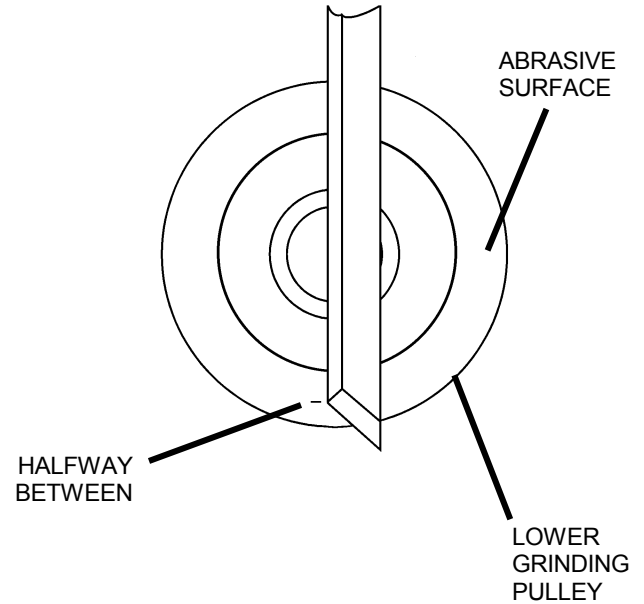
11. Check the up stop setting.

- a) Position the clevis at top dead center (TDC).
- b) Check the knife blade. The corner edge of the knife blade should be in the center of the abrasive surface of the lower grinding pulley.
- c) Adjust the up stop, if necessary.

12. Exit the C-200MT program.

13. Turn off the C-200 controller.

14. Close the cutter head front cover (page 46).



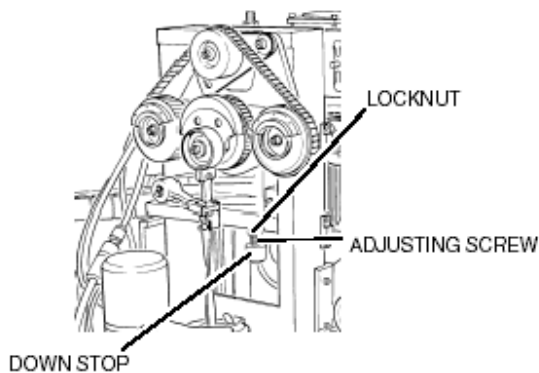
Adjust

If you find, during the up and down stop checks (page 73), that you need to adjust the head up/down stops, follow this procedure.

CAUTION: Too much penetration damages the work surface; too little keeps pieces from being cut completely.

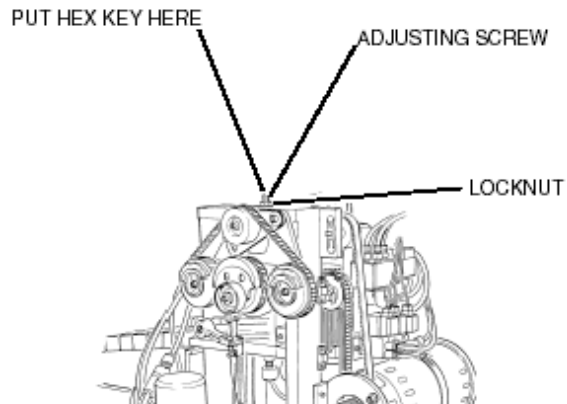
Note: Because of tolerances, bristle penetration usually cannot be set to the same value for each test point. However, all test points must be within the proper range.

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Adjust the down stop:
 - a) Loosen the locknut on the adjusting screw by turning it counterclockwise.
 - b) Turn the adjusting screw clockwise or counterclockwise to change the depth of bristle penetration.
 - c) Inspect the up/down stop settings as described on page 73.



- d) Tighten the locknut by turning it clockwise.
 - e) Repeat steps 3a through 3d until bristle penetration is set correctly.

4. Adjust the up stop.
 - a) Loosen the lock nut on the adjusting screw by turning it counterclockwise.
 - b) Put a hex key into the adjusting screw. Turn the adjusting screw clockwise or counterclockwise to change the up stop setting.
 - c) Inspect the up/down stop settings as described on page 73.
 - d) Tighten the lock nut by turning it clockwise.
 - e) Repeat steps 4a through 4d until the up stop is set correctly.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.

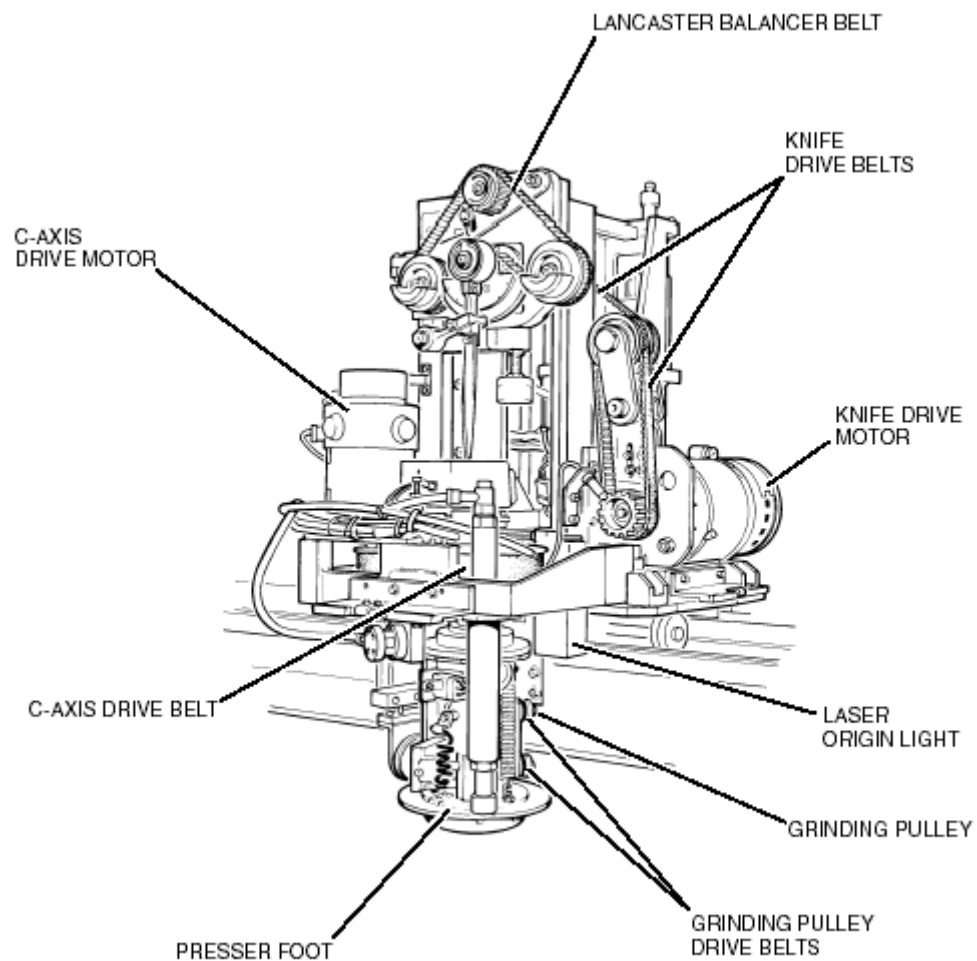


Cutter Head for GT 7250

The cutter head contains the knife and does the actual cutting of material. It moves across the cutting table on the beam.

Maintenance procedures specifically for the GT 7250 cutter head begin on the next page. Sections describing how to remove the front and back covers for both the GT 5250 and GT 7250 systems can be found on pages 46 to 48.

Note: Maintenance procedures for the GT 5250 cutter head begin on page 45.



GT 7250 Cutter Head

GT 7250 C-axis Assembly

The C-axis assembly includes the C-axis motor and drive belt. The C-axis assembly turns the knife blade to cut corners, notches, curves, and other complex shapes.

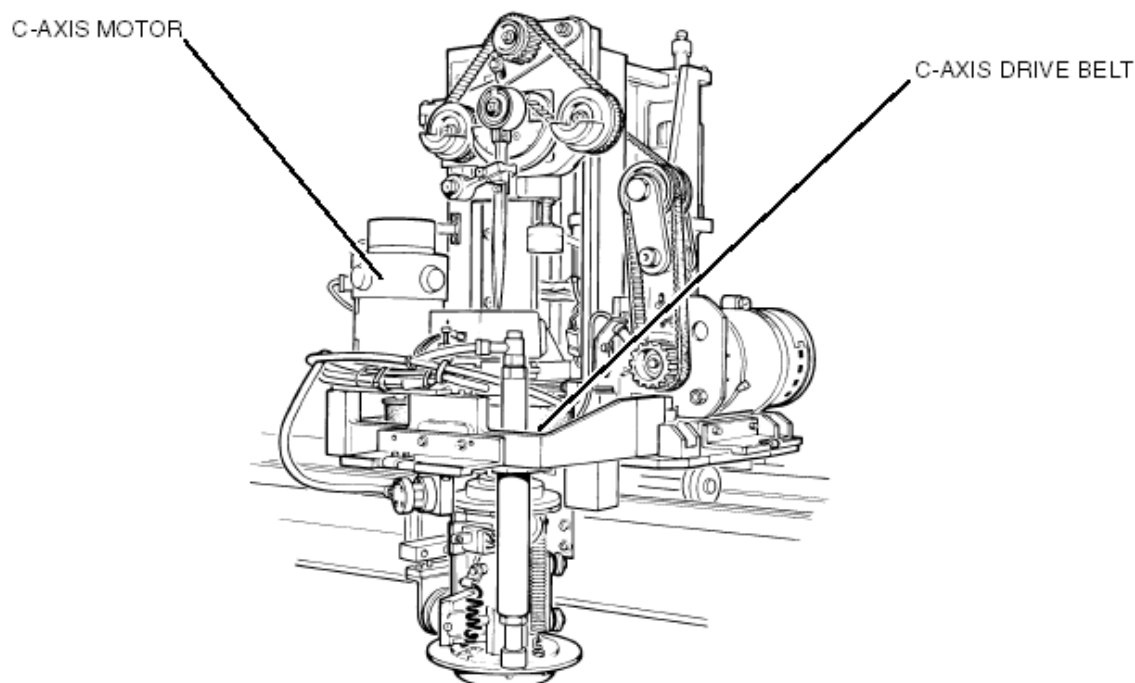
Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
appropriate standard T-handle hex key	944200406

Inspect

1. Turn off the C-200 controller.

2. Open the cutter head front cover (page 46).
3. Manually turn the C-axis assembly several times to the right and left. It should turn freely without binding or noise.
4. If you find any defects, have the assembly repaired or replaced by a Gerber qualified technician.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



GT 7250 C-axis Drive Belt

The C-axis drive belt links the C-axis motor pulley and the C-axis pulley to turn the knife blade about the C-axis. The drive belt must be in good condition and properly tensioned for accurate cutting.

Required Tools and Supplies

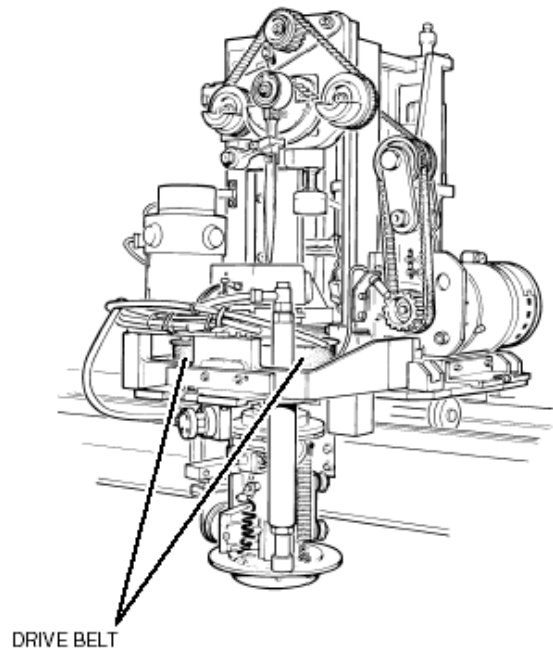
Item	Gerber Part No.
right-angle metric hex key set	945500094
spring scale, 0 - 23 kg (0 - 50 lb.)	945500252
appropriate standard right-angle hex key	
C-axis drive belt	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Inspect the drive belt for nicks, cuts, or gouges. If you find any of these defects, replace it.
4. Visually inspect the belt for proper tensioning. Adjust it, if necessary.
5. Close the cutter head cover (page 46).
6. Turn on the C-200 controller.

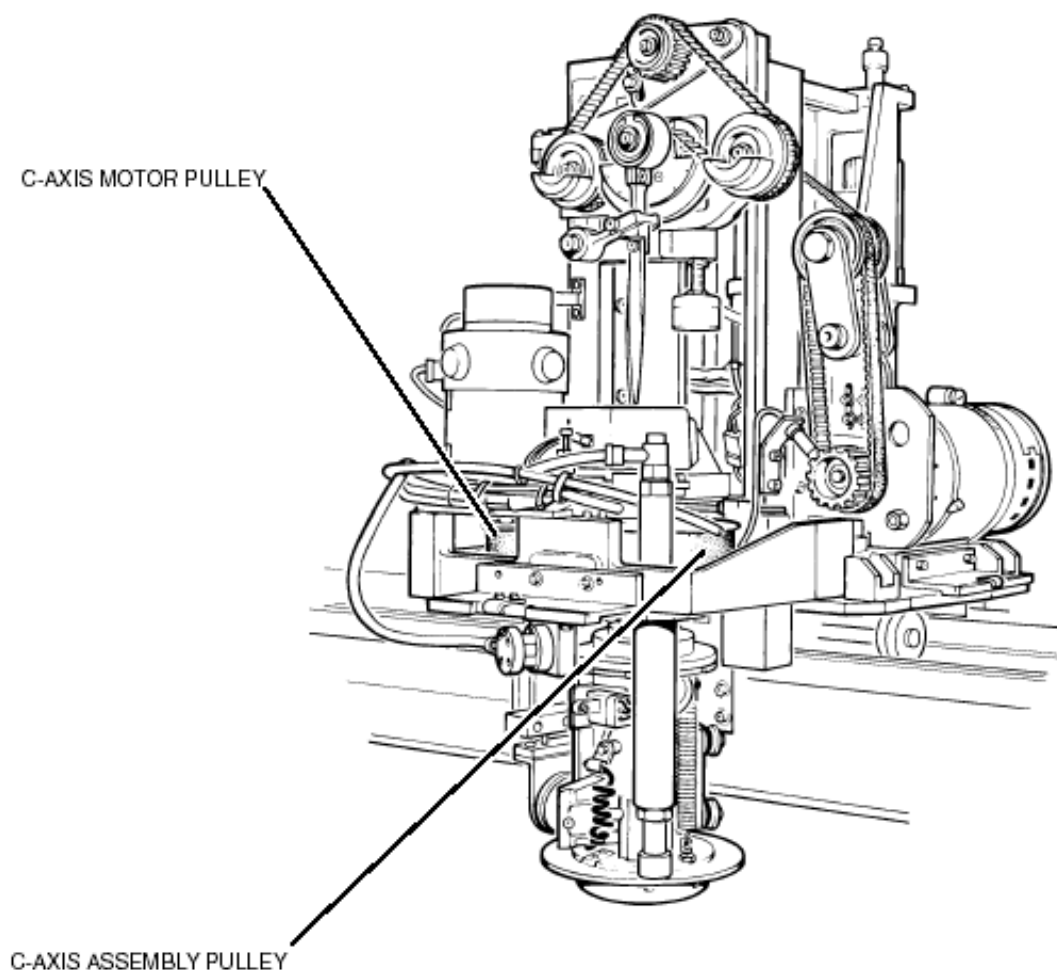
Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Loosen the three retaining screws for the C-axis motor.
4. Slide the C-axis motor toward the C-axis assembly.
5. Slip the C-axis drive belt off the pulley that is on the C-axis motor.
6. Remove the C-axis drive belt from around the C-axis.



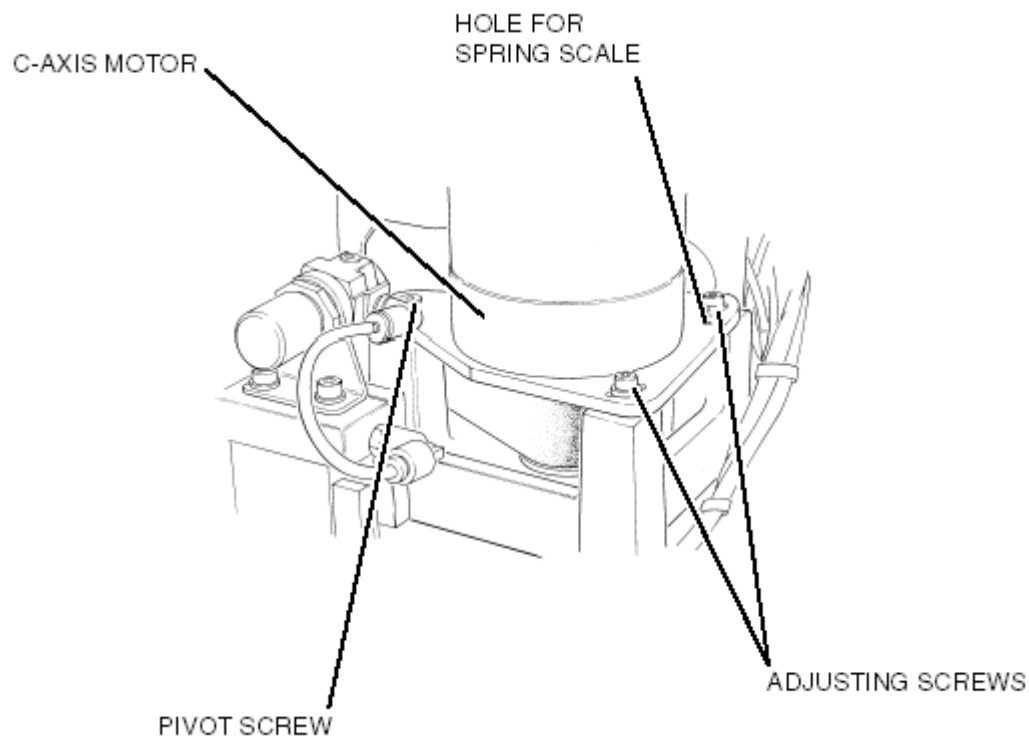
Install

1. Put the drive belt onto the pulleys on the C-axis motor and C-axis assembly.
2. Hook the end of the spring scale to the hole in the C-axis adjusting plate.
3. Pull on the spring scale until it reads 6.81 kg (15 lb.).
4. While keeping proper tension on the spring scale, tighten the screws that retain the C-axis motor.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Loosen the two C-axis motor adjusting screws.
4. Loosen the C-axis motor pivot screw.
5. Make sure that the drive belt is in mesh with the teeth on the C-axis motor pulley and C-axis pulley. Rotate the C-axis to mesh the belt with the pulley teeth, if needed.
6. Hook the end of the spring scale to the hole in the C-axis adjusting plate.
7. Pull on the spring scale until it reads 6.81 kg (15 lb.).
8. While keeping proper tension on the spring scale, tighten the screws that retain the C-axis motor.
9. Close the cutter head front cover (page 46).
10. Turn on the C-200 controller.



GT 7250 Knife Drive Belts

The knife drive belts connect the knife drive motor to the knife crankshaft through a series of pulleys. As the knife drive motor turns, the drive belt transfers its motion to the knife crankshaft. The drive belts must be in good condition and properly tensioned to give the best cutting performance.

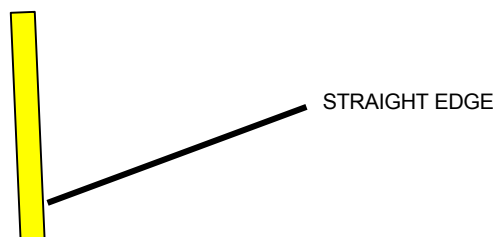
Required Tools and Supplies

Item	Gerber Part No.
belt deflection gauge	45412000
right-angle metric hex key set	945500094
appropriate standard T-handle hex key	944200406
knife drive belts	Refer to Service CD

Inspect

- Turn off the C-200 controller.
- Open the cutter head front cover (page 46).
- Check the knife drive belts for cracks, nicks, cuts, and glazing. If you see any of these defects have them replaced by a Gerber qualified technician.
- Check tension for drive belts 1 and 2.
 - Set the collar on the belt deflection gauge for 6.35 mm (0.25 in.).
 - Align a straight edge parallel with one of the knife drive belts.
 - Holding the belt deflection gauge perpendicular to the straight edge, push the belt deflection gauge until the collar meets evenly on the straight edge.
 - Measure the tension of the knife drive belt with the belt deflection gauge at each of the points shown below. The deflection should be 6.35 mm (0.25 in.) at 2.73 to 3.64 kg (6 to 8 lb.).
 - Repeat for the other knife drive belt.
- If either or both belts are tensioned incorrectly, adjust the belt tensions of both drive belts.
- Close the cutter head front cover (page 46).
- Turn on the C-200 controller.

MEASURE BELT
DEFLECTION HERE



Adjust

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

CAUTION: Drive belts 1 and 2 must be adjusted together in the proper sequence. Doing otherwise can damage the belts and bearings.

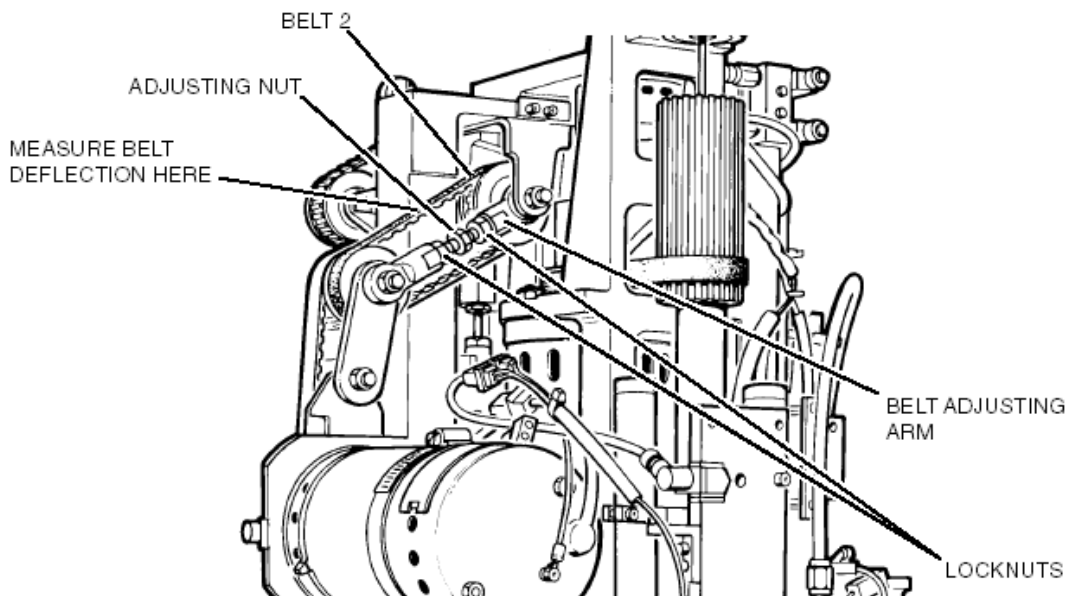
3. Adjust drive belt 2.

- a) Loosen both locknuts on the belt tensioning arm.
- b) To loosen drive belt 2, shorten the belt adjusting arm by turning the adjusting nut.

OR

- c) To tighten drive belt 2, lengthen the belt adjusting arm by turning the adjusting nut.

- d) Tighten both belt adjusting arm locknuts.
- e) Measure the tension of drive belt 2 with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 2.73 to 3.64 kg (6 to 8 lb.).
- f) If the belt tension is incorrect, repeat steps 3a through 3e until the belt is tensioned correctly.



4. Adjust drive belt 1.

- a) Loosen the four adjusting screws for drive belt 1.
- b) To loosen drive belt 1, slide the pulley mounting bracket towards the knife drive motor.

OR

- c) To tighten drive belt 1, slide the pulley mounting bracket away from the knife drive motor.

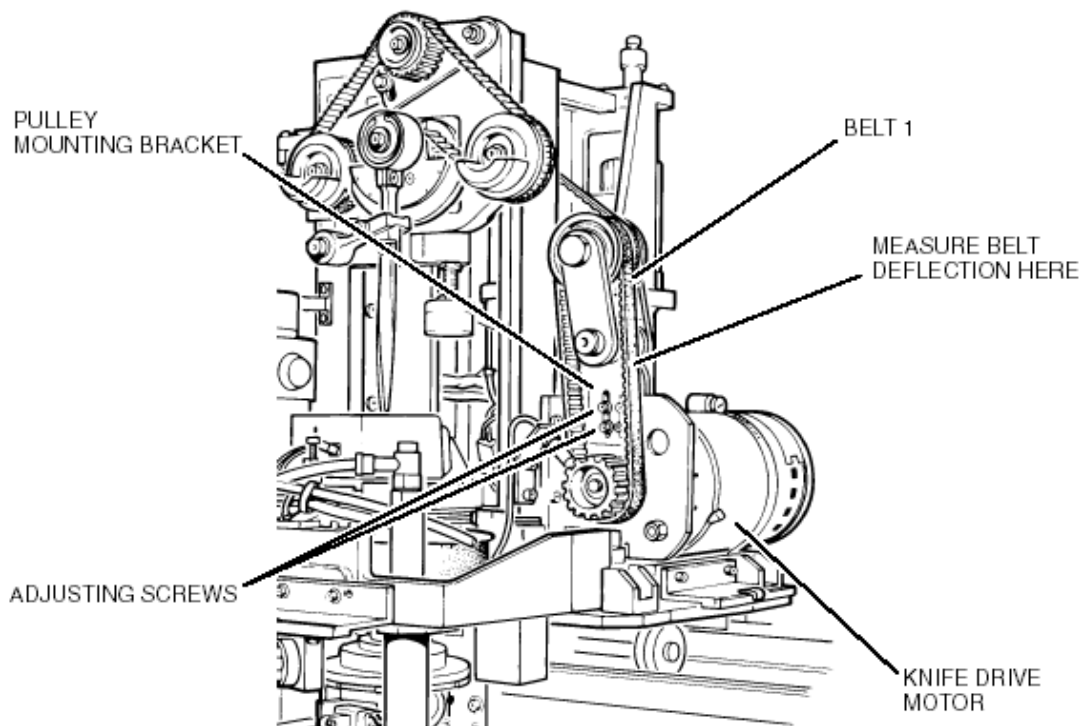
- d) Tighten the four adjusting screws for drive belt 1.

- e) Measure the tension of drive belt 1 with the belt deflection gauge at the point shown below. The deflection should be 6.35 mm (0.25 in.) at 2.73 to 3.64 kg (6 to 8 lb.).

- f) If the belt tension is incorrect, repeat steps 4a through 4e until the belt is tensioned correctly.

5. Close the cutter head front cover (page 46).

6. Turn on the C-200 controller.



GT 7250 Lancaster Balancer

The Lancaster balancer reduces vibration and noise made by the knife drive. As the knife drive motor turns, the Lancaster balancer belt and pulleys also turn. This belt must be in good condition and properly tensioned for the Lancaster balancer to work properly.

Required Tools and Supplies

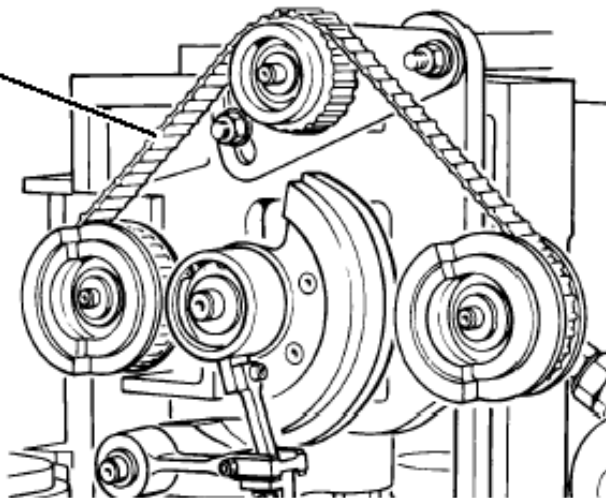
Item	Gerber Part No.
right-angle metric hex key set	945500094
spring scale, 0 - 23 kg (0 - 50 lb.)	945500252
appropriate standard T-handle hex key	
knife drive belts	Refer to Service CD

3. Check both sides of the Lancaster balancer belt for cracks, nicks, cuts, and glazing.
4. If you see any of these defects, have it replaced by a Gerber qualified technician.
5. Visually inspect the belt for proper tension.
6. If the belt is tensioned incorrectly, adjust the belt tension.
7. Close the cutter head front cover (page 46).
8. Turn on the C-200 controller.

Inspect

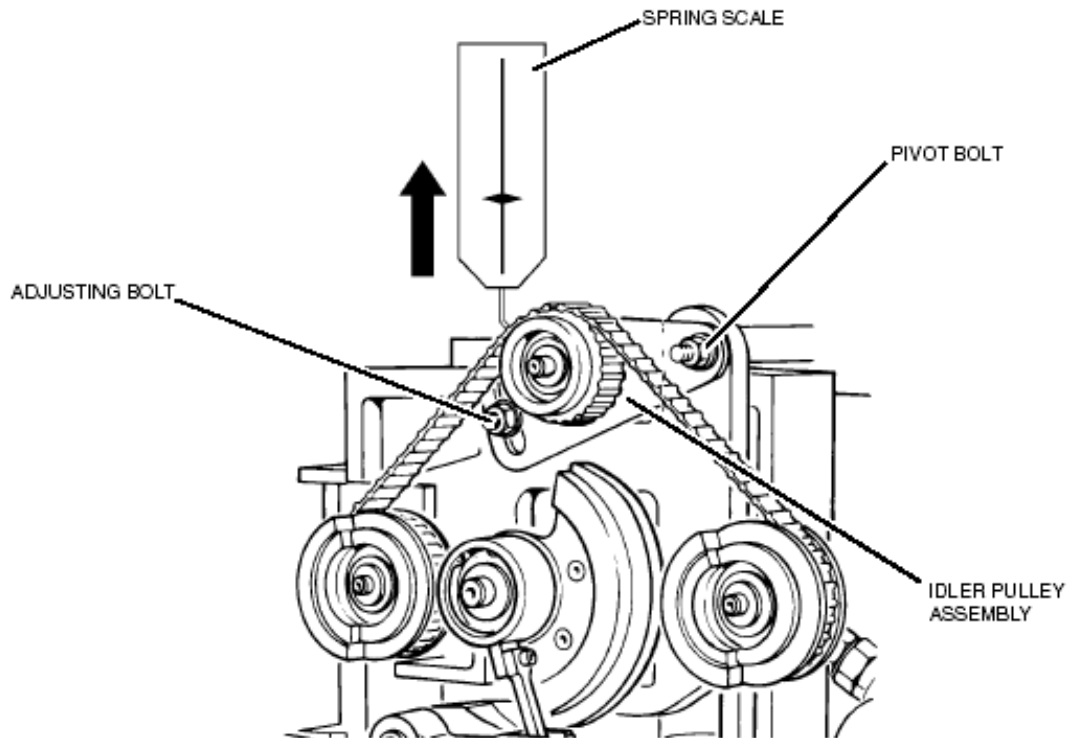
1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

LANCASTER
BALANCER BELT



Adjust

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Loosen the adjusting bolt and the pivot bolt on the idler pulley assembly.
4. Push the idler pulley assembly down to release tension on the Lancaster balancer belt.
5. Hook the end of the spring scale to the adjustable end of the idler pulley assembly.
6. Pull up on the spring scale until it reads 2.27 kg (5 lbs.).
7. Tighten the adjusting bolt and pivot bolt.
8. Close the cutter head front cover (page 46).
9. Turn on the C-200 controller.



GT 7250 Knife Blade

The knife blade cuts material. It must be sharp and within wear limits to cut properly.

WARNING!  The knife blade is very sharp. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
Petro-Lon multipurpose grease	596500005
standard T-handle hex key	944200406
screwdriver	945500298
knife blade	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Check the knife blade for excessive wear.
4. If the leading edge of the knife blade is damaged, replace the knife blade (page 88).
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



OK



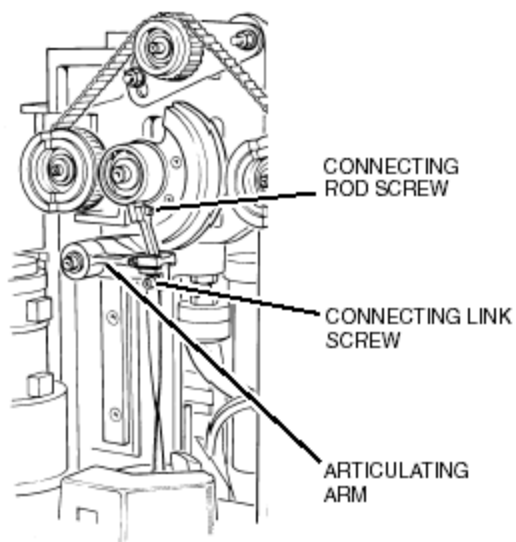
UNACCEPTABLE

Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

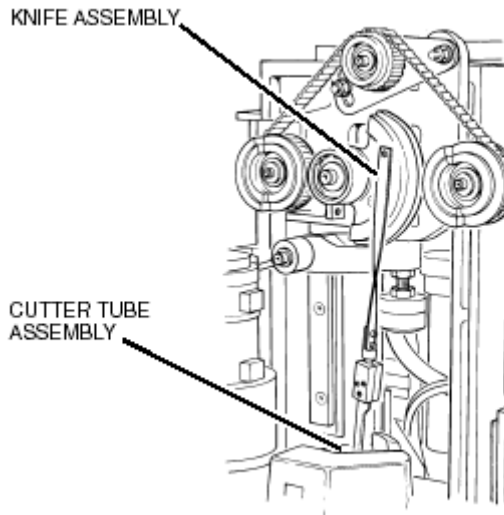
WARNING!  The knife blade is very sharp and can cause injury. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

3. Remove the connecting link screw.



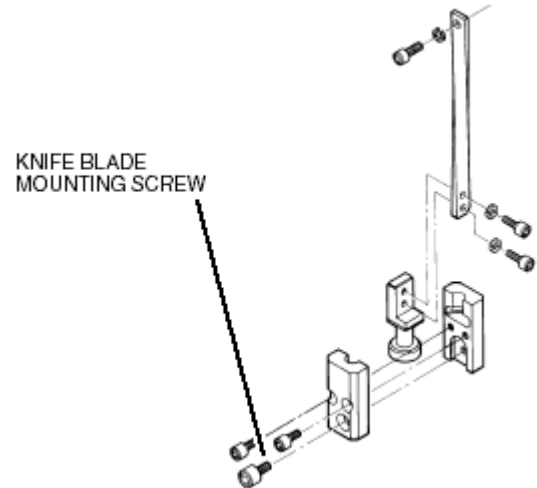
4. Remove the connecting rod screw.
5. Push the knife assembly down.
6. Move the articulating arm out of the way.

7. Pull the knife blade assembly up from the cutter tube assembly.



WARNING!  The knife blade is very sharp and can cause injury. Handle the knife blade carefully. Never hold the knife blade by its sharp end.

8. Remove the knife blade from the swivel slider assembly.
 - a) Remove the knife blade mounting screw.
 - b) Pull the knife blade from the swivel slider assembly.

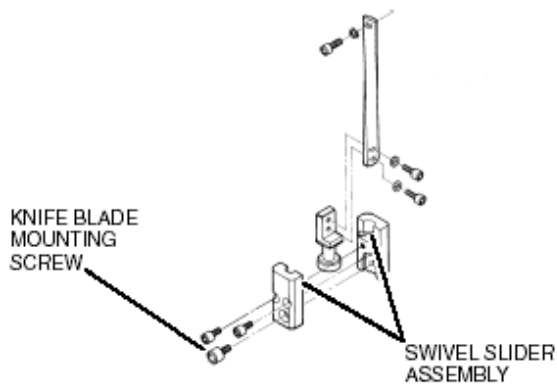


Install

1. Fill the inside of the swivel slider assembly with Petro-Lon multipurpose grease. Put enough Petro-Lon into the assembly so that, when you install the knife blade mounting screw, a small amount of Petro-Lon is squeezed out of the assembly.

Note: You can also fill the inside of the swivel assembly after you install the knife blade. To do this use a grease gun with a needle fitting to put Petro-Lon through the greasing hole in the swivel. Stop adding Petro-Lon as soon as the Petro-Lon starts to leak out of the assembly.

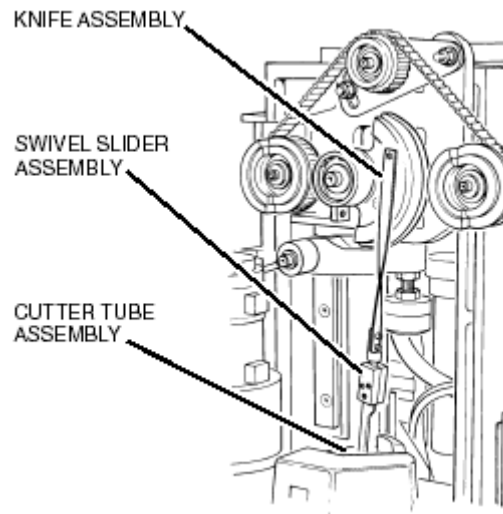
2. Put the knife blade into the swivel slider assembly.



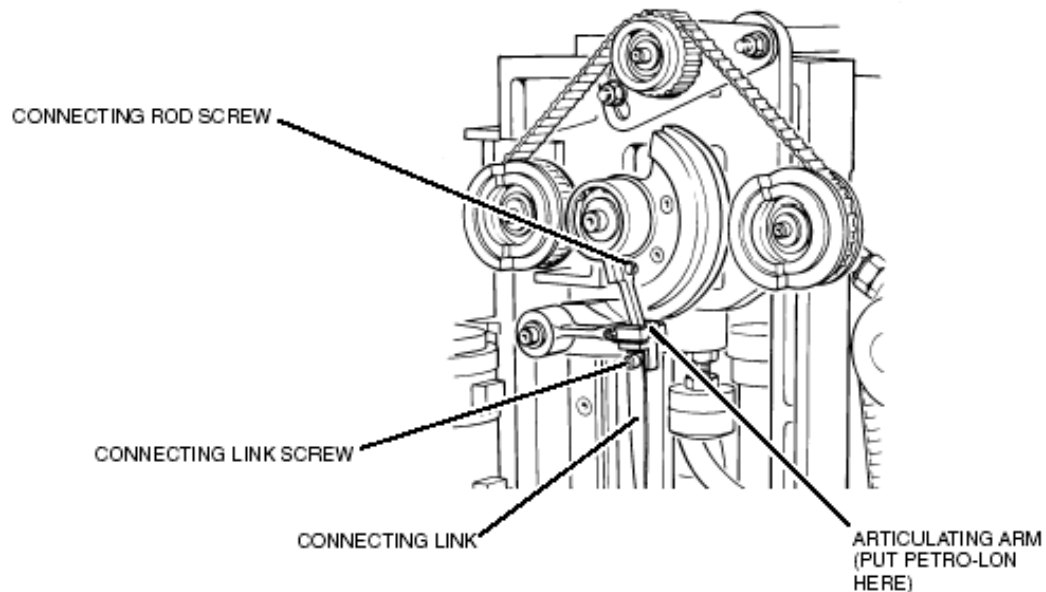
3. Install the knife blade mounting screw.

Note: The hole in the knife blade for the mounting screw is offset. You can only mount the knife blade one way.

4. Put a thin (1 mm or 0.040 in.) coat of Petro-Lon onto the outside of the swivel slider assembly.
5. Insert the knife assembly into the cutter tube assembly. Make sure the knife is straight.



6. Align the screw hole in the connecting rod with the screw hole in the block and bushing assembly.
7. Install the connecting rod screw.
8. Align the screw hole in the connecting link with the screw hole in the block and bushing assembly.
9. Install the connecting link screw.
10. Put Petro-Lon into the working surface of the articulating arm. Put in enough Petro-Lon to fill all the space in the arm. Do not overfill the arm so that Petro-Lon is squeezed onto the outside of the articulating arm.
11. Close the cutter head front cover (page 46).
12. Turn on the C-200 controller.
13. Check knife operation.
 - a) At the BCP, set **KNIFE** to **ON**.
 - b) Check that the knife moves smoothly and with no unusual noise.
 - c) At the BCP, set **KNIFE** to **OFF**.
14. Use the New Knife command to reset knife wear logic.
15. Sharpen the knife blade several times to check the position of the knife blade.



GT 7250 Presser Foot

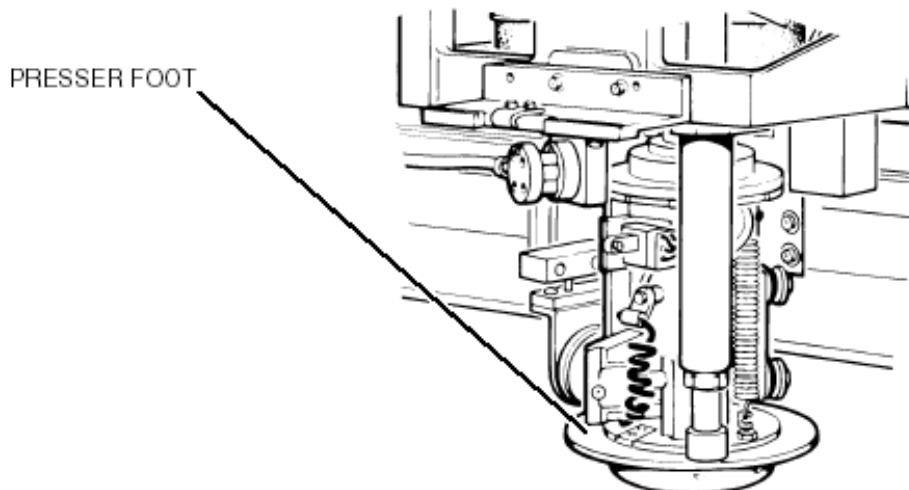
The presser foot flattens the top of the spread, or lay, as the knife blade cuts it.

Required Tools and Supplies

Item	Gerber Part No.
grease gun	944233101
clean shop rag	N/A
Mobil 1 5w-30 synthetic oil	596118001
right-angle metric hex key set	945500094
appropriate standard right-angle hex keys	945500046, 945500255

Inspect

- Turn off the C-200 controller.
- Open the cutter head front cover (page 46).
- Check the presser foot shafts for dirt and debris. Clean them if needed.
- Check the shafts for bends, nicks, and gouges. If you find any of these defects, have them repaired or replaced by a Gerber qualified technician.
- Check the knife intelligence wire for any breaks, nicks, and gouges. If you find any of these defects, have it replaced by a Gerber qualified technician.
- Check the presser foot for binding.
 - At the operator console, decrease the presser foot bowl pressure to 0 kgf/cm² (0 psi).
 - Slide the presser foot straight down. It should move smoothly without binding. Binding indicates a problem with the shafts and/or alignment of the lower blade guide.
 - If you notice any binding, have the assembly repaired or replaced by a Gerber qualified technician.
 - At the operator console, restore the presser foot bowl pressure to its original level.
- Close the cutter head front cover (page 46).
- Turn on the C-200 controller.

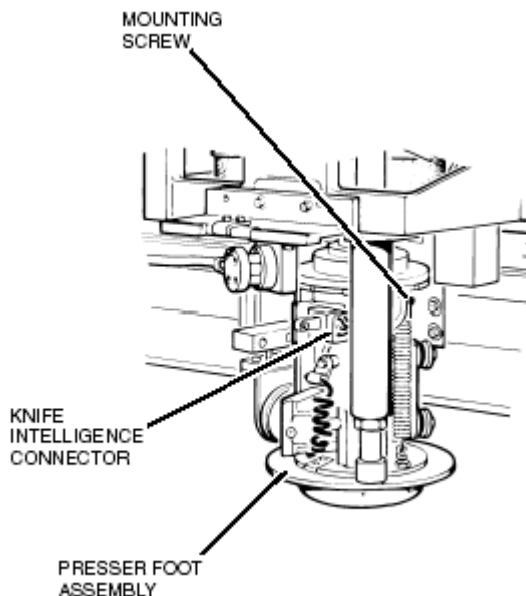


Clean/Lubricate

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Use a grease gun to lubricate the linear bearing on the shaft.

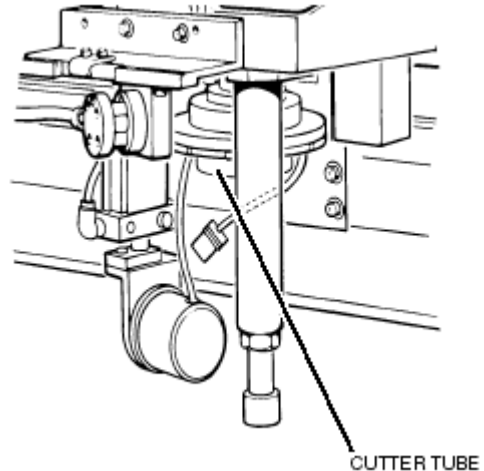
CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently, than to apply one heavy coat.

4. Remove the knife assembly (page 88).
5. Unplug the Knife Intelligence® (KI) sensor.



6. Loosen the mounting screw that holds the sharpener assembly on the cutter tube.
7. Lower the presser foot assembly and remove it from the cutter head.

8. Use a clean rag dampened with denatured alcohol to clean the inside of the cutter tube.

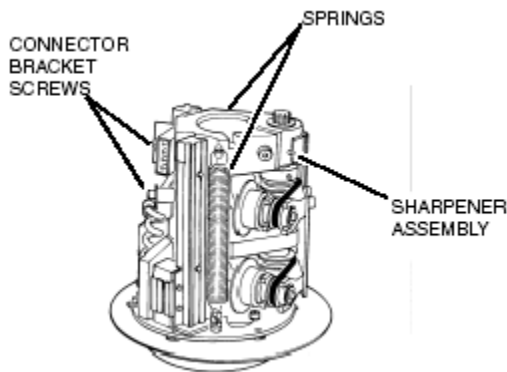


9. Remove the Knife Intelligence connector bracket screws from the presser foot assembly.

CAUTION: Do not remove the linear bearings. If you do, you may lose them.

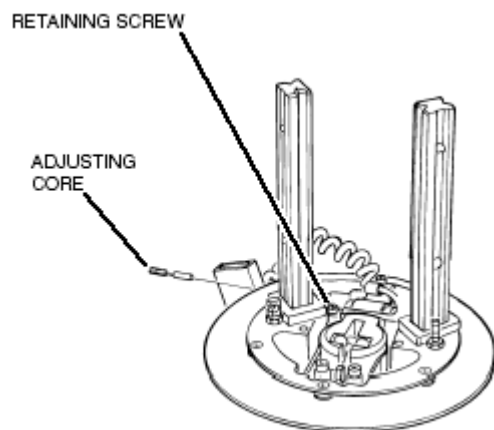
10. Remove the two springs.

11. Lift the sharpener assembly off the presser foot assembly.



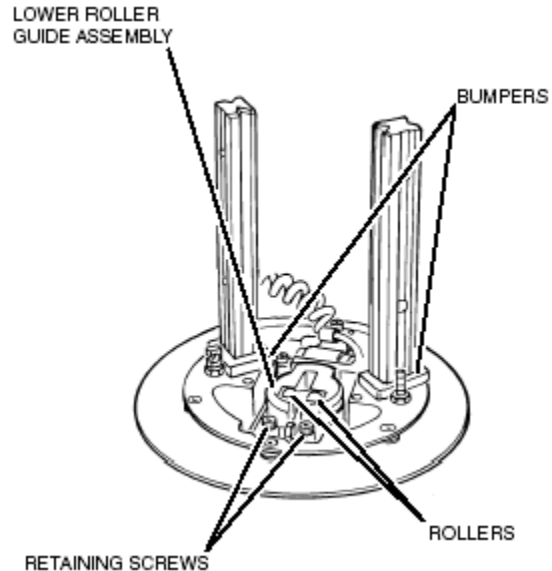
12. Loosen the retaining screw for the adjusting core.

13. Remove the adjusting core.



14. Remove the retaining screws for the lower roller guide assembly.

15. Lift out the lower roller guide assembly.



16. Dampen a lint-free clean cloth with denatured alcohol and clean the lower roller guide assembly.

17. Let the lower roller guide assembly dry completely.

18. Inspect the bumpers for wear and damage. Make sure they spin freely. If they are worn or damaged, have them replaced by a Gerber qualified technician.

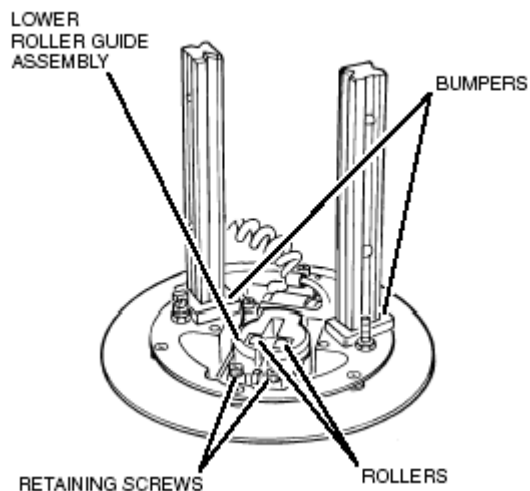
19. Inspect the rollers in the lower guide for wear and damage. Make sure they spin freely. If the rollers are worn or damaged, or do not spin easily, have them replaced by a Gerber qualified technician.

CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently, than to apply one heavy coat.

CAUTION: Only use synthetic oil. Non-synthetic oil can damage seals and bearings.

20. Lubricate the pins of the rollers with a drop of Mobil 1 5w-30 synthetic oil.

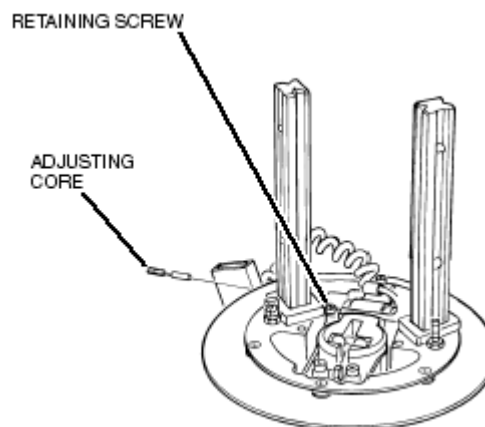
21. Put the lower roller guide assembly into the presser foot assembly.



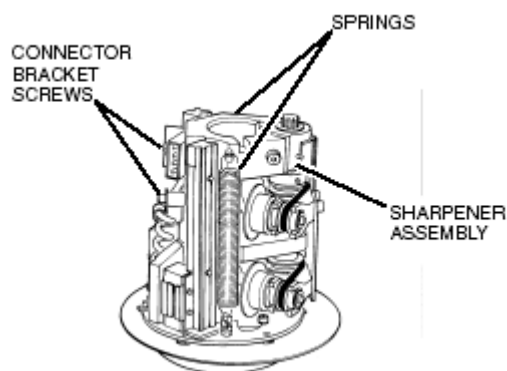
22. Install the retaining screws for the lower roller guide assembly.

23. Install the adjusting core.

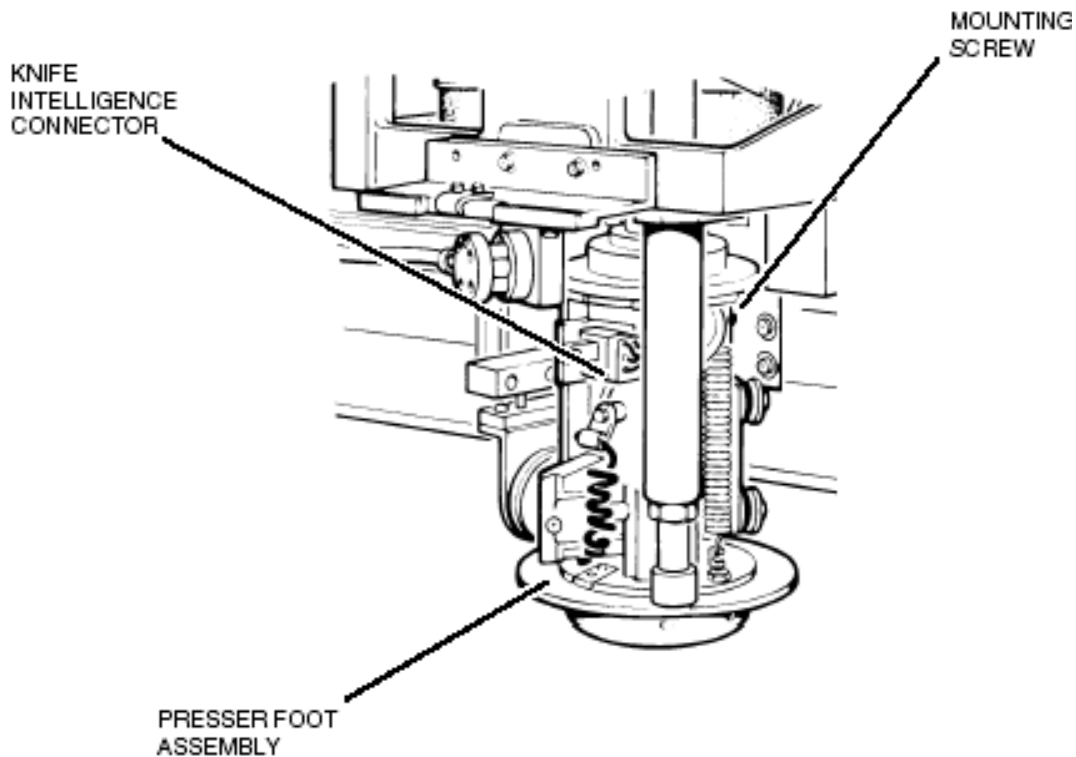
24. Install the retaining screws for the adjusting core.



25. Put the sharpener assembly into the presser foot assembly.



26. Install the Knife Intelligence connector bracket screws.
27. Install the presser foot assembly into the cutter head.
28. Tighten the mounting screw that holds the presser foot assembly on the cutter tube.
29. Plug in the Knife Intelligence sensor.
30. Install the knife blade (page 4-90).
31. Close the cutter head front cover (page 46).
32. Turn on the C-200 controller.
33. Run the C-200MT software.
34. Run the transducer alignment diagnostic. See C-200MT online help or the *C-200MT Software User's Guide* for more information about this procedure.



GT 7250 Grinding Pulley Drive Belts

The grinding pulley drive belts connect the sharpener motor to the grinding pulleys. These belts must be in good condition for the sharpener to work properly.

Required Tools and Supplies

Item	Gerber Part No.
cleaning solvent	308010101
right-angle metric hex key set	945500094
appropriate standard right-angle hex key	
grinding pulley drive belts	Refer to Service CD

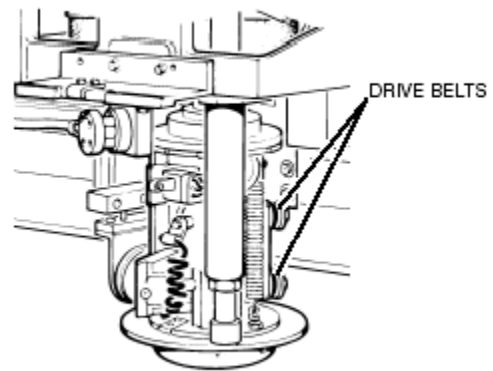
Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

3. Check both drive belts for nicks and cuts. Replace a drive belt if it has any of these defects.

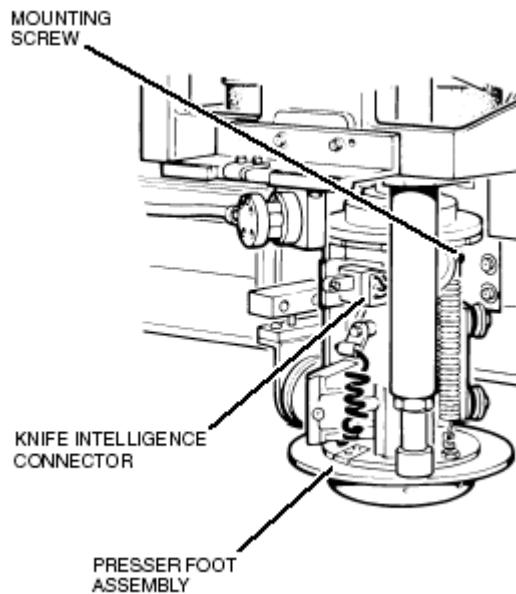
CAUTION: Do not use solvents on the drive belts. Solvents will damage the drive belts.

4. Make sure that there is no lubricant, oil, or grease on the drive belts. Replace a drive belt if it has a lubricant on it.
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.



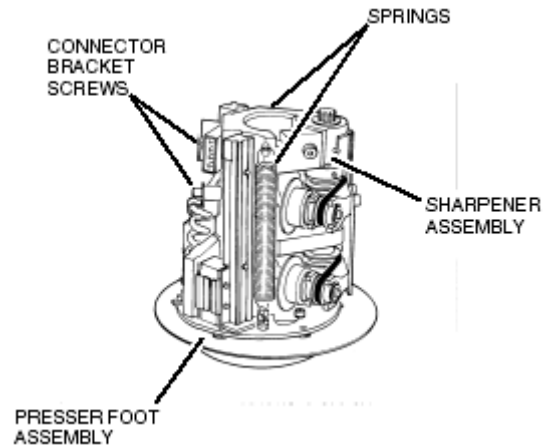
Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife assembly (page 88).
4. Unplug the Knife Intelligence sensor.
5. Remove the two springs from the presser foot assembly.
6. Loosen the screw that holds the presser foot assembly on the cutter tube.
7. Lower the presser foot assembly and remove it from the cutter head.

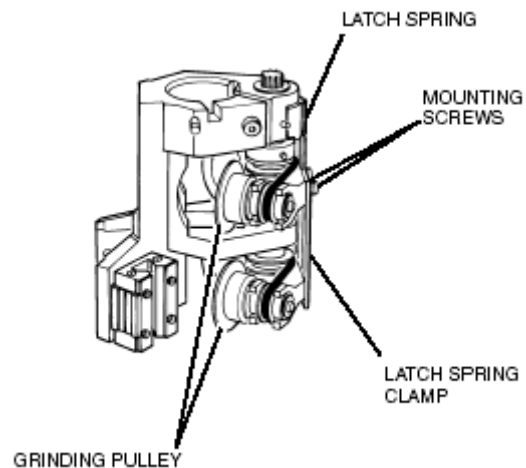


8. Remove the Knife Intelligence connector.

9. Lift the sharpener assembly off the presser foot assembly.

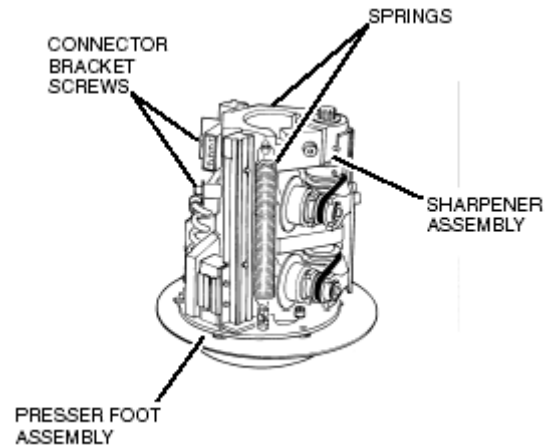
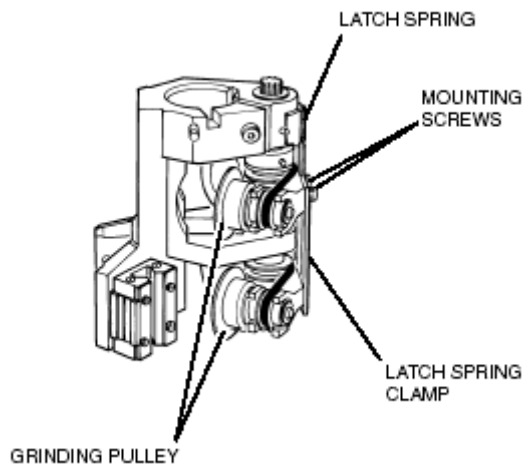


10. Remove the two screws that mount the latch spring clamp to the sharpener assembly.
11. Remove the latch spring clamp.
12. Remove the latch spring.
13. Gently stretch each drive belt and slip it off the grinding pulleys.

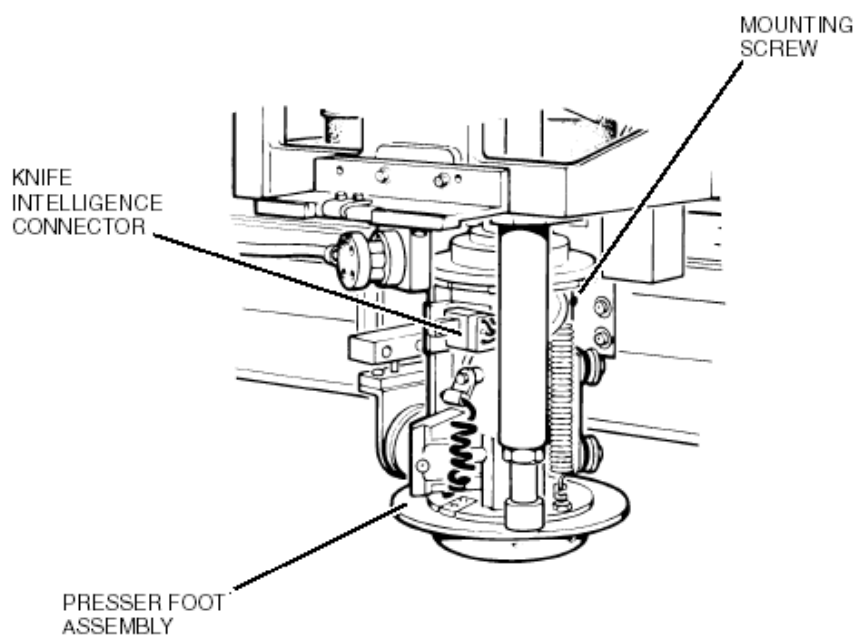


Install

1. Slip each drive belt onto the grinding pulleys.
2. Install the latch spring.
3. Install the latch spring clamp. Align the screw holes with the holes in the sharpener assembly.
4. Install the two screws that mount the latch spring clamp to the sharpener assembly.
5. Place the sharpener assembly onto the presser foot assembly.
6. Install the two springs on the presser foot assembly.



7. Install the presser foot assembly into the cutter head.
8. Tighten the screw that holds the presser foot assembly on the cutter tube.
9. Plug in the Knife Intelligence Sensor.
10. Install the knife blade (page 90).
11. Close the cutter head front cover (page 46).
12. Turn on the C-200 controller.



GT 7250 Grinding Pulley

A grinding pulley is a grit-covered metal wheel that sharpens the knife blade. Grinding pulley condition affects sharpening which, in turn, affects cutting performance.

Required Tools and Supplies

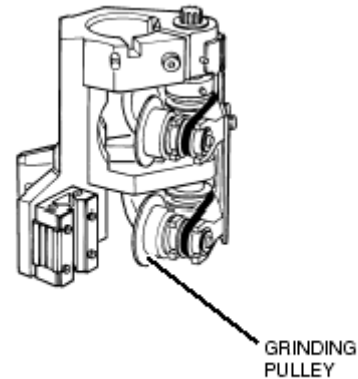
Item	Gerber Part No.
cleaning solvent	308010101
small brass brush	N/A
grinding pulley, 80 grit	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Inspect the grinding pulleys for missing grit, debris, and chips. Replace or clean the grinding pulley, as needed.
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.

Remove

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife blade (page 88).
4. Remove the grinding pulley drive belts (page 98).
5. Slide the four grinding pulleys from their supports.



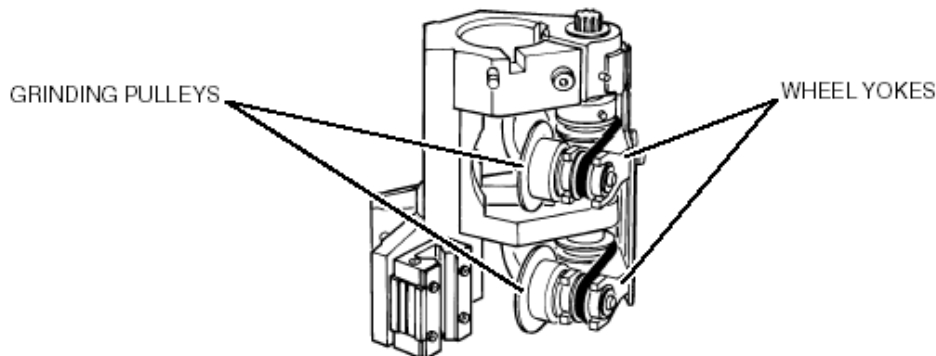
Clean

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Remove the knife blade (page 88).
4. Remove the grinding pulley drive belts (page 98).
5. Remove the grinding pulleys (page 101).
7. Using a stiff, brass brush, scrub the grinding pulley until you remove all deposits.
8. Air dry the grinding pulley until all solvent is gone.
9. Install the grinding pulley.
10. Check the wheel yokes for wear by trying to move them with your hand. If you can move the yokes easily, they are worn. Have them replaced by a Gerber qualified technician.

WARNING!  **Cleaning solvent is flammable. If it ignites, it may cause injury or death. Do not smoke or have other ignition sources near the solvent.**

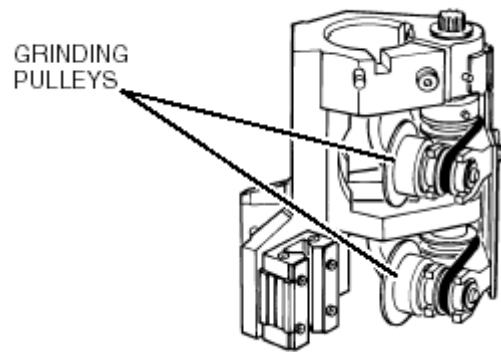
CAUTION: Do not get cleaning solvent on any bearings because it will damage them.

6. Spray the grinding pulley thoroughly with cleaning solvent.



Install

1. Slide the four grinding pulleys into their supports.
2. Install the grinding pulley drive belts (page 99).
3. Install the knife assembly (page 90).
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.



GT 7250 Elevator Flatway

The elevator flatway is a shaft on which the elevator rides up and down. As the elevator moves, it plunges and raises the knife.

Required Tools and Supplies

Item	Gerber Part No.
Petro-Lon multipurpose grease	596500005
grease gun	944233101
clean shop rag	N/A

Inspect

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Inspect the flatway for dirt and lint build-up. Clean and lubricate, if necessary.
4. Inspect the flatway for scratches, nicks, and gouges. If you see any of these problems, have it repaired or replaced by a Gerber qualified technician.

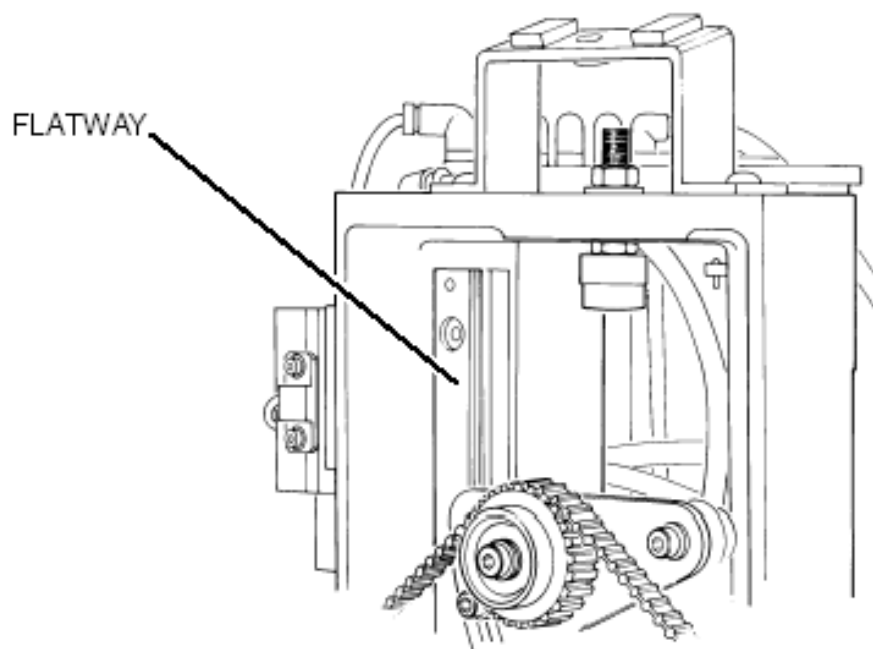
5. Close the cutter head front cover (page 46).
6. Turn on the C-200 controller.

Clean/Lubricate

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

CAUTION: Excess oil is thrown onto fabric and damages it. Use as little oil as possible. It is better to lubricate lightly and more frequently than to apply one heavy coat.

3. Use a grease gun to lubricate the grease fitting in the bearing on the flatway. Lubricate the bearing with Petro-Lon.
4. Close the cutter head front cover (page 46).
5. Turn on the C-200 controller.



GT 7250 Up/Down Stops

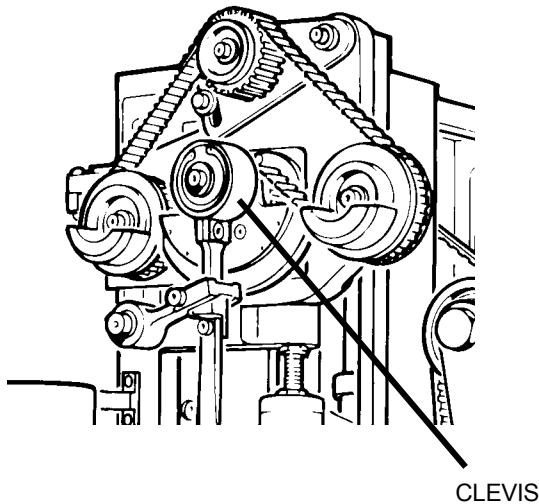
The up/down stops control the depth of knife cuts. If the down stop is set low, the knife cuts off the bristles in the BRISTLE SQUARES. If the down stop is set high, the knife does not cut bottom plies cleanly. Improper setting of the up stop causes poor sharpening.

Required Tools and Supplies

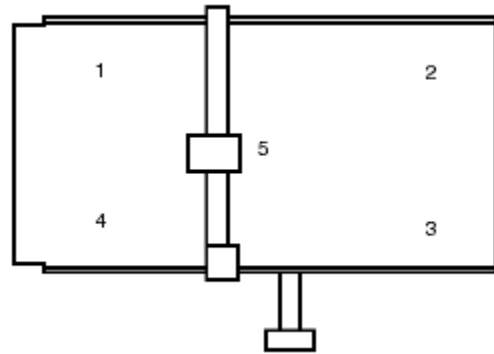
Item	Gerber Part No.
metric combination wrench set	945500093
screwdriver	945500298
appropriate standard combination wrench	944002506

Inspect

1. Turn on the C-200 controller.
2. Open the cutter head front cover (page 46).
3. Move the clevis to top dead center (TDC).



4. Cover the work surface with paper and then with plastic film.
5. You must check the knife penetration at five points on the cutting surface. Move the cutter head to test point 1 on the work surface. Test points 1 through 4 are about 30.5 cm (1 ft) from each corner. Test point 5 is at the center of the work surface.



6. Plunge the knife and presser foot bowl in the following manner:
 - a) At the BCP, turn on the **KNIFE** switch.
 - b) From the Diagnostics menu, select Input Output Test (DG23). See C-200MT online help or the *C-200MT Software User's Guide*. The monitor displays the DG 23 menu.
 - c) Use the arrow keys to highlight **Vacuum Enbl.**

- d) Hold down the CTRL key on your keyboard and press the SPACE bar. This turns on table vacuum.
- e) Use the arrow keys to highlight **Knife Sol.**
- f) Hold down the CTRL key on your keyboard and press the SPACE bar. This plunges the knife and the presser foot bowl. Note that the number in the KNIFE SOL line changes from 1 to 0.

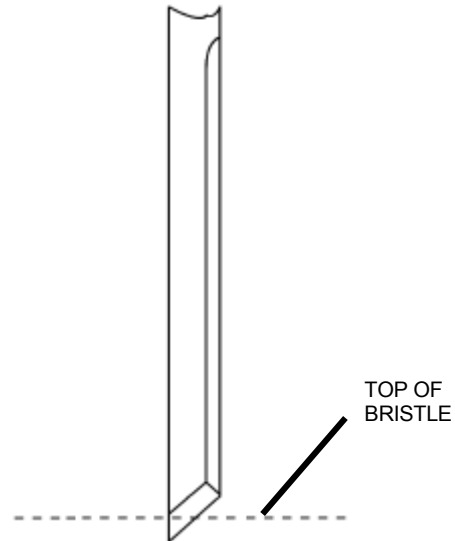
Note: Make a note of the air pressure reading on the presser foot air gauge on the operator console. You will need to reset the regulator to that pressure when you are through with this procedure.

- g) Turn off air to the presser foot bowl by releasing pressure for the regulator at the operator console. Do this by turning the regulator counterclockwise. You can now manually raise the presser foot bowl to see the knife.

7. Check the down stop setting as follows:

- a) With a pencil, mark the knife blade at the point it enters the plastic sealer.
- b) Hold down the CTRL key on your keyboard and press the SPACE bar. The knife rises.

- c) The pencil line should be 1.6 to 3.2 mm (1/16 to 1/8 inches) above the tip of the knife blade.



- d) Turn on air to the presser foot bowl by returning pressure at the regulator on the C-200 controller. Do this by turning the regulator clockwise.
8. Move the cutter head to the next test point and repeat steps 6 through 7 for test points 2 through 5. If needed, adjust the down stop.
9. On the BCP, press **VAC** to turn off table vacuum.
10. Press **Esc** to exit the DG 23 menu.

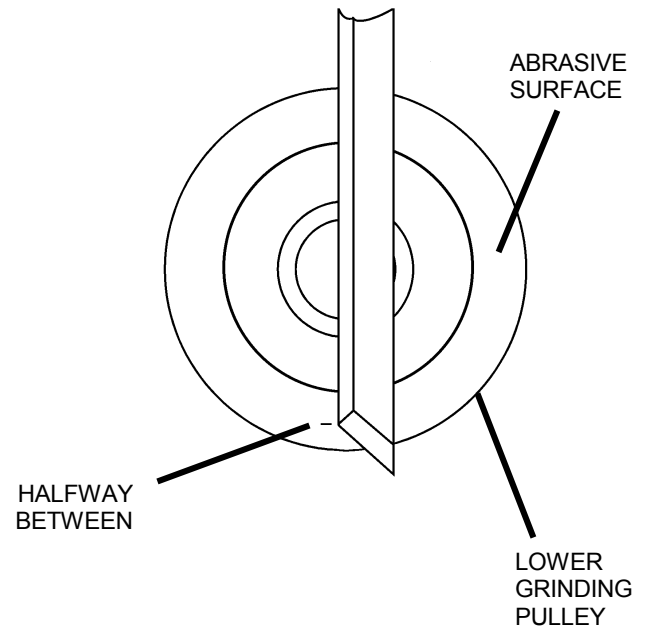
11. Check the up stop setting.

- a) Position the clevis at top dead center (TDC).
- b) Check the knife blade. The corner edge of the knife blade should be in the center of the abrasive surface of the lower grinding pulley.
- c) Adjust the up stop, if necessary.

12. Exit the C-200MT program.

13. Turn off the C-200 controller.

14. Close the cutter head front cover (page 46).



Adjust

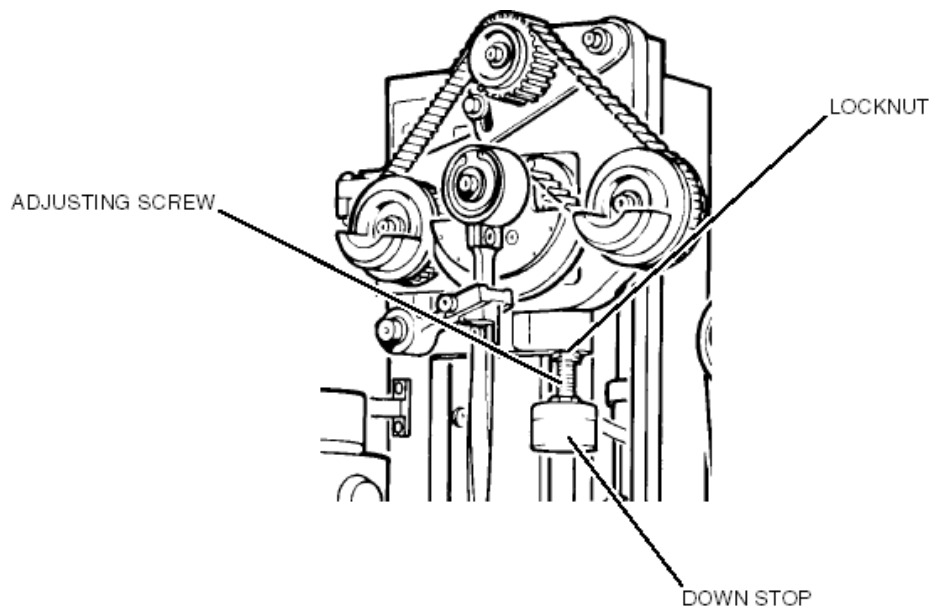
If you find, during the up and down stop checks, (previous pages) that you need to adjust the head up/down stops, follow this procedure.

CAUTION: Too much penetration damages the work surface; too little keeps pieces from being cut completely.

Note: Because of tolerances, bristle penetration usually cannot be set to the same value for each test point. However, all test points must be within the proper range.

1. Turn off the C-200 controller.
2. Open the cutter head front cover (page 46).

3. Adjust the down stop:
 - a) Loosen the locknut on the adjusting screw by turning it counterclockwise.
 - b) Turn the adjusting screw clockwise or counterclockwise to change the depth of bristle penetration.
 - c) Inspect the up/down stop settings as described on page 105.
 - d) Tighten the locknut by turning it clockwise.
 - e) Repeat steps 3a through 3d until bristle penetration is set correctly.

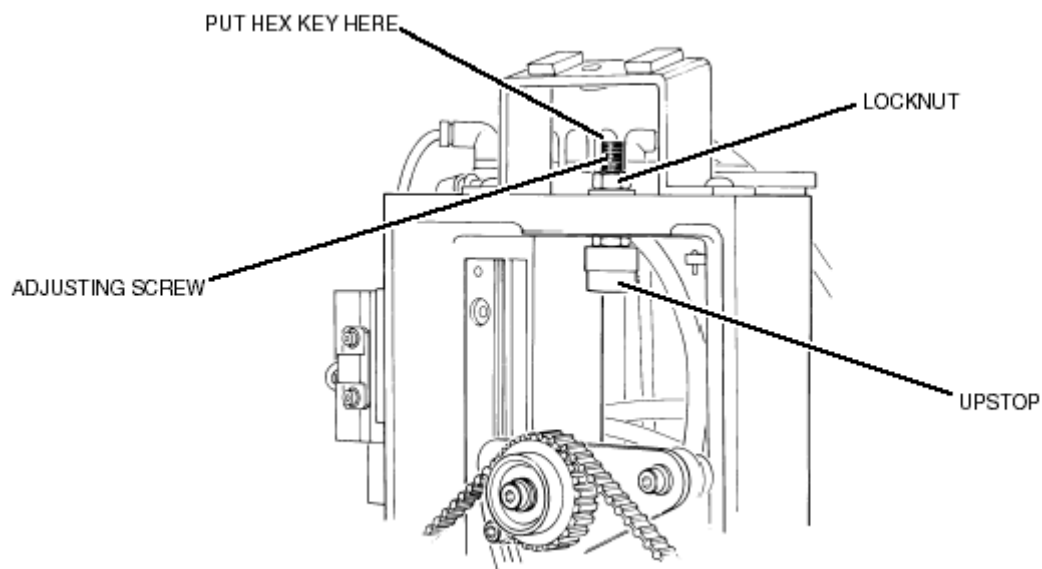


4. Adjust the up stop.

- a) Loosen the lock nut on the adjusting screw by turning it counterclockwise.
- b) Put a hex key into the adjusting screw. Turn the adjusting screw clockwise or counterclockwise to change the up stop setting.
- c) Inspect the up/down stop settings (page 105).
- d) Tighten the lock nut by turning it clockwise.
- e) Repeat steps 4a through 4d until the up stop is set correctly.

5. Close the cutter head front cover (page 46).

6. Turn on the C-200 controller.

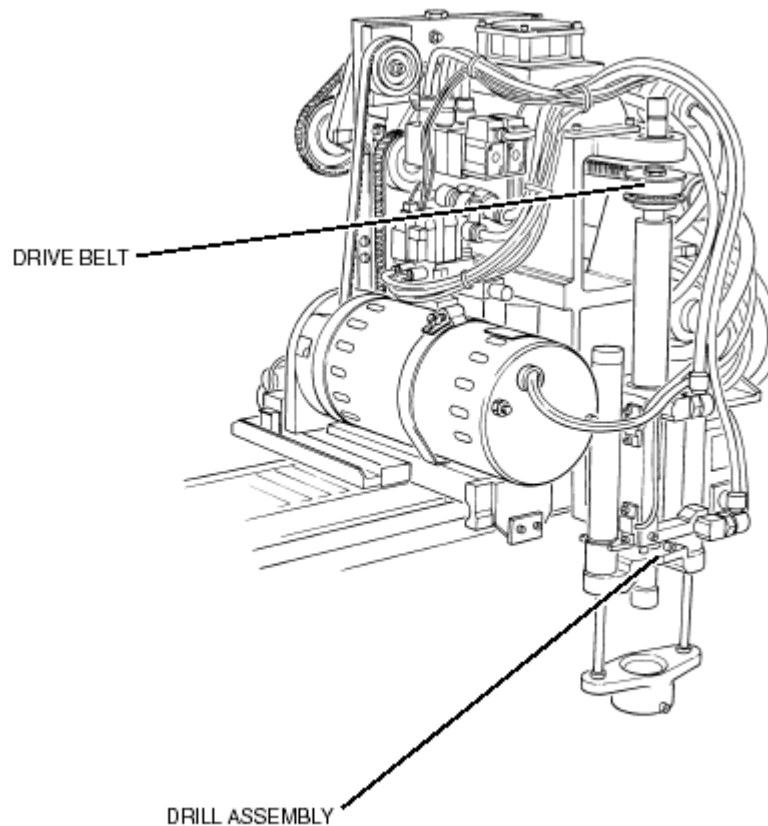


Drill Maintenance

Drill for the GT 5250

The optional drill attaches to the cutter head. It lets the GERBERcutter automatically make drill holes as it cuts the material. A dual drill option is also available. The sections beginning on the next page describe maintenance for the GT 5250 drill option.

Note: Maintenance procedures for the GT 7250 drill option begin on page 121.



Drill Option for GT 5250

GT 5250 Drill Drive Belt

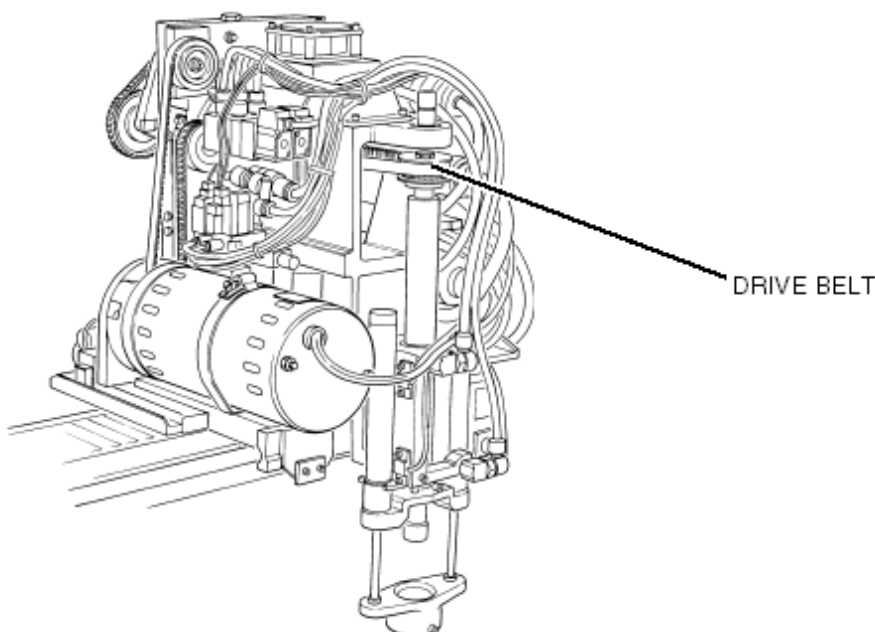
The drive belt connects the drill to the drill motor.

Required Tools and Supplies

Item	Gerber Part No.
belt deflection gauge	45412000
metric combination wrench set	945500093
appropriate standard combination wrench	944002506
appropriate standard right-handle hex key	
pliers	N/A
drill drive belt	Refer to Service CD

Inspect

1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. Check the drive belt for glazing, cuts, and nicks. Replace it, if necessary.
5. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) Adjust the drive belt, if necessary.
6. Install the cutter head back cover (page 48).
7. Install the cutter head front cover (page 46).
8. Turn on the C-200 controller.

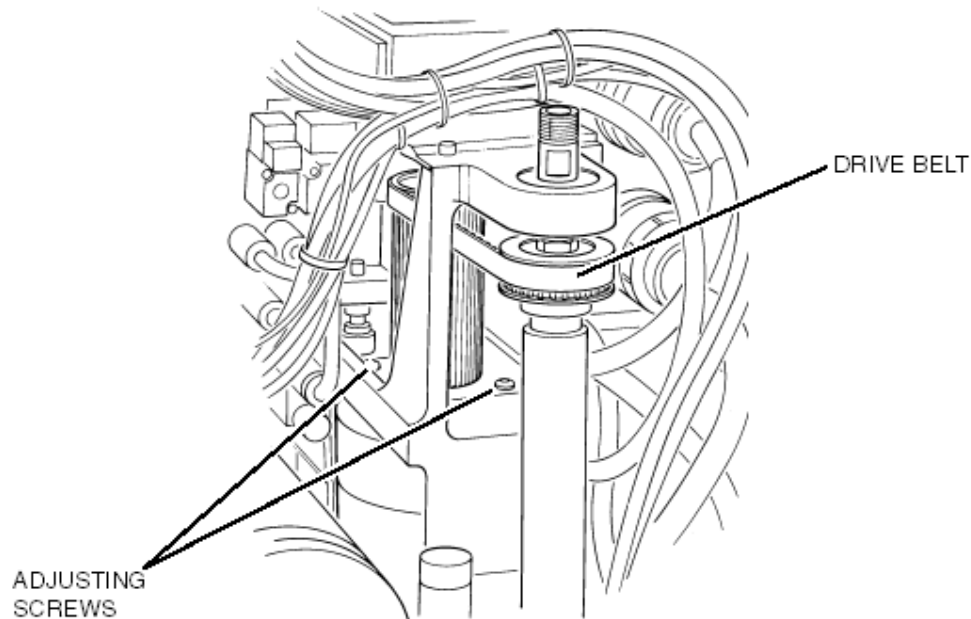


Remove

1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. Remove the drill bit and ejector rod (page 116).
5. Loosen the four adjusting screws.
6. Slide the drill motor toward the drill.
7. Slide the drive belt off the pulleys.
4. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) If the tension is not correct, loosen the adjusting screws and repeat steps 2 through 4 until the deflection is correct.
5. Install the drill bit and ejector rod (page 117).
6. Install the cutter head back cover (page 48).
7. Install the cutter head front cover (page 46).

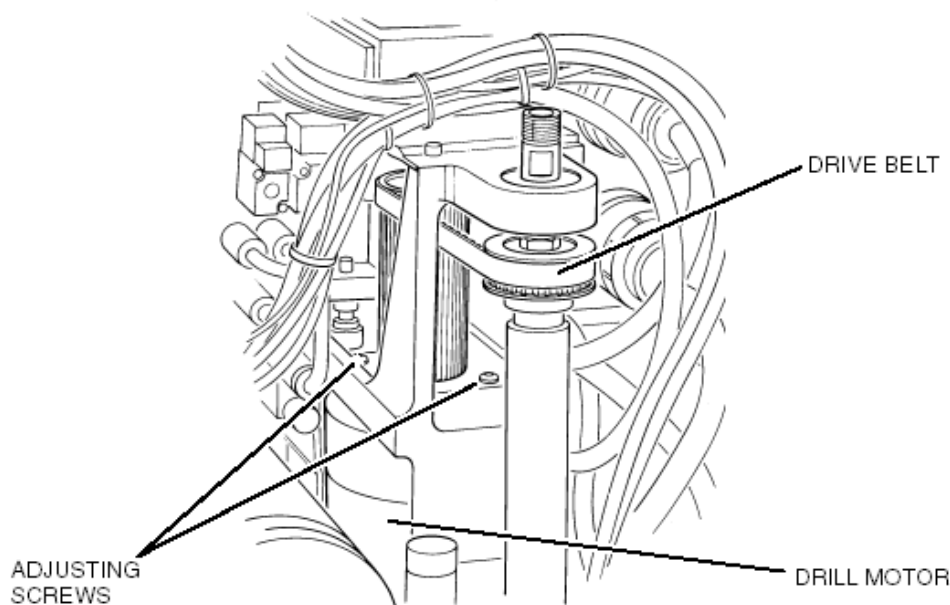
Install

1. Slide the drive belt onto the pulleys.
2. Slide the drill motor to apply the correct tension to the drive belt.
3. Tighten the adjusting screws.
8. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
 2. Remove the cutter head front cover (page 46).
 3. Remove the cutter head back cover (page 48).
 4. Loosen the drill motor adjusting screws.
 5. Slide the drill motor toward the drill to loosen the drive belt.
- OR**
6. Slide the drill motor away from the drill to tighten the drive belt.
 7. Tighten the mounting screws.
 8. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) Repeat steps 4 through 7 until the drill drive belt tension is correct.
 9. Install the cutter head back cover (page 48).
 10. Install the cutter head front cover (page 46).
 11. Turn on the C-200 controller.



GT 5250 Drill Bit

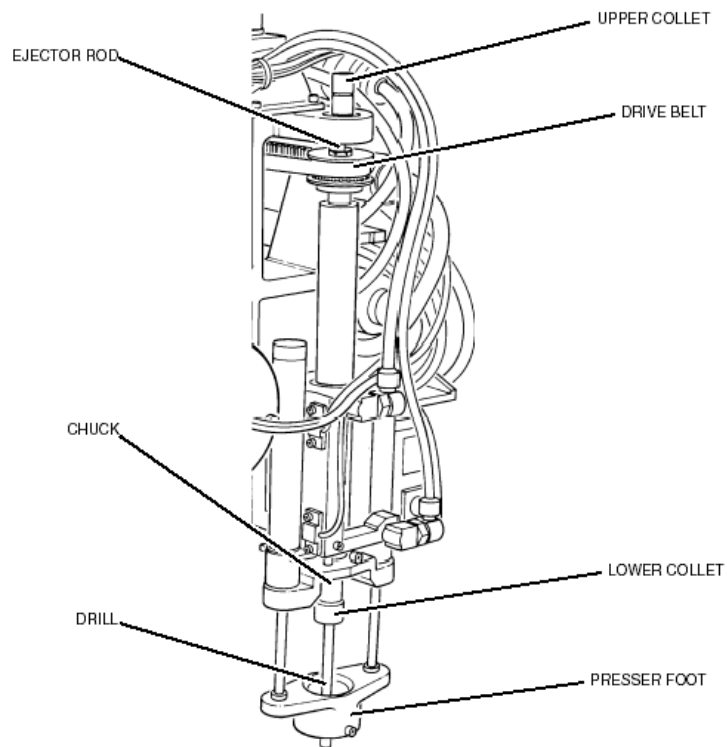
The drill bit is the part of the drill that cuts holes in the material. Drill bits are available in a variety of sizes and types.

Required Tools and Supplies

Item	Gerber Part No.
appropriate standard combination wrench	944002506
drill bits, ejector rods, collets	See page 120

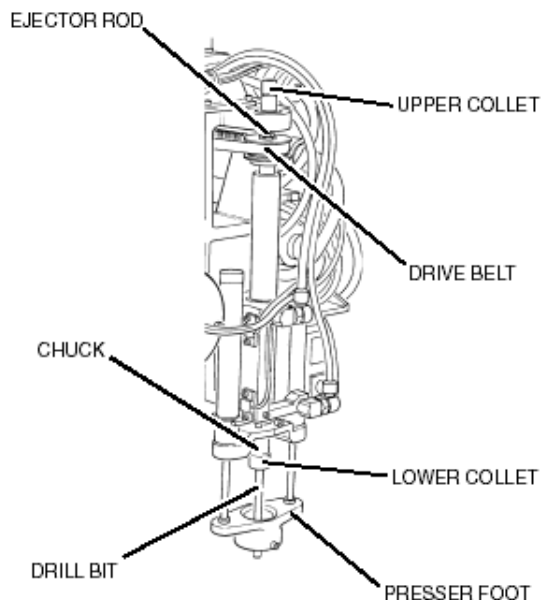
Inspect

1. Turn off the C-200 controller.
2. Remove the cutter head back cover (page 48).
3. Check the cutting surface of the drill bit for nicks, burrs, and dullness. Check that the drill bit is not bent. If necessary, replace the drill bit.
4. Install the cutter head back cover (page 48).
5. Turn on the C-200 controller.

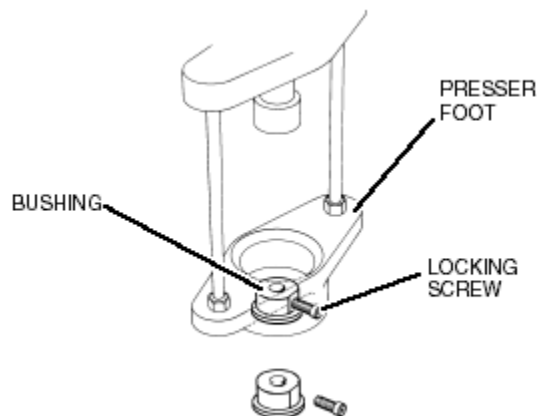


Remove

1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. If you are using a hollow bit, remove the ejector rod.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the upper collet.
 - c) Slide the ejector rod up and out through the center of the collet.

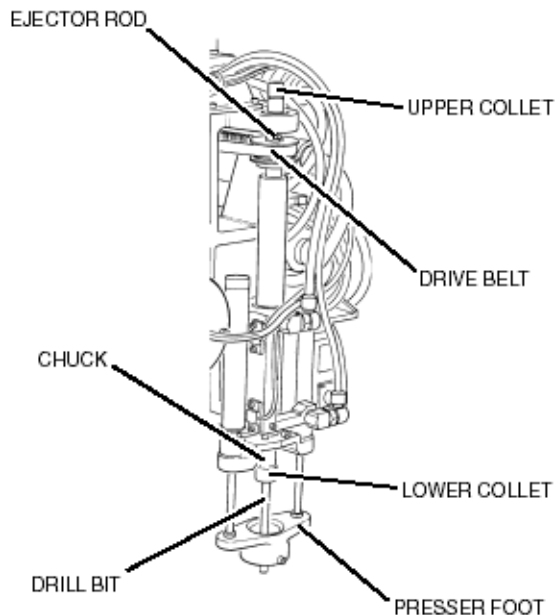


5. Remove the drill bushing.
 - a) Loosen the locking screw.
 - b) Pull the bushing out from the bottom of the presser foot assembly.
6. Remove the drill bit.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the lower collet.
 - c) Pull the drill bit down out of the collet through the presser foot assembly.



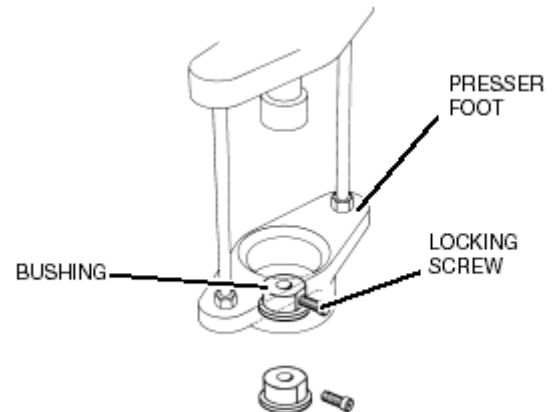
Install

1. If you are installing a new drill bit, use the table on page 120 to select the matching ejector rod and collet.
2. Select and install the new drill bit.
 - a) Slide the drill bit up through the center of the lower collet until the drill bit will not slide up any more.
 - b) Hold the chuck in place with a wrench.
 - c) Tighten the lower collet finger tight.
 - d) With another wrench, tighten the lower collet 1/4 turn more. This holds the drill in place.

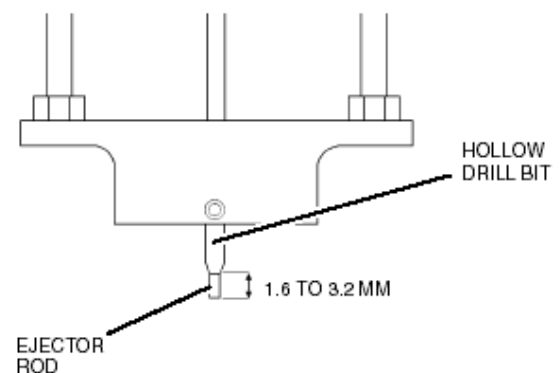


3. Install the drill bushing.
 - a) Slide the drill bushing up into place from under the presser foot assembly. The bushing fits around the drill bit.

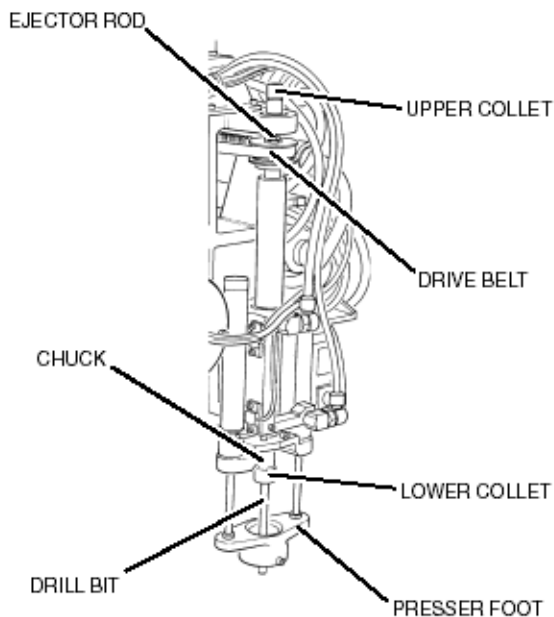
- b) Install the locking screw.



4. If you are installing a hollow bit, install the correct ejector rod (see page 120).
 - a) Make sure that the upper collet is in place and loose.
 - b) Slide the ejector rod down through the center of the collet.
 - c) Align the ejector rod so that it is 1.6 to 3.2 mm (1/16 in. to 1/8 in.) lower than the bottom of the drill with the drill up.

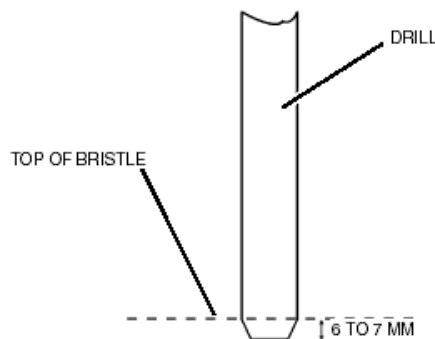


- d) Tighten the upper collet finger tight.
 - e) Use a wrench to tighten the upper collet 1/4 turn more. This holds the ejector rod in place.
5. Tighten the chuck finger-tight.
6. Use an open-end wrench to tighten the chuck 1/4 turn more.



7. Check drill penetration into the bristle.
- a) Turn on the C-200 controller.
 - b) At the BCP, turn on the **KNIFE** switch.
 - c) From the Diagnostics menu, select Input Output Test (DG23). See C-200MT online help or the *C-200MT Software User's Guide*.
The monitor displays the DG 23 menu.
 - d) Use the arrow keys to highlight **Drill Motor**.
 - e) Hold down the CTRL key on your keyboard and press the SPACE bar. This turns on the drill motor.
 - f) Use the arrow keys to highlight **Drill Sol**.

- g) Hold down the CTRL key on your keyboard and press the SPACE bar. This plunges the drill and the presser foot bowl.
 - h) With a pencil, mark the drill at the point it enters the bristle.
 - i) Hold down the CTRL key on your keyboard and press the SPACE bar. The drill rises.
 - j) Press **Esc** to exit the diagnostic.
 - k) Turn off the C-200 controller.
 - l) Measure the distance from the pencil line on the drill to the tip of the drill. The pencil line should be 6 to 7 mm (1/4 inch) above the tip of the drill.
8. If the pencil line is not the correct distance from the tip of the drill, adjust the drill penetration.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the lower collet.
 - c) Slide the drill bit up or down to decrease or increase drill penetration.
 - d) Hold the chuck in place with a wrench.
 - e) Tighten the lower collet finger tight.
 - f) With another wrench, tighten the lower collet 1/4 turn more. This holds the drill in place.
 - g) Repeat steps 7 and 8 until drill penetration is correct.
 9. Install the cutter head back cover (page 48).
 10. Install the cutter head front cover (page 46).
 11. Turn on the C-200 controller.



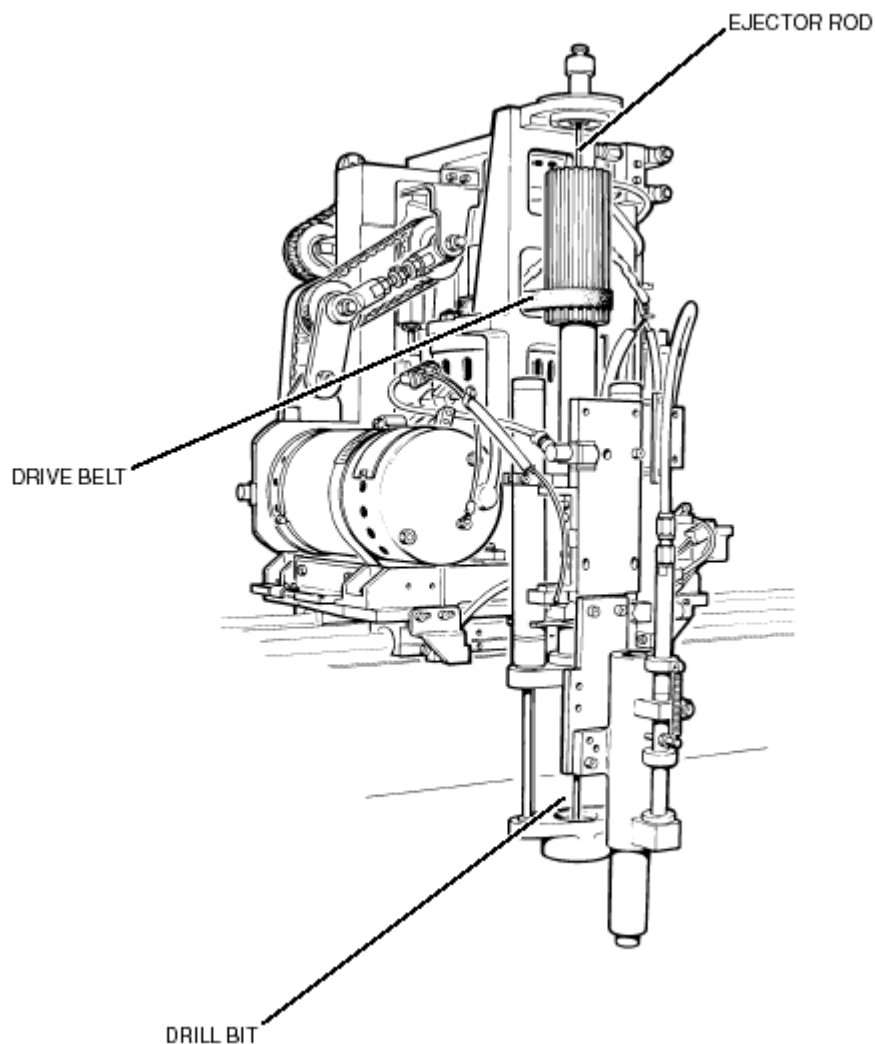
GT 5250 Standard Drill Components

Drill Bit			Accessories Part No.
Size (mm/in.)	Type	Gerber Part No.	
1.2 / 0.046	solid	57500000	55738020
1.6 / 0.062	solid	66598000	55738028
2.0 / 0.078	solid groove point	62396000	55738027
2.4 / 0.093	half-round	58244200	55738023
2.4 / 0.093	hollow	55544203	55738000
3.2 / 0.125	half-round	66600000	55738029
3.2 / 0.125	hollow	55544204	55738001
4.0 / 0.156	hollow	55543007	55738002
4.8 / 0.187	hollow	55543008	55738003
6.4 / 0.250	hollow	55545003	55738004
7.9 / 0.312	hollow	55545004	55738005

Drill for the GT 7250

The optional drill attaches to the cutter head. It lets the GERBERcutter automatically make drill holes as it cuts the material. A dual drill option is also available. The sections beginning on the next page describe maintenance for the GT 7250 drill option.

Note: Maintenance procedures for the GT 5250 drill option begin on page 111.



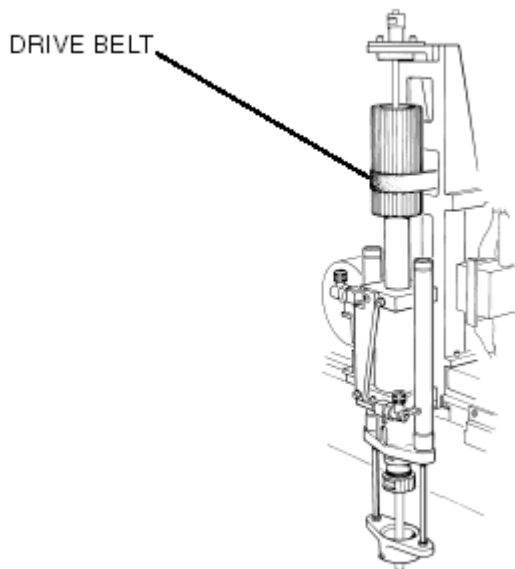
Drill Option for GT 7250

GT 7250 Drill Drive Belt

The drive belt connects the drill to the drill motor.

Required Tools and Supplies

Item	Gerber Part No.
belt deflection gauge	45412000
metric combination wrench set	945500093
appropriate metric right-angle hex key set	945500094
standard combination wrench	944002506
appropriate standard right-handle hex key	
pliers	N/A
drill drive belt	Refer to Service CD



Inspect

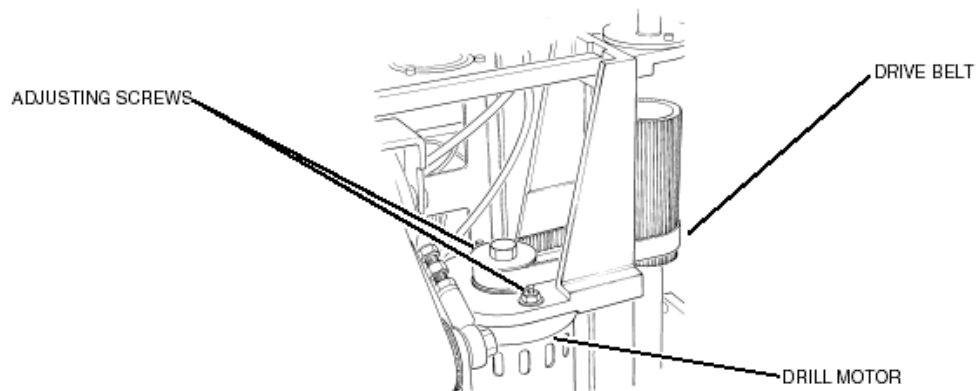
1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. Check the drive belt for glazing, cuts, and nicks. Replace it, if necessary.
5. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) Adjust the drive belt, if necessary.
6. Install the cutter head back cover (page 48).
7. Install the cutter head front cover (page 46).
8. Turn on the C-200 controller.

Remove

1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. Remove the drill bit and ejector rod (page 126).
5. Loosen the adjusting screws.
6. Slide the drill motor toward the drill.
7. Slide the drive belt off the pulleys.
4. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) If the tension is not correct, loosen the adjusting screws and repeat steps 2 through 4 until the deflection is correct.

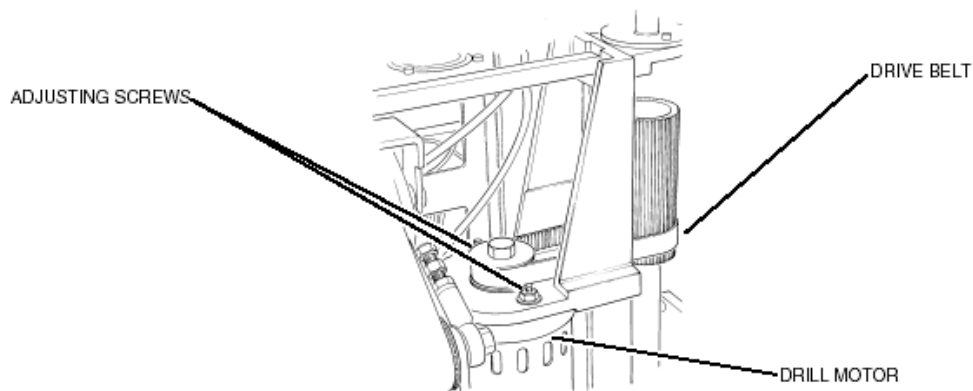
Install

1. Slide the drive belt onto the pulleys.
2. Slide the drill motor to apply the correct tension to the drive belt.
3. Tighten the adjusting screws.
5. Install the drill bit and ejector rod (page 127).
6. Install the cutter head back cover (page 48).
7. Install the cutter head front cover (page 46).
8. Turn on the C-200 controller.



Adjust

1. Turn off the C-200 controller.
 2. Remove the cutter head front cover (page 46).
 3. Remove the cutter head back cover (page 48).
 4. Loosen the drill motor adjusting screws.
 5. Slide the drill motor toward the drill to loosen the drive belt.
- OR**
6. Slide the drill motor away from the drill to tighten the drive belt.
 7. Tighten the adjusting screws.
 8. Check the drive belt for proper tension.
 - a) Measure the deflection of the drive belt midway between the drill motor pulley and the drill pulley. The belt deflection should be 6.35 mm (0.25 in.) at 0.91 to 1.82 kg (2 to 4 lb.).
 - b) Repeat steps 3 through 6 until the drill drive belt tension is correct.
 9. Install the cutter head back cover (page 48).
 10. Install the cutter head front cover (page 46).
 11. Turn on the C-200 controller.



GT 7250 Drill Bit

The drill bit is the part of the drill that cuts holes in the material. Drill bits are available in a variety of sizes and types.

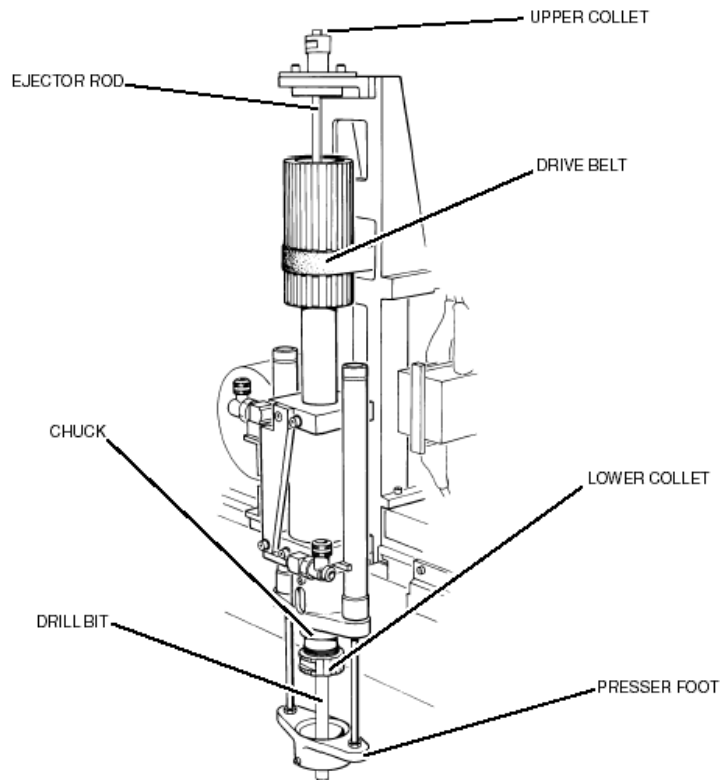
Required Tools and Supplies

Item	Gerber Part No.
appropriate standard combination wrench	944002506
drill bits, ejector rods, collets	See page 130

Inspect

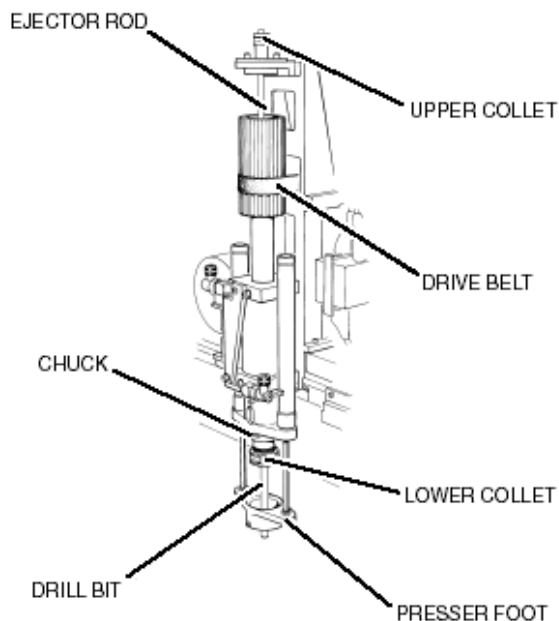
1. Turn off the C-200 controller.

2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. Check the cutting surface of the drill bit for nicks, burrs, and dullness. Check that the drill bit is not bent. If necessary, replace the drill bit.
5. Install the cutter head back cover (page 48).
6. Install the cutter head front cover (page 46).
7. Turn on the C-200 controller.

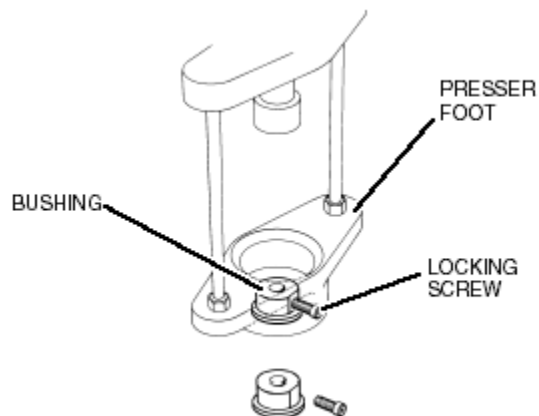


Remove

1. Turn off the C-200 controller.
2. Remove the cutter head front cover (page 46).
3. Remove the cutter head back cover (page 48).
4. If you are using a hollow bit, remove the ejector rod.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the upper collet.
 - c) Slide the ejector rod up and out through the center of the collet.

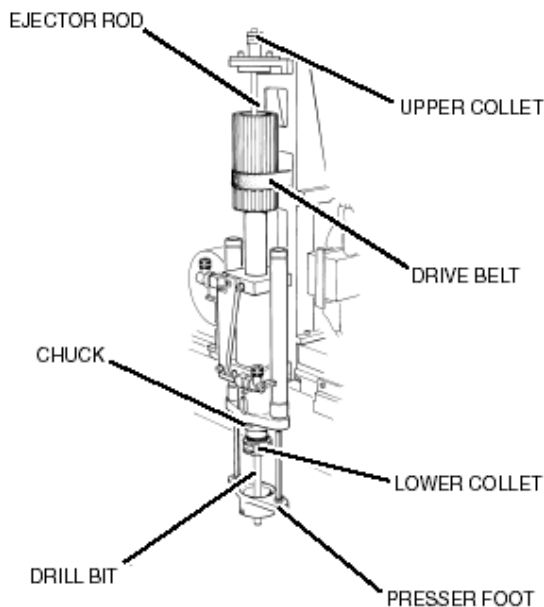


5. Remove the drill bushing.
 - a) Loosen the locking screw.
 - b) Pull the bushing out from the bottom of the presser foot assembly.
6. Remove the drill bit.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the lower collet.
 - c) Pull the drill bit down out of the collet through the presser foot assembly.



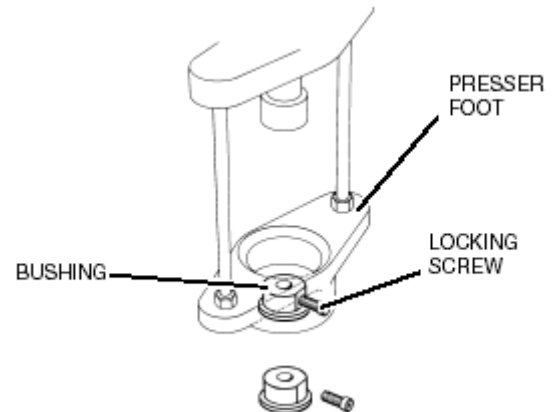
Install

1. If you are installing a new drill bit, use the tables on page 130 to select the matching ejector rod and collet.
2. Select and install the new drill bit.
 - a) Slide the drill bit up through the center of the lower collet until the drill bit will not slide up any more.
 - b) Hold the chuck in place with a wrench.
 - c) Tighten the lower collet finger tight.
 - d) With another wrench, tighten the lower collet 1/4 turn more. This holds the drill in place.

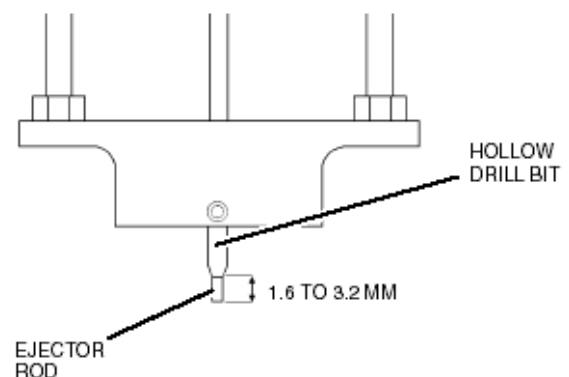


3. Install the drill bushing.
 - a) Slide the drill bushing up into place from under the presser foot assembly. The bushing fits around the drill bit.

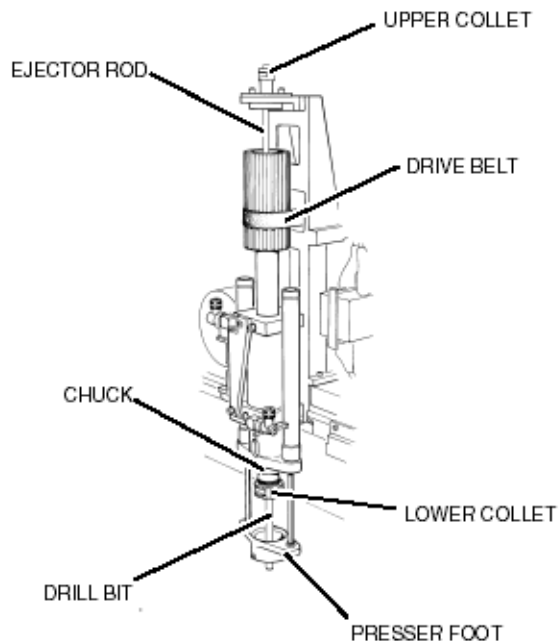
- b) Install the locking screw.



4. If you are installing a hollow bit, install the correct ejector rod (see table on page 130).
 - a) Make sure that the upper collet is in place and loose.
 - b) Slide the ejector rod down through the center of the collet.
 - c) Align the ejector rod so that it is 1.6 to 3.2 mm (1/16 in. to 1/8 in.) lower than the bottom of the drill with the drill up.

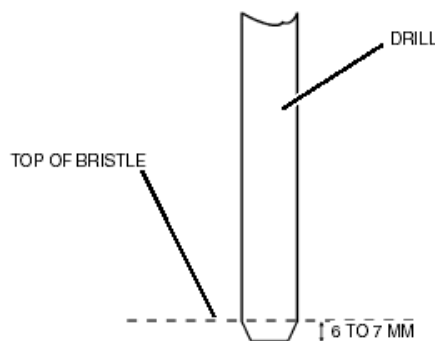


- d) Tighten the upper collet finger tight.
 - e) Use a wrench to tighten the upper collet 1/4 turn more. This holds the ejector rod in place.
5. Tighten the chuck finger-tight.
6. Use an open-end wrench to tighten the chuck 1/4 turn more.



7. Check drill penetration into the bristle.
- a) Turn on the C-200 controller.
 - b) At the BCP, turn on the **KNIFE** switch.
 - c) From the Diagnostics menu, select Input Output Test (DG23). See C-200MT online help or the *C-200MT Software User's Guide*.
The monitor displays the DG 23 menu.
 - d) Use the arrow keys to highlight **Drill Motor**.
 - e) Hold down the CTRL key on your keyboard and press the SPACE bar. This turns on the drill motor.
 - f) Use the arrow keys to highlight **Drill Sol**.

- g) Hold down the CTRL key on your keyboard and press the SPACE bar. This plunges the drill and the presser foot bowl.
 - h) With a pencil, mark the drill at the point it enters the bristle.
 - i) Hold down the CTRL key on your keyboard and press the SPACE bar. The drill rises.
 - j) Press **Esc** to exit the diagnostic.
 - k) Turn off the C-200 controller.
 - l) Measure the distance from the pencil line on the drill to the tip of the drill. The pencil line should be 6 to 7 mm (1/4 inch) above the tip of the drill.
8. If the pencil line is not the correct distance from the tip of the drill, adjust the drill penetration.
 - a) Hold the chuck in place with a wrench.
 - b) Use another wrench to loosen the lower collet.
 - c) Slide the drill bit up or down to decrease or increase drill penetration.
 - d) Hold the chuck in place with a wrench.
 - e) Tighten the lower collet finger tight.
 - f) With another wrench, tighten the lower collet 1/4 turn more. This holds the drill in place.
 - g) Repeat steps 7 and 8 until drill penetration is correct.
 9. Install the cutter head back cover (page 48).
 10. Install the cutter head front cover (page 46).
 11. Turn on the C-200 controller.



GT 7250 Drill Components

For Standard Drill

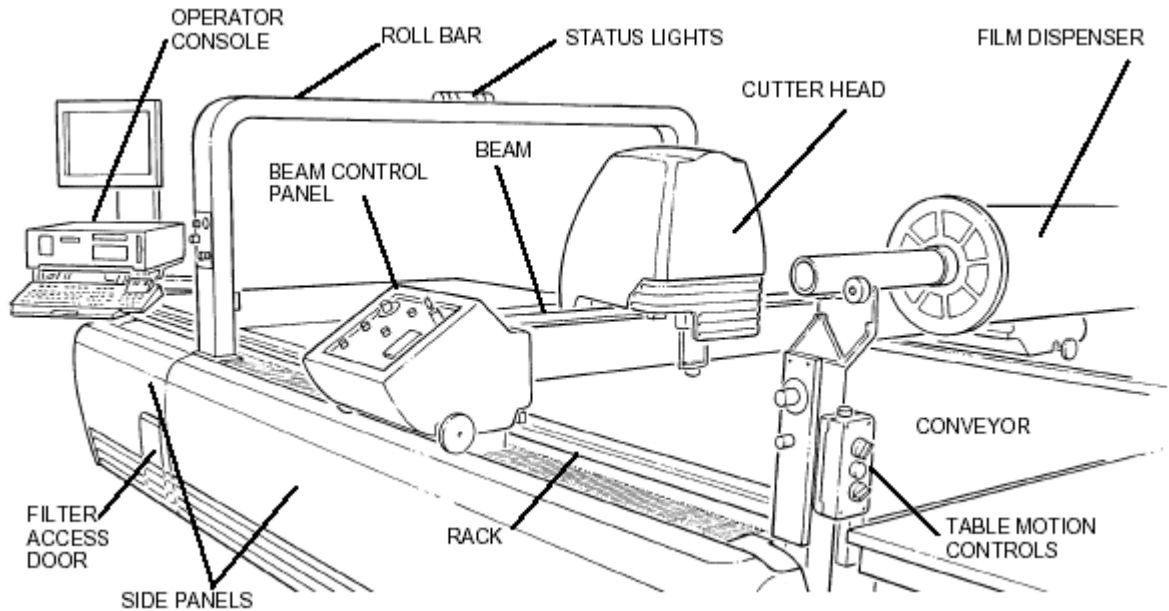
Drill Bit			Accessories Part No.
Size (mm/in.)	Type	Gerber Part No.	
1.2 / 0.046	solid	44330000	64385000
1.6 / 0.062	solid	44330001	64387000
2.0 / 0.078	solid groove point	30595003	64389000
2.4 / 0.093	half-round	30595002	64391000
2.4 / 0.093	hollow	24351001	64393000
3.2 / 0.125	half-round	30595001	64395000
3.2 / 0.125	hollow	24351002	64397000
4.0 / 0.156	hollow	24351003	64399000
4.8 / 0.187	hollow	24351004	64401000
6.4 / 0.250	hollow	44330000	64385000

For Heavy-duty Drill

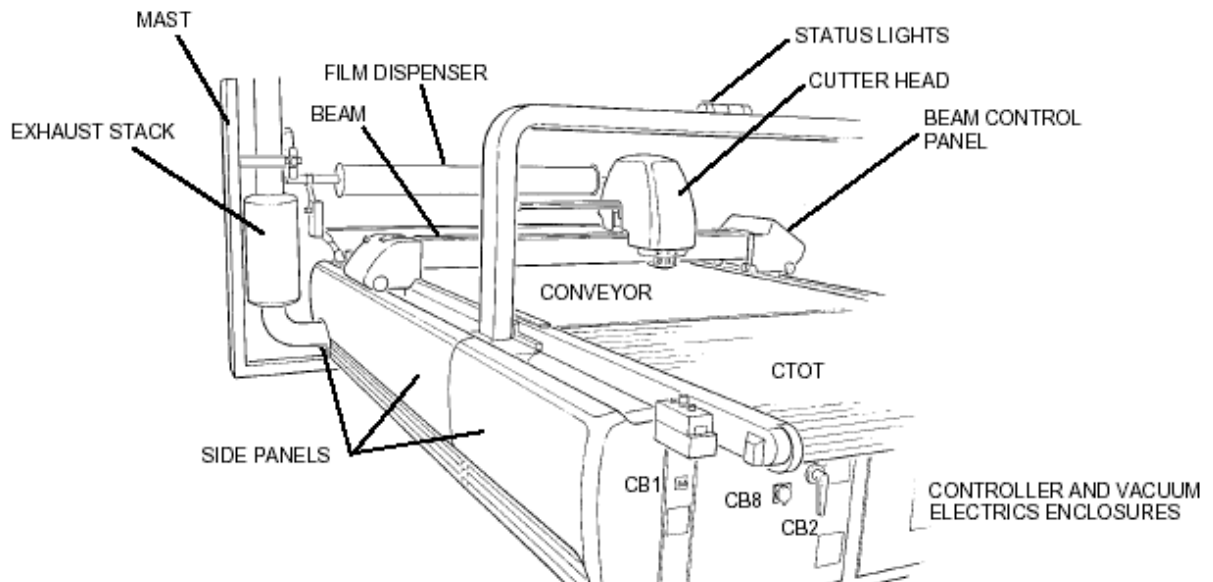
Drill Bit			Accessories Part No.
Size (mm/in.)	Type	Gerber Part No.	
1.2 / 0.046	solid	44330000	64386000
1.6 / 0.062	solid	44330001	64388000
2.0 / 0.078	solid groove point	30595003	64390000
2.4 / 0.093	half-round	30595002	64392000
2.4 / 0.093	hollow	24351001	64394000
3.2 / 0.125	half-round	30595001	64396000
3.2 / 0.125	hollow	24351002	64398000
4.0 / 0.156	hollow	24351003	64400000
4.8 / 0.187	hollow	24351004	64402000
6.4 / 0.250	hollow	72490000	64404000
7.9 / 0.312	hollow	72490001	64405000
9.5 / 0.375	hollow	62379001	62610000
12.7 / 0.500	hollow	62379002	62611000
15.0 / 0.591	hollow	62379003	62745000

Cutting Table Maintenance

The cutting table provides a work surface for cutting material. Its major assemblies include the beam, cutter head, conveyor, vacuum generator, and C-200 controller.



Operator Side from Take-on End



Remote Side from Take-off End

Racks

The racks are two, toothed tracks that run the length of the cutting table. Motors and gears in the beam drive the beam along the racks. The beam cannot travel smoothly, if the racks are dirty.

Required Tools and Supplies

Item	Gerber Part No.
clean shop rag	N/A
paint brush	N/A

Inspect

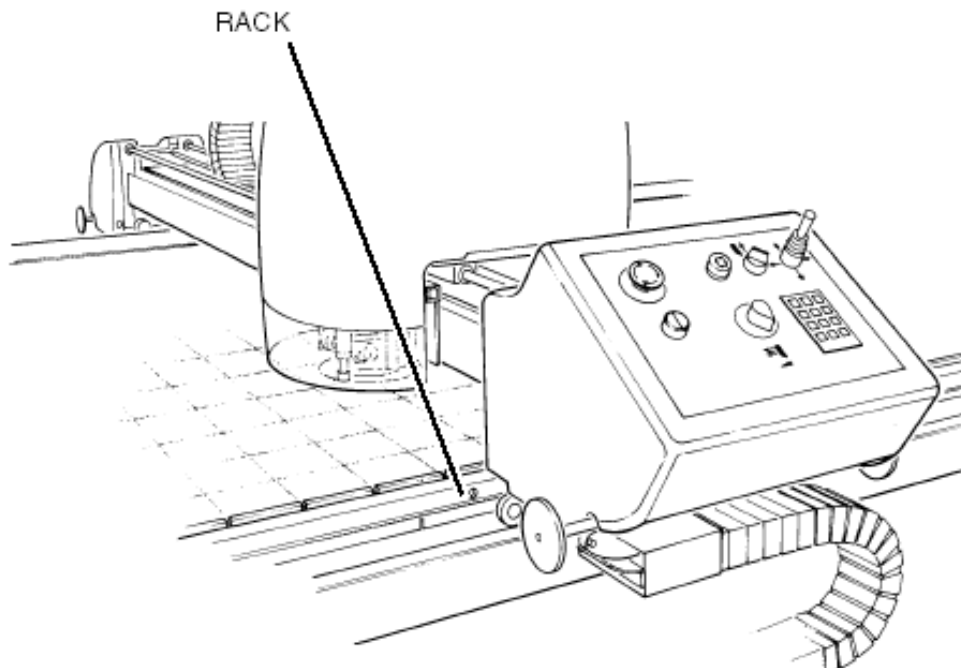
1. Turn off the C-200 controller.
2. Inspect the racks for built-up lint and dirt. Clean them, if necessary.
3. Turn on the C-200 controller.

Clean

1. Turn off the C-200 controller.
2. Remove all side panels (page 135).
3. Clean the racks with a paint brush and rag.

CAUTION: Do not lubricate the racks. Grease and oil cause lint to build up rapidly on the racks, make cleaning more difficult, and get thrown or rubbed into fabric.

4. Install all side panels (page 135).
5. Turn on the C-200 controller.



Flatways

The flatways are two, flat surfaces that run the length of the cutting table. Motors and gears in the beam drive the beam along the flatways. The beam cannot travel smoothly, if the flatways are dirty.

Required Tools and Supplies

Item	Gerber Part No.
clean shop rag	N/A
paint brush	N/A

Inspect

1. Turn off the C-200 controller.
2. Inspect the flatways for built-up lint and dirt. Clean them, if necessary.
3. Turn on the C-200 controller.

Clean

1. Turn off the C-200 controller.
2. Remove all side panels (page 135).
3. Clean the flatways with a paint brush and rag.

CAUTION: Do not lubricate the flatways. Grease and oil cause lint to build up rapidly on the flatways, make cleaning more difficult, and get thrown or rubbed into fabric.

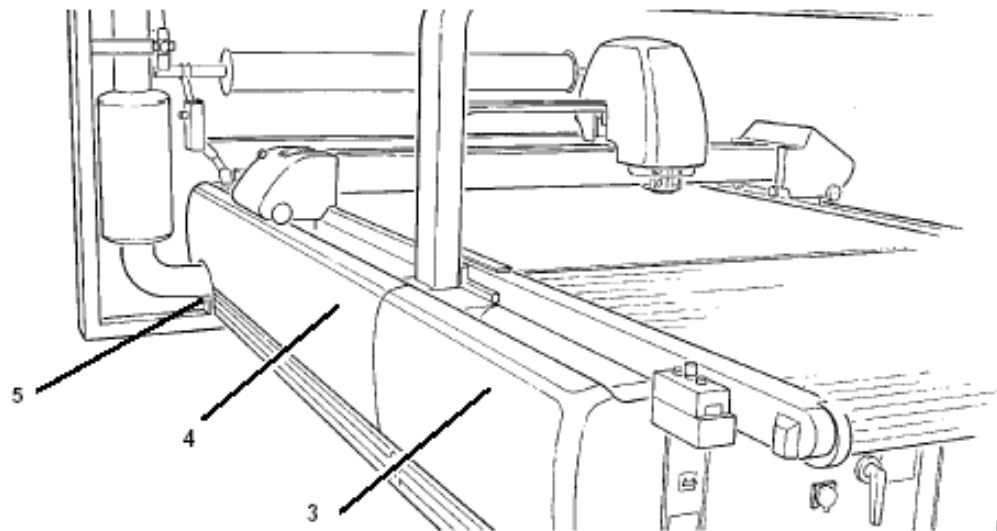
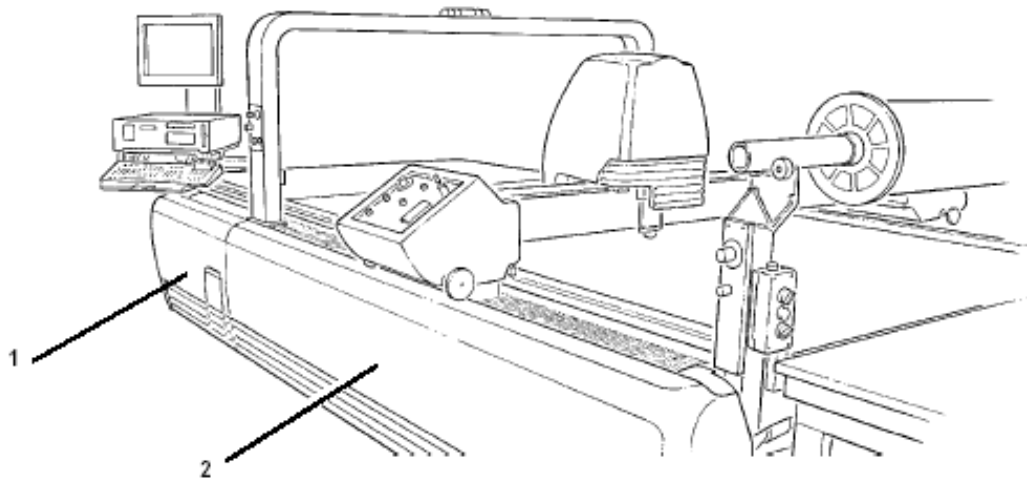
4. Install all side panels (page 135).
5. Turn on the C-200 controller.

Side Covers

The design of the side covers on your GERBERcutter provides additional protection of your investment in the cutting room environment. If a maintenance procedure requires removal of a side cover panel, follow the procedure on the next page.

There are two side cover panels on the operator side of the cutter (1 and 2 below).

There are two major side cover panels on the remote side of the cutter (3 and 4 below). An additional small panel (5 below) is located at the exhaust stack and mast.



Remove

1. Use a socket wrench to remove the bolts located along the bottom flange of each side cover that needs to be removed.

Note: If removing the side cover near the stack, first remove the small stack side cover.

2. With the help of an assistant, lift the cover away from the cutter.

Install

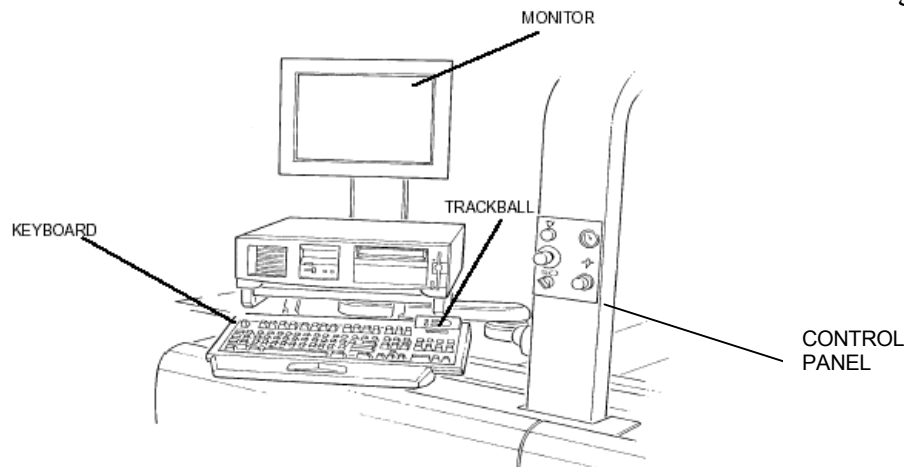
1. With the help of an assistant, lift the cover into position at the side of the cutter.
2. Replace and tighten the bolts.

Clean

1. Wipe side covers with a soft cloth dampened in water and mild detergent.

Operator Console

The operator console runs the entire GERBERcutter. It includes a PC, monitor, keyboard, trackball and control panel. Refer to the *GT Series GERBERcutters Getting Started Manual* for information on using controls at the operator console.

**Monitor**

The monitor is a flat panel screen that lets you communicate with the GERBERcutter. It displays the commands you enter and the messages the operator console sends.

Inspect

1. Inspect the screen for dust and smudges. Clean the screen, if needed.

Clean

1. Clean the screen with an anti-static glass cleaner and a clean, soft cloth.

Keyboard

The keyboard is a device with keys, like those on a typewriter, that lets you enter data. The trackball allows you to make selections on the screen. Both let you communicate with the C-200MT program.

Inspect

1. Inspect the keyboard for dirt buildup. Clean these surfaces, if needed

Clean

1. Clean the keyboard and trackball with a clean, soft cloth dampened with water and detergent.

Take-On Ramp

The take-on ramp is a smooth, flat piece of stainless steel at the take-on end of the cutting table. This lets you easily load material for cutting onto the conveyor.

Required Tools and Supplies

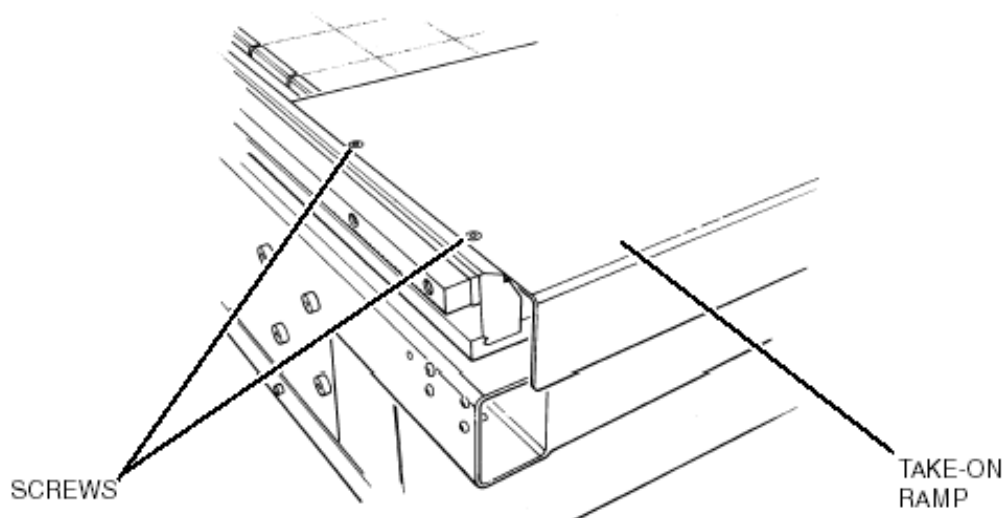
Item	Gerber Part No.
metric T-handle hex key set	945500089

Install

1. Place the take-on ramp onto the cutting table. Make sure the screw holes on the ramp align with those on the cutting table.
2. Install the screws that fasten the take-on ramp to the cutting table.
3. Turn on the C-200 controller.

Remove

1. Turn off the C-200 controller.
2. Remove the screws that fasten the take-on ramp to the cutting table.
3. Lift the take-on ramp off the cutting table.



Conveyor

The conveyor is a moving work surface for cutting material. Proper cleaning and lubrication are vital for maintaining cut quality.

BRISTLE SQUARES

The GERBER BRISTLE SQUARES® form the conveyor's work surface. They are plastic squares with bristles on the top side and a perforated base on the bottom. The BRISTLE SQUARES are installed in aluminum slats that join to make the conveyor. BRISTLE SQUARES last longer if you clean and rearrange them regularly. (Rearrange means to move the BRISTLE SQUARES to different places within each slat to spread wear evenly.)

Required Tools and Supplies

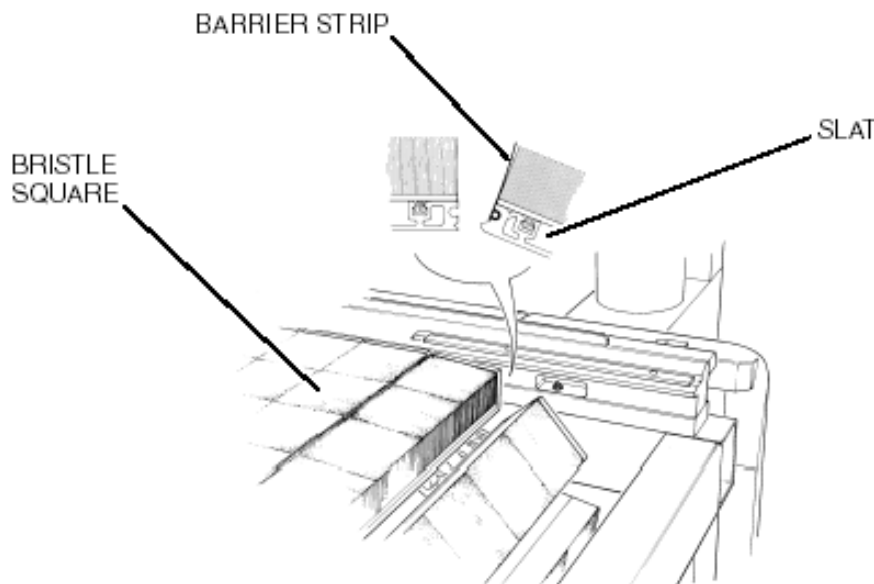
Item	Gerber Part No.
metric T-handle hex key set	945500089
slotted screwdriver	945500298
shop vacuum cleaner	N/A
table air filter	Refer to Service CD

Inspect

1. Turn on the C-200 controller.
2. Run the C-200MT program.

Note: You must run the C-200MT program so you can jog the conveyor. Jogging moves hidden parts of the conveyor into view.

3. Inspect the BRISTLE SQUARES for lint, broken bristles, and other cutting debris. Clean them, if necessary.
4. Inspect the BRISTLE SQUARES for signs of uneven wear, such as scarring or creasing of the bristle. Rearrange them, if necessary.



Clean

The optional automatic bristle cleaner or optional manual bristle cleaner are each designed to clean the BRISTLE SQUARES on your GERBERcutter.

Using the Manual Bristle Cleaner

1. Inspect the table air filters. Clean them, if necessary (page 143).
2. If your system does not have the optional automatic bristle cleaner, connect the 14-ft. vacuum hose to the vacuum port.

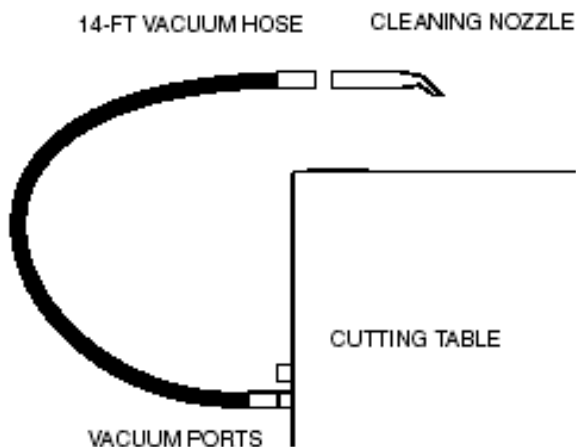
OR

3. If your system has the optional automatic bristle cleaner, connect the 14-ft. vacuum hose to the lower vacuum port.

Note: There is only one port on systems that do not have an optional automatic bristle cleaner. It is the same as the lower port on the automatic systems. It leads to the vacuum generator and produces suction.

4. Install the cleaning nozzle onto the 14-ft. vacuum hose.
5. Turn on the C-200 controller.
6. Run the C-200MT program.

7. Initialize the cutting table.
8. Turn on table vacuum. (Press **F9** on the operator console or turn on the **VAC** switch on the BCP.)
9. From the Cutter menu, turn on vacuum Energy Savings Mode.
10. Move the cleaning nozzle over the work surface to clean the BRISTLE SQUARES.
11. Jog the conveyor as needed to clean all the BRISTLE SQUARES.
12. Turn off table vacuum.
13. Inspect the table air filters. Clean them, if necessary (page 149).
14. Disconnect the 14-ft. vacuum hose from the port.
15. Remove the cleaning nozzle from the 14-ft. hose.
16. Turn off vacuum Energy Savings Mode.
17. Exit the C-200MT program.
18. Turn off the C-200 controller.

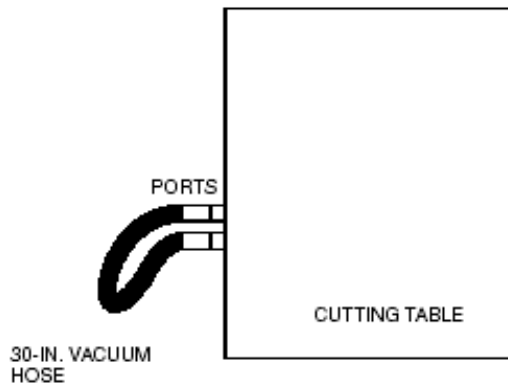


Using the Automatic Bristle Cleaner

1. Inspect the air filters. Clean them, if necessary (page 149).
2. Connect the 30-in. vacuum hose between both vacuum ports. This routes vacuum from the vacuum generator to the automatic bristle cleaner.
3. Turn on the C-200 controller.
4. Run the C-200MT program.
5. Initialize the cutting table.
6. At the operator console, press **F9** to turn on the vacuum generator.
7. From the Utilities menu, select **Bristle Cleaner** (see C-200MT online help).

The Bristle Cleaner dialog box displays.

8. If desired, click to select one or more of the Upon Completion options. Click the **Start** button.
9. When you finish, click **OK** or press **ENTER**.
10. Turn off table vacuum.
11. Inspect the table air filters. Clean them, if necessary (page 149).
12. Remove the 30-in. vacuum hose.
13. Exit the C-200MT program
14. Turn off the C-200 controller.



Rearrange

Rearrange the BRISTLE SQUARES to ensure even wear.

Note: Move the BRISTLE SQUARES with the most wear to the area of the slat that gets the least wear.

1. Turn on the C-200 controller.
2. Run the C-200MT program.

Note: You must run the C-200MT program so you can jog the conveyor. Jogging moves hidden parts of the conveyor into view.

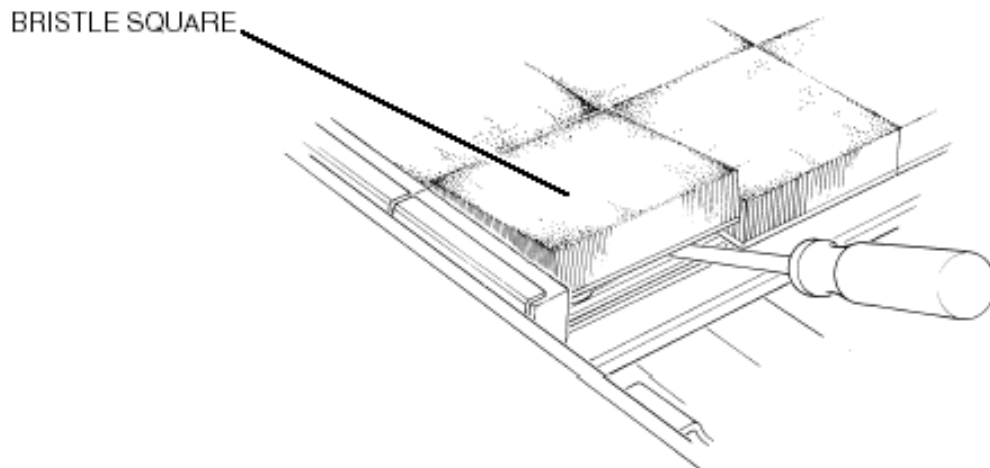
3. Remove the take-on ramp (page 136).
4. Choose one slat to work on.
5. Jog the conveyor until the slat that you want to work on is $\frac{1}{3}$ the way around the end of the conveyor. You need a space between the slat you want to work on and the neighboring slats.

WARNING!



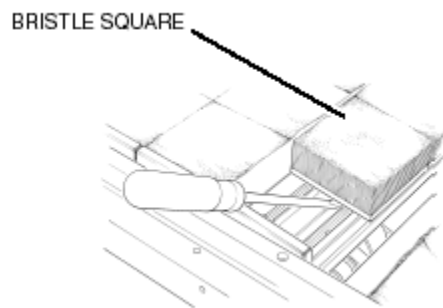
Turn off the C-200 controller before removing any BRISTLE squares. This prevents the conveyor from being jogged accidentally. Failure to do this may cause severe injury to yourself and/or damage to the equipment.

6. Turn off the C-200 controller.
7. First remove the BRISTLE SQUARE from the left end of the slat.
 - a) Insert the tip of a slotted screwdriver between the slat and the BRISTLE SQUARE.
 - b) Work the BRISTLE SQUARE loose by rotating the tip of the screwdriver.
 - c) When the BRISTLE SQUARE is loose, pull it away from the slat.



8. Remove additional BRISTLE SQUARES from the same slat.

- a) Slide the BRISTLE SQUARES along the slat so there is a space next to the one that you want to remove.
- b) Insert the blade of a screwdriver under the BRISTLE SQUARE.
- c) Use the screwdriver as a lever to pry the BRISTLE SQUARE out of the slat.



- d) Repeat steps 8a through 8c for each BRISTLE SQUARE that you want to remove.

9. Determine where you want to install each BRISTLE SQUARE.

Note: Make sure you move the BRISTLE SQUARES with the most wear to a part of the slat that gets the least wear.

10. In one slat, install all the BRISTLE SQUARES.

- a) Align each BRISTLE SQUARE horizontally on the left end of the slat.
- b) There are four clips on the back of the BRISTLE SQUARE that fit into two slots on the slat. Push the BRISTLE SQUARE firmly onto the slat with the palm of your hand. This sets the BRISTLE SQUARE in the slat.
- c) Push down on each corner of the BRISTLE SQUARE to make sure that each of the four clips is secure.
- d) Slide each BRISTLE SQUARE to the right end of the slat.
- e) Repeat steps 10a through 10d until the slat is full.

Note: If you install the BRISTLE SQUARES correctly, they will be flat and flush with the rest of the BRISTLE SQUARES on the work surface.

11. Repeat step 10a through 10e for each slat until you have installed BRISTLE SQUARES in all the slats

Slats

The slats are aluminum trays that hold the BRISTLE SQUARES. The slats are linked by the conveyor chain to form the conveyor. Each slat has several seals that must be kept in good condition to maintain vacuum during cutting. These seals are the barrier strip, tongue, groove, and vacuum seal.

Vacuum Seals

Vacuum seals may or may not be used depending on your system's needs. If vacuum seals are used, they may be used on every slat, every second slat, every third slat, or every fourth slat.

CAUTION: Before doing any slat maintenance, you must see whether vacuum seals are used and at what interval (see Vacuum Seals above). Make sure that you keep this interval when you reinstall any slat that you remove. If vacuum seals are not installed at the correct interval, cut quality and system performance may suffer.

Required Tools and Supplies

Item	Gerber Part No.
3/8 in. drive socket set (metric)	N/A
slotted screwdriver	945500298

Inspect

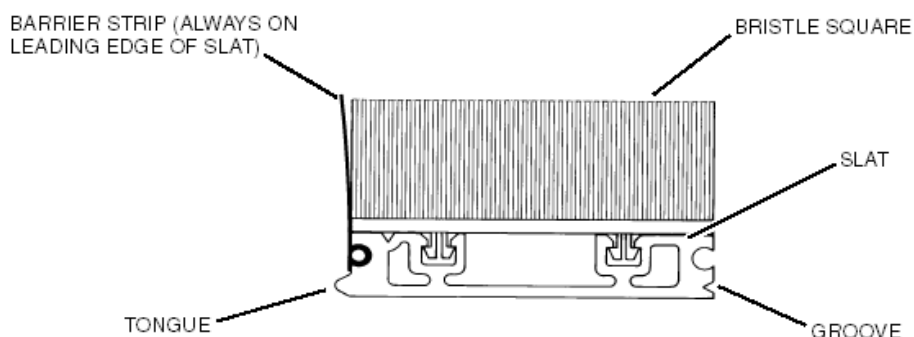
1. Remove the take-on ramp (page 136).

Note: Removing the take-on ramp lets you see the edges of the slats.

2. Turn on the C-200 controller.
3. Run the C-200MT program.

Note: You must run the C-200MT program so you can jog the conveyor. Jogging moves hidden parts of the conveyor into view.

4. Stand at the take-on end of the cutting table. From here, you can see the edges of the slats as they come from under the table.



5. Inspect the slat edges for:
 - loose, torn, or crushed foam blocks
 - loose or torn barrier strips
 - dented, gouged, or bent tongue
 - dented, gouged, or bent groove
6. If you see any of these defects, mark the problem slat with a piece of masking tape. This will help to identify the problem slat later.
7. Jog the conveyor so the next set of edges comes into view. Repeat steps 4, 5, and 6 until you have checked all the slats. If you find any of these defects, have them repaired or replaced by a Gerber qualified technician.
8. Turn off the C-200 controller.
9. Install the take-on ramp (page 136).

Remove

CAUTION: Do not jog the table if more than three adjacent slats are removed. If you need to jog the table, install every other slat and torque the mounting bolts to 677 Nmm (6 in.lbs.). Failure to install every other slat may damage the conveyor and conveyor chain.

1. Remove the take-on ramp (page 136).
2. Turn on the C-200 controller.
3. Run the C-200MT program.

CAUTION: You need to know whether your system uses vacuum seals and, if so, on which slats the vacuum seals are applied. You must make sure you install the slats with the vacuum seal at the correct interval.

4. Jog the conveyor to see whether and where slat vacuum seals are used. If vacuum seals are used, note whether they are used on every slat, every second slat, every third slat, or every fourth slat.
5. Choose the slat that you want to remove.

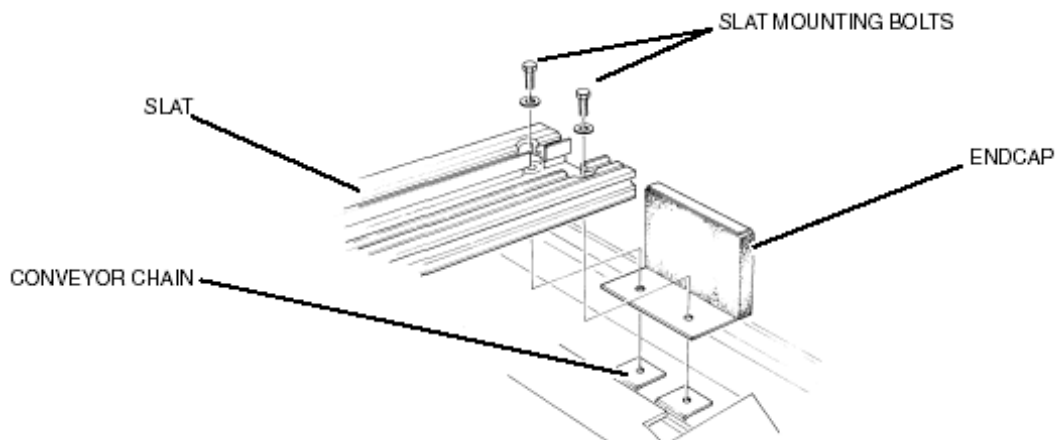
6. Jog the conveyor until the slat that you want to work on is 1/3 the way around the end of the conveyor. You need a space between the slat you want to work on and the neighboring slats.

WARNING!



Turn off the C-200 controller before removing any **BRISTLE SQUARES** or slats. This prevents the conveyor from being jogged accidentally. Failure to turn off the C-200 controller may cause severe injury to yourself and/or damage to the equipment.

7. Exit the C-200MT program.
8. Turn off the C-200 controller.
9. Remove the BRISTLE SQUARE from each end of the slat (page 140).
10. Remove the two, slat mounting bolts from each end of the slat. The slat mounting bolts fasten the slat and the endcap to the conveyor chain.
11. Lift the slat from the cutting table.
12. Lift the end caps from the table.



Install

CAUTION: Do not jog the table if more than three adjacent slats are removed. If you need to jog the table, install every other slat and torque the mounting bolts to 677 Nmm (6 in.lbs). Failure to install every other slat may damage the conveyor and conveyor chain.

1. Turn on the C-200 controller.
2. Run the C-200MT program.
3. Choose the place where you want to install a slat.

CAUTION: If vacuum seals are used on your system, they are spaced at regular intervals. Make sure that you do not change this interval when you reinstall slats.

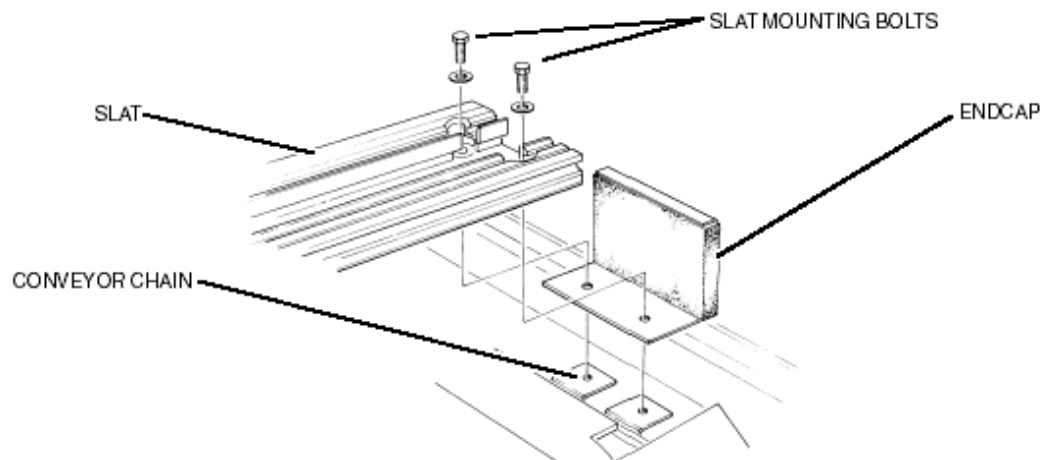
4. Check to see whether the slat you want to install should have a vacuum seal. If so, make sure that you install a slat with a vacuum seal.
5. Jog the conveyor until the place for the slat is 1/3 the way around the end of the conveyor.
6. Exit the C-200MT program.

7. Turn off the C-200 controller.

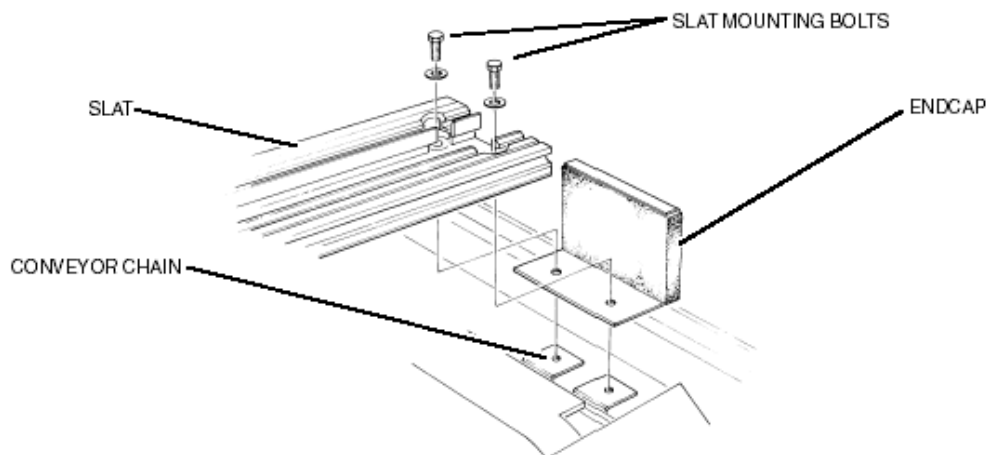
WARNING!

Turn off the C-200 controller before installing any slats. This prevents the conveyor from being jogged accidentally. Failure to turn off the C-200 controller may cause severe injury to yourself and/or damage to the equipment.

8. Put the endcap with the slotted bolt holes on the conveyor at the left side of the table.
9. Put the endcap with the round bolt holes on the conveyor at the right side of the table.
10. Align the bolt holes in the endcaps with the bolt holes in the conveyor.
11. Put the slat onto the endcaps. Align the bolt holes in the slat with those in the endcaps and conveyor.
12. Put the slat mounting bolts into the bolt holes. Each bolt should go through bolt holes in the slat, end cap, and conveyor.



13. With your hand, turn each bolt clockwise to thread it part of the way into its bolt hole.
 14. Gently push the left end cap towards the middle of the slat until it is aligned with neighboring end caps.
 15. With a torque wrench, tighten each slat mounting bolt to 677 Nmm (6 in.lbs.).
 16. Repeat steps 1 through 15 for each slat that you need to install.
 17. Jog the conveyor through two complete revolutions.
 18. Exit the C-200MT program.
 19. Turn off the C-200 controller.
 20. On each slat that you are installing, tighten the slat mounting bolts to between 17.6 and 20.3 Nm (13 and 15 ft.lbs.).
 21. Install BRISTLE SQUARES in both ends of each slat (page 140).
 22. Turn on the C-200 controller.
 23. Run the C-200MT program.
 24. Advance the conveyor to expose other slats that need to be tightened.
 25. Exit the C-200MT program.
 26. Turn off the C-200 controller.
 27. Repeat steps 20 through 26 until you have tightened all the slats.
 28. Turn on the C-200 controller.
 29. Run the C-200MT program.
 30. Slowly jog the conveyor through two revolutions and make sure that all endcaps and slats align.
- CAUTION:** If any slats or endcaps are misaligned, you must remove and reinstall them. Failure to do so can damage the conveyor and/or cause inaccurate cutting.
31. Exit the C-200MT program.
 32. Turn off the C-200 controller.
 33. Install the take-on ramp (page 136).



Drive Assembly

The drive assembly turns the conveyor. It is on the take-off end of the operator side of the cutting table.

Required Tools and Supplies

Item	Gerber Part No.
clean shop rag	N/A
Shell Omala Mineral Oil	N/A

1. Turn off the C-200 controller.
2. Remove side panels as needed to work on the drive assembly (page 135).

3. Check the oil level in the gearbox.

- a) Remove the fill plug. The oil should be up to the bottom of the fill hole.

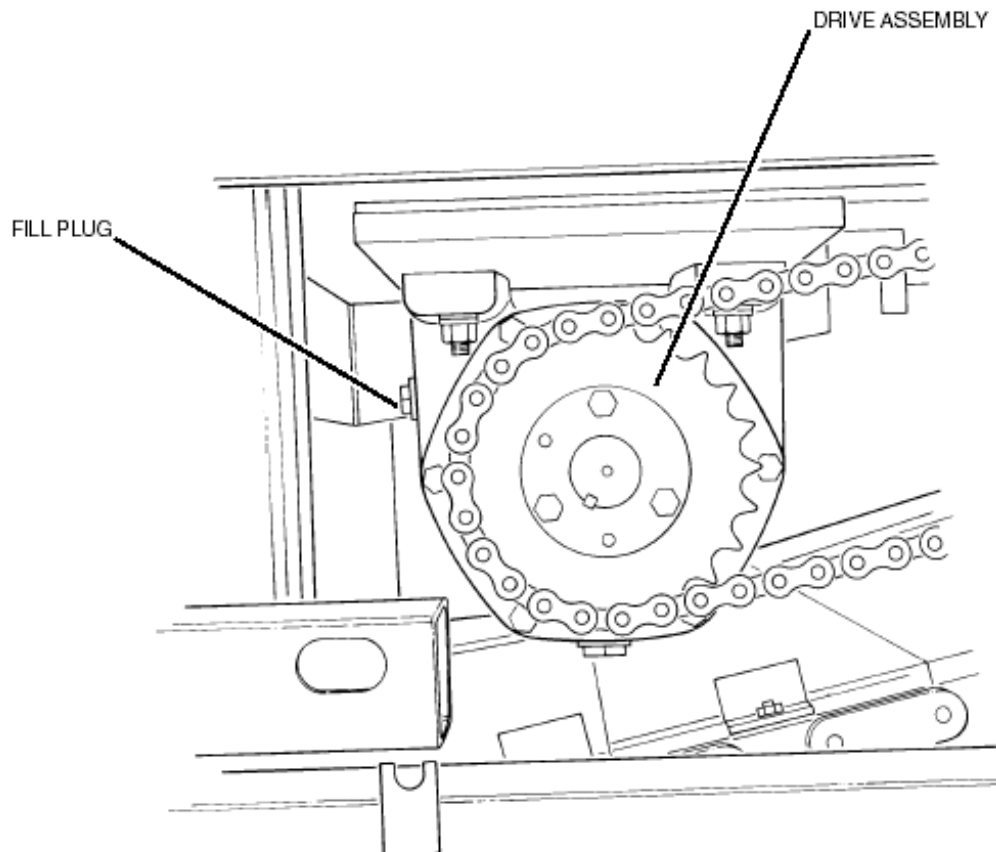
CAUTION: Use Shell Omala mineral oil. The viscosity is temperature dependent. Filling the gearbox with oils other than Shell Omala may damage the gearbox.

- b) If necessary, fill the gearbox with Shell Omala mineral oil.

- c) Install the fill plug.

4. Install any side panels that you removed (page 135).

5. Turn on the C-200 controller.



Drive Chains

The slats are bolted to two drive chains to form the conveyor. One drive chain runs under each side of the conveyor. The drive chain on the right side connects the conveyor to the drive assembly.

Required Tools and Supplies

Item	Gerber Part No.
clean shop rag	N/A
multipurpose grease	596500005

Lubricate

1. Turn off the C-200 controller.
 2. Remove all side panels (page 135).
 3. Find the left-side drive chain.
 4. Using the shop rag, clean lint and other debris from the chain.
 5. Lubricate the chain lightly with multipurpose grease.
- CAUTION:** To avoid damage to material, do not over grease. Moving equipment throws excess lubricant.
6. Find the right-side drive chain.
 7. Repeat steps 4 and 5 for the right-side drive chain.
- CAUTION:** To avoid damage to material, do not over grease. Moving equipment throws excess lubricant.
8. Install all side panels (page 135).
 9. Turn on the C-200 controller.

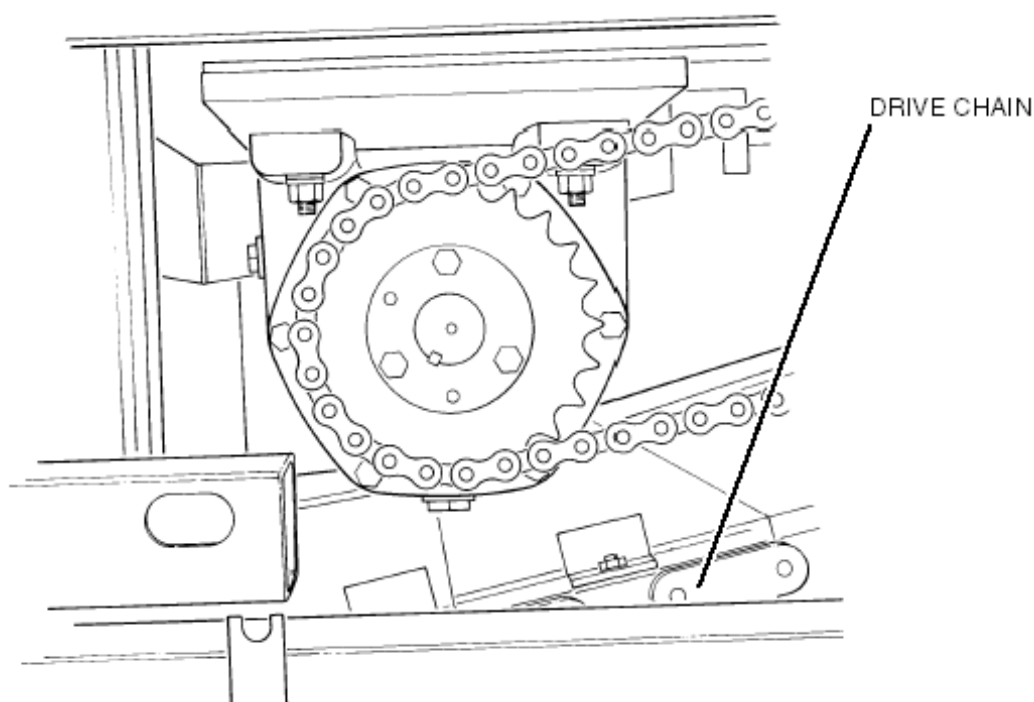


Table Air Filters

Your GERBERcutter has two high-efficiency air filters to improve air flow. If they get clogged, air pressure drops. The optional manual and automatic bristle cleaners also use these filters to collect debris.

Required Tools and Supplies

Item	Gerber Part No.
shop vacuum cleaner	N/A
air filter	see Service CD

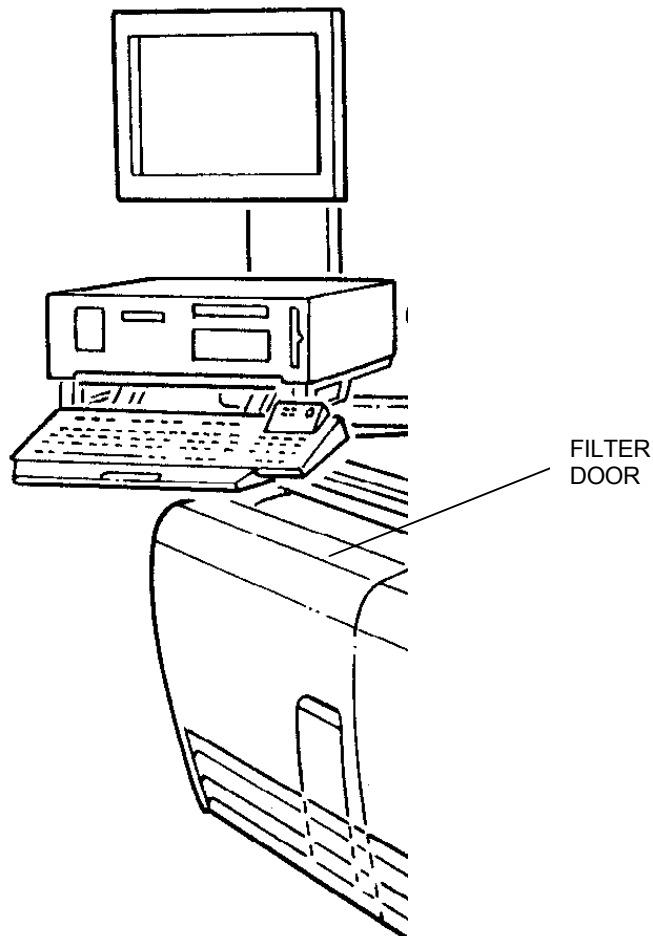
Inspect

1. Pull the filter access door down and lift out.
2. Remove the filters by sliding them out.
3. Inspect the filters and compartment for built-up lint. Clean them, if necessary.
4. Install the filters by sliding them into the compartment until they stop.
5. Replace and close the filter access door.

WARNING!



Avoid breathing lint from the filters. Wear a mask suitable for this purpose.



Clean

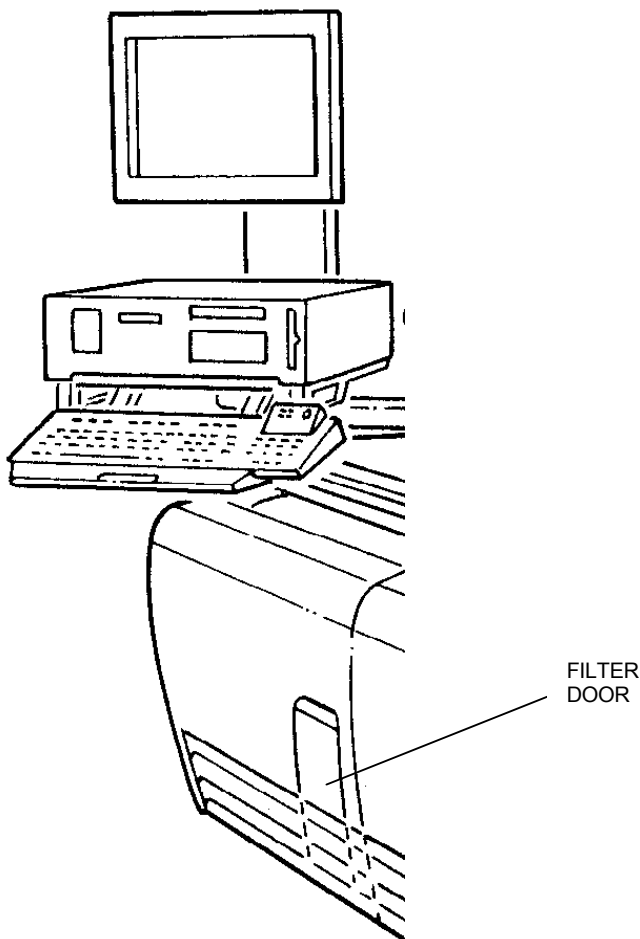
1. Turn off the C-200 controller.
2. Open and remove the filter access door.
3. Remove the filters by sliding them straight out.

CAUTION: Never use the optional manual bristle cleaner to vacuum the air filters. The optional manual bristle cleaner uses the filters to collect debris.

4. Vacuum the filters to remove heavy lint.

Note: A filter may become so plugged that you cannot clean it effectively. If this happens, install a new filter.

5. Vacuum the inside of the filter compartment to remove lint.
6. Install the filters by sliding them into the compartment until they stop.
7. Replace and close the filter access door.
8. Turn on the C-200 controller.



Vacuum Generator Assembly

The vacuum generator assembly includes the vacuum generator and the vacuum generator motor. The motor turns the vacuum generator through four drive belts.

Required Tools and Supplies

Item	Gerber Part No.
belt deflection gauge	45412000
Petro-Ion multipurpose grease	596500005
grease gun	944233101
24-inch ruler/straight-edge	N/A

Vacuum Generator Motor

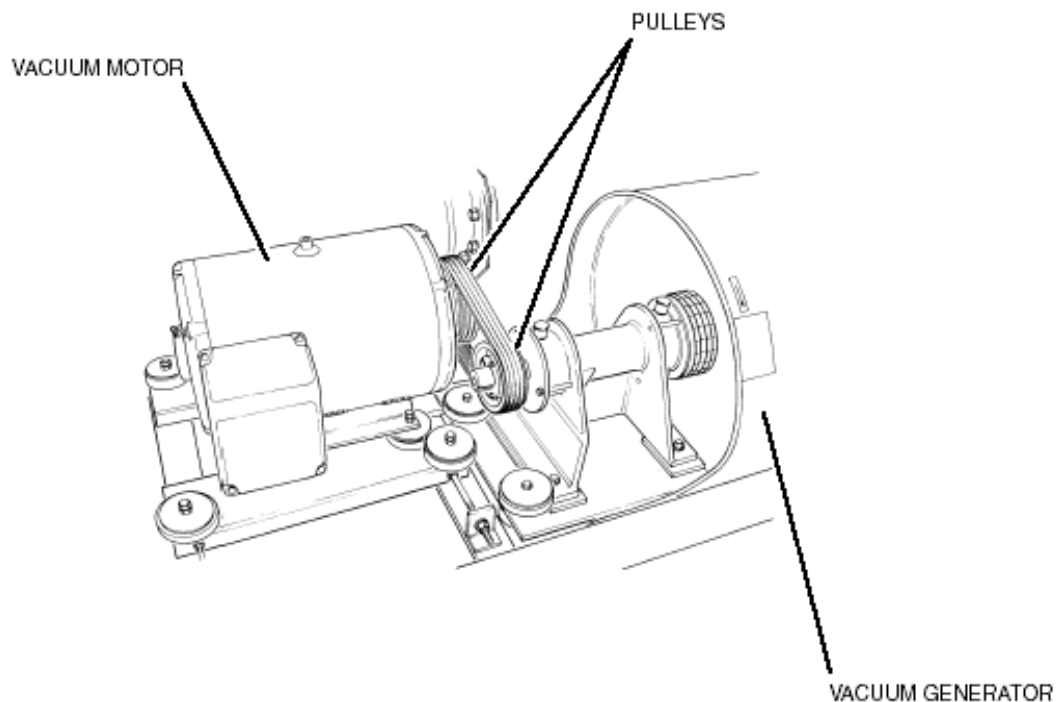
Inspect

WARNING!



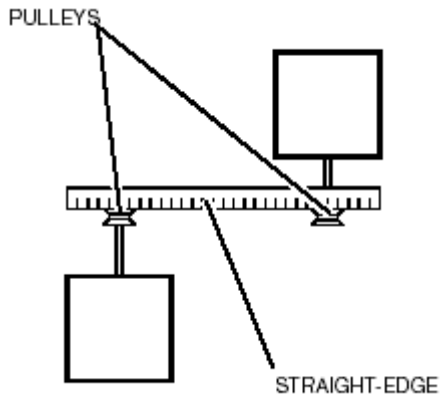
Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

1. Turn off the C-200 controller.
2. Open the CTOT (page 174).

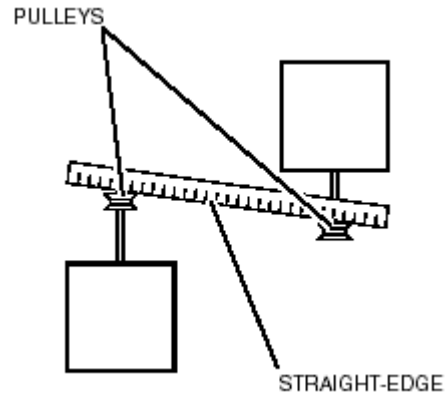
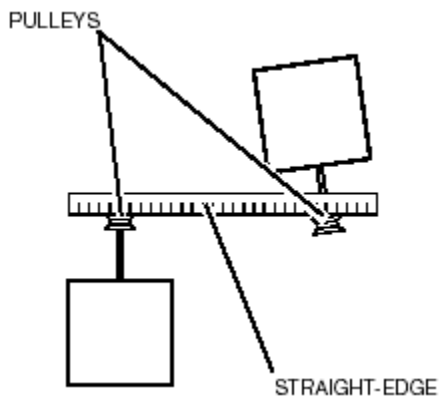


3. Check that the motor is square.

- a) Put a straight-edge across the face of the vacuum generator motor and generator pulleys. Do this both at the top and bottom of the pulleys.



- b) If the straight-edge is not flat, adjust the squareness of the vacuum motor.



OR

- c) If the straight edge is flat against the top of the pulleys but not the bottom, or if it is flat against the bottom and not the top of the pulleys, your motor may have an incorrect vertical alignment. Call your service technician.
4. Lubricate the motor as needed according to scheduled maintenance (page 154).
5. Close the CTOT (page 175).
6. Turn on the C-200 controller.

Adjust**WARNING!**

Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

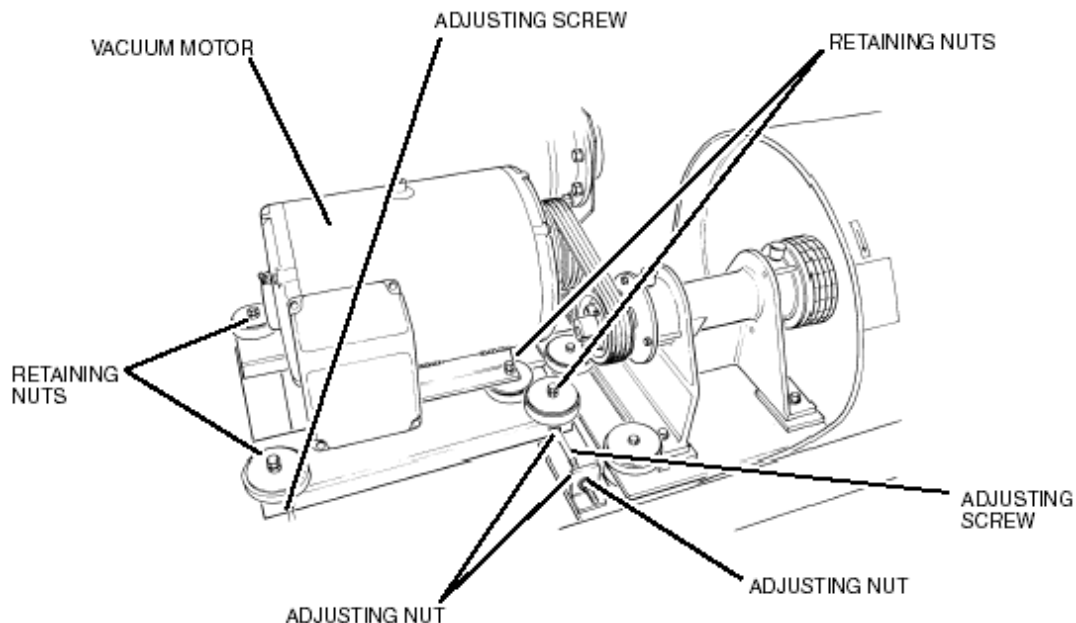
1. Turn off the C-200 controller.
2. Open the CTOT (page 174).
3. Loosen all retaining nuts on the motor mounts.
4. Turn the adjusting nuts on either adjusting screw to move the vacuum motor so it is square with the vacuum generator.
5. Inspect the vacuum motor (page 151).
6. If the vacuum motor is not square, repeat steps and until it is square.

7. Tighten all of the retaining nuts to 81.6 Nm (60 ft. lbs.).

CAUTION: Make sure the adjusting nuts are tight. Otherwise, the vacuum motor may vibrate out of position.

CAUTION: Do not move the motor while you are tightening the adjusting nuts. If you do, you must recheck the squareness of the vacuum motor.

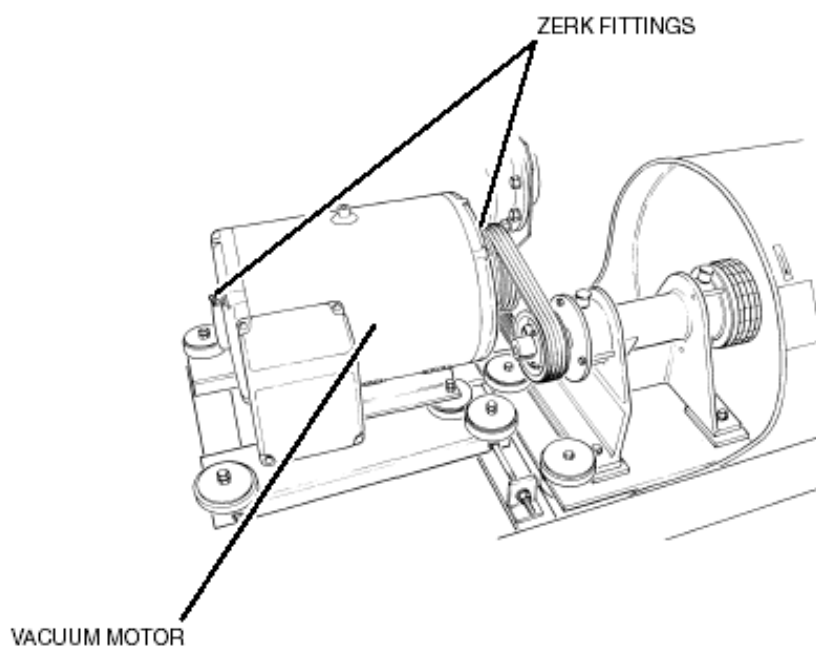
8. On each adjusting screw, tighten each adjusting nut with an open-end wrench until the nut stops moving.
9. Inspect the vacuum drive belts and adjust them if needed (page 155).
10. Close the CTOT (page 175).
11. Turn on the C-200 controller.



Lubricate

WARNING!  Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

1. Turn off the C-200 controller.
2. Open the CTOT (page 174).
3. Use the grease gun to apply multipurpose grease to the two vacuum motor zerk fittings.
4. Close the CTOT (page 175).
5. Turn on the C-200 controller.



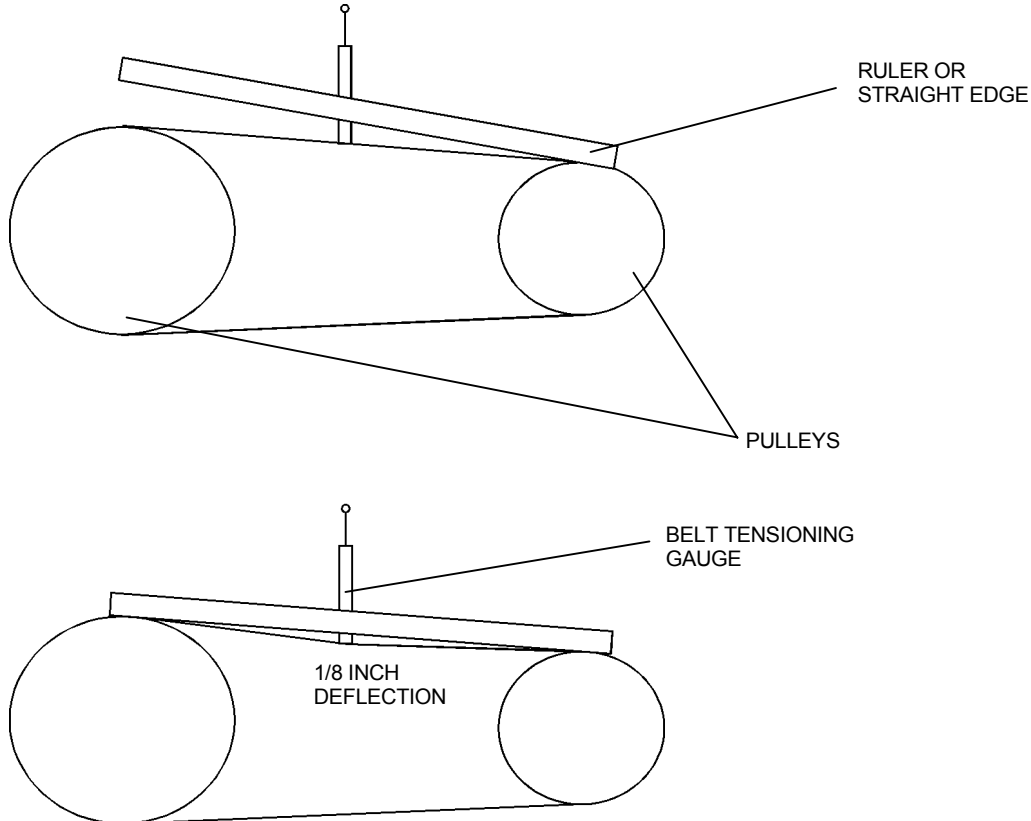
Vacuum Drive Belt

A drive belt connects the vacuum generator motor to the vacuum generator.

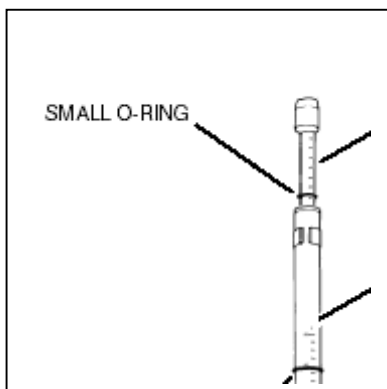
Inspect

WARNING!  **Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.**

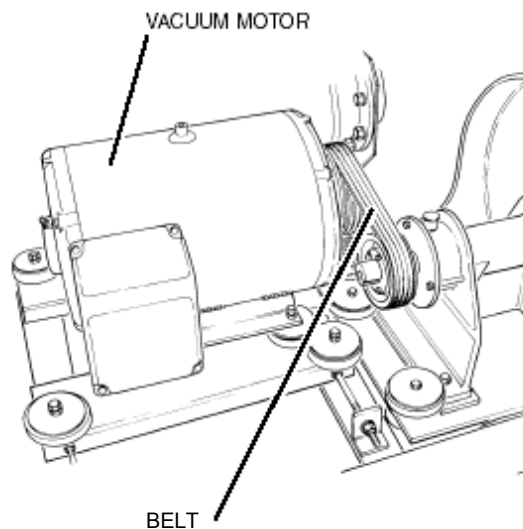
1. Turn off the C-200 controller.
2. Open the CTOT (page 174).
3. Inspect the vacuum generator drive belts for wear, stretching, and fraying. If you see any of these defects, on one or more belts, replace the belt.
4. Measure the deflection of the drive belt.
 - a) On the belt deflection gauge, move the large O-ring to 1/8 inch.
 - b) Move the small O-ring to zero on the deflection force scale.
 - c) Position a 24 inch ruler or straight edge on the belt. It must be long enough to span the two pulleys.
 - d) Put the span-scale end of the deflection gauge onto the drive belt halfway between the pulleys. The long axis of the gauge is perpendicular to the belt.
 - e) Hold only the deflection-force end of the gauge.



- f) Push the gauge into the belt until the bottom of the large O-ring is level with the edge of the ruler or straight edge.
- g) Move the tension gauge from the belt. The small O-ring should show a measurement of 11.8 kg (26 lb.) deflection force.



- 5. Close the CTOT (page 175).
- 6. Turn on the C-200 controller.



Adjust**WARNING!**

Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

1. Turn off the C-200 controller.
2. Open the CTOT (page 174).
3. Loosen all retaining nuts on the motor mounts.

CAUTION: Make sure you move the adjusting nuts on both adjusting screws by the same amount. If you do not, you might misalign the vacuum motor.

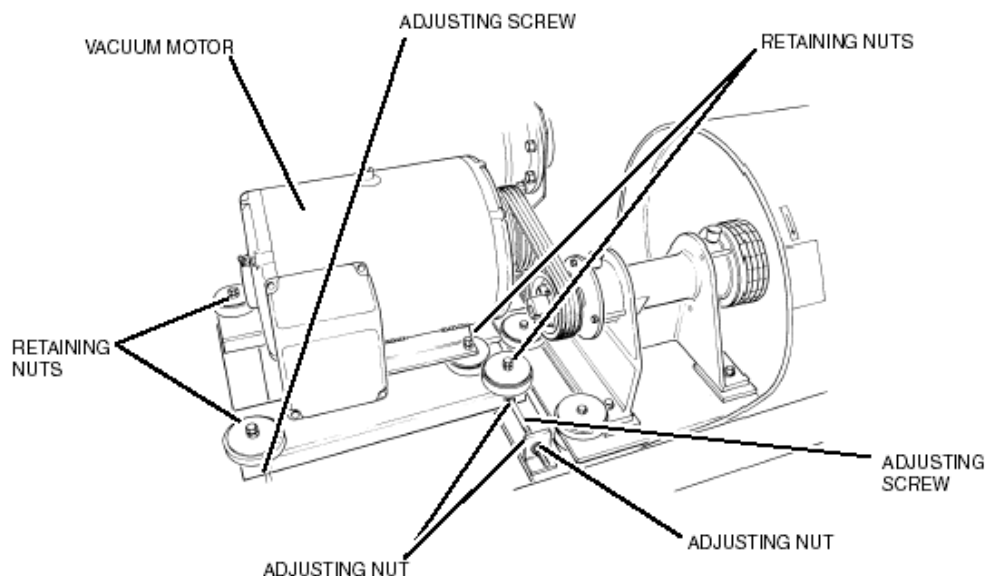
4. Turn the adjusting nuts on both adjusting screws an equal amount to move the vacuum motor.
 - a) Move the motor away from the vacuum generator pulley to tighten the drive belt.

OR

- b) Move the motor towards the vacuum generator pulley to loosen the drive belt.

CAUTION: Do not move the motor while you are tightening the adjusting nuts. If you do, you will change the belt tension.

- c) On both adjusting screws, tighten all adjusting nuts by hand.
5. Measure the deflection of the drive belt. See the diagram on page 155.
 - a) On the belt deflection gauge, move the large O-ring to 1/8 inch.
 - b) Move the small O-ring to zero on the deflection force scale.

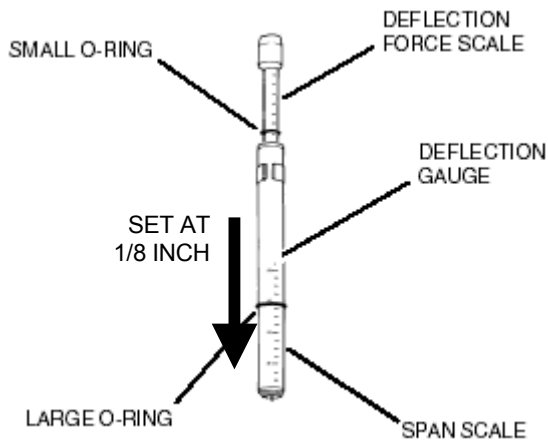


- c) Put the span-scale end of the deflection gauge onto the drive belt halfway between the pulleys. The long axis of the gauge is perpendicular to the belt.
- d) Hold only the deflection-force end of the gauge.
- e) Push the gauge into the belt until the bottom of the large O-ring is level with the edge of the ruler or straight edge.
- f) Move the tension gauge from the belt. The small O-ring should show a measurement of 11.8 kg (26 lb.) deflection force.

CAUTION: Make sure the adjusting nuts are tight. Otherwise, the vacuum motor may vibrate out of position.

CAUTION: Do not move the motor while you are tightening the adjusting nuts. If you do, you must recheck the squareness of the vacuum motor.

- 6. On each adjusting screw, tighten each adjusting nut with an open-end wrench until the nut stops moving.
- 7. Tighten all of the retaining nuts to 81.6 Nm (60 ft. lbs.).
- 8. Inspect the vacuum motor drive belts again (page 155).
- 9. Inspect the vacuum motor (page 151).
- 10. Close the CTOT (page 175).
- 11. Turn on the C-200 controller.



Remove

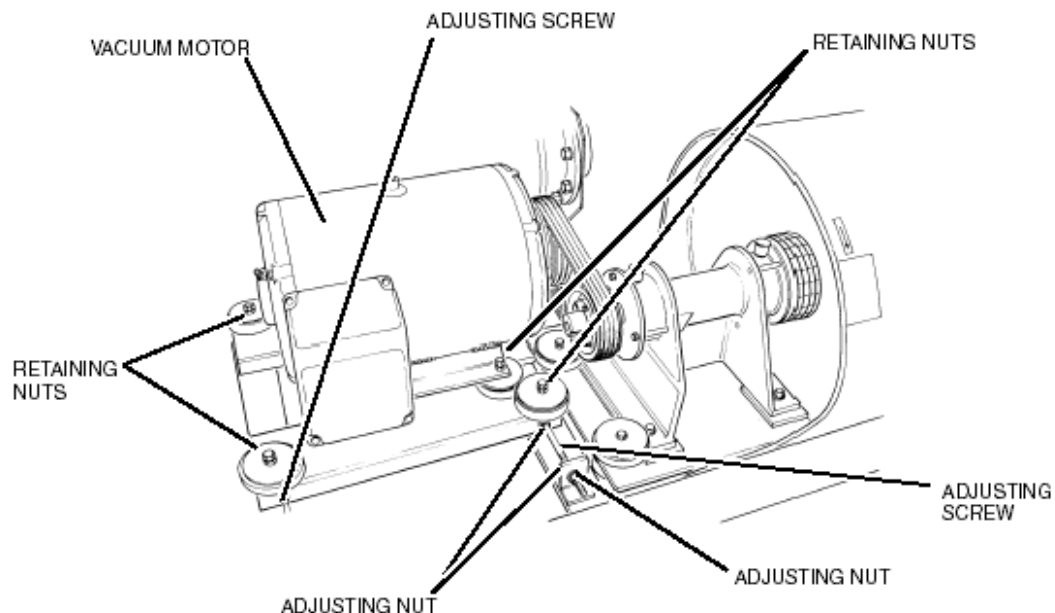
WARNING!



Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

CAUTION: Make sure you move the adjusting nuts on both adjusting screws by the same amount. If you do not, you might misalign the vacuum motor.

1. Turn off the C-200 controller.
2. Open the CTOT (page 174).
3. Loosen all eight retaining nuts on the motor mounts.
4. Turn the adjusting nuts on both adjusting screws an equal amount to move the vacuum motor towards the vacuum generator pulley.
5. Move the motor until you can remove the drive belt from the pulley.



Install

WARNING!



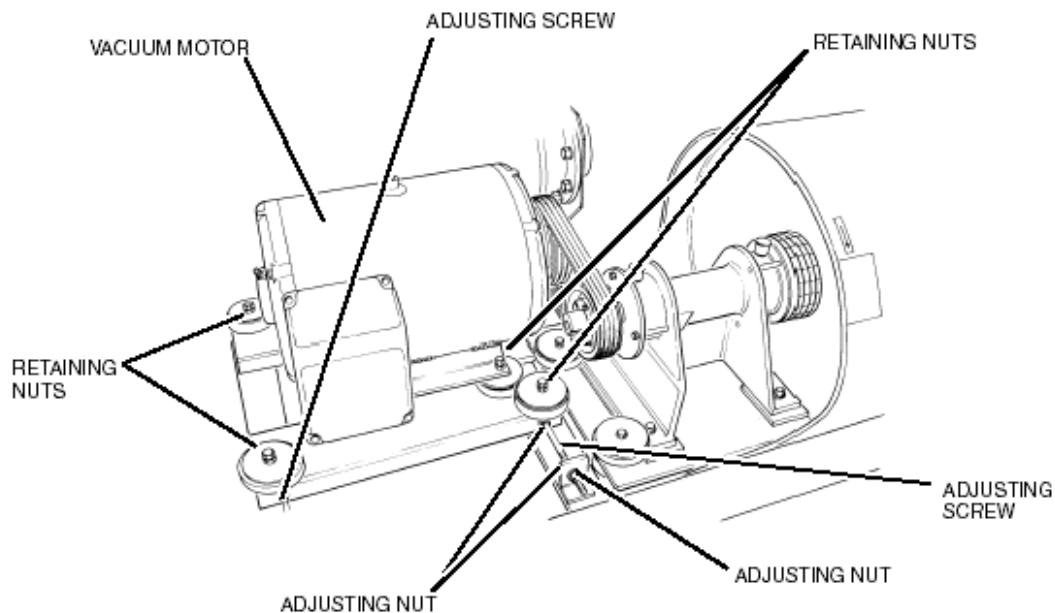
Make sure the C-200 controller is turned off. Failure to do so may result in injury or death.

1. Put the vacuum generator drive belt onto the pulleys.
2. Adjust the drive belt (page 155).

CAUTION: Make sure the adjusting nuts are tight. Otherwise, the vacuum motor may vibrate out of position.

CAUTION: Do not move the motor while you are tightening the adjusting nuts. If you do, you must recheck the squareness of the vacuum motor.

3. On each adjusting screw, tighten each adjusting nut with an open-end wrench until the nut stops moving.
4. Tighten all of the retaining nuts to 81.6 Nm (60 ft. lbs.).
5. Inspect the vacuum generator motor to make sure it is square (page 151).
6. Close the CTOT (page 175).
7. Turn on the C-200 controller.



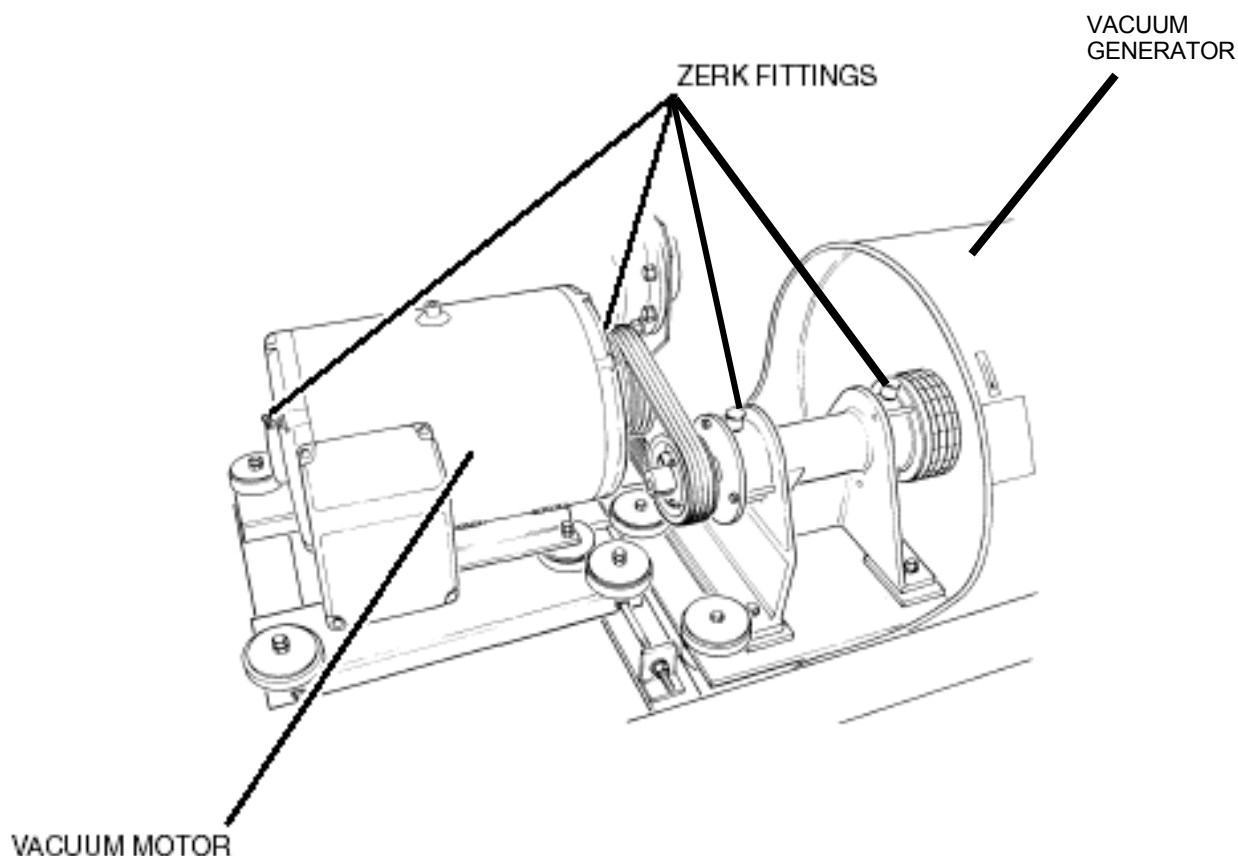
Vacuum Generator

Lubricate

WARNING!  Before working in the controller electrics enclosure, turn off the C-200 controller. This turns off power to the controller electrics enclosure. Failure to turn off the C-200 controller may result in injury or death.

1. Turn off the C-200 controller.
2. Open the CTOT (page 174).

3. Use the grease gun to apply multipurpose grease to the two vacuum generator zerk fittings.
4. Close the CTOT (page 175).
5. Turn on the C-200 controller.



C-200 Controller

The C-200 controller is the command center of the GERBERcutter. Using its software, the C-200MT tells the system how to cut material. The C-200 controller has two parts: the Front End Processor (FEP) or PC and the Motion Control Computer (MCC).

The PC is located in the operator console. The MCC is located in the controller electrics enclosure.

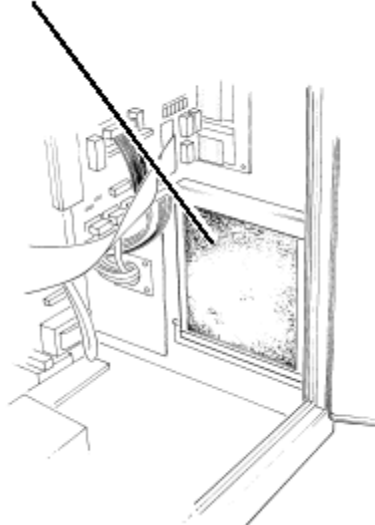
MCC Air Filter

The MCC is cooled by air that flows through a filter and into the controller electrics enclosure. You must keep this filter clean and free from lint. If this filter is clogged, air flow over the MCC is reduced and the MCC may overheat.

Required Tools and Supplies

Item	Gerber Part No.
MCC air filter	Refer to Service CD
shop vacuum cleaner	N/A

MCC AIR FILTER



Inspect

WARNING!



Before working in the controller electrics enclosure, turn off the C-200 controller with circuit breaker 1. This turns off power to the controller electrics enclosure. Failure to turn off the C-200 controller may result in injury or death and damage to the C-200 controller.

1. Turn off the C-200 controller.
2. Open the controller electrics enclosure.

WARNING!



Do not open or work in the vacuum electrics enclosure. Circuit breaker 1 does not turn off all power to the vacuum electrics enclosure. High voltages exist inside the vacuum electrics enclosure. These voltages can cause injury and death.

3. Inspect the filter for built-up lint. Clean it, if necessary.
4. Close the controller electrics enclosure.
5. Turn on the C-200 controller.

Clean

WARNING!

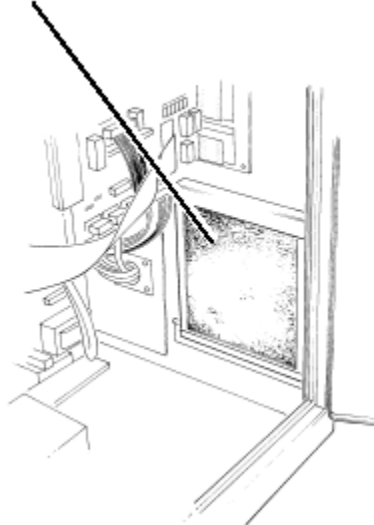
Before working in the controller electrics enclosure, turn off the C-200 controller with circuit breaker 1. This turns off power to the controller electrics enclosure. Failure to turn off the C-200 controller may result in injury or death and damage to the C-200 controller.

1. Turn off the C-200 controller.
2. Open the controller electrics enclosure.

WARNING!

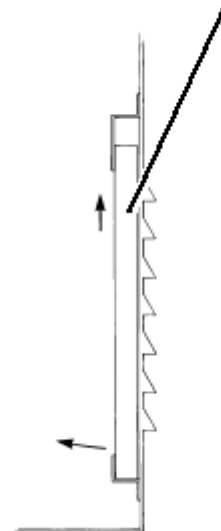
Do not open or work in the vacuum electrics enclosure. Circuit breaker 1 does not turn off all power to the vacuum electrics enclosure. High voltages exist inside the vacuum electrics enclosure. These voltages can cause injury or death.

MCC AIR FILTER



3. Inspect the filter for built-up lint. Clean it, if necessary.
4. Open the controller electrics enclosure.
5. Remove the dirty MCC air filter:
 - a) Slide the filter up.
 - b) Pull the bottom of the filter out of its bracket.
 - c) Slide the filter down to remove it completely from the bracket.

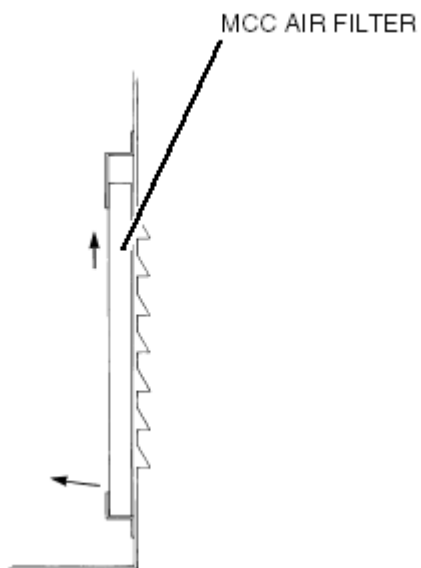
MCC AIR FILTER



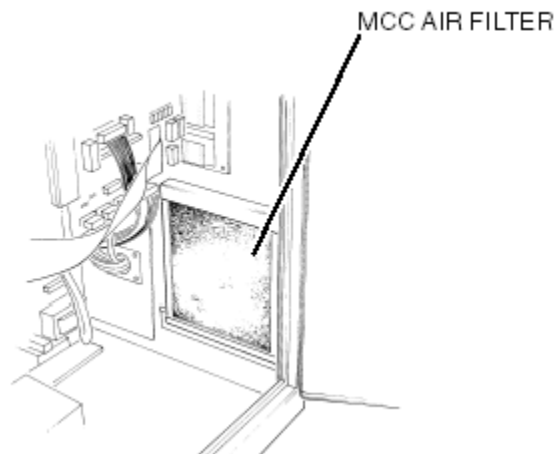
6. Vacuum the filter to remove any lint.

Note: If the filter is so plugged that you cannot clean it effectively, install a new filter.

7. Install the clean MCC air filter.
 - a) Slide the filter up into the top of its bracket.
 - b) Put the bottom of the filter flat against the side of the controller electrics enclosure.
 - c) Slide the filter down into the bracket.



8. Close the controller electrics enclosure.
9. Turn on the C-200 controller.



Mast

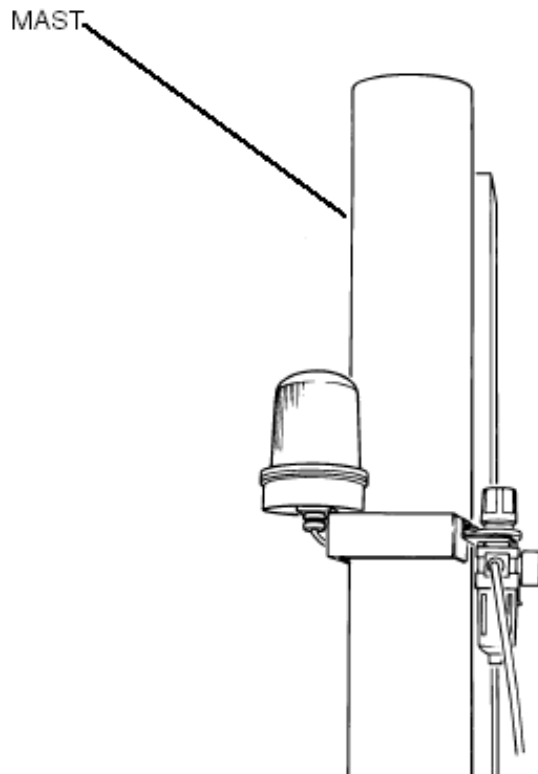
The mast is a metal pole mounted on the right side of the cutting table. It provides a convenient attachment point for the cutting table's electrical and compressed air lines. On traveling tables, the mast houses a warning light that flashes when the cutting table moves.

Electrical and Air Lines

These lines provide the GERBERcutter with power and compressed air. On traveling tables, you should check them regularly for wear.

Inspect

1. Turn off the C-200 controller.
2. Check the electrical and compressed air lines where they enter the mast. They should be free of cuts, tears, and chafing. If you see any of these defects, have them repaired or replaced by a Gerber qualified technician.
3. Turn on the C-200 controller.

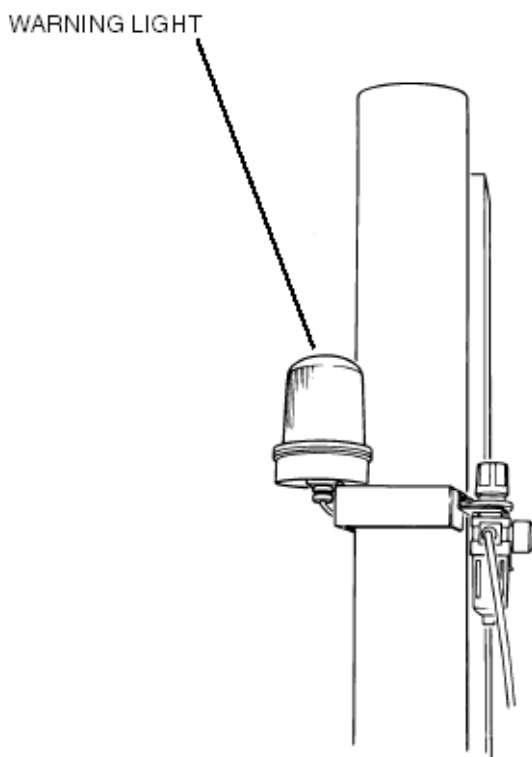


Warning Light

The warning light is a safety device. It flashes to let you know that a traveling table is moving.

Inspect

1. Turn on the C-200 controller and run the C-200MT program.
2. Using the TABLE MOVE control, move the table a short distance in FORWARD. The warning light should flash.
3. Using the TABLE MOVE control, move the table a short distance in REVERSE. The warning light should flash.
4. If the warning light does not flash, have it repaired by a Gerber qualified technician.



Film Dispenser

The film dispenser holds a roll of plastic film. You use the plastic to seal the lay for proper vacuum during cutting.

Plastic Film Roll

The roll holds the plastic film. Replace it when the film is used up. You must periodically adjust two brackets on either side of the dispenser to allow it to work properly.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
plastic film roll	Refer to Service CD

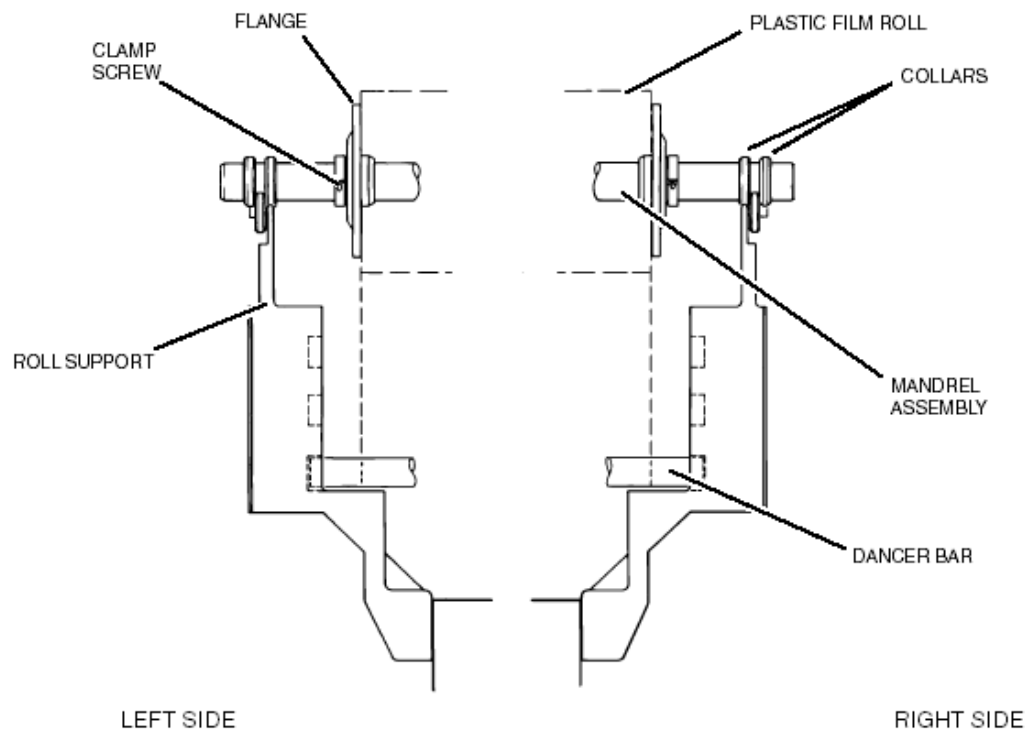
WARNING!



You will need the assistance of one person for this procedure. Failure to do this may cause severe injury to yourself and/or damage the equipment.

Remove

1. With an assistant, lift the mandrel assembly and film roll off the roll supports.
2. Loosen the clamp screw on the left flange.
3. Slide the flange off the assembly.
4. Slide the roll off the mandrel assembly.



Install

1. Slide the roll onto the mandrel assembly.

Note: Make sure the leading edge of the film faces the take-off end of the table.

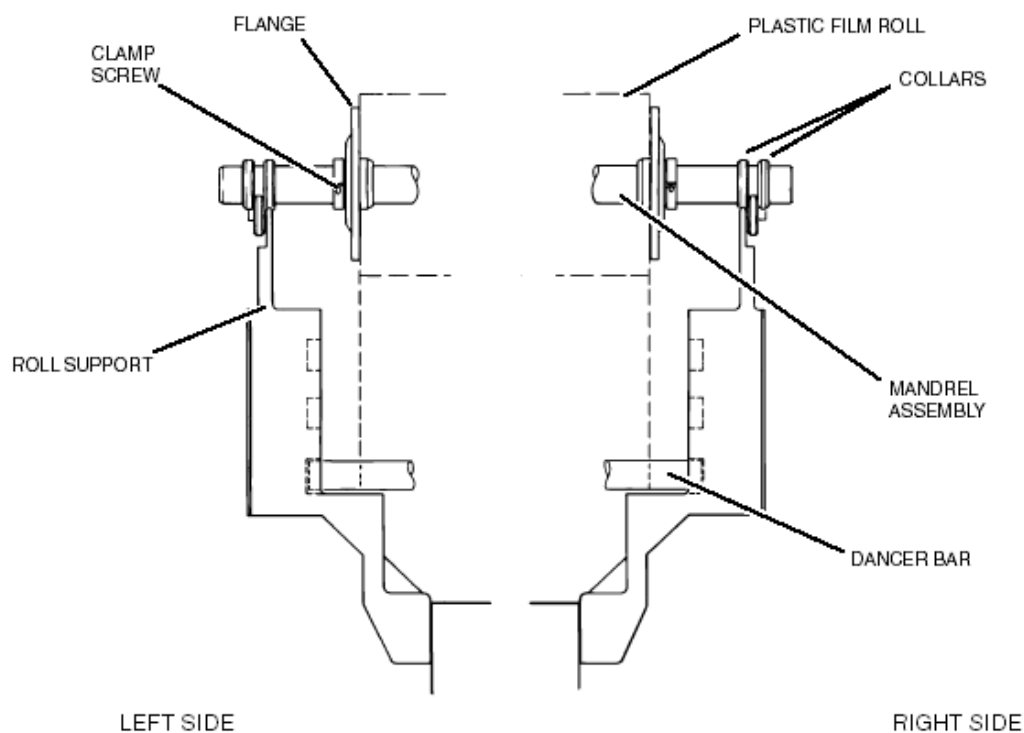
2. Slide the left flange onto the mandrel assembly.
3. Tighten the clamp screw on the flange.
4. Lower the mandrel assembly onto the supports. The collars must fit either side of the idlers on the right side of the table.

WARNING!



Improper installation may cause the mandrel to slide out of the roll supports. This could injure people or damage equipment.

5. Thread the film through the dispenser.



Resealer

The optional resealer attaches to the take-off end of the cutting table and the beam. It stretches a sheet of plastic film over the work surface during cutting. This plastic film covers and reseals the cut part of the spread to maintain vacuum.

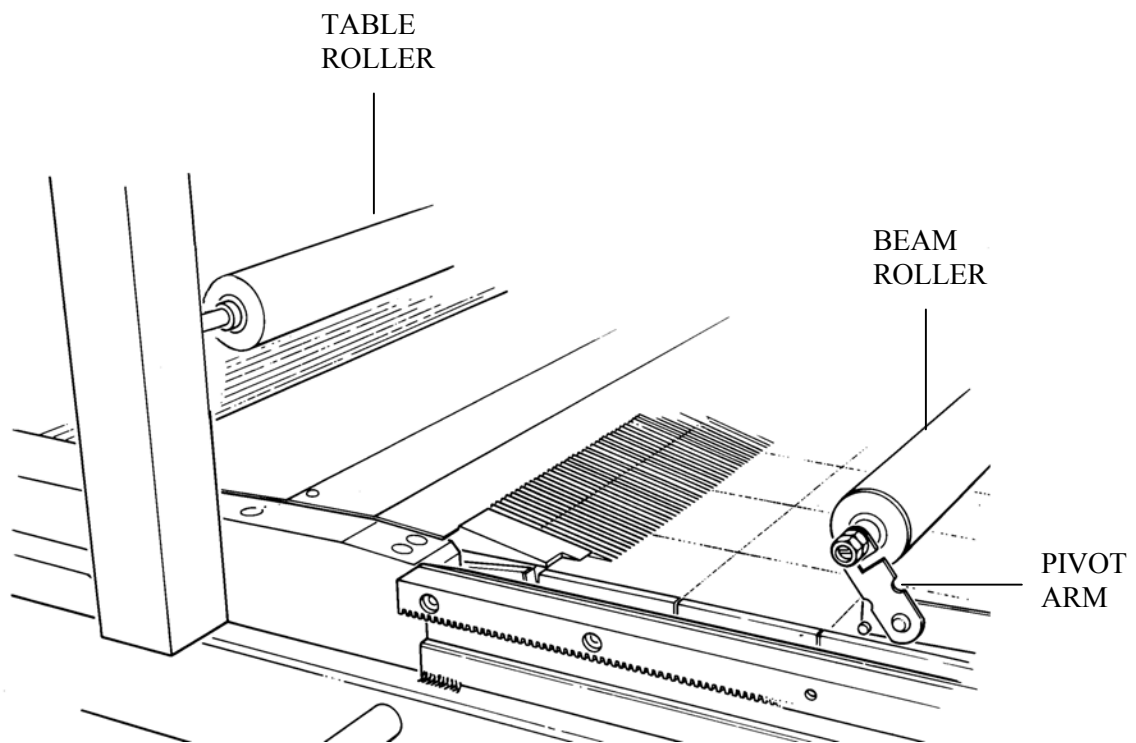
The plastic film is attached to two spring-loaded rollers (table roller and beam roller) that work like window shades. When the beam moves toward the take-on end, the rollers let out plastic film. When the beam moves toward the take-off end, the rollers take up the plastic film.

Plastic Film

The plastic film reseals the cut part of the spread as the GERBERcutter processes a cut file.

Required Tools and Supplies

Item	Gerber Part No.
metric right angle hex key set	945500079
crescent wrench	N/A
two-sided tape	N/A
plastic film roll	Refer to Service CD

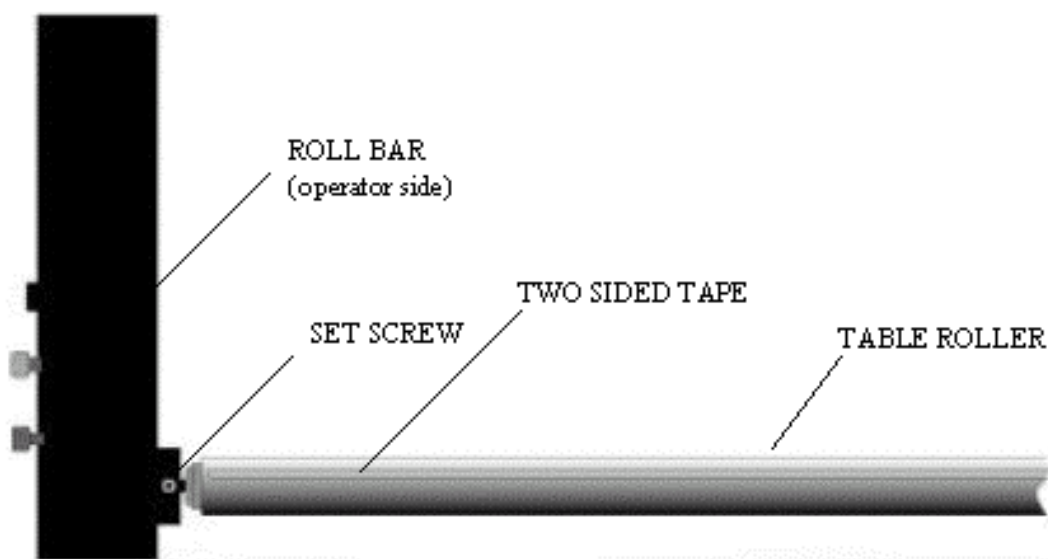


Inspect

1. Turn on the C-200 controller.
2. Run the C-200MT program.
3. Manually slew the beam to the take-on end of the cutting table. This unrolls the plastic film on the resealer.
4. Inspect the plastic film for tears, cracks, cuts, and brittle areas. Replace the plastic film, if necessary.
5. Exit the C-200MT program.
6. Turn off the C-200 controller.

Remove

1. Turn on the C-200 controller.
2. Run the C-200MT program.
3. Manually slew the beam to the take-off end of the cutting table. This rolls the plastic film onto the beam and resealer rollers.
4. Exit the C-200MT program.
5. Turn off the C-200 controller.
6. At the table roller, shown below, loosen the set screw in the side of the clamp to unlock the left shaft. This lets the rubber spring unwind inside the roller.
7. Unroll the plastic film from the resealer roller.

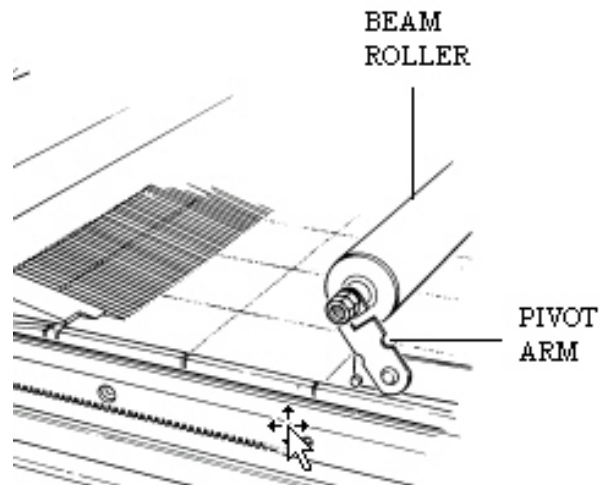


8. At the table roller, peel off the end of the plastic film. It is fastened with two-sided tape. Remove any adhesive remaining on the roller.
9. Remove the beam roller from the pivot arms.
 - a) Loosen the nut on the end of the table roller (console side) where it is attached to the pivot arm.
 - b) Lift the table roller from the pivot arm (console side) and pull the other end of the table roller from the hole in the pivot arm (remote side)
10. Unroll the plastic film from the beam roller.
11. Peel the end of the plastic film off the beam roller. It is fastened with two-sided tape. Remove any adhesive remaining on the roller.

Install

Note: This procedure requires assistance from a second person.

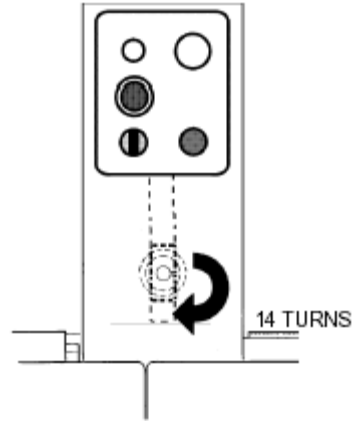
1. At the table roller, fasten one end of the plastic film to the newly installed, two-sided tape.
2. At the beam roller, fasten the other end of the plastic film to the newly installed, two-sided tape.
3. Roll half the plastic film onto the table roller and half onto the beam roller.



4. Tension the beam roller.
 - a) Have an assistant hold the beam roller and right shaft so they cannot turn.
 - b) Wind the left shaft counterclockwise 14 turns. This tensions the rubber spring inside the roller.
 - c) Slide the beam roller onto the pivot arms.
 - d) Before tightening the nut on the end of the beam roller (console side), make sure the beam roller and the plastic film are centered.
 - e) Slide the beam roller to align it with the pencil mark on the trailing arm.
 - f) Install and tighten the outermost screw on the right side of the beam roller.
 - g) Tighten the left clamp on the beam roller.

5. Tension the table roller.

- a) Have an assistant hold the beam roller and the table roller to keep them from moving.
- b) Use a crescent wrench to wind the left shaft clockwise 14 turns. This tensions the rubber spring.
- c) Tighten the set screw on the left clamp.
- d) Release the beam and resealer rollers.



6. Turn on the C-200 controller.

Note: These are initial settings that will need to be adjusted during use. If the plastic film tends to accumulate on the take-off roller, increase the number of turns on both rollers.

Conveyorized Take Off Table

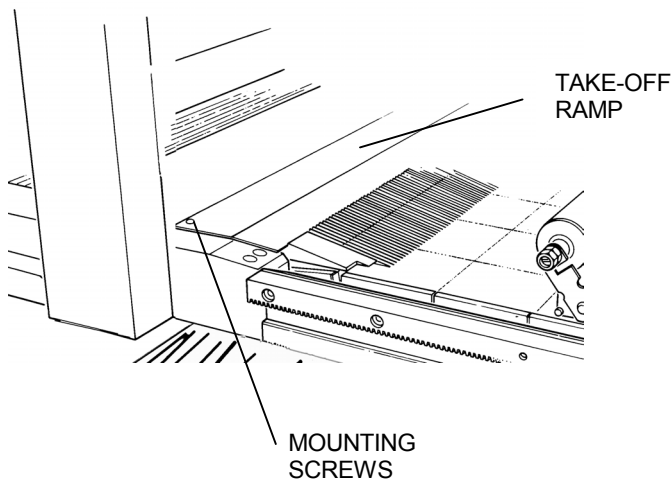
The optional conveyorized take off table (CTOT) attaches to the take-off end of the cutting table. The CTOT helps you more easily and quickly move material from the cutting table.

Required Tools and Supplies

Item	Gerber Part No.
metric T-handle hex key set	945500089
metric combination wrench set	945500093
multipurpose grease	596500005
shop rag	N/A

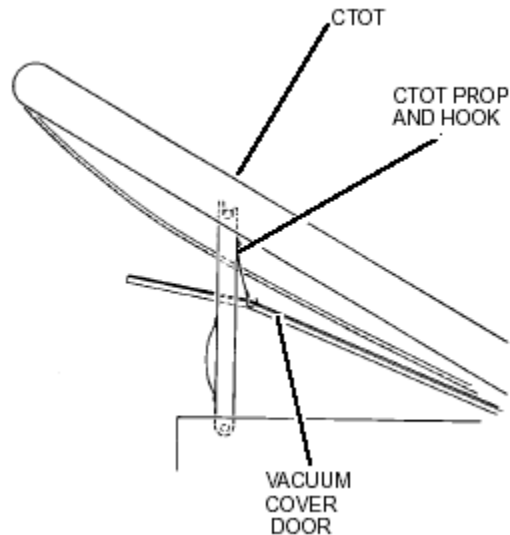
Open

1. Turn off the C-200 controller.
2. Remove the take-off ramp by removing the mounting screws.



WARNING!  The CTOT is heavy. Use two or three people to open the CTOT. Failure to use the necessary number of people may cause injury.

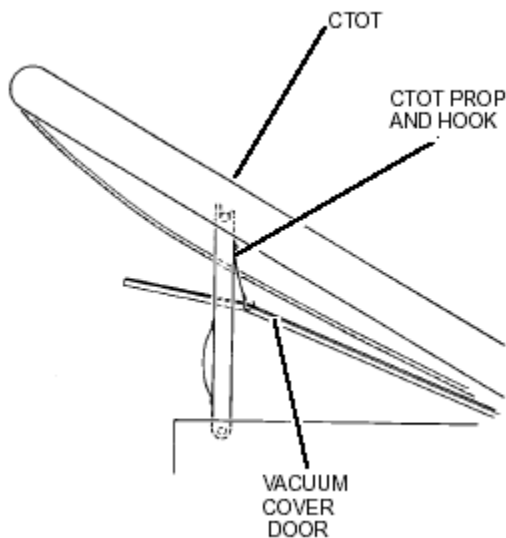
3. Lift the take-off end of the CTOT.
4. Hold the CTOT open.
5. Lift the CTOT props into place. The CTOT props are two hinged metal pieces that hold open the CTOT.
6. Lift the vacuum assembly cover.
7. Attach the hooks to hold open the vacuum assembly cover.



Close**WARNING!**

The CTOT is heavy. Use two or three people to close the CTOT. Failure to use the necessary number of people may cause injury.

1. Remove the hooks from the vacuum assembly cover and lower the cover.
2. Lift the CTOT and vacuum assembly insulation up.
3. Hold the CTOT open.
4. Lower the CTOT props. The CTOT props are two hinged metal pieces that hold open the CTOT.
5. Lower the CTOT.
6. Install the take-off ramp by installing the two mounting screws.
7. Turn on the C-200 controller.

**Drive Chain**

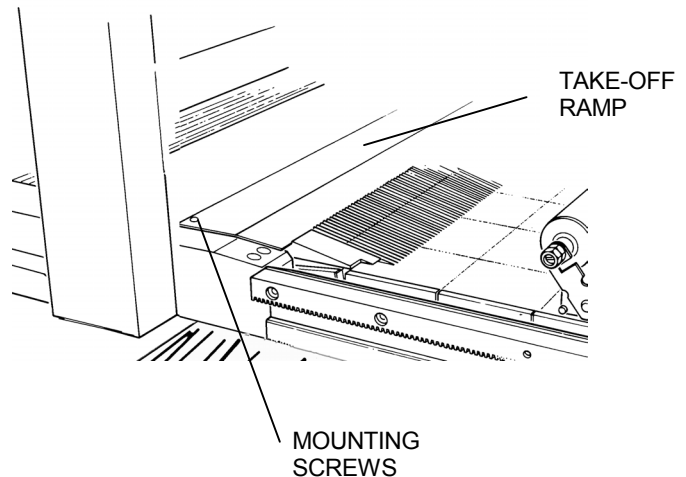
A drive chain, driven by an electric motor, turns the CTOT. The drive chain is located on the left side of the CTOT.

Lubricate

1. Turn off the C-200 controller.
2. Remove the take-off ramp by removing the mounting screws. The drive chain is under the right side of the take-off ramp when viewed from the take-off end of the cutter.

CAUTION: To avoid damage to the material, do not use too much grease. Moving equipment throws excess lubricant.

3. Lubricate the drive chain lightly with multipurpose grease.

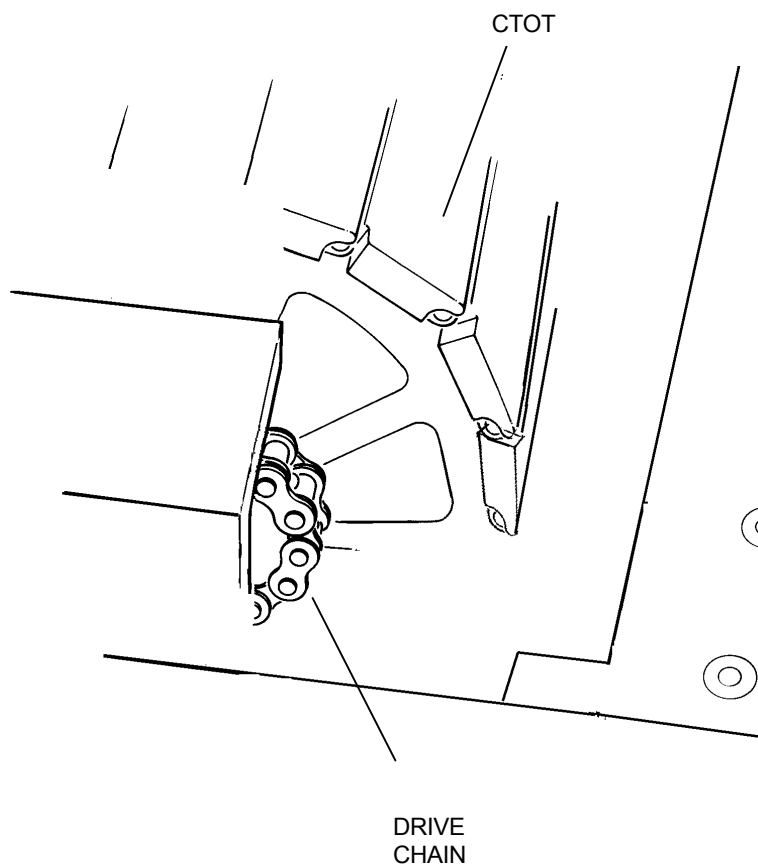


WARNING!  Make sure that personnel stay away from the CTOT and the GERBERcutter while the power is on and the CTOT is moving. Failure to do so can result in injury.

4. Turn on the C-200 controller.
5. Move the CTOT conveyor through one-half a revolution.
6. Turn off the C-200 controller.

CAUTION: To avoid damage to the material, do not use too much grease. Moving equipment throws excess lubricant.

7. Lubricate the drive chain lightly with multipurpose grease.
8. Install the take-off ramp by installing the two mounting screws.
9. Turn on the C-200 controller.



Lateral Drive

The optional lateral drive moves your GERBERcutter between several spreading tables.

Required Tools and Supplies

Item	Gerber Part No.
grease gun	944233101
multipurpose grease	596500005
shop rag	N/A
non-compressible cribbing	N/A
two fork lift trucks	N/A

Lateral Drive Wheels

There are three wheels that your GERBERcutter rides on, if you have the lateral drive option.

Lubricate

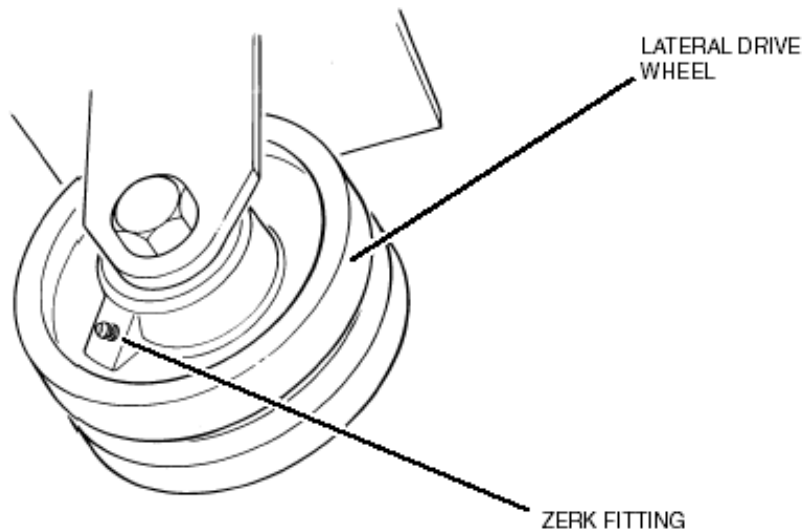
1. Turn off the C-200 controller.

WARNING!



Make sure that all power is disconnected from your GERBERcutter before you do this maintenance. Failure to do so may result in injury or death.

2. Disconnect your GERBERcutter from all power sources.
3. Remove all loose items from your GERBERcutter.
4. Move all cables and obstructions away from your GERBERcutter.



WARNING!  Each forklift that you use must be rated by its manufacturer to lift the total weight of your GERBERcutter including options. Failure to follow this precaution can result in injury or death.

5. Use at least two forklift trucks to lift your GERBERcutter. Lift your GERBERcutter at least three feet. Make sure that each forklift is rated to lift the weight of your GERBERcutter.

WARNING!  Use only cribbing material that can support all the weight of your GERBERcutter including options.

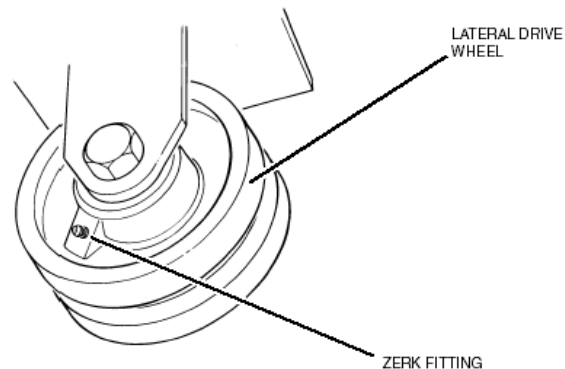
6. Place non-compressible cribbing under the frame rails of your GERBERcutter. This prevents the GERBERcutter from falling.

WARNING!  Do not remove either fork lift truck from under the GERBERcutter at any time while the GERBERcutter is off the floor. Failure to follow this precaution can result in injury or death.

7. Use the forklift trucks to lower your GERBERcutter onto the cribbing.

WARNING!  Make sure that the GERBERcutter is held securely and safely off the floor before lubricating the lateral drive wheels. If you are unsure about the safety of the precautions taken, **DO NOT** perform this maintenance. Call your service technician, if necessary.

8. Use the grease gun to lubricate each lateral drive wheel. Each wheel has a zerk fitting.
9. Use the shop rag to clean excess grease from the outside surface of the zerk fitting.
10. Use the forklift trucks to lift your GERBERcutter off the cribbing.
11. Remove the cribbing.
12. Use the forklift trucks to lower your GERBERcutter onto its track.
13. Connect your GERBERcutter to power.
14. Turn on the C-200 controller.



Troubleshooting

The C-200MT cutter control software provides the following features to assist you in troubleshooting your GERBERcutter.

- Error messages
- Diagnostics

Additional troubleshooting tools available on CD-ROM:

- GERBERservice Parts lists
- *GERBERcutter Application Manual*

This chapter briefly explains these features and tells where to find specific information.

Customer Support

If a problem still exists after troubleshooting, call Gerber Field Service Engineering for assistance. In North America, call Customer Support 800-321-2448. Outside North America, call your local service office.

Have the following information ready when you call:

- GERBERcutter you are using
- Any special equipment or software you are using, such as the drill, rescaler, or bar code reader option
- Software packages you are using; include the software version numbers
- Equipment number, if available

Error Messages

The GERBERcutter displays an error message dialog box when it detects an error condition. The system numbers these error messages for easy reference. The first digit of the error number defines the error class. The error classes range from 1 to 4, with class 1 being the least severe. PreProcessor error messages do not have an error number.

Error Message Type	Error Number Range
Class 1 FEP	1000-1999
Class 1 MCC	1000-1999
Class 2	2000-2999
Class 3	3000-3999
Class 4	4600

For a complete list of errors and solutions, refer to the C-200MT online help.

Diagnostics

Diagnostics are provided in the C-200MT program. These are tools that can be used by the person in your company who is a Gerber qualified technician to ensure cutter performance. Diagnostics used for routine maintenance include:

- Input Output Test (DG23)
- KI Calibration (DG 20)
- C Home to System Zero
- Heel to Heel Cut Test
- Beam Square Cut Test
- Knife Offset Setup Cut Test

Note: For detailed information about diagnostics, refer to the *C-200MT Software User's Guide* or C-200MT online help.

Service Parts

The GERBERservice CD-ROM provides the latest parts information for your GERBERcutter. Assembly drawings in exploded view are each provided with an index that identifies names and part numbers. Drawings are grouped by GERBERcutter model. Categories within each type are Head, Beam, Electrical, and Installation. There is also a listing for all GERBERcutters of consumables (spare parts), such as drill bits, knife blades, and belts.

Information on the CD is viewed using Internet Explorer or the Adobe Acrobat reader.

Features in the Adobe Acrobat reader include:

- Zoom for enlarging drawings
- Search for finding a part by name or number
- Print for printing out assembly drawings or part number lists

Refer to the installation instructions that come with the CD when installing the Parts documentation and the Adobe Acrobat reader.

Applications

Flexibility within the C-200MT program allows you to set up your GERBERcutter based on differences in patterns, materials, and types of manufacture. When your GERBERcutter was first installed, your Field Service representative or trainer customized setup files for your manufacturing needs at that time. To achieve optimal cutting performance for a different application you should:

- Prepare the cut file data for automated cutting. Consider fabric, size of the pieces, and cutting sequence. See Data Preparation on page 182.
- Select the most appropriate preset setup available in the C-200MT program. Your selection should be based on fabric hardness, porosity, tendency to fuse, and ply height. Refer to the *C-200MT Software User's Guide* or C-200MT online help.
- Customize vacuum settings, knife sharpening, and motion control parameters for this setup. See page 183.

Automated cutting concepts are explained briefly on the next page. For detailed applications information, refer to the *CAM Applications Manual*, which is available on the C-200MT software and GERBERservice CDs.

Automated Cutting Concepts

The GERBERcutter is an efficient, computer-aided manufacturing (CAM) system. Computer aided design (CAD) systems provide instructions in a cut file to control the cutting system. The GERBERcutter executes those commands. Parameters in the C-200MT program determine exactly how those commands are executed.

To improve cutting results consider the following concepts.

Data Preparation

Preparation of data for a GERBERcutter differs from hand cutting in that special consideration must be given to the pattern pieces and the material being cut. An operator cutting by hand intuitively compensates for a different fabric or tattered corners on a pattern piece.

Because the GERBERcutter simply follows CAD instructions, the pattern maker, digitizer, and marker maker must give consideration to some of the things mentioned below. Remember, automated cut quality is directly related to the CAD instructions.

Pattern Pieces — Before pattern pieces are digitized and stored, it is a good idea to answer the following questions based on the cutting results desired.

- Are slit, V, or both notch types going to be used?
- Is it permissible to miter corners?
- Is there an added buffer on the pattern template to compensate for hand cutting inaccuracy? If so, is it still needed?

If existing pattern pieces are torn, wrinkled, or frayed, redefine them. Make sure corners are accurate, crisp, and clean with straight lines and smooth curves.

Digitizing — Always define an accurate grain line. If the grain line is off in rectangular pieces, the parts will not nest with true common lines.

Marker Making — The marker maker should consider the placement of large and small pieces. See Vacuum Hold Down on page 183.

Vacuum Hold Down

The GERBERcutter relies upon a vacuum system to compress and stabilize the material to be cut. The size of cut pieces, material porosity, and cutting sequence affect the performance of the cutting system. You can customize a setup by changing parameters in the Vacuum Systems category.

Regularly scheduled maintenance of the vacuum system ensures optimum mechanical performance of the GERBERcutter. For more about Vacuum System features, see Appendix A.

Knife Sharpening

Material content (i.e. natural or synthetic fibers, coatings, etc.) and ply heights determine knife sharpening requirements. You can customize a setup by changing parameters in the Sharpen Parameters category. Refer to the *C-200MT Software User's Guide* or C-200MT online help.

Regularly scheduled maintenance of the sharpener assembly ensures optimum mechanical performance of the GERBERcutter.

Motion Control

Quality requirements, material, and type of manufacture must be considered when optimizing the performance of the GERBERcutter.

The C-200MT program has a built-in knowledge base and is flexible enough to meet the needs of Aerospace, Apparel, Automotive, Furniture and many other industrial applications. For example: if cutting tubular goods, you may want to use the Clipping Window or Centerline Origin features. If cutting narrow markers, Marker Stacking may speed throughput. Refer to the *C-200MT Software User's Guide* or C-200MT online help.

Glossary

axis

An imaginary line used to describe movement and location on the cutting table. The GERBERcutter utilizes three axes: X (parallel to table length), Y (parallel to table width), and C (knife direction).

beam

A support and carrier for the cutter head as it moves across the width and up/down the length of the table.

beam control panel (BCP)

A terminal at the cutting table that lets you control cutter head and beam assembly activities. The computer that processes input from the beam control panel is called the BCC.

For specific controls available on the BCP for your cutter, refer to the Operator Controls section of the Getting Started Manual or Operator's Manual for your cutter.

bite

A section of a marker that (a) fits on the cutting table and (b) can be cut at one time. Bite feeding matches cutter head movement with conveyor movement. This permits automatic cutting of a long spread of material. The spread is cut in sections that fit on the cutting table surface. Each section is called a bite.

BRISTLE SQUARES

Squares of molded bristle that install into conveyor slats or into a fixed grid. They form a work surface for cutting material.

C-axis

An imaginary, straight line that the knife rotates about.

CPI

See Path Intelligence.

Cutter Information Database (CIDB)

An optional C-200MT utility that uses a database to track time relative to cutter operation and to report this information.

command

Request to perform a function. The request can be entered at a terminal, such as the BCP at the cutting table or FEP or PC at the operator console. Also, an item you can choose in a C-200MT menu in order to use a feature in the C-200MT program.

conveyor

Moving work surface made from slats and covered with Gerber BRISTLE SQUARES.

CTOT

The Conveyorized Take-off Table (CTOT) is a device for moving cut pieces off the cutting table surface. The CTOT is made of plastic links, that move in synch with the conveyor.

cut file

Computer data file with directions on how particular pieces should be cut from a spread of material by a GERBERcutter. Also, see marker.

data

Information stored on a disk.

default

A setting that is programmed into the computer at the factory or set by Field Service when the cutter is first installed. This is used until you, or your supervisor, choose to override it and select another setting.

disk or disc

Medium used for storage of computer data. GERBERcutter systems can use 3.5-inch disks and CD-ROM discs.

disk drive

A device that lets a computer read and write data on a disk. Your cutter has a hard drive, a 3.5-inch disk drive, and a CD-ROM disc drive.

drill

An optional unit for making drill holes that attaches to the cutter head.

dry haul

The distance traveled with the knife out of the material.

dry run

To execute a cut file without plunging the knife or drill into the material.

EStop

Red button that can be pressed during an emergency to immediately stop all GERBERcutter motion. Also, a yellow paddle on the beam that when bumped stops all GERBERcutter motion.

feed rate dial/slider

Control used to set the desired cut speed.

front end processor (FEP)

A computer (PC) in the operator console that relays your commands, cut file data, and setup file data to the Motion Control Computer (MCC) for processing.

hard disk drive

A device that holds the software that runs the GERBERcutter. A hard disk drive uses rigid metal disks to store large amounts of data.

You need to manage your hard disk so it does not become too full and disorganized. For information on managing files on the hard disk, see File Manager in the C-200MT online help for cut file management or refer to documentation for AccuMark Explorer or Windows NT for data management.

hardware

Physical elements of a computer, such as the monitor and keyboard.

heel cut

Cut made by the heel of the knife. The heel is the thickest edge (following edge) of the knife blade.

home

Cutting table's (X0,Y0) coordinates.

initialize

To set the cutting table's (X0,Y0) coordinate after downloading the C-200MT software. The system uses this coordinate as a reference when processing a marker.

Knife Intelligence

Knife Intelligence® (KI) is a GERBERcutter feature that:

- Senses deflection of the knife (lateral force) caused by cutting difficult materials or high ply spreads
- Changes knife angle to compensate for this deflection

job queue

A job queue contains multiple markers or cut files to be cut sequentially. Each marker and its corresponding setup file is called a job. Refer to Queue Manager in the C-200MT online help.

Lay

One or more layers of material spread on the cutting table surface for cutting. May also be called a spread.

limit switch

Sensors that restrict beam travel across and down the length of the cutting table.

marker

A layout of pieces on a spread that provides cutting instructions. This can be created in a grading/marketing system such as AccuMark or MicroMark.

mast

A tower available with lateral drive GERBERcutter systems that holds the overhead power cables, shop air hoses, and houses a flashing warning light and alarm to alert others when the cutting table is moving. An exhaust stack for the vacuum system may also be located on the mast.

motion control computer (MCC)

A computer, or controller board, that responds to instructions sent from the front end processor, or PC, to control cutting table, beam, and head motion.

network

A set of computers connected together. For example, a GERBERcutter may be networked to a CAD grading and marking system.

offset

A condition when one item's location differs relative to another item. For example: stripe and plaid lines on fabric are offset by a certain distance.

operator console

The location on the GERBERcutter where the C-200MT program is accessed and marker processing is initiated. The operator console includes a monitor, keyboard, and computer with a hard drive, disk drive, and CD-ROM disc drive. See also Front End Processor FEP.

OpStop

A command in a cut file that causes the cutter to stop processing while the operator checks matching points and slews the cutting head as necessary.

origin

The "start" position (X0,Y0) of a marker on the table. This is set at the corner of the marker closest to the take-off end on the side of the cutting table where the operator console is located.

parameter

A setting defined in the setup file.

Path Intelligence

Path Intelligence (PI) completely or partially controls cut speed using built-in application knowledge and by measuring one or more of the following:

- Knife sharpness
- Vacuum level
- XY force

Use commands in the Diagnostics/ Path Intelligence menu to ensure that the GERBERcutter is running properly.

program

A set of instructions that allows a computer to interact with people or equipment. The programs that direct a computer are called software.

remote side

The side of the cutter opposite of the beam control panel.

right side

The right side of the cutting table as seen from the take-on end. For cutters with the RSO option, the beam control panel is located on this side of the cutter.

servo motor

An electric motor used to move the beam along the X-axis or the cutter head along the beam in the Y-axis of the cutting table.

setup

A setup is a file containing parameters that enable or disable various features, options, and modes depending on such considerations as ply height, material characteristics, or marker width.

software

Set of computer programs used to run the GERBERcutter and communicate with you. These include the C-200MT program and Windows NT.

spread

One or more layers of material spread on the cutting table surface for cutting. May also be called a lay.

stack

A pipe, mounted vertically, that vents vacuum exhaust away from the GERBERcutter. This is usually mounted on the mast.

take-off end

The end of the cutting table where cut pieces and scrap material leaves after being cut.

take-on end

The end of the cutting table where material spread for cutting is loaded on from the spreading table.

utility

A program that helps you operate the cutter, such as Queue Manager, Preprocessor, or View Marker.

X-axis

A horizontal reference line in a coordinate system. For the GERBERcutter, the X-axis runs the length of the table from the take-on end to the take-off end.

Y-axis

A vertical reference line in a coordinate system. For the GERBERcutter, the Y-axis runs the width of the table from the left side to the right side.

zone (vacuum)

An area of the cutting table separated from other areas. Baffles divide a cutting table with this feature into five or six zones. Vacuum can be applied to one or more zones at a time depending on parameter settings.

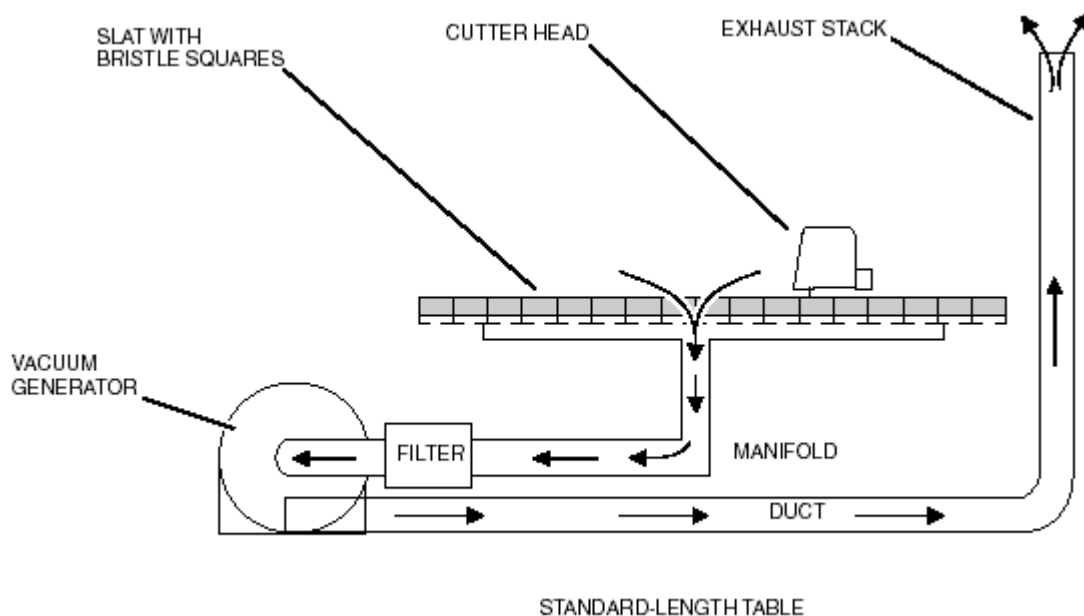
Appendix A – Vacuum Features

This chapter describes vacuum features for the GT 5250 or GT 7250 GERBERcutter. The topics included in this section are:

- How vacuum works
- Energy saving mode
- Vacuum zoning
- Low vacuum messages
- CPI low vacuum feature

How Vacuum Works

The vacuum generator plays a critical part in the cutting process. It makes a vacuum that holds the material firmly on the work surface during cutting. The vacuum generator is under the take-off end of the cutting table. The vacuum generator draws air through the BRISTLE SQUARES into the conveyor slats. From the slats, the air goes through a manifold and a filter, into the vacuum generator, and out the silencer in the exhaust stack.



Cutting Table Vacuum

There are three available GERBERcutter vacuum features that let you control where and when vacuum is applied. You can use one, two, or all of these features, in various combinations. If more than one feature is turned on, they interact.

To use these vacuum features, your service representative turns on the option and you set up certain parameters in the configuration file. You can temporarily enable/disable two of the features. The table below summarizes this information.

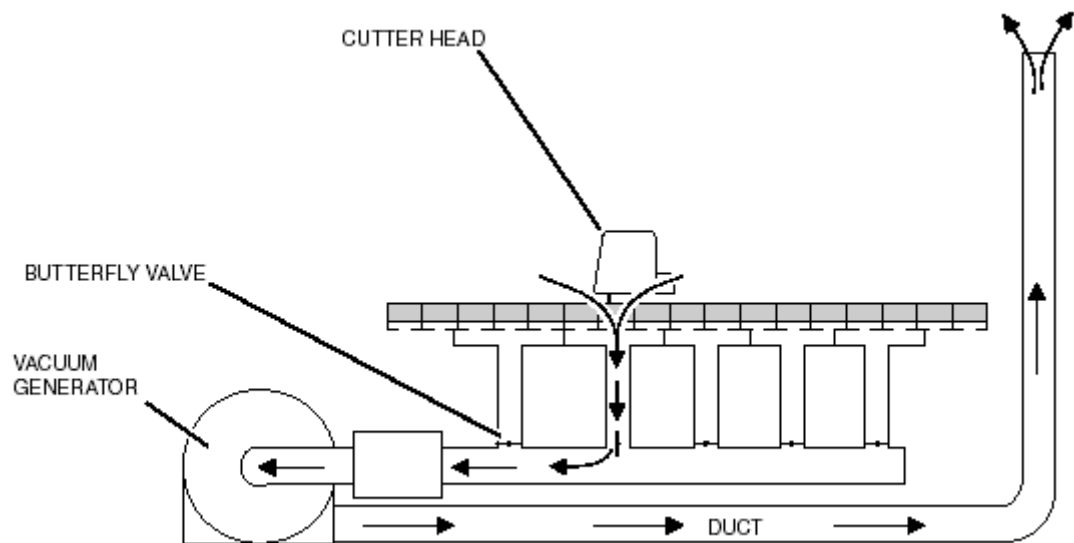
Refer to the *C-200MT Software User's Guide* or C-200MT online help for information about configuration parameters.

List of Vacuum Features

Feature Name	What It Controls	Configuration Parameters	Parameter Change and Display Menu Category
Vacuum Zoning (optional)	Where vacuum is applied	Vacuum Zoning Vacuum Zone Overlap	Vacuum System
Energy Saving Mode	When vacuum is applied	Energy Saving Mode Time Until ESM	Options & Modes
Cut Path Intelligence Vacuum Feature (optional)	Feedrate if vacuum is at caution level	PI: FR Slowdown - Low Vac	Path Intelligence

Vacuum Zoning

The optional vacuum zoning feature improves cut quality and lowers power use. This feature applies vacuum where it is needed most — the area being cut. In zoned vacuum, the cutting table is divided into five zones (six on optional long tables). Each zone receives vacuum from a duct that can be opened or closed automatically with a butterfly valve. Each butterfly valve is closed unless the cutter head is in its zone. When the cutter head enters a zone, the butterfly valve opens, applying vacuum to the area being cut.



Cutting Table Air Flow with Vacuum Zoning

If you turn off vacuum zoning (disable it), the GERBERcutter applies vacuum to the entire cutting table surface. You may want to disable vacuum zoning if you need consistent vacuum to prevent slippage. For example, you may want to disable zoning when cutting high ply spreads of slippery material.

The GERBERcutter applies vacuum in one to three zones at a time: the zone in which the cutter head is currently cutting and one or two “look-ahead” zones. The purpose of the look-ahead zones is to ensure consistent vacuum during cutting.

The GERBERcutter calculates look-ahead zones in several different ways. For more about these methods, see the next page.

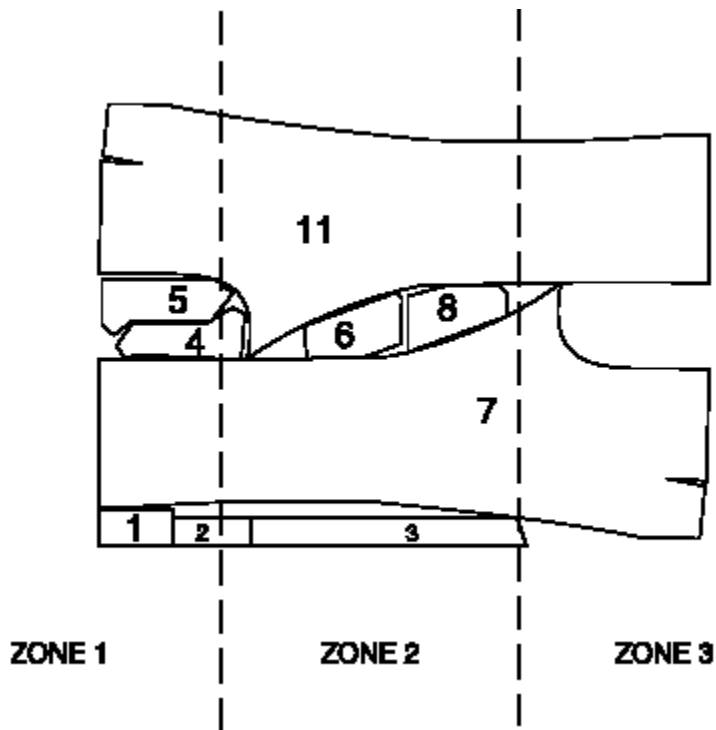
Look-Ahead Zones

When the GERBERcutter looks at the marker and sees that the cutter head will be in the next zone within a certain period of time (about 0.25 seconds), the GERBERcutter applies vacuum to that zone. For the example shown below, when the system cuts Piece 2, it will turn on vacuum to both Zones 1 and 2.

When the GERBERcutter looks at the marker and sees that the cutter will be zigzagging between the same two zones, it will apply vacuum to both zones instead of continually turning them on and off.

In the example below, when the system is cutting Piece 5, it will apply vacuum to both Zones 1 and 2. When the cutter head is cutting a piece close to the next zone, it will apply vacuum to that zone. It does this even though it will not be moving into the next zone.

In the example below, when the system is cutting Piece 8, it will apply vacuum to both Zones 2 and 3 even though the cutter head will not cut in Zone 3.



Marker with Look-Ahead Zones

Configuration Setup

To use vacuum zoning, set up two parameters in the configuration file:

- Vacuum Zoning = ENABLE
- Vacuum Zone Overlap = number of centimeters (inches)
If the cutter head is within this distance of the next zone, the GERBERcutter applies vacuum to that next zone even if the cutter head is not going to cut in that zone.

Temporary Enable/Disable

To temporarily turn this feature on or off, use the Display or Change Parameters menu (PM). Select the Vacuum System category and set up the vacuum zoning parameter:

- Vacuum Zoning = ENABLE/DISABLE

Energy Saving Mode

The energy saving mode (ESM) greatly reduces table vacuum when the GERBERcutter is idle for a certain period of time. The system displays a warning message on the operator screen thirty seconds before ESM goes into effect.

ESM reduces energy consumption in two ways. First, since ESM automatically sets vacuum requirements, the operator does not need to remember when to lower vacuum to the table. Secondly, ESM reduces vacuum without turning off the vacuum generator. Instead, it opens and closes the vacuum zone valves. This method uses less energy than turning the motor on and off.

ESM goes into effect when these three conditions are met:

- the vacuum generator is on
- no marker is being processed
- the beam, head, and conveyor are idle for a certain period of time (defined in the configuration file)

Configuration Setup

To use energy saving mode, set up two parameters in the configuration file:

- Energy Saving Mode = ENABLE
- Time Until ESM = the number of seconds the system is idle before ESM goes into effect. The operator sees a warning message 30 seconds before ESM starts.

Temporary Enable/Disable

To temporarily turn this feature on or off, use the Display or Change Parameters menu (PM). Select the Options & Modes category and set up the energy savings mode parameter:

- Energy Savings Mode = ENABLE/DISABLE

Low Vacuum Messages

The GERBERcutter has two messages which indicate vacuum is too low for cutting:

- Vacuum Low Caution message – warns that vacuum is dangerously low. If you have Cut Path Intelligence (CPI) low vacuum feature turned on when the vacuum is at caution level, the feedrate of the marker slows by a configurable amount. For more information, see Cut Path Intelligence Vacuum on page 198. If this CPI feature is not turned on, you must either increase vacuum or stop cutting.
- Vacuum Low Error message – causes the cutter to stop cutting until proper vacuum is resumed.

Configuration Setup

Two parameters in the configuration file define the vacuum level at which these messages go into effect.

- Vacuum Low Caution Level = vacuum level below which the GERBERcutter displays the caution message.
- Vacuum Low Error Level = vacuum level below which the GERBERcutter displays the error message and halts the cutting process.

Cut Path Intelligence Vacuum Feature

The optional Cut Path Intelligence (CPI) vacuum feature slows the feedrate of your marker if table vacuum is at caution level. To use this feature, you must first set the Vacuum Low Caution Level in your configuration file. See page 197.

Configuration Setup

To use the Cut Path Intelligence low vacuum feature, set up the configuration file parameter:

- PI: FR Slowdown - Low Vac = ON/OFF

Temporary Enable/Disable

To temporarily turn this feature on or off, use the Display or Change Parameters menu (PM). Select the Path Intelligence category and set up the CPI vacuum feature parameter:

- PI: FR Slowdown - Low Vac = ON/OFF

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